FIELD-EFFECT TRANSISTORS

CHAPTER 3

3.1 Introduction

Field-effect transistors (FETs) are different from the bipolar transistors[1] that we talked about in the last chapter. Broadly speaking, however, they are similar devices, which we might call *charge-control devices*: in both cases (Figure 3.1) we have a three-terminal device in which the conduction between two electrodes depends on the availability of charge carriers, which is controlled by a voltage applied to a third *control electrode*.

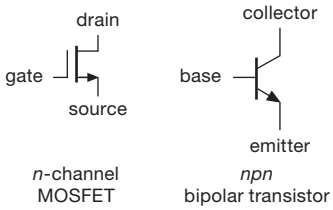


Figure 3.1. The *n*-channel MOSFET and its *npn* transistor analog.

Here's how they differ: in a bipolar transistor the collector–base junction is back-biased, so no current normally flows. Forward-biasing the base–emitter junction by $\approx$0.6 V overcomes its diode "contact potential barrier," causing electrons to enter the base region, where they are strongly attracted to the collector. Although some base current results, most of these "minority carriers" are captured by the collector. This results in a collector current, controlled by a (much smaller) base current. The collector current is proportional to the rate of injection of minority carriers into the base region, which is an exponential function of V_{BE} (the Ebers–Moll equation). You can think of a bipolar transistor as a current amplifier (with roughly constant current gain β) or as a transconductance device (Ebers–Moll: collector current programmed by base-emitter *voltage*).

In an FET, as the name suggests, conduction in a *channel* is controlled by an *electric field*, produced by a voltage applied to the *gate* electrode. There are no forward-biased junctions, so the gate draws no current. This is perhaps the most important advantage of the FET. As with BJTs, there are two polarities, *n-channel* FETs (conduction by electrons) and *p-channel* FETs (conduction by holes). These two polarities are analogous to the familiar *npn* and *pnp* bipolar transistors, respectively. In addition, however, FETs tend to be confusing at first because they can be made with two different kinds of gates (thus JFETs and MOSFETs) and with two different kinds of channel doping (leading to *enhancement* and *depletion* modes). We'll sort out these possibilities shortly.

First, though, some motivation and perspective. The FET's nonexistent gate current is its most important characteristic. The resulting high input impedance (which can be greater than $10^{14}\,\Omega$) is essential in many applications, and in any case it makes circuit design simple and fun. For applications like analog switches and amplifiers of ultra-high input impedance, FETs have no equal. They can be easily used by themselves or combined with bipolar transistors to make integrated circuits. In the next chapter we'll see how successful that process has been in making nearly perfect (and wonderfully easy to use) *operational amplifiers*, and in Chapters 10–14 we'll see how digital electronics has been revolutionized by MOSFET integrated circuits. Because many FETs using very low current can be constructed in a small area, they are especially useful for very large-scale integration (VLSI) digital circuits such as microprocessors, memory, and "application-specific" chips of the sort used in cellphones, televisions, and the like. At the other end of the spectrum, robust high-current MOSFETs (50 amps or more) have replaced bipolar transistors in many applications, often providing simpler circuits with improved performance.

3.1.1 FET characteristics

Beginners sometimes become catatonic when directly confronted with the confusing variety of FET types. That variety arises from the combined choices of polarity (*n-channel* or *p-channel*), form of gate insulation [semiconductor *junction* (JFET) or oxide *insulator* (MOSFET)], and channel doping (*enhancement* or *depletion* mode). Of the

[1] Often called BJTs, for "bipolar junction transistors," to distinguish them from FETs.

eight resulting possibilities, six *could* be made, and five actually are. Four of those five are of major importance.

It will aid understanding (and sanity), however, if we begin with one type only, just as we did with the *npn* bipolar transistor. Once comfortable with FETs, we'll have little trouble with their family tree.

A. FET *V–I* curves

Let's look first at the *n*-channel enhancement-mode MOSFET, which is analogous to the *npn* bipolar transistor (Figure 3.2). In normal operation the drain (∼collector) is more positive than the source (∼emitter). No current flows from drain to source unless the gate (∼base) is brought positive with respect to the source. Once the gate is thus "forward-biased," there will be drain current, all of which flows to the source. Figure 3.2 shows how the drain current I_D varies with drain-source voltage V_{DS} for a few values of controlling gate-source voltage V_{GS}. For comparison, the corresponding "family" of curves of I_C versus V_{BE} for an ordinary *npn* bipolar transistor is shown. Evidently there are a lot of similarities between *n*-channel MOSFETs and *npn* bipolar transistors.

Like the *npn* transistor, the FET has a high incremental drain impedance, giving roughly constant current for V_{DS} greater than a volt or two. By an unfortunate choice of language, this is called the "saturation" region of the FET (a better term is "current saturation") and corresponds to the "active" region of the bipolar transistor. Analogous to the bipolar transistor, a larger gate-to-source bias produces a larger drain current. And, analogous to bipolar transistors, FETs are not perfect transconductance devices (constant drain current for constant gate-source voltage): just as the ideal Ebers–Moll transconductance characteristic of bipolar transistors is degraded by the Early effect (§2.3.2D and §*2x.8*), there's an analogous departure from the ideal transconductance behavior for FETs, characterized by a finite drain output resistance r_o (more usually called $1/g_{os}$, see §3.3.2 and §*3x.4*).

So far, the FET looks just like the *npn* transistor. Let's look closer, though. For one thing, over the normal range of currents the saturation drain current increases rather modestly with increasing gate voltage (V_{GS}). In fact, it is approximately proportional to $(V_{GS} - V_{th})^2$, where V_{th} is the gate threshold voltage at which drain current begins ($V_{th} \approx 1.63$V for the FET in Figure 3.2); compare this mild quadratic law with the steep exponential transistor law, as given to us by Ebers and Moll. Second, there is *zero* dc gate current, so you mustn't think of the FET as a device with current gain (which would be infinite). Instead, think of the FET as a transconductance device, with gate-source voltage programming the drain current, as we did with the bipolar transistor in the Ebers–Moll treatment. Recall that the transconductance g_m is simply the ratio i_d/v_{gs} (using the convention of lowercase letters to indicate "small-signal" changes in a parameter; e.g., $i_d/v_{gs} = \delta I_D/\delta V_{GS}$). Third, the gate of a MOSFET is truly insulated from the drain-source channel; thus, unlike the situation for bipolar transistors (or JFETs, as we'll see), you can bring it positive (or negative) at least 10 V or more without worrying about diode conduction. Finally, the FET differs from the bipolar transistor in the so-called *linear* (low-voltage) region of the graph, where it behaves rather accurately like a resistor, *even for negative* V_{DS}; this turns out to be quite useful because the equivalent drain-source resistance is, as you might guess, programmed by the gate-source voltage.

B. Two examples

FETs will have more surprises in store for us. But before getting into more details, let's look at two simple switching applications. Figure 3.3 shows the MOSFET equivalent of Figure 2.5, our first saturated transistor switch. The FET circuit is even simpler, because we don't have to concern ourselves with the inevitable compromise of providing adequate base drive current (considering worst-case minimum β combined with the lamp's cold resistance) without squandering excessive power. Instead, we just apply a full-swing dc voltage drive to the cooperative high-impedance gate. As long as the switched-on FET behaves like a resistance that is small compared with the load, it will bring its drain close to ground; typical power MOSFETs have $R_{ON} < 0.1\,\Omega$, which is fine for this job.

We demonstrate this circuit in our electronics course, but we put a resistor in series with the gate. The students are surprised when they discover its resistance – 10 MΩ – which implies a "beta" of at least 100,000. They are even more surprised when they notice that the light stays on when the gate is then open-circuited: the gate voltage is held on the gate's capacitance, and will stay that way for the rest of the hour's lecture.[2] That implies that the gate current is well below a *pico*ampere!

Figure 3.4 shows an "analog switch"[3] application, which cannot be done at all with bipolar transistors. The idea here is to switch the conduction of a FET from open-circuit (gate reverse-biased) to short-circuit (gate forward-biased), thus blocking or passing the analog signal (we'll see plenty of reasons to do this sort of thing later). In this

[2] The gate capacitance "remembers" whatever voltage was last applied. So you can have it stay on, stay off, or even stay at half-brightness, with no noticeable change even with the gate floating.

[3] Also called a "linear switch."

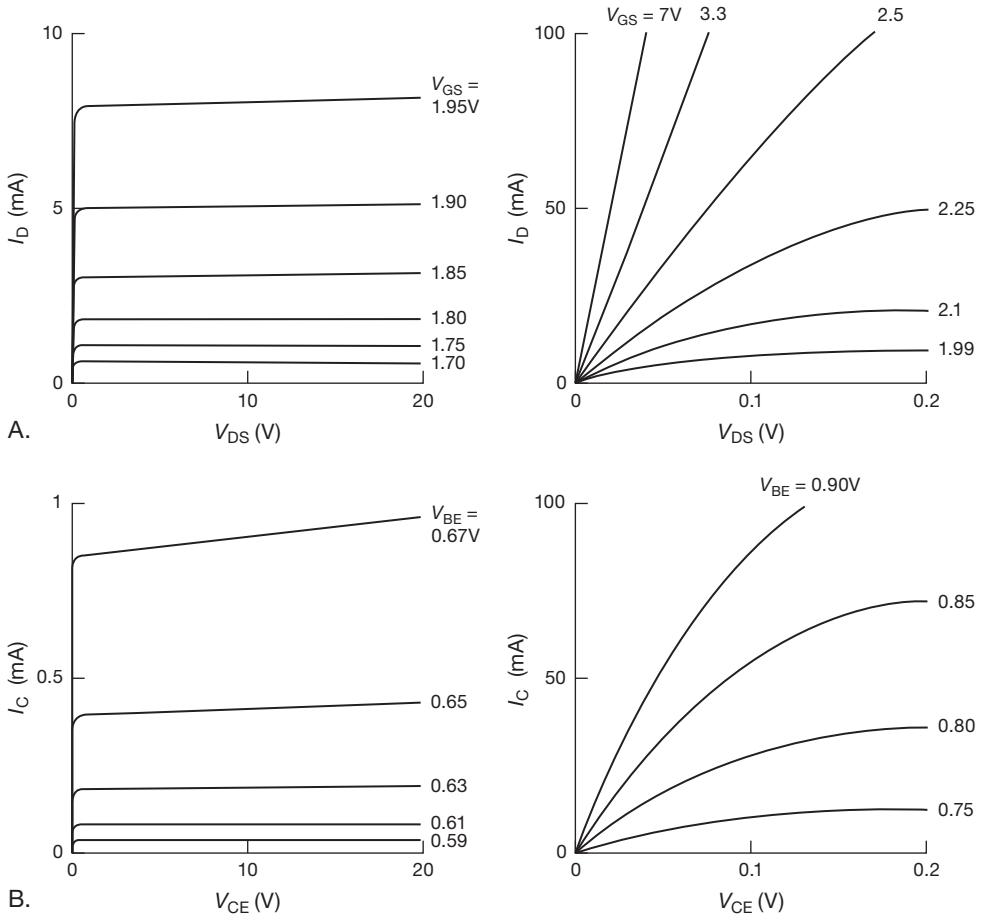


Figure 3.2. Measured MOSFET/transistor characteristic curves: A. VN0106 (similar to the popular 2N7000) n-channel MOSFET: I_D versus V_{DS} for various values of V_{GS}. B. 2N3904 npn bipolar transistor: I_C versus V_{CE} for various values of V_{BE}.

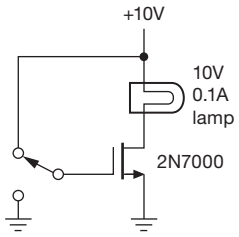


Figure 3.3. MOSFET power switch.

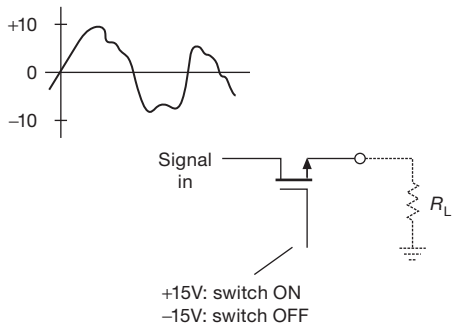


Figure 3.4. MOSFET analog (signal) switch.

case we just arrange for the gate to be driven more negative than any input signal swing (switch *open*), or a few volts more positive than any input signal swing (switch *closed*). Bipolar transistors aren't suited to this application, because the base draws current and forms diodes with the emitter and collector, producing awkward clamping action. The MOSFET is delightfully simple by comparison, needing only a voltage swing into the (essentially open-circuit) gate.[4]

[4] It's only fair to mention that our treatment of this circuit has been some-

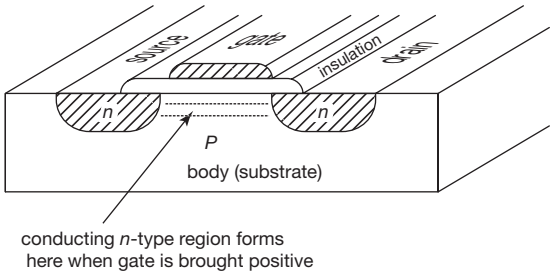


Figure 3.5. An *n*-channel "lateral" MOSFET.

3.1.2 FET types

A. *n*-channel, *p*-channel

Now for the family tree. First, FETs (like BJTs) can be fabricated in both polarities. Thus the mirror twin of our *n*-channel MOSFET is a *p*-channel MOSFET. Its behavior is symmetrical, mimicking *pnp* transistors: the drain is normally negative with respect to the source, and drain current flows if the gate is brought at least a volt or two negative with respect to the source. The symmetry isn't perfect because the carriers are holes, rather than electrons, with lower mobility and minority carrier lifetime.[5] The consequence is worth remembering – *p*-channel FETs usually have poorer performance, manifested as a higher gate threshold voltage, higher R_{ON}, and lower saturation current.[6]

B. MOSFET, JFET

In a MOSFET ("Metal-Oxide-Semiconductor Field-Effect Transistor") the gate region is separated from the conducting channel by a thin layer of SiO_2 (glass) grown onto the channel (Figure 3.5). The gate, which may be either metal or doped silicon, is truly insulated from the source-drain circuit, with characteristic input resistance $>10^{14}\,\Omega$. It affects channel conduction purely by its electric field. MOSFETs are sometimes called *insulated-gate* FETs, or IGFETs. The gate insulating layer is quite thin, typically less than a wavelength of light, and can withstand gate voltages up to ± 20 V in typical power MOSFETs (less for the small MOSFETs in low-voltage integrated circuits). MOSFETs are easy to use because the gate can swing either polarity relative to the source without any gate current flowing. They are, however, quite susceptible to damage from static electricity; you can destroy a MOSFET device literally by touching it.

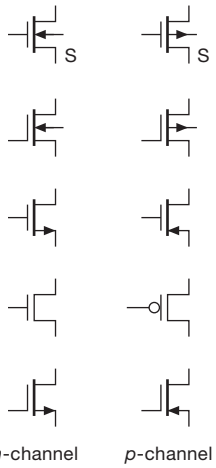


Figure 3.6. MOSFET schematic symbols.

The symbols for MOSFETs are shown in Figure 3.6. The extra terminal that is sometimes shown is the "body," or "substrate," the piece of silicon in which the FET is fabricated. Because the body forms a diode junction with the channel, it must be held at a nonconducting voltage. It can be tied to the source or to a point in the circuit more negative (positive) than the source for *n*-channel (*p*-channel) MOSFETs. It is common to see the body terminal omitted; furthermore, engineers often use the symbol with the symmetrical gate. Unfortunately, with what's left you can't tell source from drain; worse still, you can't tell *n*-channel from *p*-channel! In this book we most often use the bottom pair of schematic symbols, which, though somewhat unconventional, are unambiguous and uncluttered.[7]

In a JFET (Junction Field-Effect Transistor) the gate forms a semiconductor junction with the underlying channel. This has the important consequence that *a JFET gate should not be forward biased with respect to the channel, to prevent gate current*. For example, diode conduction will occur as the gate of an *n*-channel JFET approaches $+0.6$ V with respect to the more negative end of the channel (which is usually the source). The gate is therefore operated reverse-biased with respect to the channel, and no

what simplistic, for instance ignoring the effects of gate-channel capacitance and the variation of R_{ON} with signal swing. We'll have more to say about analog switches later.

[5] These are semiconductor parameters of importance in transistor performance.

[6] In the case of so-called "complementary pairs" (an *n*-channel and a *p*-channel part with similar voltage and current ratings), the *p*-channel part is usually built with a larger area in order to match the performance of the *n*-channel part. You can see the evidence in the datasheet in the form of greater capacitance for the *p*-channel part.

[7] In current practice, logic designers like to use the second pair up from the bottom, while power MOSFET users prefer the second pair down from the top.

current (except diode leakage) flows in the gate circuit. The circuit symbols for JFETs are shown in Figure 3.7. Once again, we favor the symbol with offset gate to identify the source (though JFETs and small integrated MOSFETs are symmetrical, power MOSFETs are quite asymmetrical, with very different capacitances and breakdown voltages).

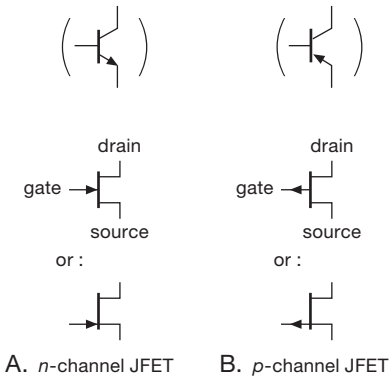


Figure 3.7. JFET schematic symbols: A. *n*-channel JFET. B. *p*-channel JFET.

C. Enhancement, depletion

The *n*-channel MOSFETs with which we began the chapter were nonconducting, with zero (or negative) gate bias, and were driven into conduction by bringing the gate positive with respect to the source. This kind of FET is known as *enhancement mode*. The other possibility is to manufacture the *n*-channel FET with the channel semiconductor "doped" so that there is plenty of channel conduction even with zero gate bias, and the gate must be reverse-biased by a few volts to cut off the drain current. Such a FET is known as *depletion mode*. MOSFETs can be made in either variety, because the gate, being insulated from the channel, can swing either polarity. But JFETs, with their gate-channel diode, permit only reverse gate bias, and therefore are made only in depletion mode.

A graph of drain current versus gate-source voltage, at a fixed value of drain voltage, may help clarify this distinction (Figures 3.8 and 3.9). The enhancement-mode device draws no drain current until the gate is brought positive (these are *n*-channel FETs) with respect to the source, whereas the depletion-mode device is operating at nearly its maximum value of drain current when the gate is at the same voltage as the source. In some sense the two categories are artificial, because the two curves are identical except for a shift along the V_{GS} axis. In fact, it is possible to manufacture "in-between" MOSFETs. Nevertheless, the distinction is an important one when it comes to circuit design.

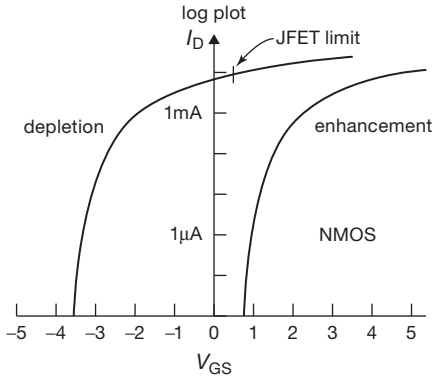


Figure 3.8. Transfer characteristics (I_D versus V_{GS}) for a JFET (depletion-mode) and a MOSFET (enhancement-mode) transistor. See also the measured curves in Figure 3.19.

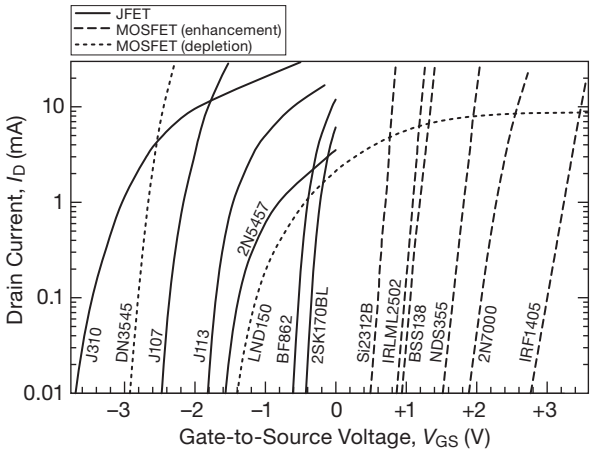


Figure 3.9. Lending some authenticity to Figure 3.8's notional sketch: measured I_D versus V_{GS} for a selection of *n*-channel FETs.

Note that JFETs are always depletion-mode devices and that the gate cannot be brought more than about 0.5 V more positive (for *n*-channel) than the source, since the gate-channel diode will conduct. MOSFETs *can* be either enhancement or depletion, but in practice the dominant species is enhancement, with a sprinkling of depletion-mode MOSFETs.[8] Most of the time, then, you need worry only about (a) depletion-mode JFETs and

[8] In the form of *n*-channel GaAs FETs, "dual-gate" cascodes for radiofrequency applications, and a selection of high-voltage depletion-mode power MOSFETs (such as the Supertex lateral LND150 or vertical DN3435, as well as offerings by six other manufacturers).

(b) enhancement-mode MOSFETs. Each come in the two polarities, *n*-channel and *p*-channel.

3.1.3 Universal FET characteristics

A family tree (Figure 3.10) and a map (Figure 3.11) of input–output voltage (source grounded) may help simplify things. The different devices (including garden-variety *npn* and *pnp* bipolar transistors) are drawn in the quadrant that characterizes their input and output voltages when they are in the active region with source (or emitter) grounded. You don't have to remember the properties of the five kinds of FETs, though, because they're all basically the same.

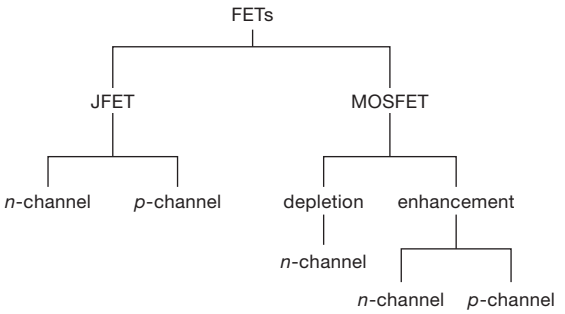


Figure 3.10. FET family tree.

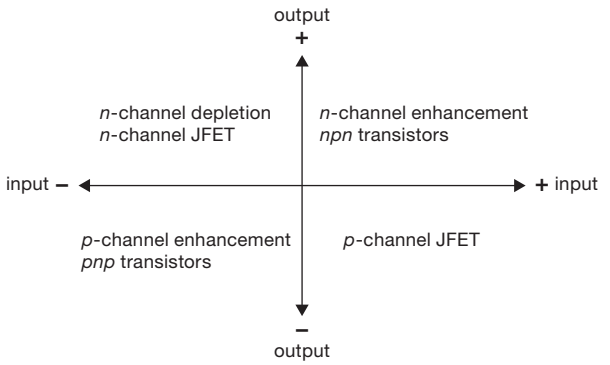


Figure 3.11. Transistor "polarity map."

First, with the source grounded, a FET is turned on (brought into conduction) by bringing the gate voltage "toward" the active drain supply voltage. This is true for all five types of FETs, as well as the bipolar transistors. For example, an *n*-channel JFET (which is necessarily depletion mode) uses a positive drain supply, as do all *n*-type devices. Thus a positive-going gate voltage tends to turn on the JFET. The subtlety for depletion-mode devices is that the gate must be (negatively) back-biased for zero drain current, whereas for enhancement-mode devices zero gate voltage is sufficient to give zero drain current.

Second, because of the near symmetry of source and drain, either terminal can act as the effective source (exception: not true for power MOSFETs, in which the body is internally connected to the source). When thinking of FET action, and for purposes of calculation, remember that the effective source terminal is always the one most "away" from the active drain supply. For example, suppose a FET is used to switch a line to ground, and both positive and negative signals are present on the switched line, which is usually selected to be the FET drain. If the switch is an *n*-channel MOSFET (therefore enhancement) and a negative voltage happens to be present on the (turned-off) drain terminal, then that terminal is actually the "source" for purposes of gate turn-on voltage calculation. Thus a negative gate voltage larger than the most negative signal, rather than ground, is needed to ensure turn-off.

The graph in Figure 3.12 may help to sort out all these confusing ideas. Again, the difference between enhancement and depletion is merely a question of displacement along the V_{GS} axis, i.e., whether there is a lot of drain current or no drain current at all when the gate is at the same potential as the source. The *n*-channel and *p*-channel FETs are complementary in the same way as *npn* and *pnp* bipolar transistors.

In Figure 3.12 we have used standard symbols for the important FET parameters of saturation current and cutoff voltage. For JFETs the value of drain current with the gate shorted to source is specified on the datasheets as I_{DSS} and is nearly the maximum drain current possible. (I_{DSS} means current from drain to source with the gate shorted to the source. Throughout the chapter you will see this notation, in which the first two subscripted letters designate the pair of terminals and the third specifies the condition.) For enhancement-mode MOSFETs the analogous specification is $I_{D(ON)}$, given at some forward gate voltage ("I_{DSS}" would be zero for any enhancement-mode device).

For JFETs the gate-source voltage at which the drain current is brought essentially to zero[9] is called the "gate-source cutoff voltage," $V_{GS(OFF)}$, or (sometimes) the "pinch-off voltage," V_P, and is typically in the range of -1 V to -5 V (positive for *p*-channel, of course). The analogous quantity is not normally specified for enhancement-mode MOSFETs;[10] instead, datasheets specify the

[9] Usually chosen to be 10 nA; a pinchoff-voltage test circuit is described in §4.3.4.

[10] We will use the symbol V_{th} to designate the analogous idealized "gate-source cutoff voltage" for MOSFETs, which we will need in some

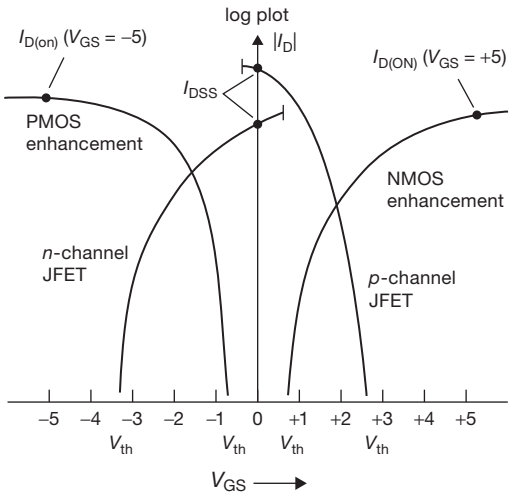


Figure 3.12. Important gate voltages and drain currents.

"gate-source threshold voltage," $V_{GS(th)}$, at which the onset of drain current has reached a small but arbitrary threshold value, typically 0.25 mA. $V_{GS(th)}$ is typically in the range of 0.5–5 V, in the "forward" direction, of course.

With FETs it is easy to get confused about polarities. For example, *n*-channel devices, which usually have the drain positive with respect to the source, can have positive or negative gate voltage and positive (enhancement) or negative (depletion) threshold voltages. To make matters worse, the drain can be (and often is) operated negative with respect to the source. Of course, all these statements go in reverse for *p*-channel devices. In order to minimize confusion, we will always assume that we are talking about *n*-channel devices unless explicitly stated otherwise. Likewise, because MOSFETs are nearly always enhancement mode, and JFETs are always depletion mode, we'll omit those designations from now on.

3.1.4 FET drain characteristics

In Figure 3.2 we showed a family of curves of I_D versus V_{DS} that we measured for a VN0106, an *n*-channel enhancement-mode MOSFET.[11] We remarked that FETs behave like pretty good transconductance devices over most of the graph (i.e., I_D nearly constant for a given V_{GS}), except at small V_{DS}, where they approximate a resistance (i.e., I_D proportional to V_{DS}). In both cases the applied gate-source voltage controls the behavior, which can be well described by the FET analog of the Ebers–Moll equation. Let's look now at these two regions a bit more closely; we'll revisit this important subject in greater detail in §3.3 and again in the advanced-topics Chapter *3x*.

Figure 3.13 shows the situation schematically. In both regions the drain current depends on $V_{GS} - V_{th}$, the amount by which the applied gate-source voltage exceeds the threshold (or pinch-off) voltage. The linear region, in which drain current is approximately proportional to V_{DS}, extends up to a voltage $V_{DS(sat)}$, after which the drain current is approximately constant. The slope in the linear region, I_D/V_{DS}, is proportional to the gate bias, $V_{GS} - V_{th}$. Furthermore, the drain voltage at which the curves enter the "saturation region," $V_{DS(sat)}$, is approximately $V_{GS} - V_{th}$, making the saturation drain current, $I_{D(sat)}$, proportional to $(V_{GS} - V_{th})^2$, the quadratic law we mentioned earlier. For reference, here are the universal FET drain-current formulas:

$$I_D = 2\kappa[(V_{GS} - V_{th})V_{DS} - V_{DS}^2/2] \quad \text{(linear region)} \quad (3.1)$$

$$I_D = \kappa(V_{GS} - V_{th})^2 \quad \text{(saturation region)} \quad (3.2)$$

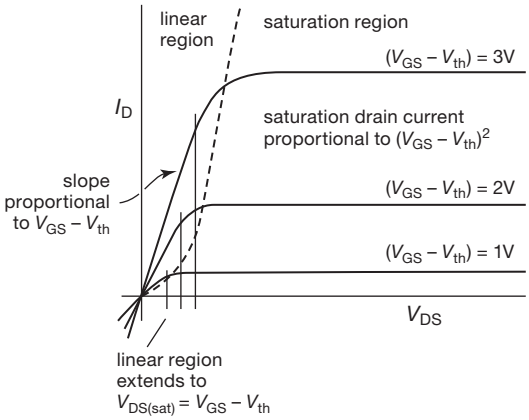


Figure 3.13. Linear and saturation regions of FET operation.

If we call $V_{GS} - V_{th}$ (the amount by which the gate-source voltage exceeds the threshold) the "gate drive," the

discussion that follows. In the electronics literature the symbol V_T is used for this quantity, called the "threshold voltage"; but we prefer to avoid the same symbol that is used for the "thermal voltage" V_T in the Ebers–Moll equation, where $V_T = kT/q \approx 25$ mV. And don't confuse V_{th} with $V_{GS(th)}$: V_{th} is obtained from extrapolating a $\sqrt{I_D}$ versus V_{GS} plot; it's not found in datasheets, but it's quite useful. By contrast, $V_{GS(th)}$ is not terribly useful, but it's the quantity you find in datasheets.

[11] The VN0106 is not widely available. It is similar to the very popular 2N7000 or BS170 (in the TO-92 package) and to the 2N7002, BSS138, or MMBF170 (in the SMT packages).

important results are that (a) the resistance in the linear region is inversely proportional to the gate drive, (b) the linear region extends to a drain-source voltage approximately equal to the gate drive, and (c) the saturation drain current is proportional to the square of the gate drive. These equations assume that the body is connected to the source. Note that the "linear region" is not really linear because of the V_{DS}^2 term; we'll show a clever circuit fix later.

The scale factor κ depends on particulars such as the geometry of the FET, oxide capacitance, and carrier mobility.[12] It has a temperature dependence $\kappa \propto T^{-3/2}$, which alone would cause I_D to decrease with increasing temperature. However, V_{th} also depends slightly on temperature (2–5 mV/°C); the combined effect produces the curve of drain current versus temperature, as shown in Figure 3.14.

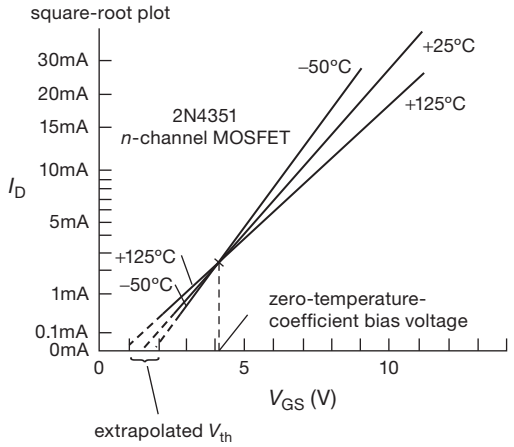


Figure 3.14. The "threshold voltage" V_{th} is found by extrapolating a square-root plot of I_D to zero drain current. The FET saturation drain current has a negative tempco in the high-current regime.

At large gate voltages the negative temperature coefficient of κ causes the drain current to decrease with increasing temperature. As a consequence, FETs of a given type, operating in this high-current regime, can often be paralleled without the external current-equalizing ("emitter-ballasting") resistors that you must use with bipolar transistors (see §3.6.3).[13] This same negative coefficient also prevents thermal runaway in local regions of the junction (an effect known as "current hogging"), which severely limits the power capability of large bipolar transistors, as we'll see when we discuss "second breakdown" and "safe operating area" in Chapter 9.

At small drain currents (where the temperature coefficient of V_{th} dominates), I_D has a positive tempco, with a point of zero temperature coefficient at some drain current in between. This effect is exploited in FET op-amps to minimize temperature drift, as we'll see in the next chapter.

A. Subthreshold region

Our expression given earlier for saturation drain current does not apply for very small drain currents. This is known as the "subthreshold" region, where the channel is below the threshold for conduction, but some current flows anyway because of a small population of thermally energetic electrons. If you've studied physics or chemistry, you probably know in your bones that the resulting drain current is exponential (with some scale factor) in the difference voltage $V_{GS}-V_{th}$.

We measured some MOSFETs over nine decades of drain current (1 nA to 1 A) and plotted the result as a graph of I_D versus V_{GS} (Figure 3.15). The region from 1 nA to 1 mA is quite precisely exponential; above this subthreshold region the curves enter the normal "quadratic" region. For the *n*-channel MOSFET (Supertex type VN01, similar to the ever-popular 2N7000) we checked out a sample of 20 transistors (from four different manufacturing runs spread over two years), plotting the extreme range to give you an idea of the variability (see next section). Note the somewhat poorer characteristics (V_{th}, $I_{D(ON)}$) of the "complementary" VP01 (similar to the popular BS250).

JFETs exhibit similar behavior, as illustrated in the measured data of Figure 3.16 (though V_{GS} is necessarily limited to reverse-bias voltage polarity, or at most to a forward-bias less than a diode drop). The quadratic region, where $I_D \propto (V_{GS}-V_{th})^2$, is seen most clearly by plotting the *square root* of drain current versus gate voltage; see Figure 3.14 and Figure 3.51 later in the chapter.

3.1.5 Manufacturing spread of FET characteristics

Before we look at some circuits, let's take a look at the range of FET parameters (such as I_{DSS} and $V_{GS(th)}$), as well as their manufacturing "spread" among devices of the same nominal type, in order to get a better idea of the FET. Unfortunately, many of the characteristics of FETs show a much greater process spread than the corresponding

[12] You'll commonly see the symbol k used here. We prefer κ, to avoid confusion with the Boltzmann constant k that figures in the Ebers–Moll equation for bipolar transistor behavior. The SPICE model for JFETs calls this parameter β (and for V_{th} it uses the parameter "VTO").

[13] Some cautions apply, most notably with ordinary ("vertical") power MOSFETs in linear applications, where they are operated at drain currents well below the region of negative temperature coefficient – see §3.5.1B and §3.6.3. In such applications (e.g., audio power amplifiers) the alternative "lateral" MOSFET is popular, owing to its stabilizing negative coefficient.

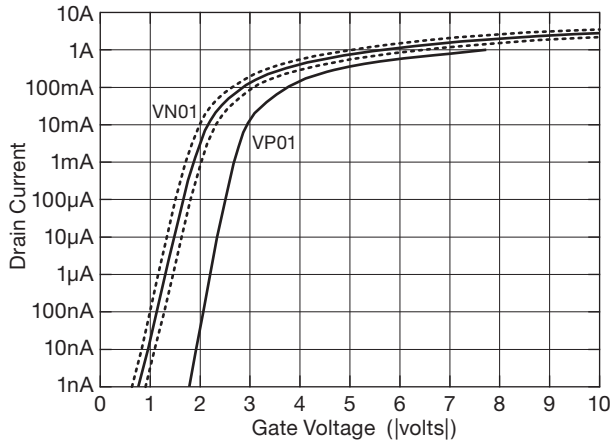


Figure 3.15. Measured MOSFET saturation drain current versus gate-source voltage. For the VN01 the dotted curves are the extreme specimens, and the solid curve is the median, from a group of 20 MOSFETs.

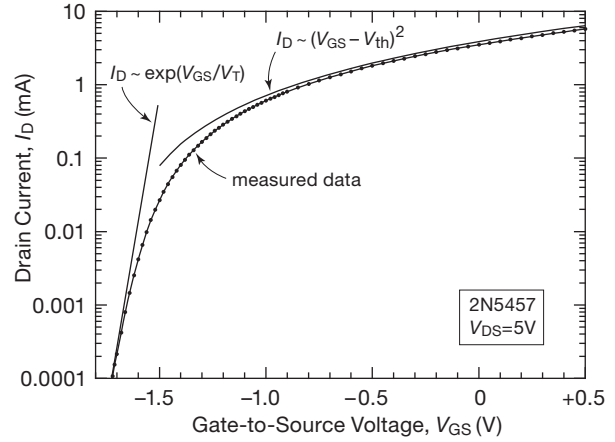


Figure 3.16. Five decades of measured drain current versus gate-to-source voltage for the n-channel 2N5457 JFET. In the subthreshold region the drain current is exponential, like a BJT, with nearly the same scale factor V_T (kT/q, or 25.3 mV at room temperature); at higher currents it becomes quadratic (the calculated curve has been offset by +10% for clarity).

characteristics of bipolar transistors, a fact that the circuit designer must keep in mind. For example, the 2N7000 (a typical n-channel MOSFET) has a specified $V_{GS(th)}$ of 0.8–3 V ($I_D = 1$ mA), compared with the analogous V_{BE} spread of 0.63–0.83 V (also at $I_C = 1$ mA) for a small *npn* bipolar transistor. Here's what you can expect:

FET Characteristics: Manufacturing Spread

Characteristic	*Available Range*	*Spread*
I_{DSS}, $I_{D(ON)}$	1 mA to 500 A	×5
$R_{DS(ON)}$	0.001Ω to 10k	×5
g_m @ 1 mA	500–3000 μS	×5
V_P (JFETs)	0.5–10 V	5 V
$V_{GS(th)}$ (MOSFETs)	0.5–5 V	2 V
$BV_{DS(OFF)}$	6–1000 V	
$BV_{GS(OFF)}$	6–125 V	

$R_{DS(ON)}$ is the drain-source resistance (linear region, i.e., small V_{DS}) when the FET is conducting fully, e.g., with the gate grounded in the case of JFETs or with a large applied gate-source voltage (usually specified as 10 V) for MOSFETs. I_{DSS} and $I_{D(ON)}$ are the saturation-region (large V_{DS}) drain currents under the same turned-on gate drive conditions. V_P is the pinch-off voltage (JFETs), $V_{GS(th)}$ is the turn-on gate threshold voltage (MOSFETs), and the BV's are breakdown voltages. As you can see, a JFET with a grounded source may be a good current source, but you can't predict very well what the current will be. Likewise, the V_{GS} needed to produce some value of drain current can vary considerably, in contrast to the predictable (≈0.6 V) V_{BE} of bipolar transistors. Figure 3.17 illustrates this latter point graphically: we measured the V_{GS} values at a drain current of 1 mA for a hundred pieces each (hey, they're pretty cheap: about $0.10 each) of three popular JFET types (the 2N5457–59 series, graded by their I_{DSS}). The spread of gate-source voltages, within each type, is about 1 V. For comparison, look at the analogous plot for BJTs in Figure 8.44; there the spread is just 10–20 mV.

A. Matching of characteristics

As you can see, FETs are inferior to bipolar transistors in V_{GS} predictability, i.e., they have a large spread in the V_{GS} required to produce a given I_D. Devices with a large process spread will, in general, have a larger offset (voltage unbalance) when used as differential pairs. For instance, typical run-of-the-mill bipolar transistors might show a spread in V_{BE} of 25 mV or so, at some collector current, for a selection of off-the-shelf transistors. The comparable "official" figure (as specified on datasheets) for MOSFETs is more like 1 V to 2 V![14] Because FETs have some very desirable characteristics otherwise, it is worthwhile putting in some

[14] In practice, we've found considerably better matching within a single batch of MOSFETs, sometimes as tightly matched as 50 mV or so. On the other hand, a more typical spread within one batch is several hundred millivolts, as illustrated later in Figure 3.41. If matching is

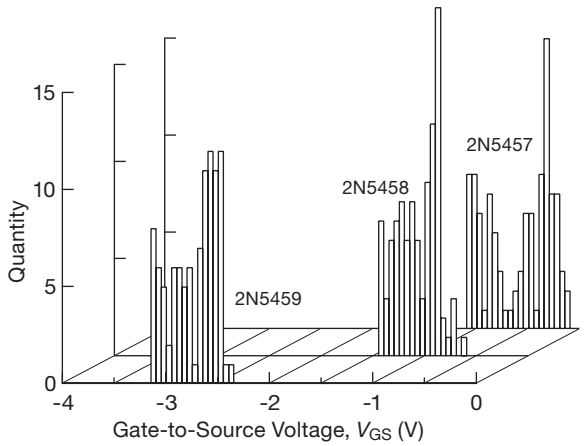


Figure 3.17. We wore our fingers to the bone (a "digital" measurement?) collecting V_{GS} values (at V_{DS}=5 V and I_D=1 mA) for 300 JFETs in the popular 2N5457–59 series. Compare with the analogous histograms in Figure 8.44.

extra effort to reduce these offsets in specially manufactured matched pairs. IC designers use techniques like interdigitation (two devices sharing the same general piece of IC real estate) and thermal-gradient cancellation schemes to improve performance (Figure 3.18).

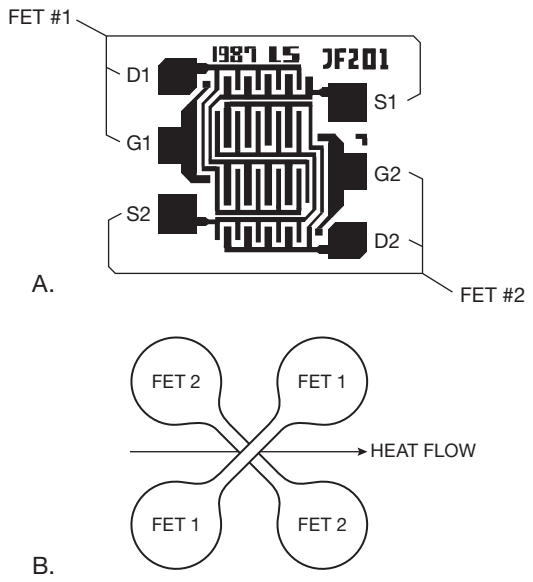


Figure 3.18. Techniques for transistor matching: A. Interdigitation (Courtesy of Linear Integrated Systems.) B. Temperature gradient cancellation.

important in some application (when several transistors are used in parallel, for example), you should measure the actual parts.

The results are impressive. Although FET devices still cannot equal bipolar transistors in V_{GS} matching, their performance is adequate for most applications. For example, the best previously available matched FET had[15] a voltage offset of 0.5 mV and tempco of 5 μV/°C (max), whereas the best bipolar pair has values of 25 μV and 0.3 μV/°C (max), roughly 20 times better.

Operational amplifiers (the universal high-gain differential amplifiers we'll see in the next chapter) are available in both flavors; you would generally choose one with bipolar innards for high precision (because of its close input-transistor V_{BE} matching), whereas a FET-input op-amp is the obvious choice for high-impedance applications (because its inputs – FET gates – draw no current). For example, the inexpensive JFET-input LF411 and LF412 that we will use as our all-around op-amp in the next chapter has a typical input (leakage) current of 50 pA and costs $0.60; the popular MOSFET-input TLC272 costs about the same and has a typical input (leakage) current of only 1 pA! Compare this with a common bipolar op-amp, the LM324, with typical input (bias) current of 45,000 pA (45 nA).[16]

3.1.6 Basic FET circuits

Now we're ready to look at FET circuits. You can usually find a way to convert a circuit that uses BJTs into one using FETs – but the new circuit may not be an improvement! For the remainder of the chapter we'd like to illustrate circuit situations that take advantage of the unique properties of FETs, i.e., circuits that work better with FETs or that you can't build at all with bipolar transistors. For this purpose it may be helpful to group FET applications into categories; here are the most important, as we see it.

High-impedance/low-current. Buffers or amplifiers for applications in which the base current and finite input

[15] Sadly, these parts are no longer available. But the art of transistor matching is alive and well in the innards of op-amps, for which the best JFET specimen has an offset and tempco of 0.1 mV and 1 μV/°C, respectively, whereas the best BJT specimen has 0.01 mV and 0.1 μV/°C, i.e., 10 times better.

[16] BJT enthusiasts would cry "foul!," and point out that you can use superbeta BJTs, combined with bias-current cancellation schemes, to bring the input current down to 25 pA; they would further point out that FET-input current (which is leakage) rises dramatically with temperature, whereas BJT-input current (which is honest bias current) is stable or even tends to decrease slightly (see Figure 3.48). FET enthusiasts would prevail, though, with the rejoinder that MOSFET-input amplifiers like the dual LMC6042 have typical input currents of 2 *femto*amps (that's 0.000002 nA!).

impedance of BJTs limit performance. Although you *can* build such circuits with discrete FETs, current practice favors using integrated circuits built with FETs. Some of these use FETs as a high-impedance front-end for an otherwise bipolar design, whereas others use FETs throughout. When available FET ICs do not provide adequate performance, a hybrid approach (discrete JFET front-end, assisted by an op-amp) can push the performance envelope.

Analog switches. MOSFETs are excellent voltage-controlled analog switches, as we hinted in §3.1.1B. We'll look briefly at this subject. Once again, you should generally use dedicated "analog switch" ICs, rather than building discrete circuits.

Digital logic. MOSFETs dominate microprocessors, memory, special-purpose VLSI, and most high-performance digital logic. They are used exclusively in micropower logic and low-power portable devices. Here, too, MOSFETs make their appearance in integrated circuits. We'll see why FETs are preferable to BJTs.

Power switching. Power MOSFETs are usually preferable to ordinary bipolar power transistors for switching loads, as we suggested in our first circuit of the chapter. For this application you use *discrete* power FETs.

Variable resistors; current sources. In the "linear" region of the drain curves, FETs behave like voltage-controlled resistors; in the "saturation" region they are voltage-controlled current sources. You can exploit this intrinsic behavior of FETs in your circuits.

Generalized replacement for bipolar transistors. You can use FETs in oscillators, amplifiers, voltage regulators, and radiofrequency circuits (to name a few), where bipolar transistors are also normally used. FETs aren't *guaranteed* to make a better circuit – sometimes they will, sometimes they won't. You should keep them in mind as an alternative.

Now let's look at these subjects. We'll adopt a slightly different order, for clarity.

3.2 FET linear circuits

A note to the reader: This section and the next (§§3.2 and 3.3) deal primarily with *JFETs*, which are well suited to linear applications such as current sources, followers, and amplifiers. If you need a low-noise amplifier with extremely high input impedance, the JFET is your friend (and maybe your *only* friend). Readers wishing to move directly to MOSFETs, starting with FET switches, may wish to skip over these JFET materials,[17] and proceed directly to §3.4 on page 171, where we launch into the MOSFET-dominated subjects of signal switching, digital logic, and power switching.

Table 3.1 JFET Mini-table[a] (see also JFET Table 3.7)

Part #	I_D Curve	I_{dss} (mA)	$V_{GS(off)}$ min (V)	$V_{GS(off)}$ max (V)	measured at 1mA V_{GS} (V)	measured at 1mA g_m (mS)	measured at 1mA G_{max}[b] (V/V)	C_{rss} typ (pF)	R_{ON} typ (Ω)
2N5484	A	1–5	–0.3	–3	–0.73	2.3	180	1	-
2N5485	B	4–10	–0.4	–4	–1.7	2.1	110	1	-
2N5486	C	8–20	–2	–6	–2.4	2.1	50	1	-
2N5457	D	1–5	–0.5	–6	–0.81	2.0	200	1.5	-
2N5458	E	2–9	–1	–7	–2.3	2.3	170	1.5	-
2N5459	F	4–16	–2	–8	–2.8	2.0	100	1.5	-
BF862	G	10–25	–0.3	–1.2	–0.40	12	250	1.9	-
J309	H	12–30	–1	–4	–1.6	4.2	300	2	50
J310	J	24–60	–2	–6.5	–3.0	4.3	100	2	50
J113	K	2–	–0.5	–3	–1.5	5.7	140	3	50
J112	L	5–	–1	–5	–3.3	5	100	3	30
PN4393	M	5–30	–0.5	–3	–0.83	6.2	100	3.5	100
PN4392	N	25–75	–2	–5	–2.6	5.4	130	3.5	60
LSK170B	P	6–12	–0.2	–2	–0.09	11	160	5	-
J110	Q	10–	–0.5	–4	–1.2	6.1	220	8	18
J107	R	100–	–0.5	–4.5	–2.6	8.2	340	35	8
J105	-	500–	–4.5	–10	–8.7	6.4	60	35	3
IF3601	S	30–	–0.04	–3	–0.24	27	1400	300	-

Notes: (a) sorted by family C_{rss}, and within each family by increasing I_{DSS}. (b) $G_{max}=g_m/g_{os}$, the maximum grounded-source voltage gain into a current source as drain load; G_{max} is proportional to V_{DS} (tabulated values are at V_{DS}=5V), and for most JFETs G_{max} is relatively constant over varying I_D.

3.2.1 Some representative JFETs: a brief tour

Table 3.1 lists a small selection of representative *n*-channel JFETs.[18] Let's take a look at what you get.

This selection includes only *n*-channel JFETs, the dominant polarity. Complements with similar characteristics are sometimes available, for example the *p*-channel 2N5460–62 for the *n*-channel 2N5457–59; see Table 3.7 on page 217 for additional examples.

Many JFETs come in families of three or four parts, graded by I_{DSS} and $V_{GS(off)}$, which alleviates somewhat

[17] You'll want to study this material, though, if you want to understand MOSFET linear amplifiers, because we address topics like the significance of an FET's transconductance and output conductance, and their variation with drain voltage and current.

[18] The expanded Table 3.7 on page 217 includes many more JFETs; later in this chapter there are analogous tables of MOSFETs (Tables 3.4a and 3.4b, pages 188–191, Table 3.5, page 206, and Table 3.6, page 210).

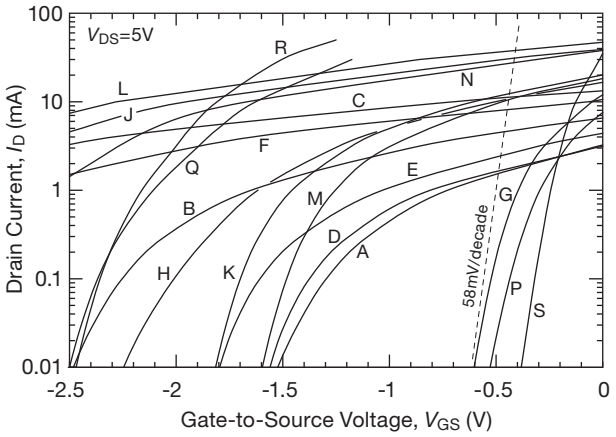


Figure 3.19. Measured drain current versus gate-source voltage for the JFETs in Table 3.1 on the previous page.

the annoying circuit design problems created by the wide spread of those parameters. But even those graded families may present a spread of as much as 5:1 (or more). Note also that JFETs intended for switching applications (the ones specifying R_{on}) may specify only a *minimum* value of I_{DSS}: what can you say, for example, about the likely value for a J110 (specified as I_{DSS} = 10 mA, minimum)? Answer: not much – our sample measured 122 mA!

In many applications (amplifiers, followers) you want lots of transconductance gain, g_m. JFET datasheets usually specify g_m at the part's I_{DSS}, but that's not terribly useful if you don't know what I_{DSS} is. Moreover, the listed g_m at I_{DSS} is afflicted with the usual specification spread, typically 5:1 or so. Unlike BJTs, for which the transconductance is predictably given by $g_m = 1/r_e = I_C/V_T$ (where $V_T = kT/q \approx 25.3$ mV), the transconductance of different JFET types can vary by an order of magnitude, even when each is operated at the same drain current. In Table 3.1 on the previous page we've listed measured values of g_m, all at a standard current of 1 mA.[19] At these currents their transconductance is much less than that of a BJT (where g_m=40 mS at 1 mA), though they compete well at very low currents (the subthreshold region). This behavior can be seen in the different slopes of the measured I_D versus V_{GS} curves of Figure 3.19.

The column labeled G_{max} lists the voltage gain when used as a grounded-source amplifier with current-source load; in that case the effective load resistance is related to a quantity called g_{os}, the output conductance seen looking into the drain with the gate voltage held constant (analogous to the Early effect in BJTs; more on this in Chapter *3x*). Here, too, there is a wide spread among JFET types.

A parameter that is important in low-level amplification is a JFET's input noise voltage, not listed here but treated in detail in Chapter 8. The standout happens to be the IF3601 (an amazing $e_n = 0.3\ \text{nV}/\sqrt{\text{Hz}}$), but your pact with the devil is the high 300 pF capacitance of the large-area junction.[20]

There's much more to say about the inhabitants of the JFET zoo, as we'll see in connection with Table 3.7 on page 217. Chapter 8 discusses JFETs in connection with noise (§§8.6 and 8.6.5), with a table of relevant parts (Table 8.2 on page 516).

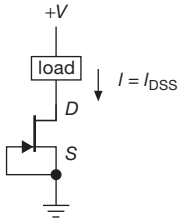


Figure 3.20. An *n*-channel JFET current sink.

3.2.2 JFET current sources

JFETs are used as current sources within integrated circuits (particularly op-amps), and also sometimes in discrete designs. The simplest JFET current source is shown in Figure 3.20; we chose a JFET, rather than a MOSFET, because it needs no gate bias (it's depletion mode). From a graph of FET drain characteristics (Figure 3.21) you can see that the current will be reasonably constant for V_{DS} larger than a couple of volts. However, because of I_{DSS} spread, the current is unpredictable. For example, the MMBF5484 (a typical *n*-channel JFET) has a specified I_{DSS} of 1–5 mA. Still, the circuit is attractive because of the simplicity of a two-terminal constant-current device. If that appeals to you, you're in luck. You can buy "current-regulator diodes" that are nothing more than JFETs with gate tied to source, sorted according to current. They're the current analog of a zener (voltage-regulator) diode. Here are the characteristics of the 1N5283–1N5314 series:[21]

[19] In the normal "quadratic" region of drain current, transconductance varies approximately as $\sqrt{I_D}$, see §3.3.3.

[20] See Table 8.2 for the IF3601 and IF3602 (dual). Runners-up in the low-noise competition are the LSK170B and the BF862, with considerably lower capacitances.

[21] Said to be available from several manufacturers. Alternatives include the MS5283, MV5283, and MX5283 series from Microsemi; the SST502–SST511 and CR160–CR470 series from Vishay; and the J500–J511, J553–J557, and U553–U557 series from InterFET. Alternative sources: Central Semiconductor and Linear Integrated Systems.

Characteristic	*Value*
Currents available	0.22–4.7 mA
Tolerance	±10%
Temperature coefficient	±0.4%/°C
Voltage range	1–2.5 V min, 100 V max
Current regulation	5% typical
Impedance	1M typ (for 1 mA device)

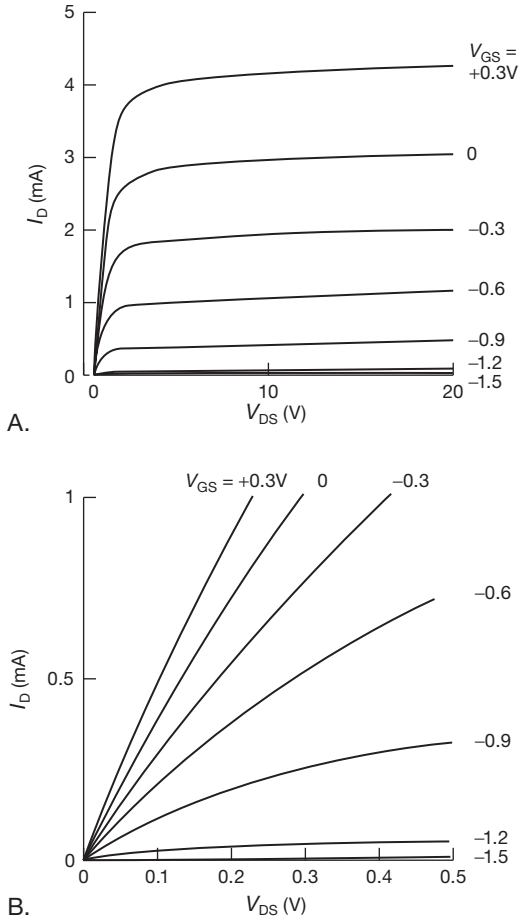


Figure 3.21. Measured JFET characteristic curves. 2N5484 *n*-channel JFET: I_D versus V_{DS} for various values of V_{GS}. See also Figure 3.47.

We measured I versus V for a 1N5294 (rated at 0.75 mA), applying 1 ms voltage pulses at 100 ms intervals to prevent heating. Figure 3.22A shows good constancy of current up to the breakdown voltage (∼145 V for this particular specimen). You can see also the effect of heating when voltage is applied continuously in a dc measurement, caused by the negative temperature coefficient of drain current. Figure 3.22B shows that the device reaches full current with somewhat less than 1.5 V across it (here both pulsed and dc curves are plotted, demonstrating negligible thermal effects with less than 0.4 mW dissipation). We'll show how to use these devices to make a cute triangle-wave generator in §7.1.3E. And we'll have much more to say about current sources in §4.2.5 and §9.3.14.

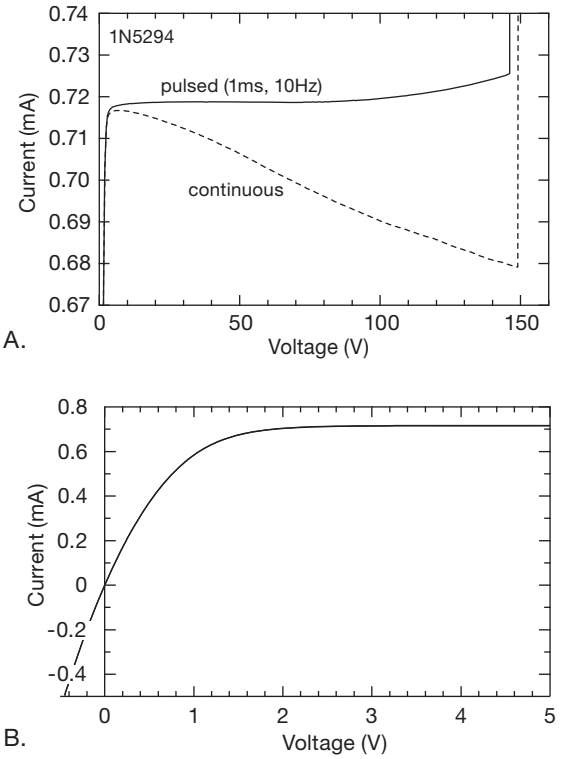


Figure 3.22. 1N5294 current regulator diode.

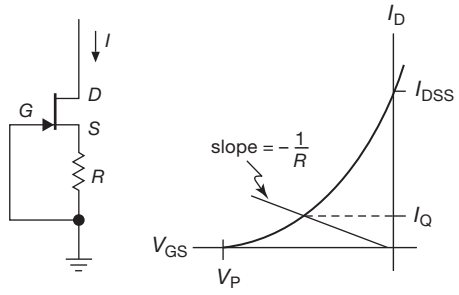


Figure 3.23. JFET current sink ($I = V_{GS}/R$) for $I_D < I_{DSS}$.

A. Source self-biasing

A variation of the previous circuit (Figure 3.23) gives you an adjustable current source. The self-biasing resistor R back-biases the gate by $I_D R$, reducing I_D and bringing the

JFET closer to pinch-off. You can estimate the value of R from the drain curves for the particular JFET. This circuit allows you to set the current (which must be less than I_{DSS}), as well as to make it more predictable. Furthermore, the circuit is a better current source (higher impedance) because the source resistor provides "current-sensing feedback" (which we'll learn about in §4.2.5A). (There's a nice demonstration of this in in the figure "Measured I_D versus V_{DS} for four JFETs" in §*3x.4.3*, where you will find drain–current versus drain–voltage curves both with and without a source self-biasing resistor.) Remember, though, that actual curves of I_D for some value of V_{GS} obtained with a real FET may differ markedly from the values read from a set of published curves, owing to manufacturing spread. (This is illustrated nicely by the examples of Figures 3.25 and 3.41 on pages 145 and 156, using actual measured drain characteristics from a batch of JFETs.) You may therefore want to use an adjustable source resistor, if it is important to have a specific current.

Exercise 3.1. Use the 2N5484 measured curves in Figure 3.21 to design a JFET current source to deliver 1 mA. Now ponder the fact that the specified I_{DSS} of a 2N5484 is 1 mA (min), 5 mA (max).

B. Example: emitter follower pull-down

Let's look at an example to explore further this problem of unpredictability of JFET zero-bias drain current, I_{DSS} (or, equivalently, the difficulty of predicting the gate-source bias needed to produce a desired drain current).

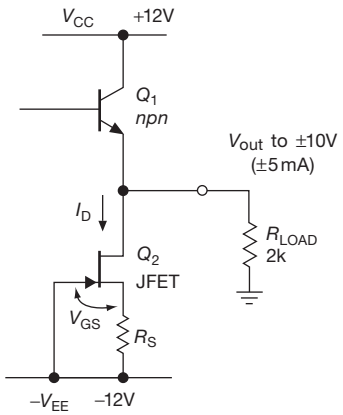


Figure 3.24. Design example: *npn* emitter follower with JFET current sink.

Figure 3.24 shows a BJT emitter follower, running between ±12 V split supplies, with a current-sinking JFET pull-down to the negative rail. We specify that the circuit must be able to deliver a full swing of ±10 V into a 2kΩ load (that's ±5 mA load current). You might at first think of using a simple pull-down resistor R_E to the −12 V rail. But the output swing requirement makes things difficult, because you would need to keep R_E less than 400 Ω (we might choose 365 Ω, a standard 1% value) to get a full negative swing; and that low resistance would produce a relatively high quiescent current (at 0 V output) of 33 mA (thus ∼400 mW quiescent dissipation both in Q_1 and in R_E), compared with the 5 mA peak current delivered to the load (recall the discussion in §2.4.1). Worse still, a resistive pulldown also greatly degrades linearity, owing to variations in the follower's r_e caused by the large variation of collector current (65 mA at the top of the swing, dropping to 0.5 mA at the bottom, thus corresponding r_e of 0.4 Ω and 50 Ω into a combined load resistance of ∼300 Ω). Finally, the small resistive pull-down resistor (compared with the minimum load resistance) undesirably reduces the circuit's input impedance by a factor of 6.

So an active current sink is the way to go. A first possibility is to choose a JFET whose minimum specified I_{DSS} is at least equal to our required 5.5 mA. Only the 2N5486 member of the 2N5484–86 family satisfies this requirement (8 mA ≤ I_{DSS} ≤ 20 mA, see Table 3.1 on page 141). But these currents are rather more than we'd like, and a part with $I_{DSS} = 20$ mA produces too much heat: the worst-case dissipation is 440 mW in either the JFET (at peak positive swing) or the BJT (at peak negative swing with no load), too much for a transistor in a TO-92 or SOT-23 package without a heatsink.

So, let's add a source resistor so we can tailor the JFET's drain current; we'll aim for a 5.5 mA minimum sink current, so that we retain 0.5 mA reserve at full negative swing. The 2N5486's minimum I_{DSS} of 8 mA guarantees that a source self-biased circuit can sink the required 5.5 mA current. Now we need only choose the source resistor R_S.

The problem is that datasheet curves of I_D versus V_{GS} (called "transfer characteristics"), when provided at all, do not show the full range of possibilities; instead they show curves typical of parts with two or three selected values of I_{DSS} within the allowable range. And sometimes all you get are tabulated limits for I_{DSS} and for $V_{GS(off)}$.[22] But you can measure some JFETs to get a sense of things. We did this, and Figure 3.25 shows measured I_D versus V_{GS} curves for seven 2N5486s from different manufacturers

[22] It *is* possible to extrapolate from published (or measured) curves by estimating k and V_{th} for the simple quadratic law $I_D = k(V_{GS} - V_{th})^2$. See the discussion in Chapter *3x*.

and batches.[23] Assuming this represents the full range of variability (it doesn't, quite, as seen from the minimum I_{DSS} of 9.2 mA) then we can swing a load line up from the origin until the lowest intersection is above $I_D = 5.5$ mA. That's an R_S of 140 Ω (shown), for which the range of drain current is 5.7 mA (minimum) to 9.5 mA (maximum).

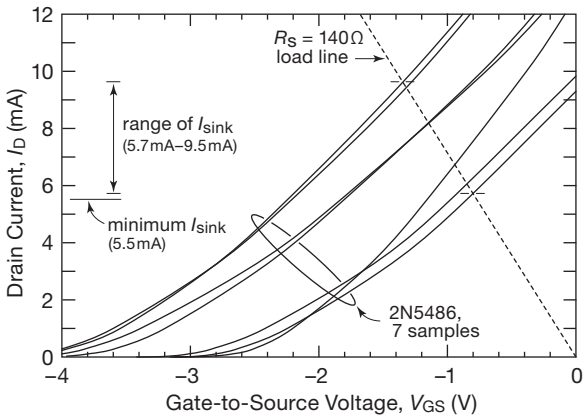


Figure 3.25. Choosing a source resistor R_S to bias a JFET current sink to produce $I_{sink} \geq 5.5$ mA.

The good news is that the circuit will work; the bad news is that the range of sink currents is nearly 2:1 (taking into account the possibility of production parts whose curves span a somewhat wider range than seen in these seven parts). But the good news, again, is that even for a JFET at the high end of the range (thus $I_{sink} \approx 10$ mA), the follower's worst-case dissipation is limited to 220 mW (at peak negative swing, no load), and the JFET's worst-case dissipation is likewise limited to 220 mW (at peak positive swing). This is well within allowable dissipation for a TO-92 transistor (350 mW at 25°C ambient).

C. Current sinks for JFET amplifiers

Stepping back a bit, one might ask whether a JFET current sink, with its 2:1 spread of quiescent current, was a good choice. True, it works. But you can do better with a simple BJT current sink, five versions of which are shown in Figure 3.26. These use more parts, but sink a predictable current. And if you *really* care about minimizing parts count, then you can always use the alternative of a JFET selected for a narrow range of I_{DSS}, with no self-biasing resistor, i.e., Figure 3.24 with $R_S = 0$ (the 2N5485 specifies I_{DSS} as 4 mA–10 mA; you might select parts from 5.5–8 mA).[24]

This example illustrates the down side of the loose drain current (and corresponding gate voltage) specifications characteristic of all JFETs. As attractive as it may seem to drop in a JFET when you need a current source, it's problematical. But JFETs come into their own when you need an amplifier with high input impedance and low noise – although the loosey-goosey specs are still challenging, the results are worth the bother. We'll see examples presently.

D. Imperfect current source

A JFET current source, even if built with a source resistor, shows some variation of output current with output voltage; i.e., it has finite output impedance, rather than the desirable infinite Z_{out}.[25] The measured curves of Figure 3.21, for example, suggest that, over a drain voltage range of 5–20 V, a 2N5484 shows a drain current variation of 5% when operated with gate tied to source (i.e., I_{DSS}). This might drop to 2% or so if you use a source resistor. An elegant solution is the use of a cascode transistor to suppress drain voltage variations in the current-setting transistor. This can be used both for BJT current sources (it's shown in §*2x.3*) and for JFET current sources, as shown in Figure 3.27. The idea (as with BJTs) is to use a second JFET to hold constant the drain-source voltage of the current source. Q_1 is an ordinary JFET current source, shown in this case with a source resistor. Q_2 is a JFET of larger I_{DSS}, connected "in series" with the current source. It passes Q_1's (constant) drain current through to the load, while holding Q_1's drain at a fixed voltage – namely the gate-source voltage that makes Q_2 operate at the same current as Q_1. Thus Q_2 shields Q_1 from voltage swings at its output; since Q_1 doesn't see drain voltage variations, it just sits there and provides constant current. If you look back at the Wilson mirror (Figures 2.61, 3.26D), you'll see that it uses this same voltage-clamping idea.

You may recognize this JFET circuit as the "cascode," which is normally used to circumvent Miller effect (§2.4.5). A JFET cascode is simpler than a BJT cascode,

[23] Making a mere cameo appearance here, compared with their full performance in Figures 3.55 and 3.56 and associated discussion).

[24] An expensive alternative is to use a pre-sorted two-terminal "current regulator diode" like those in the footnote 3.2.2 on page 142. These appear to be a vanishing species, and the range of currents is quite limited. (A double complaint, reminiscent of the dialog "The food there is so bad." "Yes, and such small portions.")

[25] This is important also for JFET amplifiers (§§3.2.3A and 3.3.2). For further details look at the discussion in Chapter *3x*.

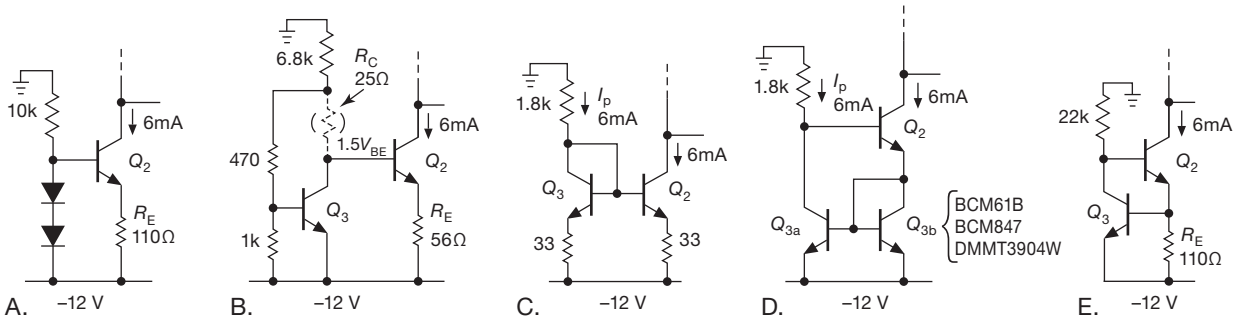


Figure 3.26. Alternatives to the JFET pulldown of Figure 3.24. A. Classic BJT current sink, base biased to $\sim 2V_{BE}$; you can substitute a red LED for the diode pair. B. Q_3 creates a "$1.5V_{BE}$" base bias for current sink Q_2; adding optional resistor R_C, chosen equal to Q_3's r_e, cleverly compensates for the latter's change in V_{BE} with supply-voltage variations (i.e., I_C variations through the 5.6k resistor). This configuration is useful if the current-sink output must operate very close to the negative rail – down to a few hundred mV, say, if configured as a "$1.25V_{BE}$" bias. C. Current mirror with $\approx$200 mV emitter ballasting (needed to equalize collector currents in view of V_{BE} mismatch, and to suppress the Early effect output-current variations). D. Wilson mirror with matched pair; no emitter resistors needed. E. "Ring-of-two" current source. See Figure 2.32 for other current source circuits.

however, because you don't need a bias voltage for the gate of the upper FET: because it's depletion mode, you can simply connect the upper gate to the lower source terminal (compare with Figure 2.84); Q_2's gate-to-source voltage at the operating current (set by Q_1 with its R_S) then sets Q_1's drain-to-source operating voltage: $V_{DS1} = -V_{GS2}$. A nice additional benefit is that the resultant circuit is a *two-terminal* current source.

It is important to realize that a good bipolar transistor current source will give far better predictability and stability than a JFET current source. Furthermore, the op-amp-assisted current sources we'll see in the next chapter are better still. For example, a FET current source might vary 5% over a typical temperature range and load-voltage variation, even after being set to the desired current by trimming the source resistor, whereas an op-amp/transistor (or op-amp/FET) current source is predictable and stable to better than 0.5% without great effort.

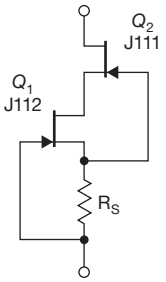


Figure 3.27. Cascode JFET current sink.

3.2.3 FET amplifiers

Source followers and common-source FET amplifiers are analogous to the emitter followers and common-emitter amplifiers made with bipolar transistors that we talked about in the previous chapter. However, the absence of dc gate current makes it possible to realize very high input impedances. Such amplifiers are essential when dealing with the high-impedance signal sources encountered in measurement and instrumentation. For some specialized applications you may want to build followers or amplifiers with discrete FETs; most of the time, however, you can take advantage of FET-input op-amps. In either case it's worth knowing how they work.

With JFETs it is convenient to use the same self-biasing scheme as with JFET current sources (§3.2.2), with a single gate-biasing resistor to ground (Figure 3.28); MOSFETs require a divider from the drain supply, or split supplies, just as we used with BJTs. The gate-biasing resistors can be quite large (a megohm or more), because the gate leakage current is measured in picoamps to nanoamps.

A. Transconductance

The absence of gate current makes *transconductance* (the ratio of output current to input voltage: $g_m = i_{out}/v_{in}$) the natural gain parameter for FETs. This is in contrast to bipolar transistors in the last chapter, where we at first flirted with the idea of current gain, or beta (i_{out}/i_{in}), then introduced the transconductance-oriented Ebers–Moll model: it's useful to think of BJTs either way, depending on the application.

FET transconductance can be estimated from the

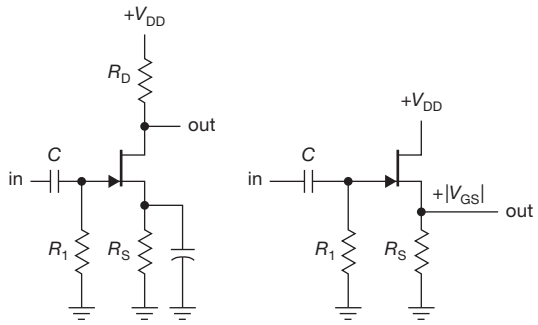


Figure 3.28. Common-source amplifier and source follower. For both configurations the source voltage is above ground, because of the source current flowing through R_S, with a quiescent point $V_S = V_{GS} = R_S I_D(V_{GS})$.

characteristic curves, either by looking at the increase in I_D from one gate-voltage curve to the next on the family of curves (Figures 3.2 or 3.21), or, more simply, from the slope of the I_D versus V_{GS} transfer characteristics curve (Figures 3.15 or 3.51). The transconductance depends on drain current (we'll see how, shortly) and is, of course,

$$g_m(I_D) = i_d/v_{gs}.$$

(Remember that lowercase letters indicate quantities that are small-signal variations.) From this we get the voltage gain,

$$G_{voltage} = v_d/v_{gs} = -R_D\, i_d/v_{gs}$$

i.e.,

$$G = -g_m R_D, \tag{3.3}$$

just the same as the bipolar-transistor result in §2.2.9, with load resistor R_C replaced with R_D. Typically, small-signal FETs have transconductances in the neighborhood of 10 millisiemens[26] (mS) at a few milliamps.[27] Because g_m depends on drain current, there will be some variation of gain (nonlinearity) over the waveform as the drain current varies, just as we have with grounded emitter amplifiers (where $g_m = 1/r_e$, proportional to I_C).

In the following discussion we'll be using the concept of FET gate drive, $V_{GS} - V_{th}$. Recall that V_{th} is the extrapolated gate threshold voltage we discussed in §§3.1.3 and 3.1.4.

The variation of g_m with drain current is easy to calculate and highly useful when designing JFET followers and amplifiers. For operation above subthreshold ($I_D > I_{DSS}/25$, say), we've seen that the drain current is quadratic in the gate drive

$$I_D = \kappa(V_{GS} - V_{th})^2, \tag{3.4}$$

from which the transconductance ($g_m = i_d/v_{gs} = \partial I_D/\partial V_{GS}$) is seen to be

$$g_m = 2\kappa(V_{GS} - V_{th}) = 2\sqrt{\kappa I_D}. \tag{3.5}$$

In other words, in the "quadratic region" of drain current, g_m is proportional to gate drive, increasing approximately linearly from pinchoff to its specified value at I_{DSS}; alternatively, you can say that it is proportional to the square root of drain current.[28] This is a helpful rule, particularly because the datasheets specify g_m only at its maximum value, at I_{DSS}; we'll use it shortly.[29]

As an example, if (as is often the case) you're operating a JFET in its quadratic region and you want to estimate the transconductance at some drain current I_D, then if you know g_m at some other drain current I_{D0} (which may be I_{DSS}) you can exploit the square root dependence on drain current in eq'n 3.5 to find, simply

$$g_m/g_{m0} = (I_D/I_{D0})^{\frac{1}{2}}. \tag{3.6}$$

FETs in general have considerably lower transconductance than bipolar transistors,[30] which makes them less impressive as amplifiers and followers; we treat this in more detail in §*3x.2*. However, their outstanding characteristic of extremely low input (gate) current, often of order a picoampere or less, makes it worthwhile to develop circuit solutions that circumvent the problems of low gain (e.g., current source as drain load), or that enhance their effective transconductance ("transconductance enhancer").

At this point it's helpful to see some JFET amplifier examples.

[26] Formerly millimhos, or m℧.

[27] This is substantially less than that of a BJT at the same current; the latter has $g_m = 40$ mS at 1 mA, and 200 mS at 5 mA, for example. There's further discussion in §3.3.3 and §*3x.2*.

[28] Be careful about signs: in these equations V_{th} and V_{GS} are negative (for *n*-channel JFETs), but V_{th} is more negative, thus a positive value for g_m. As long as you respect signs, these expressions work for *n*-channel or *p*-channel and for enhancement or depletion modes. Note that the value κ is not given on datasheets, but can be determined empirically for a given part type and manufacturer. Generally speaking, within a given batch or type of JFET you'll find variations in V_{th}, with κ being relatively constant. Thus a measurement of I_{DSS} and V_{th} allows you to calculate κ from eq'n 3.4, under the assumption that the quadratic region of drain current extends all the way to I_{DSS} (it usually does).

[29] See §*3x.2* for further discussion of transconductance versus drain current.

[30] Except in the low drain-current ("subthreshold") region; see Figure 3.54 and analogous figures in §3x.2.

B. JFET amplifier configurations

Figure 3.29 shows the basic configurations for a JFET common-source amplifier stage. In circuit A the JFET is running at its I_{DSS}, with R_D sized small enough so that the drain is at least a volt or two above ground for the maximum specified I_{DSS}. (This is often an annoying constraint, given the loose ratio of specified $I_{DSS(max)}/I_{DSS(min)}$ – commonly 5:1 for most JFETs, see Tables 3.1 on page 141, 8.2 on page 516, and 3.7 on page 217; presently we'll see ways to handle this awkward situation.) The resistor across the input can be very large – 100 MΩ or more – with an input blocking capacitor (for an ac-coupled amplifier); or it can be omitted altogether for a dc-coupled signal near ground. For this circuit the ideal voltage gain is $G=g_mR_D$, where g_m is the transconductance at the operating drain current; it is analogous to the BJT grounded-emitter amplifier of Figure 2.44.[31]

To illustrate actual component values and performance we've chosen the exemplary BF862, because of its high transconductance (45 mS typical at I_{DSS}) and tight I_{DSS} spec (10–25 mA); it happens also to be a low-noise part, as we'll see in Chapter 8. The drain resistor R_D is sized to maintain a minimum 2.5 V across Q_1 (for specified $I_{DSS(max)}$); the typical voltage gain is then $G=-g_mR_D\approx-13$ (inverting).

By adding a source resistor R_S circuit B lets you run at a drain current less than I_{DSS}, in the manner of Figures 3.23 and 3.25. But the source degeneration reduces the gain, to $G=-R_D/(R_S+1/g_m)$. This is analogous to the degenerated BJT common-emitter amplifier of Figure 2.49 (but with simpler self-biasing because the gate-source junction is back-biased), with $1/g_m$ replacing r_e (you can think of $1/g_m$ as an "intrinsic source impedance" of the JFET, analogous to the intrinsic emitter resistance r_e of the BJT).[32] Illustrating with the same BF862, we aim for a drain current of 2 mA by choosing a source self-bias resistor $R_S=200\,\Omega$, estimating that ∼0.4 V of gate back-bias is about right.[33] Estimating that $g_m\approx20$ mS at this drain current,[34] we arrive at an estimated voltage gain (from the above equation) of $G\approx-8$.

Circuit C bypasses the source resistor at signal frequencies, so you can run at the same dc drain current as circuit B, but with the higher gain as in circuit A (where g_m is the transconductance at the actual drain current, here reduced from the I_{DSS} of circuit A); it's analogous to the BJT circuit of Figure 2.48. You can throttle back the gain by adding a gain-setting resistor $R_{S'}$ in series with the capacitor (circuit D), for a signal-frequency gain of $G=-R_D/(R_S\|R_{S'}+1/g_m)$; this is analogous to the BJT circuit of Figure 2.50. Or you can step on the throttle by adding a second stage of voltage gain, as in circuit E, with the second stage common-emitter amplifier multiplying the first-stage gain of any of the previous single-stage circuits by a factor R_CI_C/V_T (i.e., R_C/r_e), where as always $V_T=kT/q\approx25$ mV. This approximation assumes Q_2 is driven by a voltage source, i.e., $R_D\ll\beta r_e$; most of the gain of the combination comes from the BJT, with its high transconductance. The similar-looking circuit F creates a three-terminal hybrid creature, with the BJT's g_m contributing to achieve a high effective overall transconductance; in this configuration the BJT is a "transconductance enhancer." This configuration is closely analogous to the BJT complementary Darlington (Sziklai) of Figure 2.77, and is treated in more detail in §*3x.2*.

C. Adding a cascode

The last four circuits show how to implement a drain-clamping cascode to the common-source stage. This elegant configuration is usually featured as a way to circumvent the Miller effect (the effective multiplication of drain-gate capacitance by the stage's voltage gain); that's the way it was presented in Figure 2.84, where it made its first appearance. Here it indeed accomplishes that (it's a "Miller killer"), which is helpful in keeping the input impedance high. But it's better than that: (a) it also lets you keep the drain-to-source voltage low (avoiding the precipitous rise in "impact-ionization" gate current, §3.2.8); and (b) by clamping the drain-source voltage it circumvents the "g_{os} effect" (finite output impedance r_o, caused by I_D dependence on V_{DS}); the voltage gain is then not degraded, and is simply $G=g_mR_D$. This latter benefit is reminiscent of the use of a cascode transistor in the Wilson current mirror (Figure 2.61); it's an "Early killer." There's more detail in Chapter *3x* (§*3x.4*), including experimental results from four JFET exemplars whose g_m and g_{os} were measured; then their gain predictions were compared with measured amplifier gains (both with and without cascode).

In circuit G the BJT's base bias sets the JFET's

[31] However, owing to the finite output impedance of the JFET (called r_o, or $1/g_{os}$), the drain load resistor effectively sees a parallel resistance of r_o, so the gain is reduced to $G=g_m(R_D\|r_o)$; this has a negligible effect for the component values here. This is analogous to the Early effect in BJTs, and becomes important for large values of R_D, or (especially) when R_D is replaced by a current source. Lots of discussion in §*3x.4*.

[32] This time we're ignoring the JFET's finite output impedance; see §*3x.4* if you're curious why and for much more detail (if it's detail you really want . . . be careful what you wish for).

[33] Our confidence is bolstered, a bit, by having measured a sample's I_D versus V_{GS}.

[34] We measured this, too.

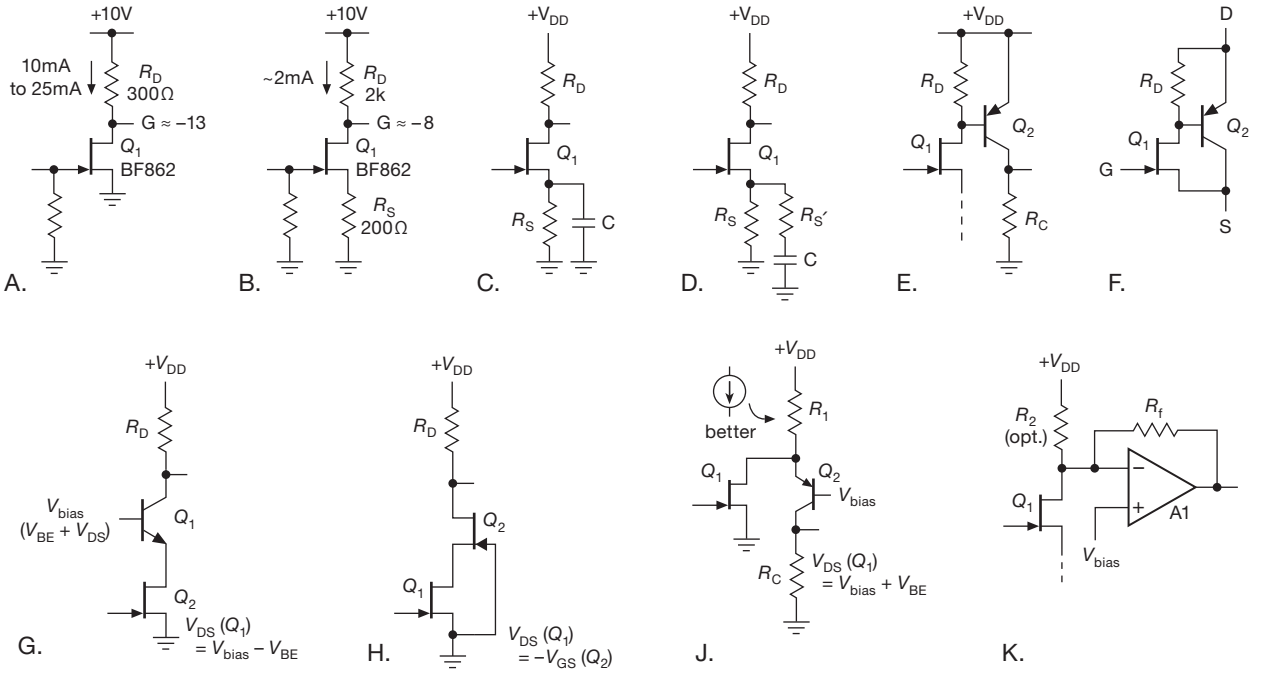


Figure 3.29. Simple single-supply JFET common-source amplifiers.

drain-to-source operating voltage. It's simpler to use a second JFET (Q_2 in circuit H), which must be chosen to have a larger V_{GS} back-bias than Q_1 at the same drain current, though the generally loose gate voltage specifications make this an uncertain proposition. Circuit J is an "inverted cascode," in which Q_1's drain current variations divert current from Q_2; it's a helpful circuit to know about when you find yourself bumping into the positive rail with a conventional cascode. Finally, in circuit K an op-amp transimpedance stage (current-to-voltage; see §4.3.1) substitutes for the drain-clamping cascode transistor Q_2: feedback through R_f maintains its inverting input (the "−" input) at the bias voltage while producing an output voltage $V_o = -I_D R_f + V_{bias}$; the optional resistor R_2 lets you add an offset to reposition the quiescent output voltage according to your whim.

The circuits in Figure 3.29 operate from a single positive supply voltage; they're simple, but, given the characteristically loose I_{DSS} and V_{GS} specifications of JFETs, they suffer from significant uncertainty of operating current. If you have available a negative supply voltage as well, there are several ways to rig things up to ensure predictable biasing. Look at Figure 3.30A, where the operating current of the *n*-channel JFET is set by the source pulldown resistor, $I_D = -(V_- + V_{GS})/R_S$, or approximately V_-/R_S for negative supply voltages large compared with

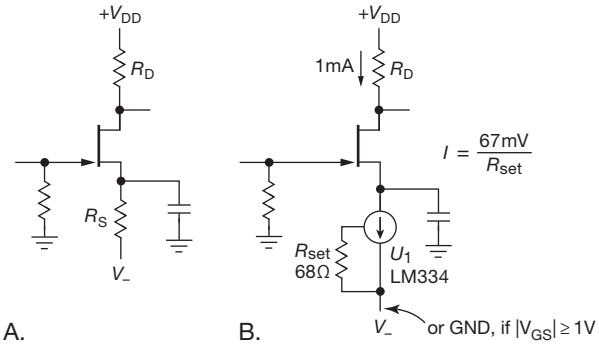


Figure 3.30. A negative supply rail allows predictable source-pulldown biasing of JFET common-source amplifiers.

the JFET's gate-source voltage. As in Figure 3.29C, the bypass capacitor lets signal frequencies partake of the full JFET gain, i.e., $G_V = -g_m R_D$, where g_m is the transconductance at the operating current. A more elegant solution is the use of a current-sink pulldown, as in Figure 3.30B. The LM334 is an inexpensive (~$0.50) resistor-programmable current source (see §9.3.14B), here configured for 1 mA ($I \approx 0.067/R_{set}$). With this circuit there's no uncertainty about the operating current (it does not depend on V_{GS}); better still, the LM334 operates down to 1 V drop, so you can operate with a single positive supply if the JFET's

minimum specified gate-source voltage at the programmed current is at least a volt.[35]

D. Series-feedback ("current-feedback") pair

The JFET amplifiers just illustrated have admirably high input impedance, but they suffer from rather low (and not terribly predictable) gain. Bipolar transistors give you predictable gain, and lots of it; but you pay the price in terms of input current. You can have the best of both worlds, though, by combining a JFET front-end of modest (and not terribly predictable) gain with some serious second-stage gain. That way you get the ultra-low input current (high input impedance) of a JFET, but with enough overall open-loop circuit gain so that negative feedback can close the loop to produce predictable gain.

The next circuit is a low-power (660 μA) battery-operated amplifier. We'll explore it in more detail than usual, introducing some new concepts along the way. Figure 3.31 shows the first of several JFET amplifier examples that exploit the JFET's ultra-low input current, combined with an additional gain stage (and feedback) to achieve predictable and stable voltage gain. It is similar to the bipolar series-feedback pair illustrated in Figure 2.92: Q_1 is a common-source amplifier, with BJT Q_2 providing second-stage voltage gain to the output (via follower Q_3, whose base–emitter drop across R_3 sets Q_2's collector current). That provides the needed voltage gain (which the JFET's low g_m is unable to provide). Negative feedback closes the loop via voltage divider R_6 and $R_5 \| R_1$ (at signal frequencies) and biasing via R_6 and R_1 at dc. This configuration is variously known as "series feedback" or "current feedback."

The spread of specified I_{DSS} (or, equivalently, of $V_{GS(off)}$) creates a problem in any JFET design. To deal with that, we choose a JFET with a tight $V_{GS(off)}$ specification (-1.2 to -2.7 V), and we run it at a drain current well below I_{DSS} (10 mA minimum) so that the gate-source voltage is close to $V_{GS(off)}$. The feedback path sets the signal gain. With some careful thought (and some juggling and iteration), the same feedback path can be made to establish the (dc-coupled) bias condition.

Here's how it goes. The gate is at ground; we start by assuming a source voltage of approximately 1.7 V, and choose R_1 for 500 μA. Of that, about 300 μA comes from the JFET's drain current (Q_2's V_{BE} across R_2); thus

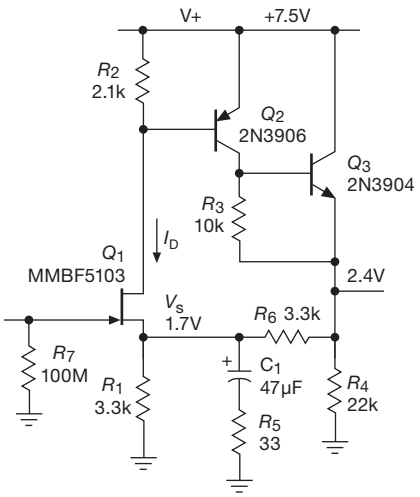


Figure 3.31. JFET–BJT series-feedback pair.

$\sim$200 μA is coming from R_6. This puts the output at about +2.4 V and sets Q_3's emitter current at about 300 μA (110 μA through R_4, plus 200 μA through R_6, minus 60 μA from R_3).

That's the self-consistent situation, under the assumed V_{GS} of -1.7 V. For a different V_{GS}, the dc output voltage would change accordingly; it could range from +1.3 to +4 V over the extreme specified range of gate threshold voltage. This would degrade the maximum possible output swing, but that would usually be OK for an amplifier handling small signals (if not, R_1 could be selected for different V_{GS} ranges of JFET parts).[36]

The nominal gain at signal frequencies is approximately 100, set by R_5 (blocked by C_1): $G = 1 + R_6/(R_5 \| R_1)$. The low-frequency -3 dB point is at 100 Hz (where the reactance of C_1 equals R_5). The high-frequency -3 dB point is not so easily calculated, but a SPICE model puts it at approximately 800 kHz (it measured 720 kHz in our breadboard, where there are some added parasitic capacitances). The latter is due primarily to the *RC* rolloff of Q_1's output signal impedance of 2.1 kΩ (i.e., R_2) driving Q_2's input capacitance of $\sim$4 pF, with the latter greatly magnified by the Miller effect.

For large signal driving impedances R_{sig} the amplifier's

[35] The effective capacitance of an LM334 current source/sink is 10 pF, small enough to ignore for most purposes. We calculated this from the slew-rate plot in the datasheet. TI's application note LB-41 has additional useful information about the LM334.

[36] A better way to ensure proper biasing is to replace R_1 with a 0.5 mA current sink. A JFET comes to mind (we've got JFETs on the mind!), but, given their unpredictable dc characteristics, a much better choice would be one of the BJT current sinks illustrated in Figure 3.26. Another way of handling this thorny problem is to use a slow feedback loop to stabilize I_D to a desired value less than the specified minimum I_{DSS}.

bandwidth is reduced,[37] owing to an input capacitance of ∼5 pF; see Figure 3.32. This is due primarily to the JFET's drain-to-gate capacitance (plus wiring capacitance), given that the source terminal is bootstrapped by feedback. There are numerous tricks to deal with this effect (if more bandwidth is desired), including a cascode in the JFET's drain (which can be bootstrapped to further suppress its input capacitance) and a cascode in the Q_2 BJT gain stage. Some of these techniques are discussed in Chapter *3x*.

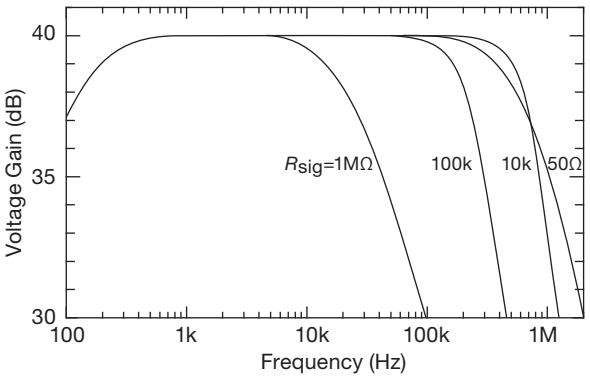


Figure 3.32. Measured gain versus frequency for the amplifier of Figure 3.31. The f_{3dB} with R_{sig}=1 MΩ shows that C_{in}=7 pF.

Design equations and design hints

Collecting it together in one place:

$$G = 1 + \frac{R_6}{R_5 \parallel R_1} \approx 1 + \frac{R_6}{R_5} \quad \text{(ac gain)},$$

$$G_{OL} = g_{m1} R_2 \, g_{m2} R_3 \, g_{m3} (R_4 \| R_6), \quad \text{(open-loop gain}^{38}\text{)}$$

$$I_D = \frac{V_{BE2}}{R_2} \approx \frac{0.7}{R_2} \quad \text{(JFET bias)},$$

$$I_{C2} = \frac{V_{BE3}}{R_3} \approx \frac{0.65}{R_3} \quad (Q_2 \text{ bias}),$$

$$V_{out} = V_S \left(1 - \frac{R_6}{R_1}\right) + \frac{R_6}{R_2} V_{BE2} + \frac{R_6}{R_1} |V_{EE}| \quad \text{(output bias)}.$$

For single-supply operation the last term is zero. Basically, R_2 sets I_D, and the ratio R_6/R_1 sets V_{out}. For single-supply operation (V_{EE}=0) use a small value for R_6 if the JFET has substantial V_{GS} at its operating current and larger values for R_6 for lower V_{GS} parts. The latter is tricky, because the "leverage" of R_6/R_1 can push V_{out} all over the map. Choose R_4 to help set I_{C3} after you've dealt with V_{out}. It may be necessary to select R_1 to go with batches of parts with similar V_{GS}. A negative V_{EE} supply helps with biasing and also permits output swings both sides of ground.

Another way to handle the uncertainty in V_{out} is to apply a positive bias to the gate, as in Figure 3.33A. This adds a positive offset of V_B at the source terminal (whose voltage is now $V_S = V_B - V_{GS}$, where V_{GS} is negative for an *n*-channel JFET), making V_{GS} less important, and thus a smaller fractional uncertainty in V_S. While you're at it, you can easily bootstrap the gate bias divider,[39] as in Figure 3.33B, to raise the input impedance.

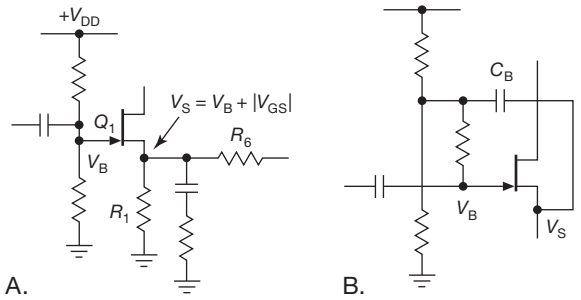


Figure 3.33. A. A positive bias at the gate of Q_1 in Figure 3.31 improves the predictability of V_{out}. B. Add a bootstrap to raise R_{in}.

E. Simple "hybrid" JFET amplifier

With the assistance of an op-amp (the magnificent centerpiece of Chapter 4) you can do wonders. Put simply, an op-amp is a "very high-gain difference amplifier in a bottle," intended to be fodder for feedback as the universal core of pretty much any analog circuit. It's *pure engine*: a turbocharged Harley unicycle with dual intakes. This example and the next show a couple of ways to use an op-amp's properties in support of a JFET amplifier. Look first at Figure 3.34. Here we've chosen the excellent 2SK170B (with LSK170B as a second source) for the front-end: it has lots of transconductance (about 25 mS at its I_{DSS} of 6–12 mA), along with very low noise voltage (∼1 nV/$\sqrt{\text{Hz}}$). We run it at zero gate voltage; and we deal with the 2:1

[37] Though paradoxically for R_{sig} values of a few kΩ it is *extended* somewhat, owing to some response "peaking."

[38] This expression overestimates the open-loop gain by neglecting the gain-limiting Early effect in Q_2, the stage where most of the circuit's overall gain resides. The measured 2N3906 Early voltage V_A≈25 V (§*2x.8*) implies a maximum voltage gain of the Q_2Q_3 stage of ∼1000 (compared with its ideal G≈2500), thus an overall open-loop gain of ∼5000. This is ample for the modest ×100 closed-loop gain.

[39] You can, of course, bootstrap the gate resistor even when the gate is biased at ground, via R_7 in Figure 3.31.

spread of specified I_{DSS} (a tighter specification than provided for most JFETs) by choosing the drain load resistor R_D small enough to avoid dc saturation even at I_{DSS} (max). The actual drain voltage is unimportant because we use ac coupling to the second stage (via C_1). Ignoring for the moment the second stage (and setting $R_g = 0$), the front-end voltage gain would be $G = g_m R_D$, or roughly $G \approx 25$, with perhaps $\pm 25\%$ uncertainly from JFET manufacturing process variations.

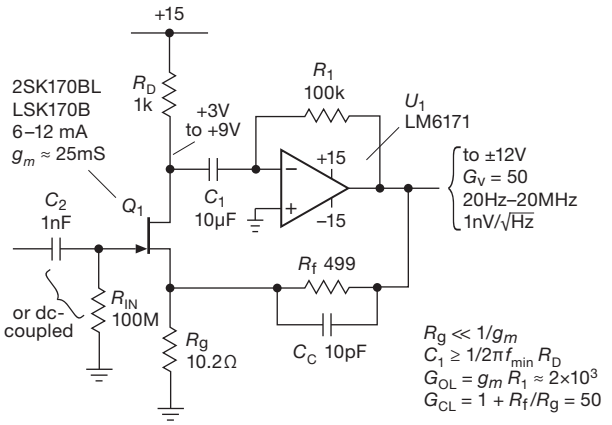


Figure 3.34. Hybrid JFET amplifier: high-Z, low-noise, wideband amplifier. A response peak (for $R_{sig} \sim 1$kΩ) can be tamed by adding a 10–20 pF capacitor across the input, see Figure 3.35.

But the load seen by the drain (at signal frequencies) is in fact the low-impedance input of the second stage, which is an op-amp configured here as a current-to-voltage ("transresistance") converter (see §4.3.1C). Its "gain" (ratio of output voltage to input current, thus units of resistance) is just R_1, making the overall open-loop gain $G = g_m R_1$ (again assuming no feedback, and $R_g = 0$). So, for the circuit values shown, the open-loop gain $G_{OL} \approx 2500$.

Now we close the loop via R_f, subtracting from the input a fraction $R_g/(R_g + R_f)$, for an ideal closed-loop gain $G_{CL} = 1 + R_f/R_g = 50$. The loop gain (ratio of open-loop to closed-loop gains) is about 50, adequate for good linearity and predictability of gain (see §2.5.3). Note the low resistance of R_g: it should be small enough so that the open-loop gain is not much reduced (thus $R_g < 1/g_m$); and it should also be small enough so that its Johnson noise contribution is insignificant (§8.1). For a JFET transconductance of 25 mS, the first constraint limits R_g to somewhat less than 40 Ω; and for this JFET's noise voltage of ~ 1 nV/$\sqrt{\text{Hz}}$ the second constraint limits R_g to somewhat less than 25 Ω. Thus we choose the low $\sim 10\,\Omega$ value. With the values shown, the open-loop gain is reduced by $\sim 20\%$, the input noise voltage is increased by $\sim 8\%$, and the feedback network loads the op-amp to ± 20 mA at full swing.[40]

The op-amp here was chosen for its wide bandwidth (it falls to unity gain at ~ 100 MHz), so the circuit's closed-loop gain rolls off at about 20 MHz,[41] as seen in the measured data of Figure 3.35. A bonus is the impressive output drive capability: up to 100 mA, and full ± 10 V swing to nearly 10 MHz. The small compensation capacitor C_c enhances stability: with no compensation we measured 5 dB of peaking at 16 MHz; adding C_c resulted in an insignificant 0.1 dB peaking at 10 MHz and a high-frequency rolloff of -3 dB at 22 MHz.[42]

An alternative circuit that exploits the JFET's high input impedance is an input JFET source follower (§3.2.6) driving a stage of voltage gain. That's a perfectly good configuration, particularly if a current sink is used for the source pulldown. But the circuit of Figure 3.34 excels in achieving both lower noise and better linearity.

3.2.4 Differential amplifiers

Thus far we've dealt with the JFET's uncertain I_D versus V_{GS} relationship by restricting ourselves to ac-coupled amplifier designs. But we can do better: a JFET matched pair lets us make dc-coupled amplifiers of respectable performance. And of course their very low input current means that these circuits can serve as high-input-impedance front-end stages for bipolar differential amplifiers, as well as for the important op-amps and comparators we'll meet in the next chapter. As we mentioned earlier, the substantial V_{GS} offsets of FETs will generally result in larger input voltage offsets and offset drifts than with a comparable amplifier

[40] If you're not happy with that, you could double or triple R_f and R_g, at the expense of slightly higher amplifier noise. See §8.6 for much more on low-noise JFET design.

[41] When driven with a low signal impedance. With a higher impedance signal source, the rolloff is dominated by the circuit's input capacitance: Q_1 has a drain-to-gate capacitance C_{rss} of 6 pF, thus the observed -3 dB bandwidth of ~ 400 kHz with a 100kΩ source. Happily, the larger gate-to-source capacitance ($C_{iss} \approx 30$ pF) is bootstrapped into insignificance by feedback. "And what of the evil Miller effect?" you wonder. That's suppressed here, because the op-amp holds its $(-)$ input fixed (a "virtual ground") via feedback through R_1; see §4.3.1C.

[42] In any feedback circuit there's the possibility of some "peaking" in the response at some frequency (or, in the worst case, a full-blown oscillation). This circuit exhibits modest peaking for input impedances in the range of a few kΩ, as seen in Figure 3.35. Its amplitude can be tamed by adding ~ 10–20 pF of input shunt capacitance, with some consequent reduction of bandwidth.

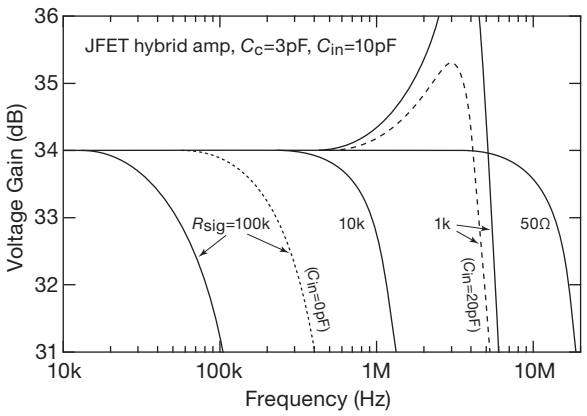


Figure 3.35. Measured gain versus frequency for the amplifier of Figure 3.34. The solid curves are with a 10 pF shunt capacitor across the input, a compromise value for good performance over a wide range of input signal impedances (omit the capacitor for high source impedance, increase it for the worst-case $R_{sig} \approx 1k\Omega$).

constructed entirely with bipolar transistors, but of course the input impedance will be raised enormously.

Figure 3.36 shows the simplest configurations, analogous to the simple BJT differential amplifiers of Figures 2.63 and 2.67. The differential gain of the classic long-tailed pair in Figure 3.36A (defined as $\Delta V_{out}/\Delta V_{in}$, with $R_1 = R_2$ and differential output as shown) is just $G{=}g_m R_1$; the common-mode rejection is greatly improved with a current sink substituting for the source resistor R_S. A drawback of this circuit is the uncertainty in gain (owing to the uncertainty in transconductance); and the gain is modest, owing to the limited transconductance of JFETs in general. You can circumvent the gain limitation by replacing the drain load resistor(s) with a current mirror, as in Figure 3.36B. This circuit fragment is not bias stable, however: it must be accompanied by a following stage that is configured to provide dc feedback.

These circuits suffer also from the Miller effect (§2.4.5), whose multiplying action on the feedback capacitance C_{rss} acts to increase the effective input capacitance, thus (in combination with the signal's source impedance) reducing the bandwidth. The current mirror in circuit (B) clamps Q_1's drain, but the Miller effect is present still at Q_2's input.[43] As with BJT amplifier stages, an effective method of eliminating the Miller effect is the use of a cascode transistor (either JFET or BJT) in the drain(s), a desirable configuration discussed later and in Chapter *3x* in the section "Bandwidth of the Cascode.". By clamping the drain voltage, the cascode also eliminates a reduction in gain seen in simple circuits like that of Figure 3.36A: that's because the JFET's drain-current depends somewhat on drain voltage (the upward slope in plots of drain-current versus drain-source voltage, which you can think of as a finite output impedance), which can reduce the ideal $g_m R_1$ gain by as much as 25%.

For this and other reasons the cascode is highly recommended, even when bandwidth is not an issue. Curious about those "other reasons"? Consider this: the gate current in JFETs, normally down in the picoamperes, rises precipitously with drain-to-source voltage – see Figure 3.49 on page 164 where picoamps become *micro*amps! A cascode lets you clamp the drain to a low operating voltage, suppressing this effect. This is nicely illustrated in the figure toward the end of §*3x.4* in Chapter *3x*.

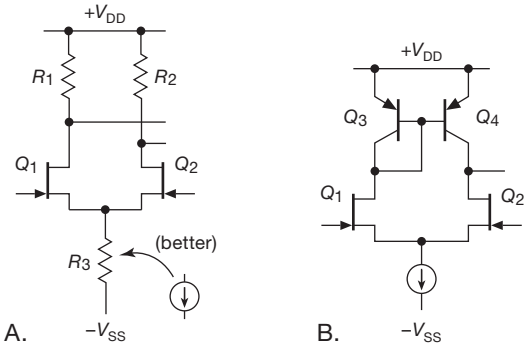


Figure 3.36. Simplest JFET differential amplifiers.

A. Example: a dc-coupled hybrid JFET amplifier

In the earlier hybrid amplifier design example (Figure 3.34) we dodged the issue of unpredictable JFET drain current by settling for an ac-coupled amplifier. That's fine for something like an audio or RF amplifier; but sometimes you'd like response all the way down to dc.

You can achieve that by exploiting a matched-pair JFET differential input stage in a fully dc-coupled arrangement, as in Figure 3.37. The overall configuration is a common-source differential amplifier with the input signal connected to one side. Its differential voltage output drives a wideband op-amp, whose output (divided by the gain-setting voltage divider) provides negative feedback to the other terminal of the input pair. As with the previous circuit, the closed-loop gain is $G_{CL}{=}1 + R_f/R_g{=}50$, with a small loop compensation capacitor C_c chosen for best response without peaking.

The devil's in the details, which, if you're lucky,

[43] Unless that output drives a transimpedance stage that clamps its voltage, as in Figure 3.31 or 3.34.

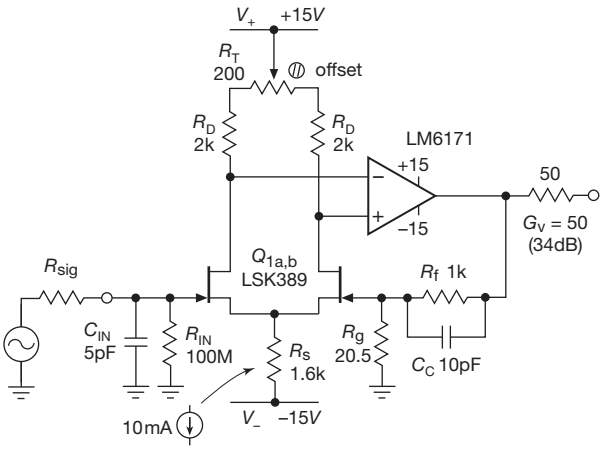


Figure 3.37. An op-amp closes the loop around a JFET matched pair to create a dc-coupled wideband low-noise amplifier with high input impedance. Replace source bias resistor R_S with a current sink (see Figure 3.26) for enhanced supply rejection.

may come together in a symphonic harmony. This circuit worked out nicely; thus our enthusiasm for a bit of discussion. Here goes.

The LSK389 is a very low-noise ($\sim$1 nV/$\sqrt{\text{Hz}}$) monolithic matched JFET pair,[44] available in three I_{DSS} grades. We chose the midcurrent –B suffix part (with a specified I_{DSS} between 6 mA and 12 mA), and we force each JFET of the pair to run at 5 mA by sinking 10 mA from the pair of source terminals. That puts the drains at +5 V (the op-amp enforces equality), with an open-loop gain (from single-ended input to differential output) of approximately $G{=}g_m(R_D + 0.5R_T){\approx}40$. To this the op-amp contributes its substantial open-loop gain (90 dB, i.e., $\times$30,000), in a scary-looking configuration whose stability might seem seriously to be in doubt. But not to worry – the $\div$50 divider in the return loop safely limits the loop gain,[45] with any tendency toward instability easily tamed with appropriate choice of C_c.

According to the datasheet, the input pair Q_{1ab} has "Tight Matching." But that's on the scale of JFETs, not BJTs – here that "matching" is a whopping $\pm$20 mV maximum (about 100$\times$ the matching of a good BJT pair), which the circuit's $G = 50$ would amplify to an output offset of 1 V! Hence the offset trim R_T, with enough range to balance the worst-case input offset.

Our initial design included a bypass capacitor from the drain of the input transistor to ground, to suppress the Miller effect. A nice thought, but the reality is that the Miller effect is almost entirely absent owing to the op-amp's clamping effect on the drain pair. And the bypass capacitor introduces two problems: it unbalances the transistor pair, so the circuit is sensitive to noise on the positive supply rail; and it introduces a phase shift within the loop, causing some undesirable peaking, thus requiring a larger value of C_c and therefore reduced bandwidth.

Now for the feedback stability "compensation." In a single-ended circuit like Figure 3.34 you can put a small capacitor across either R_1 or across R_f. Here, though, we want to maintain input-stage symmetry, so C_c has to go across R_f. On the bench we found that 10 pF was required to eliminate peaking in the frequency response, when tested with a low-level sinewave input; Figure 3.38 shows the data.

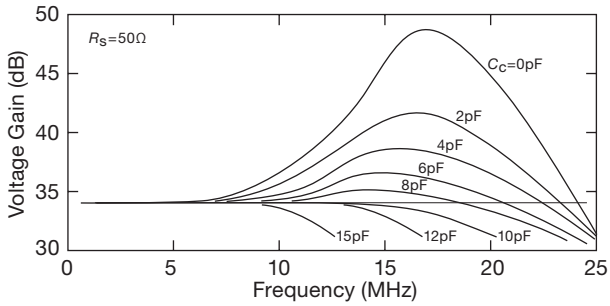


Figure 3.38. Selecting the compensation capacitor C_c for the amplifier of Figure 3.37. A value of 8 or 10 pF works well.

A final bit of compensation is the shunt capacitor C_{in} across the input to suppress some modest peaking that is seen for signal sources of $\sim$1kΩ impedance (R_{sig}). Here 5 pF did the job nicely (though it increased the circuit's input capacitance to $\sim$20 pF). Figure 3.39 shows the measured gain versus frequency for the finished circuit (with the component values shown in Figure 3.37), for nine values of R_{sig} (spanning four decades of resistance).

As with the previous circuit, the LM6171 op-amp provides a full $\pm$10 V output swing to nearly 10 MHz. The 50Ω resistor in series with the output ensures stability into capacitive loads; it also provides "back termination" into 50 Ω coax cable (see the Appendix H on Transmission Lines). This op-amp is not particularly quiet ($e_n{\sim}12$ nV/$\sqrt{\text{Hz}}$), but that's good enough: the circuit's input-stage gain of $\sim$40 reduces the op-amp's noise contribution to $\sim$0.3 nV/$\sqrt{\text{Hz}}$ when referred to the input. The amplifier's overall noise is $\sim$2 nV/$\sqrt{\text{Hz}}$.[46]

[44] A replacement for Toshiba's legendary (and discontinued) 2SK389.

[45] To approximately the same value as that of the op-amp alone, were it connected as a stable unity-gain follower.

[46] If it's *quiet* you want, you can substitute the large-die IF3602 dual

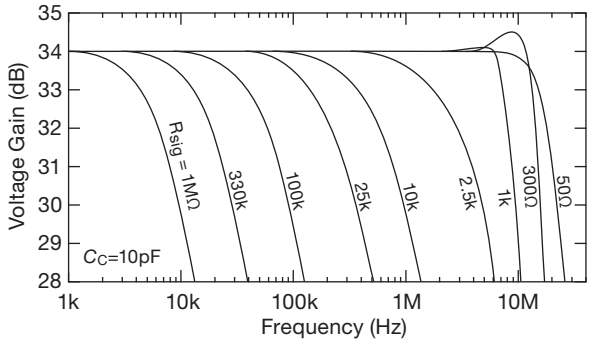


Figure 3.39. Measured gain versus frequency for the amplifier of Figure 3.37, for a range of values of signal source impedance R_{sig}.

This is a pretty good amplifier! With a bit of refinement (primarily substituting a low-noise current sink for R_S), two such amplifiers, combined in a so-called "instrumentation amplifier" (INA) configuration, will outperform any available integrated INA in terms of both noise and speed; see Figure 8.49 in §8.6.3.

B. Comparison with JFET-input op-amps

We can hear it already: "Yeah, yeah... you guys like to show off your clever circuits, with lots of discrete components. But nowadays that kind of circuit art is obsolete, because you can get all that performance, and more, in readily available integrated circuits – op-amps, in particular."

But, *can* you? Table 3.2 lists the currently available op-amps with JFET inputs that have a chance of competing. How does their performance compare with that of our dc-coupled hybrid amplifier of Figure 3.37? Let's see...

Output swing Only the first three op-amps can swing over the full ±15 V range; OK, but...

Bandwidth ...the fastest of those "high-voltage" op-amps has 80 MHz GBW, thus for G=50 a bandwidth of less than 2 MHz; and the two fastest op-amps, which can match our amplifier's bandwidth, can swing only ±4 V or so. By contrast, our amplifier's GBW is 4 GHz (40 times the op-amp's f_T=100 MHz), and thus excess gain (e.g., 400 at 10 MHz) with lower resulting distortion.

Noise Our amplifier's noise of $\sim 2\,\text{nV}/\sqrt{\text{Hz}}$ is 6 dB quieter than the best of Table 3.2's offerings.

Cost $5–10 for the op-amp solution, roughly the same for the better-performing hybrid amplifier ($2.50 for the LM6171, $3.25 for the LSK389 dual JFET)

JFET. That will reduce the input noise to $\sim 0.7\,\text{nV}/\sqrt{\text{Hz}}$, but with a greatly increased input capacitance (about 300 pF!). (And thermal noise from the feedback network will degrade this unless R_g is decreased to ~5 Ω; see §8.1 and Figure 8.80A.)

Table 3.2 Selected Fast JFET-input Op-amps[a]

Part #	Supply Voltage range (V)	Supply I_Q typ (mA)	I_{bias} 25°C typ (pA)	e_n 1kHz typ (nV/√Hz)	GBW typ (MHz)	Slew Rate typ (V/μs)	Cost qty 25 ($US)
OPA604A	9–50	5	50	10	20	25	2.93
OPA827A	8–40	5	15	4	22	28	9.00
ADA4637	9–36	7	1	6	80[d]	170	10.12
OPA656	9–13	14	2	7[b]	230	290	5.59
OPA657	9–13	14	2	7	1600[d]	700	10.01
ADA4817	5–10.6	19	2	4[c]	1050	870	4.93

Notes: (a) candidates for wideband low-noise amplifier. (b) low e_nC noise: C_{in}=2.8pF. (c) lowest e_nC noise: C_{in}=1.5pF. (d) decomp, G_{CL} >7.

The contest so far? The hybrid amplifier is winning, in terms of the combined performance metric of bandwidth, output swing, and noise voltage. But we're not done yet...

Offset voltage The op-amps win, here, with out-of-the-box V_{os} values of 2 mV (max) for the three fastest parts; the hybrid amp requires manual trimming of its 20 mV worst-case untrimmed offset, if better is needed.

Parts count A win, again, for the op-amps.

Input capacitance Just 1.5 pF for the ADA4817, versus 10 pF or more for the hybrid (the price we paid for 2× lower noise).

Input current 20 pA (max) for the ADA4817 (but that's a low-voltage part), versus 200 pA for the hybrid (*unfair!* – that's specified at a large negative bias, $V_{GS} = -30$ V)

The verdict? A split decision: the JFET op-amp solution is simple and can deliver plenty of speed (or plenty of swing, but not both), along with untrimmed accuracy and very low-input capacitance (thus low "e_nC" noise; see Chapter 8). The hybrid approach delivers speed and swing and lowest noise voltage; but it requires manual trim, it's more complicated, and it has more input capacitance. Note also that an op-amp is a more flexible building block in general, providing, for example, a wide common-mode input-voltage range that our hybrid circuit does not have; but that is not needed here, because the input is always close to ground (owing to the circuit's gain of 50).

3.2.5 Oscillators

In general, FETs have characteristics that make them useful substitutes for bipolar transistors in almost any circuit that can benefit from their uniquely high input impedance and low bias current. A particular instance is the use of a JFET amplifier stage to implement a high-stability *LC* or crystal oscillator; we'll show examples in §7.1.5D.

3.2.6 Source followers

Because of the relatively low transconductance of FETs, it's often better to use a FET "source follower" (analogous to an emitter follower) as an input buffer to a conventional BJT amplifier, rather than trying to make a common-source FET amplifier directly. You still get the high input impedance and zero dc input current of the FET, and the BJT's large transconductance lets you achieve high single-stage gain. Furthermore, discrete FETs (i.e., those that are not part of an integrated circuit) tend to have higher inter-electrode capacitance than BJTs, leading to greater Miller effect (§2.4.5B) in common-source amplifiers; the source-follower configuration, like the emitter follower, has no Miller effect.

FET followers, with their high input impedance, are commonly used as input stages in oscilloscopes as well as in other measuring instruments. There are many applications in which the signal-source impedance is intrinsically high, e.g., capacitor microphones, pH probes, charged-particle detectors, or microelectrode signals in biology and medicine. In these cases, a FET input stage (whether discrete or part of an integrated circuit) is a good solution. Within circuits there are situations in which the following stage must draw little or no current. Common examples are analog "sample-and-hold" and "peak detector" circuits, in which the level is stored on a capacitor and will "droop" if the next amplifier draws significant input current. In all these applications the advantage of negligible input current of a FET more than compensates for its low transconductance, making source followers (or even common-source amplifiers) attractive alternatives to the bipolar emitter follower.

Figure 3.40 shows the simplest source follower, which ideally should produce an accurate replica of the input waveform while drawing essentially zero input current. Let's figure out important things like its quiescent operating point, its exact voltage gain, its output impedance, and the voltage offset from input to output.

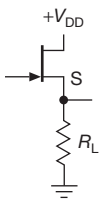


Figure 3.40. N-channel JFET source follower. Unlike the npn BJT emitter follower (in which the output trails the input by $V_{BE} \approx 0.6$ V), the output is here more *positive* than the input.

A. Quiescent operating point

The JFET source follower is not as simply analyzed as the analogous BJT emitter follower, in which the emitter voltage simply trails the base voltage by a relatively constant (and predictable) $V_{BE} \approx 0.6$ V. That is because the FET has less abrupt (and far less predictable) transfer characteristics (I_D versus V_{GS}) – the same issue we've just wrestled with in connection with the JFET current source (§3.2.2) and the JFET amplifier (§3.2.3).

We could use the same iterative approach here, seeking the quiescent source voltage V_S (and therefore $V_{GS} = V_S$) that produces a source current I_S (and therefore $I_D = I_S$) consistent with that V_S. And we could do this by interpolating between curves like those in Figure 3.21A (a family of I_D versus V_{DS} for several V_{GS}'s), or by sliding up and down a transfer characteristic curve like that in Figure 3.41 (a single curve of I_D versus V_{GS}, at some fixed V_{DS}), until we find the point at which $I_D R_L = -V_{GS}$.

But there is an elegant graphical method, used extensively during the days of vacuum tubes, that lets you find the operating point immediately: the method of "load lines."

B. Load lines

To find the operating point for the source follower in Figure 3.40, we simply notice that the load resistor R_L imposes its rules on allowable V_{GS} versus I_S, namely, Ohm's law: $I_S R_L = -V_{GS}$. We can plot this constraint on the same graph as the transfer curve of Figure 3.41, as a straight line with slope $-1/R_L$; note that it goes "backward,"

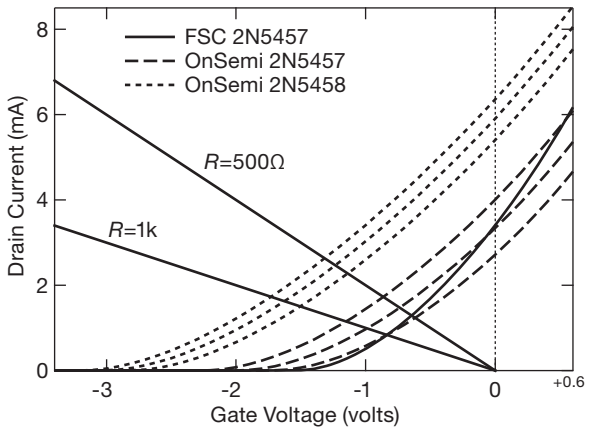


Figure 3.41. Measured transfer curves for a set of 2N5457 and 2N5458 n-channel JFETs at V_{DS}=10 V. These measurements extend beyond I_{DSS}, with V_{GS} taken 0.6 V into positive territory. The OnSemi curves show the parts with the lowest, middle, and highest I_{DSS} from a batch of ten each.

because $V_S = -V_{GS}$. The operating point has to be consistent with this constraint and simultaneously with the transfer characteristic of the JFET. In other words, the operating point is the intersection of the two curves. In this case, with $R_L = 1\text{k}$, the quiescent point is at $V_S = +1.6\,\text{V}$ (and, from the lowest 2N5458 curve, $I_D = 1.6\,\text{mA}$).

Lest one be tempted to fall too quickly in love with this technique, we hasten to point out that the characteristic curves for a particular type of JFET exhibit a large spread. For the 2N5458 illustrated in Figure 3.41, for example, the specification allows I_{DSS} to be anywhere between 2 and 9 mA (and the pinchoff voltage $V_{GS(off)}$ can range from -1.0 to $-7.0\,\text{V}$). In practice, it is rare to find devices at the extremes, and there tends to be good consistency in a single manufacturing batch (as indicated by the date code stamped on the parts); for example, by measuring a batch of 10 2N5458's (Figure 3.41) we determined that the quiescent point in this circuit would range from 1.52 to 1.74 V.

C. Output amplitude and voltage gain

We can figure out the output amplitude, as we did for the emitter follower in Section 2.3.3, using the transconductance. We have

$$v_s = R_L i_d$$

since i_g is negligible; but

$$i_d = g_m v_{gs} = g_m(v_g - v_s),$$

so

$$v_s = \left[\frac{R_L g_m}{(1 + R_L g_m)}\right] v_g.$$

That is, the gain is

$$G = \frac{1}{1 + \dfrac{1}{g_m R_L}}. \tag{3.7}$$

For $R_L \gg 1/g_m$ it is a good follower ($v_s \approx v_g$), with gain approaching, but always less than, unity. We are not near that limit in this example, in which the measured value g_m=1.9 mS implies a voltage gain of G_V=0.66 into the 1 kΩ load, far from the ideal of unity gain. Furthermore, the variation of transconductance over the signal swing results in undesirable nonlinearity. One solution is to use a JFET with higher transconductance, or (better) add a BJT transconductance enhancer (Figure 3.29F and §*3x.2*). But in situations in which the external load impedance is high, an elegant solution is to use a current sink as an active load, as we'll see presently (§3.2.6F).

D. Input impedance

Our hope that JFET source followers have infinite input impedance is largely fulfilled, but they do have some gate leakage current (see §3.2.8) and input capacitance (see Table 3.7 on page 217). Gate leakage can become a problem at drain-gate voltage greater than about 5 V (Figure 3.49), so be sure to check the JFET's datasheet and, if necessary, consider adding a cascode to limit V_{DG}.

A follower's frequency response when driven by signals of high source impedance is limited by input capacitance, $f_{3dB} = 1/2\pi C_{in}$, where $C_{in} = C_{iss} + C_{rss} + C_{stray}$. The gate-source capacitance C_{iss} is generally about two to five times higher than the gate-drain capacitance C_{rss}, but fortunately it's bootstrapped by the follower action and is effectively reduced to $(1 - G_V)C_{iss}$. If you follow our advice (below) so that G_V is nearly 1.0, only the JFET's C_{rss} remains to limit the bandwidth. But it's possible to bootstrap the drain and reduce the effect of C_{rss} by a factor of 5. This leaves C_{stray} as "the last man standing" to limit the bandwidth – but you may be able to knock him down too by "guarding" most of the input wiring capacitance (i.e., using the follower's output signal to drive the cable's shield; see the discussion of signal guarding in §5.15.3).

E. Output impedance

The preceding equation for v_s is precisely what you would predict if the source follower's output impedance were equal to $1/g_m$ (try the calculation, assuming a source voltage of v_g in series with $1/g_m$ driving a load of R_L). This is exactly analogous to the emitter-follower situation, in which the output impedance was $r_e = 25/I_C$, or $1/g_m$. It can be easily shown explicitly that a source follower has output impedance $1/g_m$ by figuring the source current for a signal applied to the output with the gate grounded (Figure 3.42). The drain current is

$$i_d = g_m v_{gs} = g_m v,$$

so

$$r_{out} = v/i_d = 1/g_m, \tag{3.8}$$

typically a few hundred ohms at currents of a few milliamps.[47]

In general, FET source followers aren't nearly as stiff as emitter followers. The exception is at very low currents,

[47] In practice, a more convenient way to measure the follower output impedance is to inject a signal *current* and measure the resulting source voltage, as in Figure 3.42B. Get the current from a signal generator, with a series resistor R_{sig} much larger than r_{out}, taking care to keep v_{out} small, say ∼50 mV; then the equation in the figure gives you r_{out}.

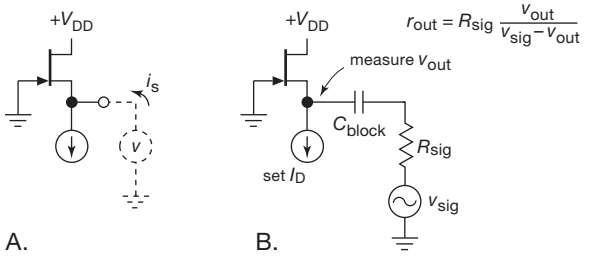


Figure 3.42. Calculating source-follower output impedance.

in the subthreshold region, where the transconductance of some JFETs approaches that of a BJT operated at the same current; see Figure 3.54 on page 168.

In this example, with $g_m = 1.9\,\text{mS}$, the impedance looking back into the JFET's source is $r_{out} = 525\,\Omega$, which combines with the parallel 1k source load resistor to produce an output impedance of 345 Ω, quite a bit higher than the analogous value of $r_e = 16\,\Omega$ for a BJT operating at the same 1.6 mA.

We were able to calculate the voltage gain and output impedance reasonably accurately in this example because we took the trouble to measure the I_D versus V_{GS} characteristic curves. It's worth pointing out, however, that the manufacturer's datasheet for the 2N5458 gives us little help here: it gives no characteristic curves for the 2N5458, only for the lower-current part (2N5457); and for the 2N5458 it specifies g_m only at I_{DSS}, where it gives a range of 1.5 mS to 5.5 mS. From these limits, along with the I_{DSS} and $V_{GS(OFF)}$ limits above, we would not be able to form a good estimate of the operating transconductance, because the operating point with a fixed value of source load resistor is ill-determined. We could do better by assuming that we adjust R_S to make I_D=1.6 mA, say; then, using the fact that $g_m \propto \sqrt{I_D}$, the specified limits of I_{DSS} and of g_m (at I_{DSS}) guarantee that g_m lies in the range 0.6 mS to 4.9 mS.[48] Our measured value of g_m falls nicely within this range, being rather close to the geometric mean of these limits.

There are two drawbacks to this circuit.

1. The relatively high output impedance means that the output swing may be significantly less than the input swing, even with high load impedance, because R_L alone forms a divider with the source's output impedance. Furthermore, because the drain current is changing over the signal waveform, g_m and therefore the output impedance will vary, producing some nonlinearity (distortion) at the output. The situation is improved if FETs of high transconductance are used, of course, but a combination FET–BJT follower (or FET–BJT "g_m enhancer," Figure 3.29F) is often a better solution.
2. Because the V_{GS} needed to produce a certain operating current is a poorly controlled parameter in FET manufacture, a source follower has an unpredictable dc offset, a serious drawback for dc-coupled circuits.

(An additional problem is caused by the fact that an FET's drain current depends to some degree on drain-to-source voltage. You might call this the "g_{os} effect," which also acts to reduce the gain from the ideal $G = 1$. It is discussed later in §§3.3.2 and in the Chapter *3x* section "Bandwidth of the Source Follower with a Capacitive Load.")

Perhaps this is a good place to pause and realize that many of the circuits we've been considering would be easier to implement, and would work better, if we had access to a negative supply voltage. But often that's not the case; so, in the spirit of real-world circuit design constraints (and as a useful learning exercise) we're slogging through the extra difficulties posed by single-supply JFET follower design. But if you do have a negative supply available, by all means use it!

F. Active load

The addition of a few components improves the source follower enormously. Work with us here, as we take it in stages (Figure 3.43).

First we replace the load resistor (called R_S in Figure 3.43A) with a (pull-down) current sink (circuit B). (You can think of this as the previous case, with infinite R_S.) The constant source current makes V_{GS} approximately constant, thus reducing nonlinearities. A nice trick (circuit B′) has a BJT follower doing double duty, both providing low output impedance while sinking a (roughly) constant current of V_{BE}/R_B.

We still have the problem of unpredictable (and therefore nonzero) offset voltage (from input to output) of V_{GS} (or $V_{GS} + V_{BE}$, for circuit B′). Of course, we could simply adjust I_{sink} to the particular value of I_{DSS} for the given FET (in the first circuit) or adjust R_B (in the second). This is a poor solution for two reasons: (a) it requires individual adjustment for each FET; and (b) even so, I_D may vary by a factor of two over the normal operating temperature range for a given V_{GS}.

A better circuit uses a matched FET pair to achieve zero offset (circuit C). Q_1 and Q_2 are a matched pair, on a single chip of silicon, for example the excellent LSK389 (see

[48] In fact, one can narrow that estimate somewhat, because g_m and I_{DSS} are correlated: a JFET sample with unusually high g_m will lie at the high end of the I_{DSS} distribution as well.

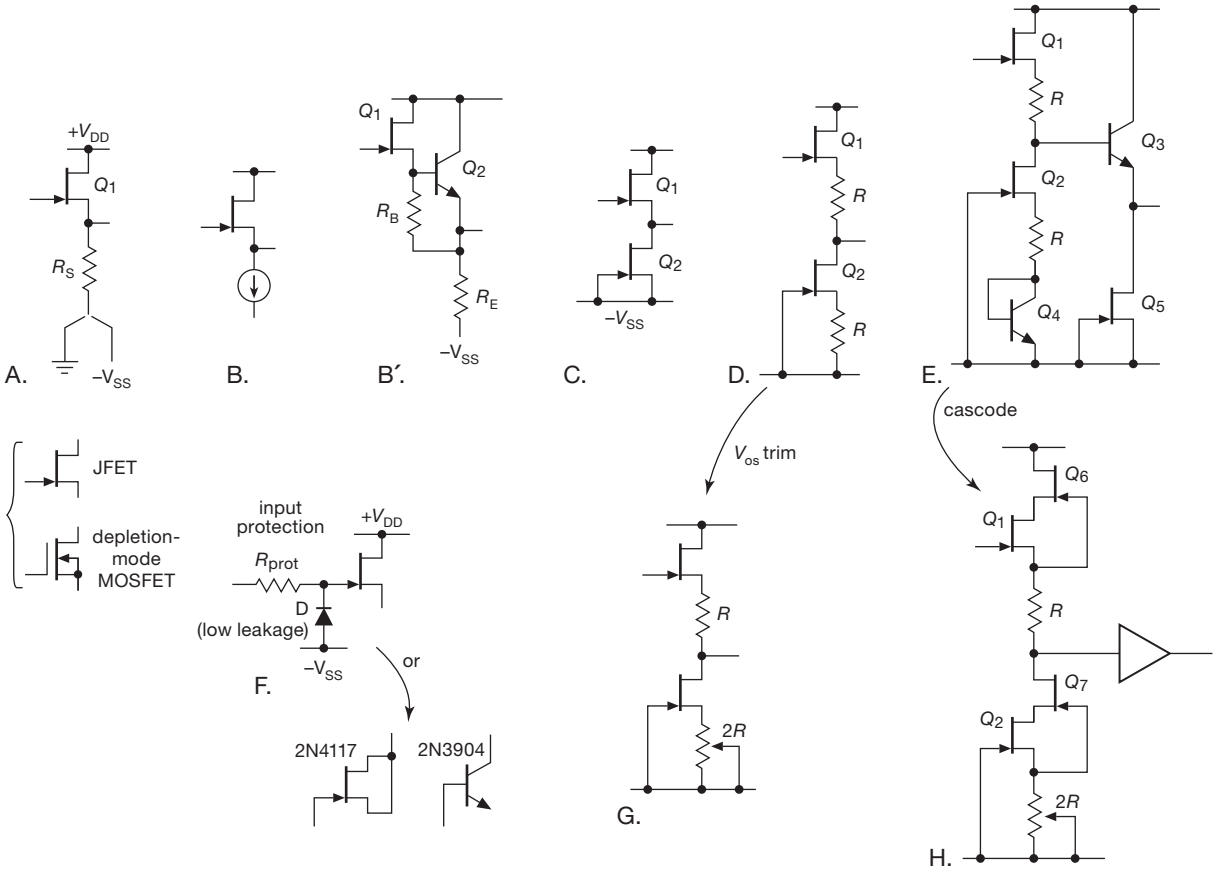


Figure 3.43. JFET unity-gain source followers – from simplest to best.

Table 3.7 on page 217), with Q_2 sinking a current of I_{DSS}, i.e., its drain-current corresponding to V_{GS}=0. But the JFETs are matched, so V_{GS}=0 for both transistors: voilà, Q_1 is a follower with zero offset. Because Q_2 tracks Q_1 in temperature, the offset remains near zero, independent of temperature.

You usually see the preceding circuit with source resistors added (circuit D). A little thought should convince you that the upper resistor R is necessary and that equal-value resistors guarantee that $V_{out} = V_{in}$ if Q_1 and Q_2 are matched. This circuit modification gives better I_D predictability, it allows you to set the drain current to some value less than I_{DSS}, and the source degeneration gives improved linearity. The variation of circuit G lets you trim the (already small) residual offset voltage caused by imperfect matching of the Q_1Q_2 pair; the LSK389, for example, specifies a worst-case mismatch (at 1 mA drain current) of $\Delta V_{GS} = 20$ mV.[49]

Circuit E adds a BJT output follower (Q_3), with a JFET current sink pull-down (Q_5). Transistor Q_4 adds a compensating V_{BE} in Q_2's source to maintain approximately zero dc offset from input to output.

Circuits A–D all share a problem, namely, the drain-to-source voltage across Q_1 varies directly with input signal. This can cause several undesirable effects. For example, imagine that circuit C is run between ± 10 V supply rails and that the input signal swings between +5 and -5 V.

[49] But a tricky "gotcha": the offset specification assumes equal V_{DS} for the two JFETs, but in circuit D there is a V_{DS} mismatch that depends on the input signal voltage relative to the supply rails. To estimate the resulting offset voltage of the follower you need to know the JFETs' output conductance (g_{os}), which causes an input–output offset in the follower that is proportional to the V_{DS} mismatch. That parameter is not specified on this JFET's datasheet, but from our measurements (see Table 3.7 on page 217) we know that $g_{os} \approx 100\,\mu$S, which causes a follower offset of $\Delta V = \Delta V_{DS}/G_{max}$; here that amounts to $\approx$60 mV for a 10V difference in V_{DS}, quite a bit larger than the 20 mV maximum untrimmed offset of the JFET pair (when the V_{DS}'s are balanced). The cure? Circuit H, a cascode in each JFET to hold each V_{DS} constant. Thunderous applause, yet again, for the remarkable cascode.

At the positive signal peak, Q_1 has less than 5 V from drain to source, while Q_2 has more than 15 V. Because a FET's drain current (at fixed V_{GS}) varies slightly with drain-to-source voltage (discussed in §§3.3.2 and *3x.4*), consequences here are a departure from strict unity gain and (worse) a potential nonlinearity; another consequence is that the input gate current can rise dramatically at drain-to-source voltages greater than 5 V (see Figure 3.49 on page 164), seriously degrading the otherwise admirably low input current.

An excellent solution to these problems (and others!) is the cascode configuration, as in circuit H. Here we've added JFETs Q_6 and Q_7, which need not be matched, but which must be chosen to have a V_{GS} larger than the minimum desired V_{DS} of Q_1 and Q_2. The cascode transistors bootstrap V_{DS} of Q_1 and Q_2 to a voltage equal to the V_{GS} of Q_6 and Q_7 while passing through the drain currents. So Q_1 and Q_2 operate at constant (and low) V_{DS}, with the cascode transistors taking up the slack as the signal swings, thus addressing both problems described in the preceding paragraph. The results are dramatic, as we'll see presently in a low-distortion "case study."

A further improvement on these JFET follower circuits is the addition of a *pnp* transconductance enhancer in Q_1's drain circuit (as in Figure 3.29, where the greatly increased transconductance rescues an otherwise mediocre amplifier stage); this is particularly helpful if the follower is to drive a relatively low load impedance. This is fleshed out in more detail in §*3x.2*.

JFETs can handle plenty of forward gate current, but they are easily damaged by reverse breakdown. When that possibility exists, it's a good idea to add gate protection, as in circuit F. The series resistor R_{prot} limits current through the clamp diode D (which should be a low-leakage part like the 1N3595, if low input current is important). You can use the base–collector junction of an ordinary BJT, or the gate-channel diode of a JFET; see the plot of measured diode reverse leakage currents in §*1x.7*. But there's a compromise here: a large value of R_{prot} safely limits the clamp current, but it introduces excessive Johnson (thermal) noise, a serious issue in low-noise applications. The use of a depletion-mode MOSFET current limiter solves this problem elegantly; see §5.15.4 for details.

Note that the JFETs in these examples can be replaced with depletion-mode MOSFETs, which are available with voltage ratings to 1000 V; in that case it's necessary to protect the gate against both forward and reverse overvoltages greater than ±20 V.

In a further variation of these circuits, you can use the output signal to drive an inner "guard" shield in order to effectively eliminate the effects of shielded-cable capacitance, which would otherwise be devastating for the high source impedances that you might see with this sort of high-impedance input buffer amplifier.

G. Case study: low-distortion JFET follower

To explore quantitatively the improvement you get with a current-sink pull-down, and further with a cascode arrangement, we wired up the three follower circuits in Figure 3.44, each with an LSK389 dual JFET; these correspond to circuits A, D, and H of Figure 3.43. To seriously challenge these circuits' linearity, we drove each with a clean 1 kHz sinewave,[50] at signal amplitudes that pushed precariously close to the supply rails.

The simple resistor pull-down circuit (Figure 3.44A) exhibited an expected dc offset (about 0.25 V at the quiescent point), with measured distortion (Figure 3.45) going from 0.02% (at 1 Vrms) to about 0.14% (at 5 Vrms). That's pretty decent performance, especially given that this circuit is entirely open-loop (no feedback); it's better than we expected. The distortion was almost entirely second harmonic (i.e., at $2f_{in}$).

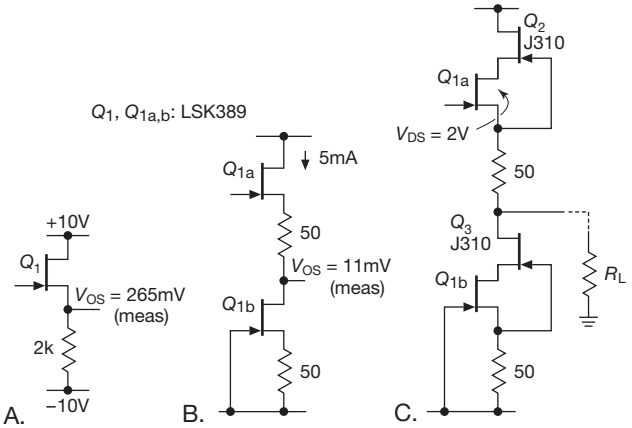


Figure 3.44. Three candidates for the JFET follower low-distortion medal-of-honor.

Adding an LSK389 matched JFET current sink with source degeneration (Figure 3.44B) made a nice improvement: a dc offset of about 10 mV and a measured distortion reduced by a factor of ten (20 dB), with the distortion now almost entirely third harmonic ($3f_{in}$). We're in serious audiophile territory here. Finally, adding a cascode (the

[50] From an SRS DS360 "Ultra-low distortion function generator": distortion less than 0.0003%. We measured the output distortion with a ShibaSoku 725B distortion analyzer.

J310's V_{GS} is much larger than that of Q_1, so that the latter runs at $V_{DS} \approx 2$ V) improves the linearity by another 20 dB, bumping into the measurement floor of our modest apparatus.[51] The low drain-to-source voltage across Q_{1a} imposed by the cascode also ensures a low input gate current.[52]

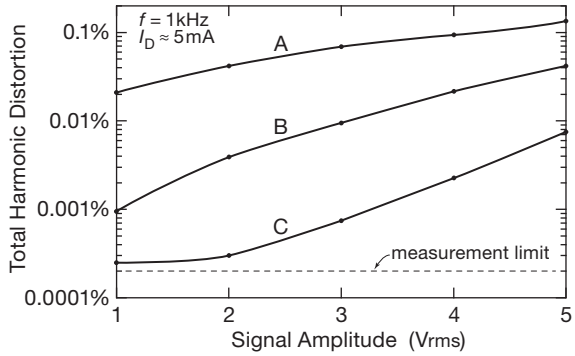


Figure 3.45. Measured distortion versus signal amplitude for the JFET followers of Figure 3.44, with $R_L = 1$ M.

3.2.7 FETs as variable resistors

Figure 3.21 showed the region of JFET characteristic curves (drain current versus V_{DS} for a small family of V_{GS} voltages), both in the normal ("saturated") regime and in the "linear" region of small V_{DS}. We showed the equivalent pair of graphs for a MOSFET at the beginning of the chapter (Figure 3.2). The I_D-versus-V_{DS} curves are approximately straight lines for V_{DS} smaller than $V_{GS} - V_{th}$, and they extend in both directions through zero, i.e., the device can be used as a voltage-controlled resistor for small signals of either polarity. From our equation for I_D versus V_{GS} in the linear region (§3.1.4, eq'n 3.1) we easily find the ratio (I_D/V_{DS}) to be

$$\frac{1}{r_{DS}} = 2\kappa\left[(V_{GS} - V_{th}) - \frac{V_{DS}}{2}\right]. \tag{3.9}$$

The last term represents a nonlinearity, i.e., a departure from resistive behavior (resistance shouldn't depend on signal voltage). However, for drain voltages substantially less than the amount by which the gate is above threshold ($V_{DS} \to 0$), the last term becomes unimportant, and the FET behaves approximately like a resistance:

$$r_{DS} \approx 1/[2\kappa(V_{GS} - V_{th})]. \tag{3.10}$$

Because the device-dependent parameter κ isn't a quantity you are likely to know, it's more useful to write r_{DS} as

$$r_{DS} \approx r_{G0}(V_{G0} - V_{th})/(V_G - V_{th}), \tag{3.11}$$

where the resistance r_{DS} at any gate voltage V_G is written in terms of the (known) resistance r_{G0} at some gate voltage V_{G0}.

Exercise 3.2. Derive the preceding "scaling" law.

From either formula you can see that the conductance $(=1/r_{DS})$ is proportional to the amount by which the gate voltage exceeds threshold. Another useful fact is that $r_{DS} = 1/g_m$, i.e., the channel resistance in the *linear* region is the inverse of the transconductance in the *saturated* region. This is a handy thing to know, because either g_m or r_{DS} is a parameter nearly always specified on FET data sheets.

Exercise 3.3. Show that $r_{DS} = 1/g_m$ by finding the transconductance from the saturation drain-current formula in §3.1.4.

Typically, the values of resistance you can produce with FETs vary from a few tens of ohms (as low as 0.001 Ω for power MOSFETs) all the way up to an open circuit. A typical application might be an automatic-gain-control (AGC) circuit in which the gain of an amplifier is adjusted (by means of feedback) to keep the output within the linear range. In such an AGC circuit you must be careful to put the variable-resistance FET at a place in the circuit where the signal swing is small, preferably less than 200 mV or so.

The range of V_{DS} over which the FET behaves like a good resistor depends on the particular FET and is roughly proportional to the amount by which the gate voltage exceeds threshold. Typically you might see nonlinearities of about 2% for $V_{DS} < 0.1(V_{GS} - V_{th})$, and perhaps 10% nonlinearity for $V_{DS} \approx 0.25(V_{GS} - V_{th})$. Matched FETs make it easy to design a ganged variable resistor to control several signals at once. You can also find some JFETs specifically intended for use as variable resistors (e.g., the InterFET 2N4338–41 series, and VCR series), with nominal ON-resistances specified at some V_{GS} (usually 0V).

A. Linearizing trick

It is possible to improve the linearity and, simultaneously, the range of V_{DS} over which a FET behaves like a resistor, by a simple compensation scheme. Look at the expression 3.9 for $1/r_{DS}$; you can see that the linearity will be nearly

[51] Oscillator courtesy of eBay; distortion analyzer courtesy of the MIT Flea Market.

[52] These impressively low distortions were measured into a high impedance. If you want to drive a substantial load, you may need to add a "g_m enhancer" to Q_{1a}; see §§*3x.2* and *3x.4* in Chapter *3x*.

perfect if you can add to the gate voltage a voltage equal to one half the drain-source voltage. Figure 3.46 shows two circuits to do exactly that.

In the first, the JFET forms the lower half of a resistive voltage divider, thus forming a voltage-controlled attenuator (or "volume control"). R_1 and R_2 improve the linearity by adding a voltage of $0.5V_{DS}$ to V_{GS}, as just discussed. The JFETs shown have an ON-resistance (gate grounded) of 60 Ω(max), giving the circuit an attenuation range of 0 to 40 dB. In the second circuit, the JFET's controllable resistance forms the lower leg of the gain-setting feedback divider in an op-amp noninverting voltage amplifier (the voltage gain is G=[10k/R_{FET}]+1).

The linearization of r_{DS} with a resistive gate-divider circuit, as above, is remarkably effective. In Figure 3.47 we've compared actual measured curves of I_D versus V_{DS} in the linear (low-V_{DS}) region for FETs, both with and without the linearizing circuit. The linearizing circuit is essential for low-distortion applications with signal swings of more than a few millivolts. We used it for amplitude control in the oscillator circuit of Figure 7.22, where the JFET was combined with a series resistor to create a low-distortion gain trimmer; the measured distortion of just 0.0002%.

When considering FETs for an application requiring a gain control, e.g., an AGC or "modulator" (in which the amplitude of a high-frequency signal is varied at an audio rate, say), it is worthwhile to look also at "analog-multiplier" ICs. These are high-accuracy devices with good dynamic range that are normally used to form the product of two voltages. One of the voltages can be a dc control signal, setting the multiplication factor of the device for the other input signal, i.e., the gain. Analog multipliers exploit the g_m-versus-I_C characteristic of bipolar transistors [$g_m = I_C$(mA)/25 siemens], in a configuration known as the "Gilbert cell," using matched arrays to circumvent problems of offsets and bias shifts. At very high frequencies (100 MHz and above), passive "balanced mixers" are often the best devices to accomplish the same task.

It is important to remember that a FET in conduction at low V_{DS} behaves like a good resistor all the way down to zero volts from drain to source (and even a bit beyond, in the opposite quadrant). There are no diode drops, sat-

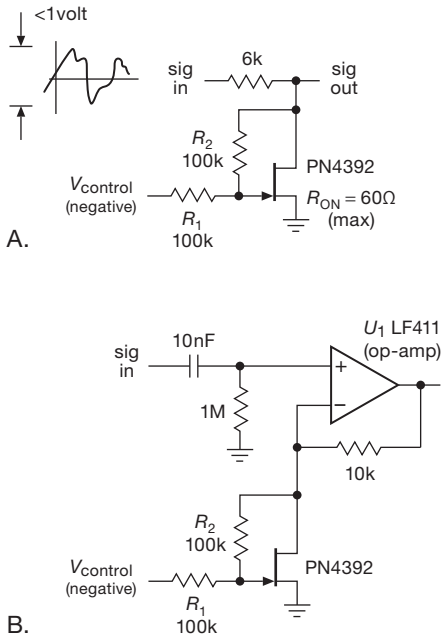


Figure 3.46. Linearizing the JFET variable resistor.

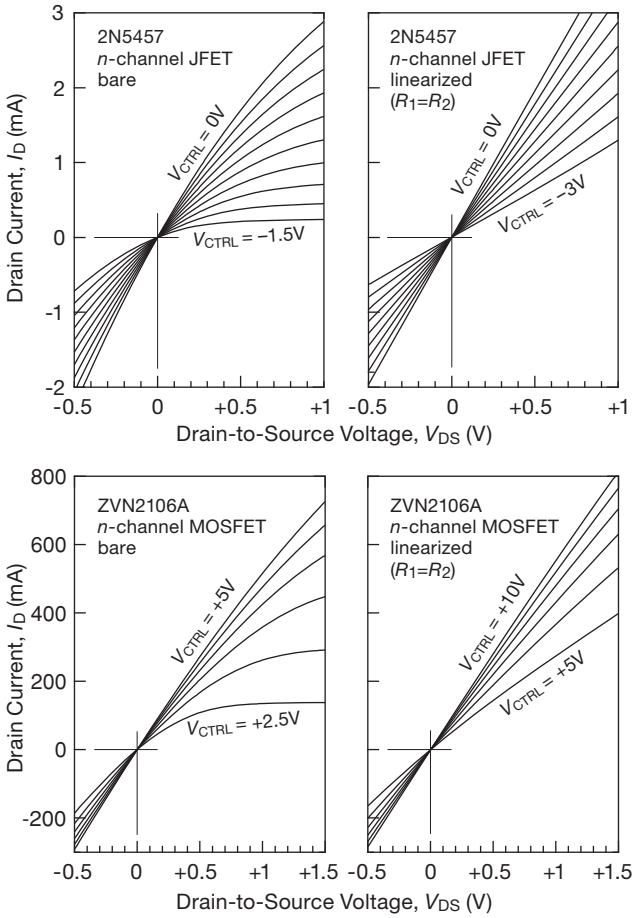


Figure 3.47. Measured curves of I_D versus V_{DS} for both (top) a JFET and (bottom) a MOSFET, showing the linearizing effect of a resistor pair (as in Figure 3.46). Note the relatively high current scale for the MOSFET.

uration voltages, or the like to worry about. We will see op-amps and digital logic families (CMOS) that take advantage of this nice property, giving outputs that saturate cleanly to the power supplies.

3.2.8 FET gate current

We said at the outset that FETs in general, and MOSFETs in particular, have essentially zero steady-state gate current. This is perhaps the most important property of FETs, and it was exploited in the high-impedance amplifiers and followers in the previous sections. It will prove essential, too, in applications to follow – most notably analog switches and digital logic.

Of course, at some level of scrutiny we might expect to see some gate current. It's important to know about gate current, because a naive zero-current model is guaranteed to get you in trouble sooner or later. In fact, finite gate current arises from several mechanisms. (a) Even in MOSFETs the silicon dioxide gate insulation is not perfect, leading to leakage currents in the picoampere range. (b) In JFETs the gate "insulation" is really a back-biased diode junction, with the same impurity and junction leakage-current mechanisms as ordinary diodes. (c) Furthermore, JFETs (*n*-channel in particular) suffer from an additional effect known as "impact-ionization" gate current, which can reach astounding levels. (d) Finally – and most important for high-speed circuits – both JFETs and MOSFETs have *dynamic* gate current, caused by ac signals driving the gate capacitance; this can cause Miller effect, just as with bipolar transistors.[53] We'll deal with this important topic later, in §§3.5 and 3.5.4.

In most cases gate input currents are negligible in comparison with BJT base currents. However, there are situations in which a FET may actually have *higher* input current. Let's look at the numbers.

A. Gate leakage

The low-frequency input impedance of a FET amplifier (or follower) is limited by gate leakage. JFET datasheets usually specify a breakdown voltage, BV_{GSS}, defined as the voltage from gate to *channel* (source and drain connected together) at which the gate current reaches 1 μA. For smaller applied gate-channel voltages, the gate leakage current, I_{GSS}, again measured with the source and drain connected together, is considerably smaller, dropping quickly to the picoampere range for gate-drain voltages well below breakdown. With MOSFETs you must never allow the gate insulation to break down; instead, gate leakage is specified as some maximum leakage current at a specified gate-channel voltage. Integrated circuit amplifiers with FETs (e.g., FET op-amps) use the misleading term "input bias current," I_B, to specify input leakage current; it's usually in the picoampere range.

The good news is that these leakage currents are in the picoampere range at room temperature. The bad news is that they increase rapidly (in fact, exponentially) with temperature, roughly doubling every 10°C. By contrast, BJT base current isn't leakage, it's bias current, and in fact tends to *decrease* slightly with increasing temperature. The comparison is shown graphically in Figure 3.48, a plot of input current versus temperature for several IC amplifiers (op-amps). The FET-input op-amps have the lowest input currents at room temperature (and below), but their input current rises rapidly with temperature, crossing over the curves for amplifiers with carefully designed BJT input stages like the LM10 and LT1012. These BJT op-amps, along with "premium" low-input-current JFET op-amps like the OPA111 and OPA627, are fairly expensive. However, we also included everyday "jellybean" op-amps like the bipolar LM358 and JFET LF411/2 in the figure to give

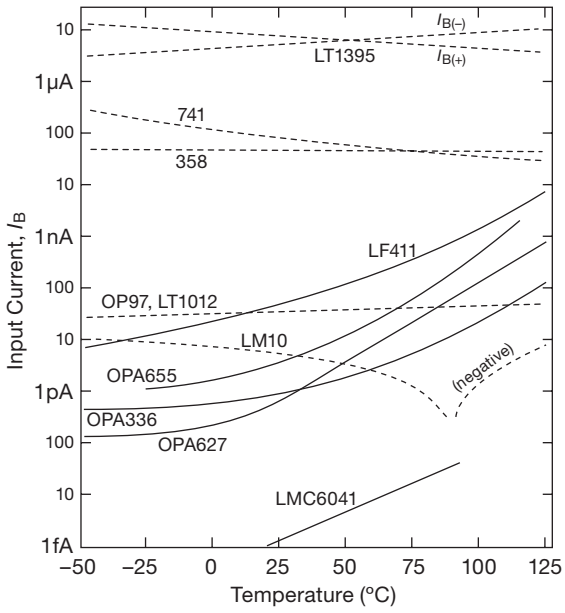


Figure 3.48. The input current of a FET amplifier is gate leakage, which doubles every 10°C. The FET-input amplifiers in this plot (solid lines) are easily spotted by their characteristic upward slope.

[53] In extreme cases, for example in high-voltage power switching, it can require *amperes* of gate drive current to switch a large MOSFET on the time scale of nanoseconds. This is not a trivial effect!

an idea of input currents you can expect with inexpensive (less than a dollar) op-amps.

B. JFET impact-ionization current

In addition to conventional gate leakage effects, *n*-channel JFETs suffer from rather large gate leakage currents when operated with substantial V_{DS} and I_D (the gate leakage specified on datasheets is measured under the unrealistic conditions that both $V_{DS}=0$ and $I_D=0$!). Figure 3.49 shows what happens. The gate leakage current remains near the I_{GSS} value until you reach a critical drain-gate voltage, at which point it rises precipitously. This extra "impact-ionization" current is proportional to drain current, and it rises exponentially with voltage and temperature. The onset of this current occurs at drain-gate voltages of about 25% of BV_{GSS}, and it can reach gate currents of a microamp or more. Obviously a "high-impedance buffer" with a microamp of input current is worthless. That's what you would get if you used a BF862 as a follower, running 1 mA of drain current from a 20 volt supply.

This extra gate leakage current afflicts primarily *n*-channel JFETs, and it occurs at higher values of drain-gate voltage. Some cures are to (a) operate at low drain-gate voltage, either with a low-voltage drain supply or with a cascode, (b) use a *p*-channel JFET, where the effect is much smaller, or (c) use a MOSFET. The most important thing is to be aware of the effect so that it doesn't catch you by surprise.

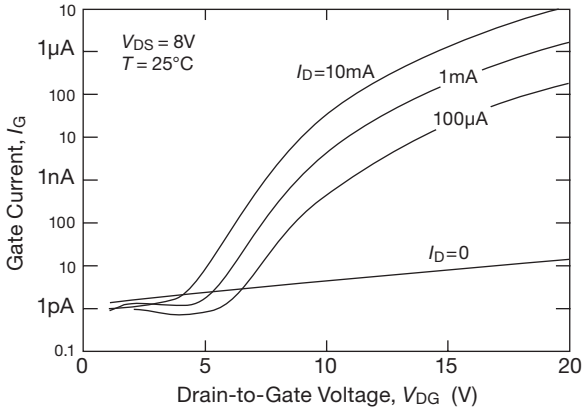


Figure 3.49. JFET gate leakage increases disastrously at higher drain-gate voltages and is proportional to drain current, as seen in the datasheet curves for the otherwise excellent BF862 *n*-channel JFET.

C. Dynamic gate current

Gate leakage is a dc effect. Whatever is driving the gate must also supply an *ac* current, because of gate capacitance. Consider a common-source amplifier. Just as with bipolar transistors, you can have the simple effect of input capacitance to ground (called C_{iss}), and you can have the capacitance-multiplying Miller effect (which acts on the feedback capacitance C_{rss}). There are two reasons why capacitive effects are more serious in FETs than in bipolar transistors. First, you use FETs (rather than BJTs) because you want very low input current; thus the capacitive currents loom relatively larger for the same capacitance. Second, FETs often have considerably larger capacitance than equivalent bipolar transistors.

To appreciate the effect of capacitance, consider a FET amplifier intended for a signal source of 100k source impedance. At dc there's no problem, because the picoampere currents produce only microvolt drops across the signal source's internal impedance. But at 1 MHz, say, an input capacitance of 5 pF presents a shunt impedance of about 30k, seriously attenuating the signal. In fact, *any* amplifier is in trouble with a high-impedance signal at high frequencies, and the usual solution is to operate at low impedance (50 Ω is typical) or use tuned *LC* circuits to resonate away the parasitic capacitance. The point to understand is that the FET amplifier doesn't look like a $10^{12}\,\Omega$ load at signal frequencies.

As another example, imagine switching a 5 amp high-voltage load with a power MOSFET (there aren't any high-power *JFETs*), in the style of Figure 3.50. One might naively assume that the gate could be driven from a digital logic output with low current-sourcing capability, for example the so-called 4000-series CMOS logic, which can supply an output current of the order of 1 mA with a swing from ground to +10 V. In fact, such a circuit would be a disaster, because 1 mA of gate drive into the 200 pF average feedback capacitance of the IRF740 would stretch the output switching speed to a leisurely 50 μs.[54]

Even worse, the dynamic gate currents ($I_{gate}=C\,dV_D/dt$) would force currents back into the logic device's output, possibly destroying it by means of a perverse effect known as "SCR latchup" (more of which in Chapters 10 and 11);

[54] Our model is unacceptably crude, because feedback capacitance varies rapidly with drain voltage (see §3.5.4A). It's OK to use a constant value for feedback capacitance C_{rss} for small-signal calculations; but for a switching application like this, you need to go to the datasheet for values of *gate charge*, which take into account the nonlinear behavior of the capacitances. In this example the datasheet specifies $Q_G \approx 40$ nC of gate charge, producing a switching time of $t = Q_G/i = 40\,\mu$s at a (dynamic) drive current i of 1 mA.

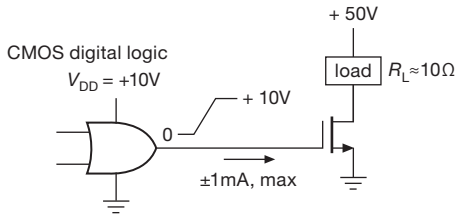


Figure 3.50. Dynamic gate current example: driving a fast-switching load.

for this and other reasons a series resistor (not shown in the figure) is usually added between the driving device and the MOSFET's gate. Bipolar power transistors have somewhat lower capacitances and therefore somewhat lower dynamic input currents (but still in the same ballpark); but when you design a circuit to drive a 5 amp power BJT, you're *expecting* to provide a few hundred milliamps or so of base drive (via a Darlington or whatever), whereas with a FET you tend to take for granted low input current. In this example – in which you would have to supply a couple of *amperes* of gate drive current to bring about the 25 ns switching speed that the MOSFET is capable of – the ultra-high-impedance FET has lost some of its luster.

Exercise 3.4. Estimate the switching times for the circuit of Figure 3.50, with 1 amp of gate drive current, assuming either (a) an average feedback capacitance of 200 pF, or (more accurately) (b) a required gate charge of 40 nC.

3.3 A closer look at JFETs

In §3.1.4 we introduced the landscape of FET operating regions: for drain voltages of a volt or more (to get beyond the "linear" resistive region) there is the conventional operating region in which the saturation drain current[55] I_D is proportional to $(V_{GS} - V_{th})^2$ and (at much lower drain currents) the subthreshold region in which I_D is exponential in V_{GS}.

That's the simple picture. Because JFETs are the devices of choice for accurate or low-noise (or both) circuits with high input impedance, it's worth looking more closely at their idiosyncrasies, ideally with measurements of actual devices.

We've undertaken an exhaustive (and exhausting!) review of most of the available JFETs, collecting sample batches of each, often from multiple manufacturers. Table 3.7 on page 217 includes most of them, with some measured parameters (I_{DSS}, and g_{os}, g_m, and V_{GS} at a useful drain current) alongside the datasheet specifications.[56] See also Table 8.2 on page 516, which lists a nice selection of low-noise JFETs.

Advanced JFET topics are treated in Chapter *3x*; here we discuss a few essential subjects – JFET operating regions (including the often neglected *subthreshold* region), JFET transconductance, and JFET capacitance.

3.3.1 Drain current versus gate voltage

A persistent issue with JFET circuit design is *parameter spread*. This is nicely illustrated in Figures 3.51 and 3.52, where we've plotted measured I_D versus V_{GS} for six samples of the 2N5457 *n*-channel JFET (three from each of two manufacturers[57]), and three samples of the related 2N5458 (from the same 2N5457–59 family). In each case we chose the parts with the highest, lowest, and median values of measured I_{DSS} in a batch of ten parts. For these measurements we ventured into positive gate voltages (up nearly to a diode drop, the onset of gate conduction), well beyond the usual zero-bias limit; nothing terrible happens, but in general this is a practice that should be avoided.

Let's look more closely at the data for these nine parts to understand different aspects of their performance and the impact on JFET circuit designs.

A. The quadratic region

The *linear* plot is what you usually see in datasheets and in textbooks. That scaling reveals well the quadratic behavior of drain current at a significant fraction of the zero-bias drain current (I_{DSS}), where JFETs are most often operated [eq'n 3.4]. You can see also the variation of I_{DSS} among samples (the specified range for the half-dozen 2N5457 samples is indicated by a vertical bar), as well as the somewhat steeper slope for the Fairchild 2N5457 samples. The slope is, of course, just the transconductance gain, $g_m = dI_D/dV_{GS}$, increasing linearly with V_{GS} [eq'n 3.5] in this regime of where the drain current is quadratic in $V_{GS} - V_{th}$.

[55] The term "saturation" can be confusing: for FETs it is used to denote *current* saturation, the region of drain voltages more than a volt or so where the drain current is approximately constant. By contrast, for BJTs the term "saturation" denotes *voltage* saturation (an ON switch), in which the collector voltage is close to zero. It never hurts to add the qualifying adjective (though people seldom do).

[56] See also Table 8.2 for noise parameters. These JFET tables represent several months' work – of comparing datasheet specifications, of checking distributors' stock, and of lab measurements. We found it revealing and rewarding, and we hope it is of use to the reader.

[57] We used dashed lines for the Fairchild parts, to untangle the crowd.

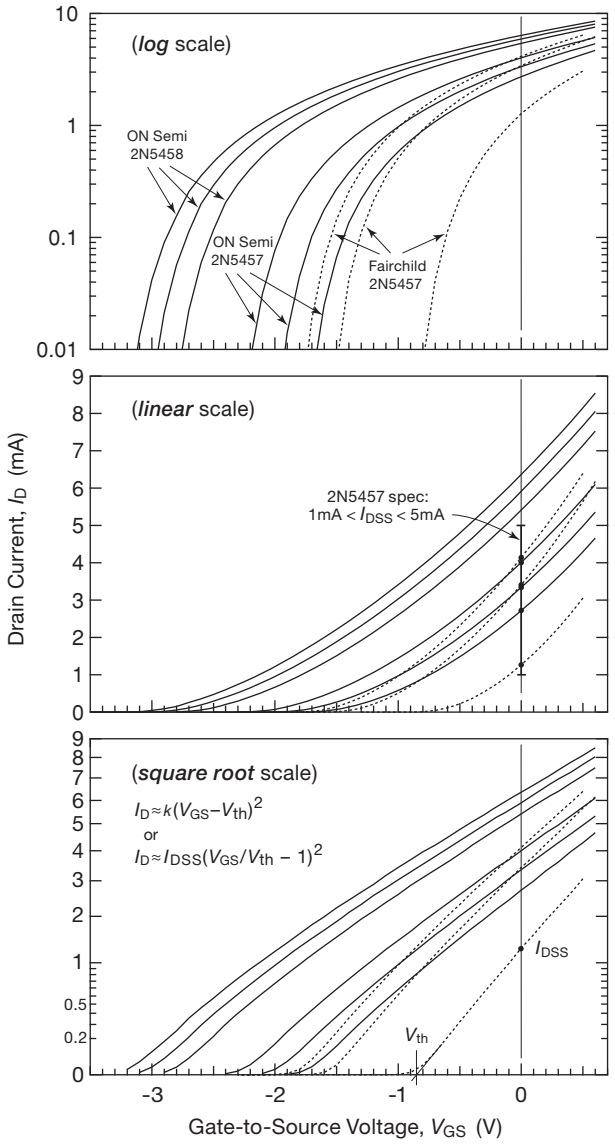


Figure 3.51. Drain current versus gate voltage for nine parts from the 2N5457–9 n-channel JFET family, operating with V_{DS}=5 V. The same data is plotted on linear, log, and square-root axes. The ON Semi and Fairchild parts both meet the I_{DSS} spec (middle panel) despite their substantially different curves. Note that the measurements extend beyond I_{DSS}, to $V_{GS}+0.6$ V.

Finding V_{th}: square-root plot

Look next at the *square-root* plot, which gives the quadratic region a chance to stretch out its limbs. The extrapolation to zero drain current defines the threshold voltage V_{th}. At that voltage the current is not *zero* – it's just down near the top of the subthreshold region.

B. The subthreshold region

Finally, the *log* plot greatly expands the low-current region. The curves bend over toward a straight-line (thus exponential) behavior at the lowest currents – this is the subthreshold region, which is explored six orders of magnitude more deeply in Figure 3.52.

C. The deep subthreshold region

From the extended curves in Figure 3.52 we can see that JFETs continue to do their stuff down to *pico*amp drain currents, conforming accurately to an exponential drain-current law (analogous to the BJT's Ebers–Moll), which can be written as

$$I_D = I_0 \exp(V_{GS}/nV_T), \tag{3.12}$$

with the same $V_T = kT/q \approx 25$ mV as in Ebers–Moll (but with an added fudge factor n). The measured data in Figure 3.52 corresponds to a near-unity value for n ($n = 1.05$). In other words, at very low drain currents, a JFET has nearly the same transconductance as a BJT operating at that same collector current (as confirmed in the measured transconductance data of Figure 3.54 on page 168).[58] Note that the Fairchild parts (in a gesture of egalitarian spirit) are no longer anomalous – unlike their behavior in the quadratic region (where they were the transconductance champs), at low currents they exhibit the same transconductance (slope) as the other parts.

3.3.2 Drain current versus drain-source voltage: output conductance

Drain current (at constant V_{GS}) *does* have some dependence upon drain-to-source voltage, in contrast to the idealized picture that you usually see (and to the promulgation of which we are guilty parties: see §3.1.4 and Figure 3.13). You can think of this effect as analogous to Early effect in BJTs (§2*x.8*), or, equivalently, as a finite output impedance r_o (or, more commonly, finite output *conductance* $g_{os}=1/r_o$) seen at the drain terminal when the gate-source voltage is held constant. This effect limits the maximum gain of a grounded-source amplifier configured with

[58] But not much bandwidth! For instance, at I_D=10 pA a 2N5457 would have a paltry gain–bandwidth product f_T of just 140 Hz ($f_T=g_m/2\pi C_{iss}$; see "Bandwidth of the Cascode, BJT vs FET" in Chapter *3x*). It's nice to know that JFETs (like BJTs) work well at very low currents, a useful characteristic for micropower and nanopower applications. But don't forget that device capacitances loom large at low currents, so current-starved designs are slower than circuits running at normal currents.

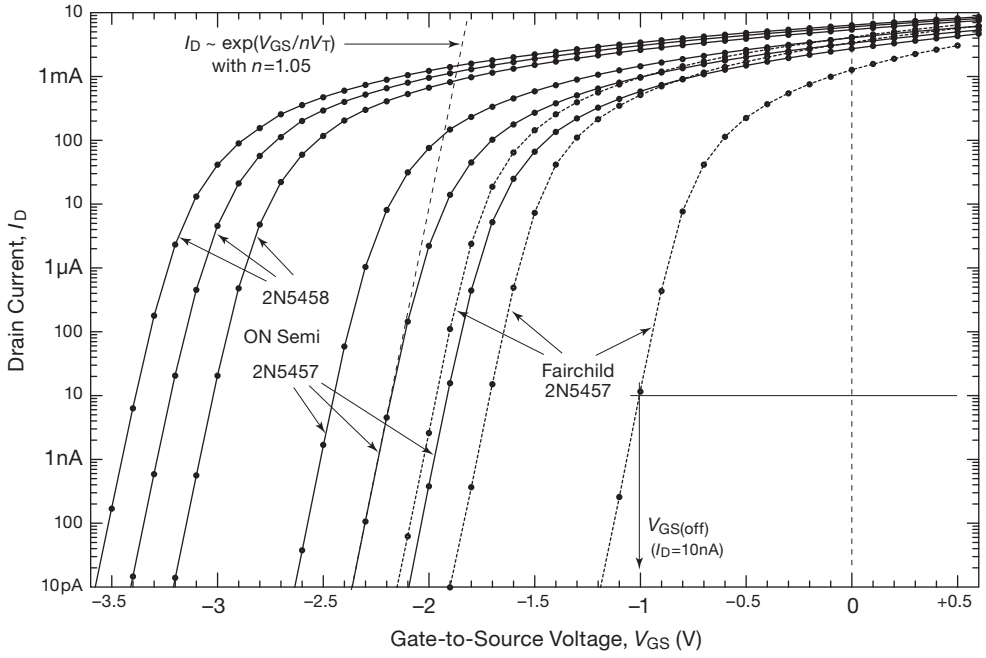


Figure 3.52. Log plot of measured drain currents (dots) versus gate voltage for the same JFETs as in Figure 3.51. JFETs work fine at 10 pA, far below the arbitrary 10 nA drain current that conventionally defines the gate-source cutoff voltage $V_{GS(off)}$.

a current source as drain load (that's the G_{max} parameter in Table 3.1 on page 141), and you need to take it into account if your gain approaches G_{max}. It also further degrades the performance of a source follower (already degraded by low $g_m R_L$; see §3.2.6) if G_{max} is comparable to or less than $g_m R_L$. These consequences are discussed in detail in Chapter *3x* (§*3x.4*); but they are important enough to be worthy of mention here.

A. Degraded gain and linearity in the common-source amplifier

This "g_{os} effect" limits the maximum gain of the common-source amplifier, by effectively putting a resistance $r_o = 1/g_{os}$ across the drain load impedance. For a simple resistive drain load R_D that reduces the gain from the ideal $G = g_m R_D$ to $G = g_m(R_D \parallel r_o)$, or

$$G = g_m R_D \frac{1}{1 + g_{os} R_D} \tag{3.13}$$

A further unwelcome consequence is some nonlinearity, owing to the dependence of g_{os} upon drain voltage. See §*3x.4* if you want to know more about this.

B. Gain error in the source follower

The "g_{os} effect" also acts to reduce the gain of the source follower from the ideal G=1. This is most noticeable with relatively light loads, where $R_L \gg 1/g_m$ (and therefore where you would expect the voltage gain to be very close to unity). Taking this effect into account, the voltage gain of the simple JFET follower of Figure 3.40 becomes

$$G = \frac{1}{1 + \dfrac{1}{g_m R_L} + \dfrac{1}{G_{max}}} \tag{3.14}$$

where G_{max} is the ratio of drain conductance to transconductance ($G_{max} = g_{os}/g_m$) at the operating voltage and current. (We defined G_{max} because it's roughly constant with current[59] and therefore more useful than the current-dependent r_o and g_{os}). Thus, for example, you need both $g_m R_L$ and G_{max} greater than 100 for <1% gain error – or you need to use some tricks, such as an active load, a cascode connection, or a "g_m enhancer." Looking back to the tutorial source follower progression in Figure 3.43, that is the reason for the use of an active load and a cascode connection in circuit H. Table 3.1 lists measured values of G_{max} for common JFETs. For further enlightenment read the exposé in §*3x.4*.

[59] Looking more closely, both g_m and g_{os} depend on drain current, approximately proportional to $\sqrt{I_D}$ (§§3.3.3, *3x.2*); their ratio G_{max} is relatively flat with drain current, but approximately proportional to drain-source voltage (§§3.3.4, *3x.4*).

3.3.3 Transconductance versus drain current

Recall that transconductance ($g_m \equiv i_d/v_{gs}$, the change of drain current with gate-to-source voltage) is the natural measure of FET gain. It is analogous to the Ebers–Moll view of bipolar transistors, where $g_m = qI_C/kT = I_C/V_T$. Generally speaking, at any given drain current an FET will have lower transconductance than a BJT. Starting at the lowest drain currents (recall §3.3.1 and Figure 3.52), the transconductance in the FET subthreshold region is proportional to drain current (like a BJT), but somewhat less than a BJT operating at the same current; see Figure 3.53. If you write $g_m = I_D/nV_T$, typical JFETs have n values in the range of 1.05 to 3, as can be seen in the measured data of Figure 3.54 (and similar expanded plots in Chapter *3x*, see §*3x.2*).

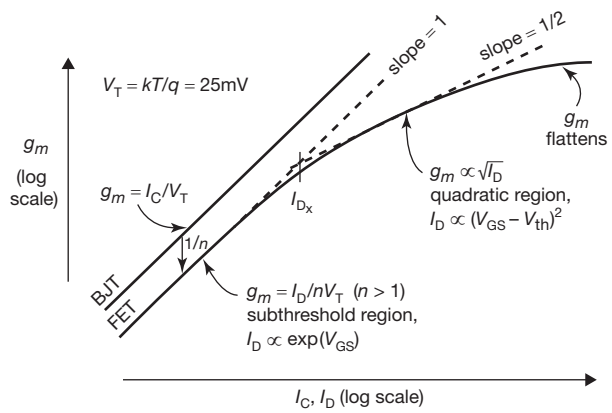


Figure 3.53. Transconductance versus drain current: in the subthreshold region an FET's transconductance is proportional to current (like a BJT); at normal currents where $I_D \propto (V_{GS} - V_{th})^2$ it goes like $\sqrt{I_D}$; then at still higher operating currents it flattens toward a maximum.

At higher currents, in the FET's "quadratic region" of drain current where $I_D \propto (V_{GS} - V_{th})^2$, the transconductance becomes proportional to the square root of drain current, falling away below the BJT's increasing ($\propto I_C$) transconductance. At still higher currents the transconductance flattens further. For JFETs the story ends here; but for MOSFETs (where you can bring the gate voltage high enough to reach constant maximum drain current), the transconductance heads back down, for example with the LND150 whose I_D versus V_{GS} curve is plotted in Figure 3.9.

It's not hard to measure transconductance directly – see §*3x.3* for circuit details of one simple method (where the secret sauce is a cascode stage to clamp the drain voltage). We used that circuit to measure the transconductance of some sixty JFETs, at drain currents going from 1 μA to 30 mA in half-decade steps.

Figure 3.54 shows measured transconductance versus drain current, for a few representative JFETs.[60] At higher currents the transconductance is approximately proportional to the square root of drain current (thus proportional to $V_{GS} - V_{th}$). And, even though these curves go down "only" to 1 μA, most of the JFETs (the four upper curves) are already well into the sub-threshold region (with transconductance proportional to drain current) down at that current.

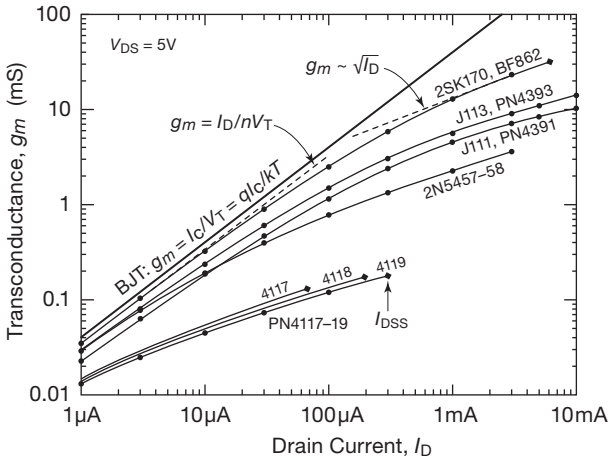


Figure 3.54. Measured low-frequency transconductance for some representative n-channel JFETs. See also the analogous plot in §3x.2.

So, what's going on with the PN4117–19? Well, these are very small JFETs (you can tell by their very low I_{DSS} values), so even at a drain current of 1 μA the current *density* is high enough to bring them over into their quadratic region. Put another way, in the preceding discussion we should have been referring to "current density" (rather than simple drain current) as the measure of whether a given JFET is in its subthreshold region or its quadratic region.

Note that this holds also for MOSFETs: a large power MOSFET (with drain current ratings in the hundreds of amps) will be running in its subthreshold region if used in a linear circuit application.[61] As we'll see in §3.6.3, this has an important effect on thermal stability when multiple MOSFETs are connected in parallel in power applications, because in the subthreshold region MOSFETs have a *positive* thermal coefficient of drain current with temperature;

[60] See the rich thicket of measured curves in the analogous plot in §3x.2.

[61] SPICE-lovers beware: the simulation models for power MOSFETs are nearly useless in the subthreshold region.

this can cause serious mischief when not fully appreciated by the circuit designer.

A. Transconductance within a JFET family

From Figure 3.54 and the preceding discussion it would appear that you cannot really predict a JFET's transconductance at any given operating current; that graph shows a 50:1 variation of g_m at a given current. Worse still, the data of Figure 3.51 suggests that you cannot even predict with any reasonable certainty the operating current of a given JFET part type for a given gate bias (or vice-versa).

As it turns out, the situation is not so bleak. Within a family (or extended family) of similar JFETs the transconductance depends primarily (and predictably) upon drain current, even though the corresponding gate voltage may be all over the map. Take a look at Figure 3.55, which shows measured drain current curves for a varied selection of seven 2N5486 JFETs (manufactured over a period of 35 years[62]), along with a sample of lower-current siblings (2N5484 and 2N5485) and a related cousin (SST4416, also featured for RF applications, and with similar low capacitance and comparable range of I_{DSS}).

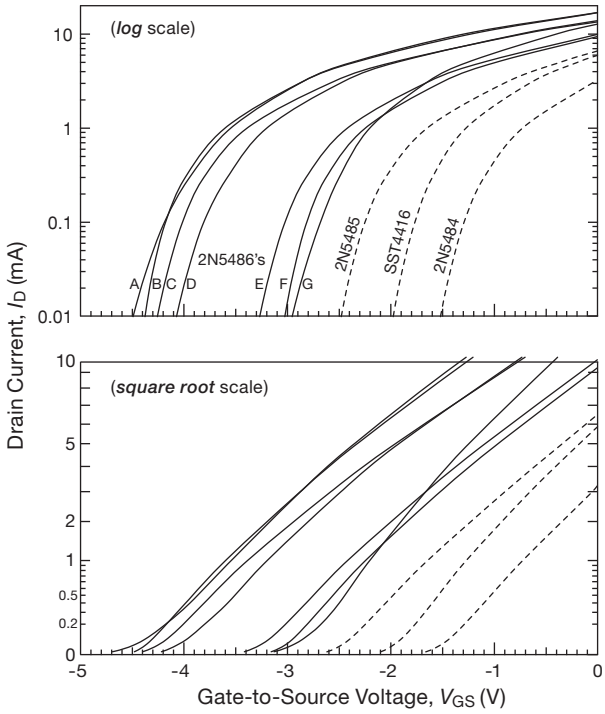


Figure 3.55. Measured drain current versus gate-source voltage for a collection of seven 2N5486 JFETs of varying vintages and manufacturers (along with three smaller JFET types, indicated with dashed lines), plotted on both log and square-root axes, illustrating the wide spread of threshold voltages V_{th} (the straight-line extrapolation of the latter's curves). Compare the large spread seen here with Figure 3.56, where the same set of JFETs exhibit only minor variations in transconductance at any given drain current.

These measured curves exhibit a wide variation of behavior for this selection of parts (there's a 3:1 spread of $V_{GS(off)}$, and a 5:1 spread of I_{DSS}), suggesting a similar unpredictability of transconductance gain.[63] But that suggestion would be misleading. When you measure their transconductances, you get the curves of Figure 3.56: within this deliberately varied selection of ten related parts there's at most a ±20% peak variation of transconductance at a given drain current.

The moral: within a JFET family of *similar part types*, the drain current (at whatever gate voltage is needed to get that current) is reasonably predictive of the transconductance. And the practical consequence is that a JFET amplifier circuit will have reasonably predictable gain if it is arranged with bias feedback to set the drain current to a desired value (which should be chosen no greater than the specified minimum I_{DSS}, unless you are willing to sort your incoming parts).

But a piece of advice: for better gain predictability in JFET amplifiers it's a good idea (a) to use some source degeneration (§*3x.4*), preferably in combination with a

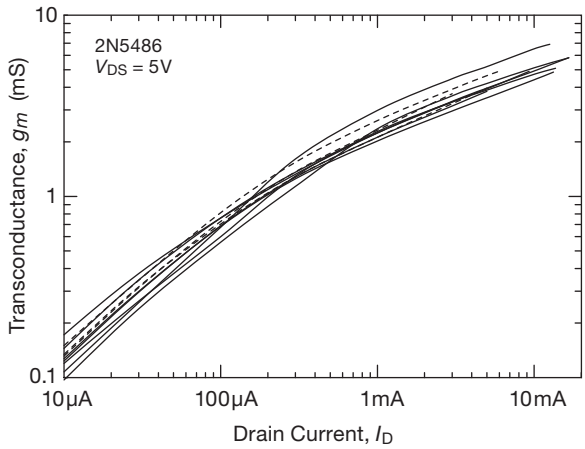


Figure 3.56. Measured transconductance versus drain current for the same set of JFETs as in Figure 3.55.

[62] A, Intersil, date code 7328 (28th week of 1973); B, Central Semiconductor, contemporary; C and E, Fairchild, date code BF44; D, Vishay SST5486, contemporary; F, Vishay, date code 0536; G, Motorola, 1990 vintage.

[63] There are some helpful correlations, though: the gate threshold voltage V_{th} is predictive of the zero-bias drain current I_{DSS}.

transconductance-boosting enhancer circuit, or (b) to use overall feedback with enough minimum loop gain (with worst-case g_m) to ensure the gain accuracy you need (as in Figures 3.31, 3.34, and 3.37).

3.3.4 Transconductance versus drain voltage

The transconductance of a FET at a given drain current is relatively independent of drain *voltage*, except for drain-to-source voltages below a volt or two. This contrasts with its strong dependence on drain *current*. See the discussion in Chapter *3x* (§*3x.2*, with measured data).

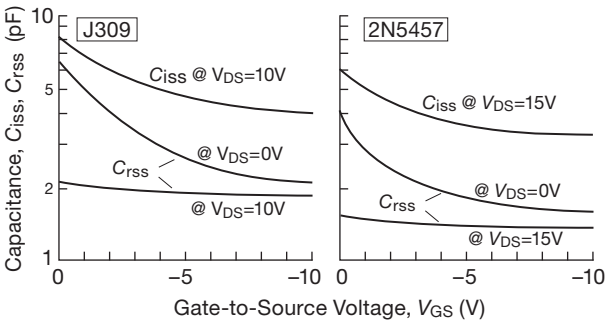


Figure 3.57. Input and feedback capacitance of JFETs.

3.3.5 JFET capacitance

Just as with bipolar transistors (and, later, MOSFETs), the capacitance seen between the terminals of a JFET depends on the (reverse) bias; this is often called a "nonlinear capacitance" and decreases markedly with increasing reverse-bias. Figure 3.57 shows datasheet plots of feedback and input capacitances for two common *n*-channel JFETs. These values – just a few picofarads – are typical of small-signal JFETs and quite a bit smaller than you see in power MOSFETs (compare with Figure 3.100 on page 197). JFETs are usually symmetrical, but owing to the larger reverse bias, the gate-drain capacitance is smaller than the gate-source capacitance.[64] That's a good thing, of course, because the Miller effect multiplies the effect of C_{rss} in a common-source amplifier without cascode.

Manufacturers generally provide plots of capacitance versus V_{GS}, and sometimes also versus V_{DS} or V_{DG}. But you get only one set of plots for a family of parts, even though their I_{DSS} specs may range over a factor of ten or more; one is left wondering if their capacitances are correlated with their I_{DSS} values, thus suggesting that capacitance data should be considered only a rough guide to reality.

3.3.6 Why JFET (versus MOSFET) amplifiers?

We've devoted considerable space in this chapter to *junction* FETs (JFETs), a topic that often receives minimal discussion in standard references, in which the emphasis is on insulated-gate FETs (i.e., *MOS*FETs). We'll soon shift the focus toward the latter, for good reasons: small integrated MOSFETs are dominant in (a) low-voltage and low-power analog circuits (op-amps, portable electronics, RF circuits, and the like); (b) analog switching; (c) logic circuits, microprocessors, and memory; and, in the form of discrete packaged power transistors, (d) power switching and linear power applications. These are all hot topics in contemporary electronics, and MOSFETs are the dominant species of transistor on the planet, by a huge margin.

Before moving on, though, it is worth emphasizing that JFETs are the device of choice for analog circuits that demand a combination of high input impedance, low noise, and good accuracy; this in addition to their niche applications as two-terminal current sources, voltage-variable resistors, and constant-R_{ON} analog switches.

The previous examples of amplifiers, followers, and current sources illustrate many of the JFET's virtues. Going beyond this first look at JFETs, Chapter *3x* includes some advanced material, and Chapter 8 discusses the important topic of noise, complete with measured data. To get a preview of the latter, look at Figure 3.58, where we've plotted the measured noise voltage density of three favorite "jellybeans": the 2N3904 *npn* bipolar transistor, the 2N5457

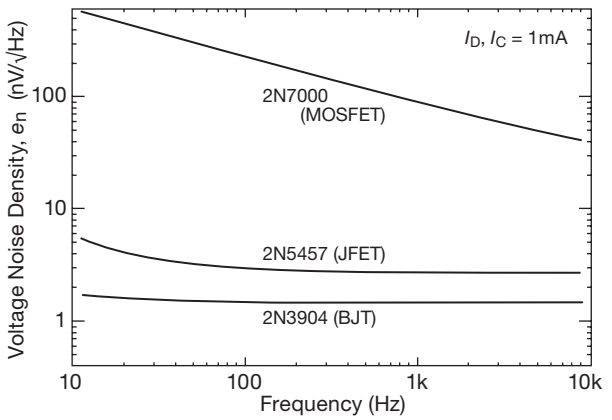


Figure 3.58. Noise voltage of three popular jellybeans, illustrating the poor low-frequency noise properties of MOSFETs.

[64] By "gate" and "source" we mean the pins your circuit uses for those functions, not the actual labels on the pins.

JFET and the 2N7000 MOSFET. These parts make no pretense of premium membership; they cost a pittance,[65] and they're not aimed at low-noise applications. But they illustrate an important trend: MOSFETs are inherently noisy at low frequencies, by as much as 40 dB relative to their BJT and JFET cousins.[66] Don't use them for low-level audio applications – but power MOSFETS can be used for the muscular output stage of an audio amplifier.

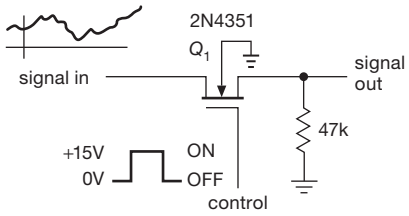


Figure 3.59. nMOS analog switch, with body terminal and diode shown.

3.4 FET switches

The two examples of FET circuits that we gave at the beginning of the chapter were both *switches*: a logic-switching application and a linear signal-switching circuit. These are among the most important FET applications, and they exploit the FET's unique characteristics: high gate impedance and bipolarity resistive conduction clear down to zero volts. In practice you usually use MOSFET integrated circuits (rather than discrete transistors[67]) in all digital logic and linear-switch applications, and it is only in power switching applications that you ordinarily resort to discrete FETs. Even so, it is essential (and fun!) to understand the workings of these chips; otherwise you're almost guaranteed to fall prey to some mysterious circuit pathology.

3.4.1 FET analog switches

A common use of FETs, particularly MOSFETs, is as analog switches. Their combination of low ON-resistance (all the way to zero volts), extremely high OFF resistance, low leakage currents, and low capacitance makes them ideal as voltage-controlled switch elements for analog signals. An ideal analog, or linear, switch behaves like a perfect mechanical switch: in the ON state it passes a signal through to a load without attenuation or nonlinearity; in the OFF state it is an open circuit. It should have negligible capacitance to ground and negligible coupling from the signal applied to the control input.

Let's look at an example (Figure 3.59). Q_1 is an *n*-channel enhancement-mode MOSFET, and it is nonconducting when the gate is grounded or negative. In that state the drain-source resistance (R_{off}) is typically more than 10,000 M, and no signal gets through (though at high frequencies there will be some coupling through drain-source *capacitance*; more on this later). Bringing the gate to +15 V puts the drain-source channel into conduction, typically 20–200 Ω in FETs intended for use as analog switches. The gate signal level is not at all critical, as long as it is sufficiently more positive than the largest signal (to maintain R_{ON} low), and it could be provided from digital logic circuitry (perhaps using a FET or BJT to generate a full-supply swing) or even from an op-amp running from a +15 V supply. Swinging the gate negative (as from a bipolarity op-amp output) doesn't hurt, and in fact has the added advantage of allowing the switching of analog signals of either polarity, as will be described later. Note that the FET switch is a bidirectional device; signals can go either way through it. Ordinary mechanical switches work that way, too, so it should be easy to understand.

The circuit as shown will work for positive signals up to about 10 V; for larger signals the gate drive is insufficient to hold the FET in conduction (R_{ON} begins to rise), and negative signals would cause the FET to turn on with the gate grounded (it would also forward-bias the channel–body junction). If you want to switch signals that are of both polarities (e.g., signals in the range −10 V to +10 V), you can use the same circuit, but with the gate driven from −15 V (OFF) to +15 V (ON); the body terminal should then be tied to −15 V.

With any FET switch it is important to provide a load resistance in the range of 1k to 100k in order to reduce capacitive feedthrough of the input signal that would otherwise occur during the OFF state. The value of the load resistance is a compromise: low values reduce feedthrough, but they begin to attenuate the input signal because of the voltage divider formed by R_{ON} and the load. Because R_{ON} varies over the input signal swing (from changing V_{GS}), this attenuation also produces some undesirable nonlinearity. Excessively low load resistance appears at the switch input, of course, loading the signal source as well. Several possible solutions to this problem (multiple-stage switches, R_{ON}

[65] Roughly $0.02, $0.05, and $0.04, respectively, in 1000-piece quantities.

[66] According to John Willison, this phenomenon may be associated with intermittent trapping and release of charges on the insulated gate.

[67] It's hardly your choice – you are *forced* to, because discrete small-signal MOSFETs are a vanishing breed (perhaps qualifying for protected species status).

cancellation) are shown in §3.4.2 and in the discussion of rail-to-rail op-amps in Chapter *4x*. An attractive alternative is to use a second FET switch section to connect the output to ground when the series FET is off, thus effectively forming a single-pole double throw (SPDT) switch (more on this in the next section).

A. CMOS linear switches

Frequently it is necessary to switch signals that may go nearly to the supply voltages. In that case the simple *n*-channel switch circuit just described won't work, because the gate is not forward-biased at the peak of the signal swing. The solution is to use paralleled complementary MOSFET (CMOS) switches (Figure 3.60). The triangular symbol is a digital inverter, which we'll discuss shortly; it inverts a HIGH input to a LOW output, and vice versa. When the control input is high, Q_1 is held ON for signals from ground to within a few volts of $+V_{DD}$ (where R_{ON} starts increasing dramatically). Q_2 is likewise held ON (by its grounded gate) for signals from $+V_{DD}$ to within a few volts of ground (where its R_{ON} increases dramatically). Thus signals anywhere between $+V_{DD}$ and ground are passed through with low series resistance (Figure 3.61). Bringing the control signal to ground turns off both FETs, providing an open circuit. The result is an analog switch for signals all the way from ground to $+V_{DD}$. This is the basic construction of the classic 4066 CMOS "transmission gate." It is bidirectional, like the switches described earlier: either terminal can be the input.

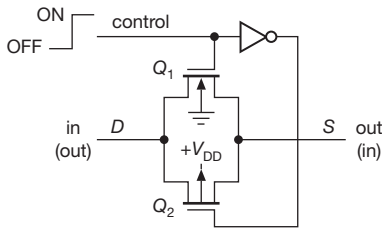


Figure 3.60. CMOS analog switch.

There is a variety of integrated circuit CMOS analog switches available, with different operating voltage ranges and with various switch configurations (e.g., several independent sections with several poles each). Going back to the prehistoric, there's the classic CD4066 "analog transmission gate," which belongs to the original 4000 series of digital CMOS logic and that acts as an analog switch for signals between ground and a single positive supply.[68]

[68] In that role it's happy to switch digital signals as well, thus its family membership.

More commonly, though, you'll choose a dedicated analog switch IC, for example a member of the industry-standard DG211 family. These parts (and their many variations; see Table 3.3 on page 176) are particularly convenient to use: they accept logic-level (0V = LOW, >2.4V = HIGH) control signals, they will handle analog signals to ±15 V (compared with only ±7.5V for the 4000 series), they come in a variety of configurations, and they have relatively low ON resistance (25 Ω or less for some members of these families, and down to a fraction of an ohm for lower voltage switches). Analog Devices, Intersil, Maxim, Vishay–Siliconix, and other manufacturers offer a nice range of analog switches.

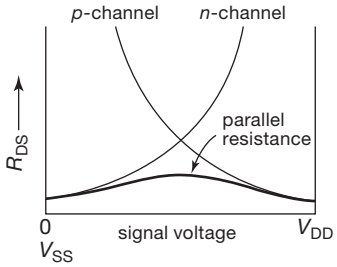


Figure 3.61. ON-resistance of the CMOS analog switch.

B. JFET analog switches

Although most available analog switches are built with a pair of paralleled MOSFETs of complementary polarities – the CMOS architecture just described – it is possible to build analog switches with JFETs, and there are some advantages.

The basic circuit (Figure 3.62) uses a single *n*-channel JFET, Q_1, as the analog switch. Its conduction is controlled by a transistor switch, Q_2, which pulls the gate down to a large negative voltage (−15 V, say) to cut off conduction in the JFET (switch OFF). Shutting off Q_2 lets the gate float to the source voltage, putting the (depletion-mode) JFET into full conduction (switch ON). The gate resistor R_1 is made deliberately large so that the output signal line does not see significant loading in the OFF state; its value is a compromise, because a larger resistance incurs a longer turn-on delay. With a low-impedance signal source (e.g., the output of an op-amp) it may be preferable to put the resistor on the input side (i.e., feed the signal in from the right).

Because there is only a single *n*-channel JFET, this switch cannot accept input signals all the way to the negative supply: signal voltages that come closer to the negative

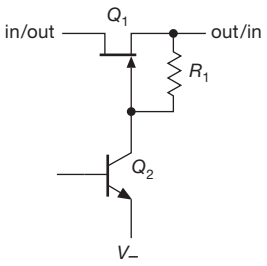


Figure 3.62. JFET analog switch.

supply than the FET's $V_{GS(off)}$ will bring it back into conduction.[69] There is no such limitation at the positive end.

A nice feature of this JFET analog switch is the constancy of R_{ON} with signal level: because the gate stays at the source voltage, there is *no* variation of R_{ON} with signal voltage; the JFET does not even know that the signal is varying! This nice characteristic is on display in Figure 3.63, which compares a plot of R_{ON} versus V_{sig} for a JFET analog switch (SW06) with that of the DG211 CMOS switch.[70]

In practice it's inconvenient to have to drive the switch control with a signal near the negative supply rail (and doubly so for CMOS switches). Instead, you would probably use a level-shifting circuit so a logic-level input that goes between 0 V and +3 V, say, would activate the switch. Figure 3.64 shows a simple way to do that, using comparators with "open-collector" outputs (§12.3) to drive the gates of discrete JFET switches. For *integrated* analog switches, this sort of control signal circuitry is ordinarily built in. The SW06 JFET analog switch (and the nearly extinct DG180–189 family from Vishay–Siliconix) includes such drivers, along with some other elegant tricks.[71]

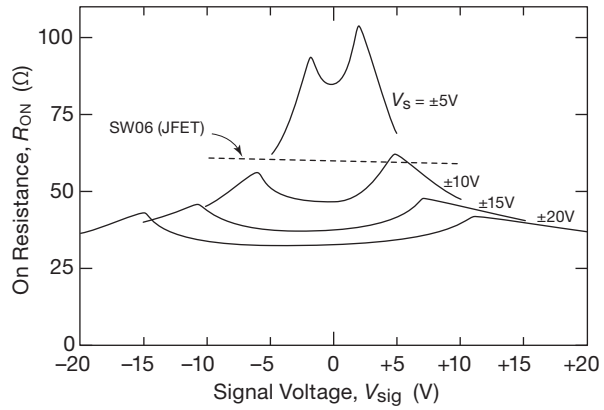


Figure 3.63. Contrasted with a CMOS analog switch like the DG211 (solid curves, shown for different supply voltages), the JFET analog switch maintains admirably flat R_{ON} over signal level.

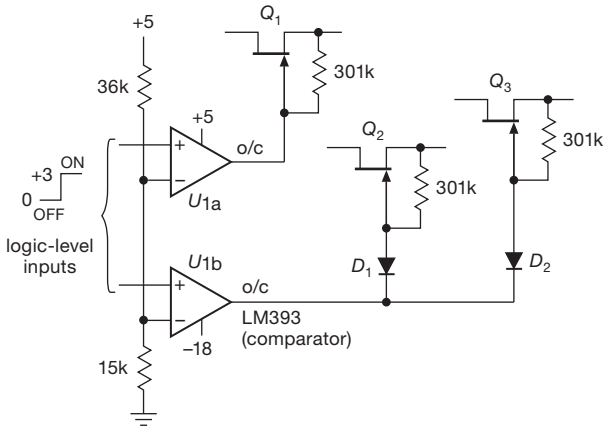


Figure 3.64. A comparator with open collector outputs, powered from +5 V and −18 V, converts a 0-to-3 V logic-level input into a JFET gate drive that swings to −18 V. This method is used in some of Agilent's digital multimeters (see "Designs by the Masters," 13.8.6) to accommodate analog signals over a full ±12 V range. The added diodes allow a control signal to drive more than one JFET switch.

JFET analog switches are inherently more rugged than their CMOS cousins, for which performance-degrading protection circuits are required for robustness against overvoltage faults. They do suffer from high charge injection, however (see §3.4.2E). Despite their nice features, integrated JFET switches and multiplexers (see next subsection) are almost extinct, with fine examples like the SW-01, SW-7510, and MUX-08 series from Precision Monolithics (now Analog Devices) gone forever (but happily the SW06 lives on!).

C. Multiplexers

A nice application of FET analog switches is the "multiplexer" (or MUX), a circuit that allows you to select any of several inputs, as specified by a digital control signal.

[69] In fact, you've got to stay a bit further away from the actual $V_{GS(off)}$: recall (§3.1.3) that it's defined as the gate-source voltage that results in a small (but nonzero) drain current, usually $I_D = 10$ nA.

[70] The slight variation of R_{ON} is caused by substrate effects: the SW06 is an integrated circuit, built on a silicon substrate; so the JFET and associated components have some inkling of the absolute signal level. If even that small variation is unacceptable, you can rig up a discrete implementation (as in Figure 3.62); you'll see such circuitry used in some of Agilent's accurate digital multimeters.

[71] Such as an internal MOSFET switch that disconnects the gate resistor (R_1 in Figure 3.62) when the switch is OFF, to eliminate circuit loading.

The analog signal present on the selected input will be passed through to the (single) output. Figure 3.65 shows the basic scheme. Each of the switches SW0 through SW3 is a CMOS analog switch. The "select logic" decodes the address and *enables* (jargon for "turns on") the addressed switch only, disabling the remaining switches. Such a multiplexer is usually used in conjunction with digital circuitry that generates the appropriate addresses (lots more in Chapters 10 and 11). A typical situation might involve a data-acquisition instrument in which a number of analog input voltages must be sampled in turn, converted to digital quantities, and used as input to some computation.

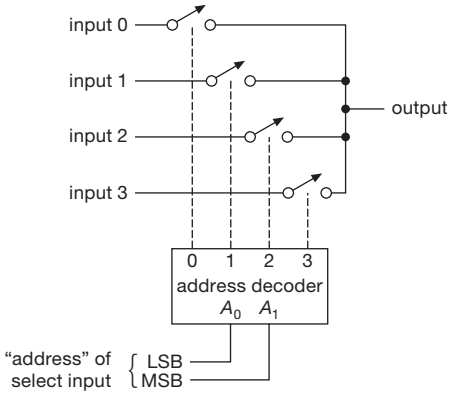


Figure 3.65. Analog multiplexer.

Because analog switches are bidirectional, an analog multiplexer such as this is also a "demultiplexer": a signal can be fed into the "output" and will appear on the selected "input." When we discuss digital circuitry in Chapters 10 and 11, you will see that an analog multiplexer such as this can also be used as a "digital multiplexer–demultiplexer," because logic levels are, after all, nothing but voltages that happen to be interpreted as binary 1s and 0s.

Typical of analog multiplexers are the industry-standard DG408–09 and DG508–09 series (and their many improved versions), 8- or 16-input MUX circuits that accept logic-level address inputs and operate with analog voltages up to ±15 V. The 4051–4053 devices in the CMOS digital family are analog multiplexers–demultiplexers with up to 8 inputs, but with 15 Vpp maximum signal levels; they have a V_{EE} pin (and internal level shifting) so that you can use them with bipolarity analog signals and unipolarity (logic-level) control signals. We are especially fond of the low-voltage '4053 series, with three SPDT switches. Our interest is evidently shared by others, with a large number of interesting parts available – see Table 13.7 (4053-style SPDT Switches) on page 917.

D. Other analog switch applications

Voltage-controlled analog switches form essential building blocks for some of the op-amp circuits we'll see in the next chapter – integrators, sample-and-hold circuits, and peak detectors. For example, with op-amps we will be able to build a "true" integrator (unlike the approximation to an integrator we saw in §1.4.4): a constant input generates a linear ramp output (not an exponential), etc. With such an integrator you must have a method to reset the output; a FET switch across the integrating capacitor does the trick. We won't try to describe these applications here. Because op-amps form essential parts of the circuits, they fit naturally into the next chapter. Great things to look forward to!

3.4.2 Limitations of FET switches

Analog switches aren't perfect – they have nonzero resistance when ON, and nonzero leakage when OFF, as well as capacitive feedthrough and charge injection during changes in switch state. You can see some of the variety in Table 3.3 on page 176). Let's take a look at these limitations.

A. Voltage range and latchup

Analog switches and multiplexers come in three broad voltage ranges: (a) "standard" parts (which might better be called "high-voltage"), which handle signals over the full traditional op-amp voltage range of ±15 V; (b) reduced-voltage parts ("midvoltage") that can handle ±7.5 V (or 0 to +15 V) signals; and (c) low-voltage parts intended for applications in which the signal swing does not exceed ±3 V (or 0 to +6 V). In all cases, analog switches operate properly (and with specified R_{ON}) with input signals that go all the way to the positive and negative supply voltages (with the exception of JFET switches like the SW06, for which the operating signal voltage range does not reach the positive supply).

However, input signals *beyond* the supply rails are another story. All CMOS integrated circuits have some form of input protection circuit, because otherwise the gate insulation is easily destroyed (see §3.5.4H on handling precautions). The usual protection network is shown in Figure 3.66. Although it may use distributed diodes, the network is equivalent to clamping diodes to V_{SS} and to V_{DD}, combined with resistive current limiting. If you drive the inputs (or outputs) more than a diode drop beyond the supply voltages, the diode clamps go into conduction, making the inputs (or outputs) look like a low impedance to the respective supplies. Worse still, the chip can be driven into "SCR latchup," a terrifying (and destructive)

condition we'll describe in more detail in §10.8. For now, all you need to know about it is that you don't want it! SCR latchup is triggered by input currents (through the protection network) of roughly 20 mA or more. Thus you must be careful not to drive the analog inputs as much as a diode drop beyond the rails.[72] This means, for instance, that for most parts you must be sure the power supply voltages are applied before any signals that have significant drive-current capability; alternatively, you can use series diodes in the supply lines, so that input signals applied before the dc power do not produce input current.

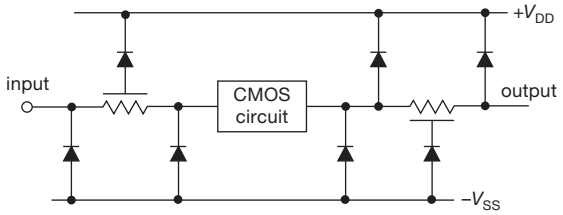


Figure 3.66. CMOS input–output protection networks. The series resistor at the output is often omitted.

The trouble with diode–resistor protection networks is that they compromise switch performance by increasing R_{ON}, shunt capacitance, and leakage. A different approach makes use of "dielectric isolation" to eliminate SCR latchup without the serious performance compromises inherent in traditional protection networks. Both methods result in a "protected" (or *fault-protected*) analog switch, in which you may safely overdrive the inputs without damage. Note, however, that the output does not follow the input beyond the rails.[73]

For example, the MAX4508 multiplexer adds fault protection to the standard DG508A 8-input analog multiplexer, making it tolerant to ±30 V input swings; it has an $R_{ON(typ)}$ of 300 Ω. The AD7510DI-series of "Protected Analog Switches" from Analog Devices uses dielectric isolation to achieve input signal fault protection to ±25 V beyond the power supplies, while maintaining a respectable 75 Ω $R_{ON(typ)}$ in the normal operating signal range. Watch out, though – fault protection is the exception in the analog switch arena, and most analog switch ICs are not forgiving!

Maxim offers a nice outboard solution that you can put in front of an unprotected switch (or any other analog component), in the form of its multichannel "signal-line protector" ICs (the three- and eight-channel MAX4506–07).[74] These accept input signal swings to ±36V (whether powered or unpowered), are free of latchup regardless of power sequencing, and they pass through signals that are properly within the supply limits while clamping their outputs to the supply rails (which can be ±8 V to ±18 V split supplies, or a +9 to +36 V single supply) when there's an input beyond the rails. They even have the grace to open-circuit the input when thus overdriven – see Figure 3.67. The price you pay (above their literal $4 and $6 cost, respectively) is an ON-resistance in the range of 50–100 Ω (depending as usual on total supply voltage) and an input capacitance of 20 pF (thus a rolloff at ∼100 MHz).

B. ON-resistance

CMOS switches operated from a relatively high supply voltage (±15 V, say) will have low R_{ON} over the entire signal swing, because one or the other of the transmission FETs will have a forward gate bias of at least half the

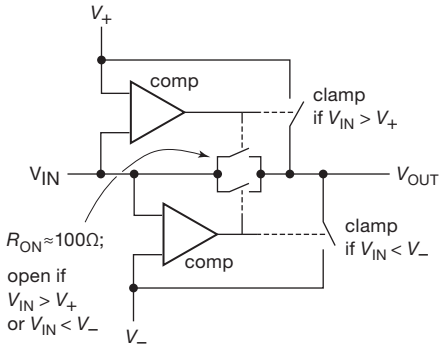


Figure 3.67. Maxim's MAX4506–7 fault-protected "Signal-Line Protectors" prevent signal swings beyond the rails, both clamping the output and disconnecting the input when overdriven. The series analog switches are normally ON, but both are switched OFF if V_{IN} goes beyond either rail, controlled by logic shown in the more complete MAX4508 datasheet.

[72] This prohibition goes for *digital* CMOS ICs as well as the analog switches we have been discussing.

[73] With a few exotic exceptions, for example the MAX14778 "Dual ±25V Above- and Below-the-Rails 4:1 Analog Multiplexer." This puppy, which runs on a single +3 to +5 V supply, is not only fault protected to ±25 V, it operates properly with signal voltages over that full range! How do they do it?! Turns out they have included an on-chip "charge-pump" voltage converter (§3.4.3D) to power the guts. Even more remarkable, this device has a very low ON-resistance (1.5 Ω) that is remarkably flat with signal voltage (0.003 Ω) over the full ±25 V range. And it can handle up to 300 mA of signal current. This device is intended for applications in which external large-swing signals need to be switched by a circuit with only a low-voltage single supply. Sadly for us experimenters, it comes only in the tiny TQFN (Thin Quad Flat-pack, No leads) package, requiring a reflow oven for soldering onto a circuit board.

[74] Analog Devices offers a similar single-channel part, the ADG465, in a convenient SOT23-6 package.

Table 3.3 Analog Switches

Part #[a]	Config[b] SPST (NO)	SPST (NC)	SPDT[u]	MUX[v]	Supply split (±V)	Supply single (+V)	R_{on} typ @ V_{supply} (Ω)	R_{on} @ V_{supply} (V)	Q_{inj} typ (pC)	Cap typ, ON (pF)	Logic[d]	Control[e]	Packages SOT-23	DIP	SOIC	other	Price qty 25 ($US)	Comments
high voltage																		
MAX4800-02	-	-	-	8:1	40 to 100	c	22	±100	600	36	V_L	S	-	-	-	28	16.18	*really* HV![1]
MAX326-27	4	4	-	-	5 to 18	10 to 30	1500	±15	2	6	T	P	-	16	16	16	6.76	low leakage[2]
MAX4508-09	-	-	-	84	4.5 to 20	9 to 36	300	±15	2	28, 22	T	P	-	16	16	-	6.78	OV to ±30V
MAX354-55	-	-	-	84	4.5 to 18	4.5 to 36	285	±15	80	28, 14	T	P	-	16	16	-	8.43	OV to ±25V
DG508-09	-	-	-	84	5 to 20	10 to 36	180	±15	2	18, 11	T	P	-	16	16	16	2.56	low leakage[3]
ADG1211-13	w	w	-	-	5 to 15	10 to ?	120	±15	0.3	2.6	T	P	-	-	-	16	4.63	low *C*, Q_{inj}
ADG1221-23	x	x	-	-	5 to 16.5	5 to 16.5	120	±15	0.1	3	T	P	-	-	-	10	3.04	low flat Q_{inj}
AD7510-12DI	4	4	2	-	5 to 15	-	75	±15	30	17	T	P	-	16	-	20	13.90	OV rails±25V
SW06	4	-	-	-	12 to 18	-	60	±15	-	15	T	P	-	16	16	20	4.30	JFET, flat R_{ON}
DG441-42	4	4	-	-	4.5 to 22	5 to 24	50	±15	2	16	V_L	P	-	16	16	16	2.05	1µA I_{supply}
DG211-12	4	4	-	-	4.5 to 22	5 to 22	45	±15	1	16	V_L	P	-	16	16	16	2.08	ADG211-12
DG408-09	-	-	-	84	5 to 20	5 to 30	40	±15	20	37, 25	T	P	-	16	16	16	3.06	ADG408-09
ADG417-19	1	1	1	-	5 to 20	5 to 20	25	±15	3	30	V_L	P	-	8	8	8	2.72	DG417-19
MAX317-19	1	1	1	-	4.5 to 20	10 to 30	20	±15	3	30	V_L	P	-	8	8	-	3.34	
DG411-13	w	w	-	-	4.5 to 20	10 to 30	17	±15	5	35	V_L	P	-	16	16	20	2.70	ADG411-13
DG447-48	1	1	-	-	4.5 to 20	7 to 36	13	±15	10	30	T	P	-	-	6	-	1.14	
ADG5412-13	y	y	-	-	9 to 22	9 to 40	10	±15	240	60	T	P	-	-	-	16	5.66	no latchup[4]
DG467-68	1	1	-	-	4.5 to 20	7 to 36	5	±15	21	76	T	P	-	-	8	-	0.75	ADG467-68
mid-voltage																		
DG4051-53	-	-	-	842	2.5 to 5	2.7 to 12	66	±5	0.25	3.4	V+	P	-	-	16	16	1.35	Table 13.7
74HC4051-53	-	-	-	842	2.5 to 5	2 to 10.5	40	±5		5, 25	V+	P	-	16	16	16	0.41	Table 13.7
MAX4541-44	x	x	1	-	-	2.7 to 12	30	+5	1	13, 20	T	P	8	8	8	8	1.33	
ISL5120-23	x	x	1	-	-	2.7 to 12	19	+5	3	28	T	P	8,6	-	8	-	1.71	
ISL43210	-	-	1	-	-	2.7 to 12	19	+5	3	28	T	P	6[s]	-	-	-	1.33	
ADG619-20[k]	-	-	1	-	2.7 to 5.5	2.7 to 5.5	7	+5	6	95	T	P	8	-	-	8	2.56	
low-voltage																		
ADG708-09	-	-	-	84	2.5	1.8 to 5.5	3	+5	3	96, 48	T	P	-	-	16	-	3.01	
ADG719	-	-	1	-	-	1.8 to 5.5	2.5	+5	-	27	T	P	6	-	-	-	1.76	SC70='749
MAX4624-25[k]	-	-	1	-	-	1.8 to 5.5	0.65	+5	65	100	T	P	6	-	-	-	2.10	
ADG884	-	-	2	-	-	1.8 to 5.5	0.3	+5	125	300	T	P	-	-	-	10	2.42	(5)
ISL84467	2xDPDT			-	-	1.8 to 5.5	0.55	+3	126	102	V+	P	-	-	16	16	1.19	80mΩ match
ISL84714	-	-	1	-	-	1.8 to 3.6	0.44	+3	20	100	V+	P	6	-	-	-	1.24	5mΩ match
NLAS52231	-	-	2	-	-	1.65 to 4.5	0.38	+3	53	85	V+	P	-	-	-	8	0.90	(6)
ISL43L110-11	1	1	-	-	-	1.1 to 4.7	0.25	+3	72	160	V+	P	5[s]	-	-	-	0.65	lowest R_{ON}
T-switch, RF																		
MAX4565-67	y	y	2	-	2.7 to 6	2.7 to 12	46	±5	25	6	T	P	-	16,20		16,20	4.74	video[7]
DG540-42	z	z	-	-	+15 & –3		30	nom	25	14	T	P	-	16	16	20	5.00	
AD8170, 74			1	4:1	4 to 6	-	NA	-	NA	1.1	T	P	-	8,14		-	5.11[n]	MUX+amp[8]
ADG918-19	-	-	1	-	-	1.65 to 2.75	-	-	-	1.6	V+	P	-	-	-	8	2.52	RF[9]
crosspoint																		
AD75019	16x16				4.5 to 12	9 to 25	150	±12	-	10 m	T	S	-	-	-	44	26.20	20 MHz
ADG2188	8x8				4.5 to 5.5	8 to 12	34	±5	3	9.5	V_L	I	-	-	-	32	9.37	200 MHz
MAX4359-60	4x4, 8x8				4.5 to 5.5	-	x1 buf	-	NA	8	T	P, ser	-	40	24	44	9.68[n]	35 MHz
MAX9675	16x16				2.5 to 5.5	-	x1,x2 buf	-	(g)	5	V_L	S	-	-	-	100	24.14	110 MHz
MAX4550, 70	dual 4x2				2.7 to 5.5	2.7 to 5.5	43	±5	7	11[o]	V+	S or I	-	2	-	28	6.39	audio[10]

Notes: (a) listed within categories by decreasing R_{ON}; all are CMOS except SW06; parts in ***bold italic*** are widely used "jellybeans." (b) numerals represent quantity of each switch type in a single package; letters refer to explanatory footnotes for successively numbered parts. (c) V_{neg} at least –15V, V_{pos} at least +40V, total no more than 200V. (d) T=TTL thresholds; V_L=external logic threshold supply; V+="CMOS" threshold, depends on positive analog supply voltage. (e) P=parallel logic inputs; I=I^2C serial; S=SPI serial. (g) 50mV glitch. (h) 0.1dB to 14MHz, –95dB xtalk at 20KHz & R_L=10k, THD+N=0.014% (R_L=1k, *f*=1kHz). (k) second p/n is make-before-break. (m) min or max. (n) higher p/n is ≈50% more. (o) switch OFF. (s) SC-70. (u) SPDT are break-before-make unless noted otherwise. (v) 84=8:1 & dual 4:1; 842=8:1, dual 4:1, & triple 2:1. all dual 4:1 have single 2-bit address. (w) 4xNC, 4xNO, 2 each. (x) 2xNO, 2xNC, 1 each. (y) 4xNO, 2 each. (z) 4xNO, 4xNO, 2 each.

Comments: (1) Supertex HV2203. (2) 1pA typ. (3) 3pA typ; I_S=10µA typ. MAX308-09 is similar. (4) 8kV HBM ESD. (5) 0.4Ω@V_S=+3V; 400mA; –3dB/50Ω@18MHz. (6) low R_{ON}, e.g. speaker switch. (7) –3dB@350MHz, –90dB xtalk@10MHz. (8) 250 MHz; ext gain set. (9) –3dB at 4GHz, -30dB xtalk at 4GHz. (10) clickless.

supply voltage. However, when operated with lower supply voltages, the switch's R_{ON} value will rise, the maximum occurring when the signal is about halfway between the supply and ground (or halfway between the supplies, for dual-supply voltages). Figure 3.68 shows why. As V_{DD} is reduced, the FETs begin to have significantly higher ON-resistance (especially near $V_{GS} = V_{DD}/2$), because enhancement-mode FETs can have a $V_{GS(th)}$ of at least a few volts, and a gate-source voltage of as much as 5 to 10 volts may be required to achieve low R_{ON}. Not only will the parallel resistances of the two FETs rise for signal voltages between the supply voltage and ground, but also the peak resistance (at half V_{DD}) will rise as V_{DD} is reduced, and for sufficiently low V_{DD} the switch will become an open circuit for signals near $V_{DD}/2$.

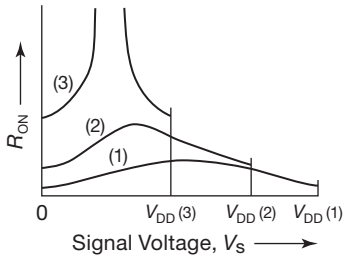


Figure 3.68. CMOS analog switch R_{ON} increases at low supply voltage.

There are various tricks used by the designers of analog switch ICs to keep R_{ON} low and approximately constant (for low distortion) over the signal swing. For example, the original 4016 analog switch used the simple circuit of Figure 3.60, producing R_{ON} curves that look like those in Figure 3.69. In the improved 4066 switch, the designers added a few extra FETs so that the *n*-channel body voltage follows the signal voltage, producing the R_{ON} curves of Figure 3.70. The "volcano" shape, with its depressed central

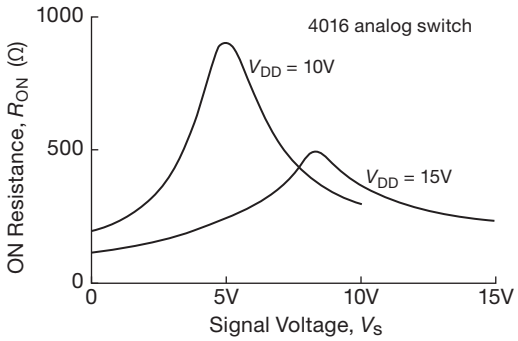


Figure 3.69. ON-resistance for 4016 CMOS switch.

R_{ON}, replaces the "Everest" shape of the 4016. Improved switches, like the industry standard DG408–09, intended for serious analog applications, succeed even better, with low and flat R_{ON} curves that deviate less than 10% or so over the signal voltage range. This is often achieved at the expense of increased "charge transfer" (see the later section on *glitches*).

Looking through manufacturers' selection tables for analog switches, you'll find standard-voltage units with R_{ON} as low as several ohms and flatness to a few tenths of an ohm; low-voltage switches can be found with R_{ON} as low as 0.25 Ω, and flatness of 0.03 Ω. This static performance comes at a real cost though, namely, high capacitance and high charge injection (see discussion below, and Table 3.3 on the preceding page). If your application requires low distortion into moderate load impedances, the better approach may be to choose a switch with excellent "ON-resistance flatness" spec ($R_{FLAT(ON)}$), and accept a higher overall R_{ON} with its lower capacitance.

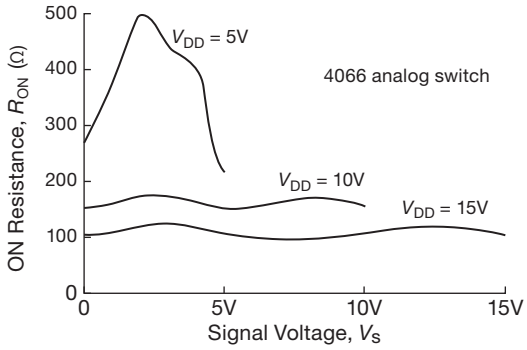


Figure 3.70. ON-resistance for the improved 4066 CMOS switch; note change of scale from previous figure.

Keep in mind, too, that in some cases you can finesse the problem altogether with a different choice of circuit configuration, as in Figure 3.71, which shows three approaches to a circuit that selects one of two input signals. Circuit A's gain is $R_2/(R_1 + R_{ON})$, so a variation of R_{ON} with signal amplitude produces changes of gain, and thus nonlinearity. Circuit B is better, because the switch output is held at ground by feedback around the op-amp; but the ON-resistance still reduces the gain somewhat, degrading the circuit's precision. Circuit C is blissfully unaware of R_{ON}, owing to the op-amp's very high input impedance; it's the most linear and precise of all.

This lesson can be applied to other circuit configurations as well. As an example, take a look at Figure 3.84 on page 183, where an analog multiplexer is used to select an amplifier's overall voltage gain. In the circuit of

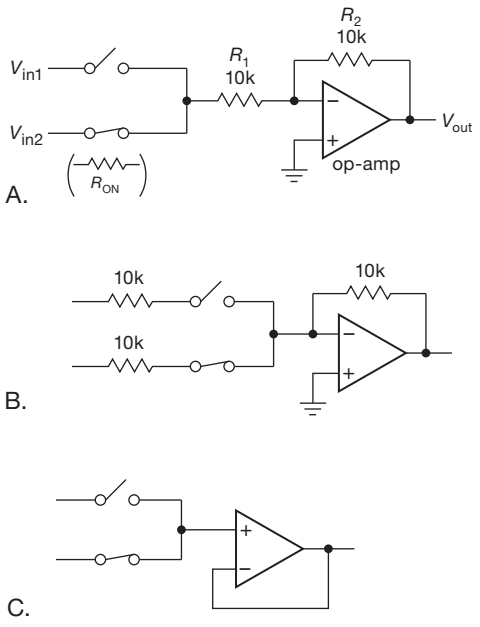


Figure 3.71. Finessing R_{ON} variation in analog switches: three ways to select between a pair of input signals, with an op-amp to buffer the output.

Figure 3.84A the multiplexer's R_{ON} is in series with the selected resistor, and represents an error term (both in gain and nonlinearity); by contrast, in the circuit of Figure 3.84B the switch's R_{ON} is irrelevant, owing to the op-amp's essentially infinite ($>10^{12}\,\Omega$) input impedance.

Another trick exploits the use of two identical (or closely-similar) JFET switches to largely cancel the effects of R_{ON}. See Chapter *4x*'s section "JFET linear switch with R_{ON} compensation" for an explanation and illustration of this elegant technique.

C. Speed

High-voltage FET analog switches have ON-resistances R_{ON} generally in the range of 20 to 200 Ω.[75] In combination with substrate and stray capacitances, this resistance forms a lowpass filter that limits operating speeds to frequencies of the order of 10 MHz or less (Figure 3.72). FETs with lower R_{ON} tend to have larger capacitances (up to 50 pF or more), so no gain in speed results (unless the designer has made other design tradeoffs). Much of the rolloff is due to protection components – current-limiting series resistance and capacitance of shunt diodes.

[75] As we remarked, you can get switches with lower R_{ON}, as low as 0.25 Ω, at the expense of some combination of increased capacitance, increased charge injection, and reduced operating voltage range.

However, low-voltage analog switches do better in terms of bandwidth (as is usually the case with smaller-geometry semiconductors): a contemporary ±2.5 V analog switch like the popular ADG719 has 2.5 Ω of ON-resistance, 27 pF of capacitance, and 400 MHz bandwidth. There is also a class of analog switches and multiplexers targeted specifically at video and RF applications. These include both passive ("unbuffered") MUX/switches, and MUX/switches combined with an amplifier ("active" or "buffered"). Active MUX/switches operate on +5 V or ±5 V supplies and have fixed voltage gains of either ×1 or ×2 (the latter are intended for driving a 50 Ω or 75 Ω transmission line through a series matching resistor, which attenuates the output by a factor of 2); in some cases you set the gain with an external resistor pair. An example of the latter is the AD8174 4-input multiplexer, with a bandwidth of 270 MHz at gains of +1 or +2 (at higher gains the bandwidth drops, e.g., to 55 MHz at G=+10).

For specialized applications you can get some really fast analog switches, for example the ADG918–19 listed in Table 3.3 on page 176, which is usable to 2 GHz (it's down 3 dB at 4 GHz). Parts like these are used in wireless applications, for example, to switch between two signal sources in "diversity reception" or to route gigahertz signals through a choice of filter paths. To reduce crosstalk, these wideband switches usually employ a T-switch topology (see Figure 3.77 in the next subsection).

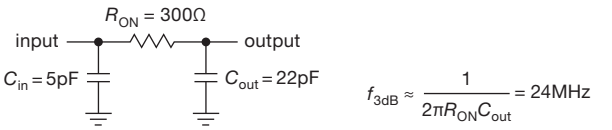


Figure 3.72. The parasitic *RC*'s of a CMOS switch limit the analog signal bandwidth.

D. Capacitance

FET switches exhibit capacitance from input to output (C_{DS}), from channel to ground (C_D, C_S), from gate to channel, and from one FET to another within one IC package (C_{DD}, C_{SS}); see Figure 3.73. Let's look at the effects.

C_{DS}: capacitance from input to output

Capacitance from input to output causes signal coupling in an OFF switch, rising at high frequencies. Figure 3.74 shows the effect for the popular DG211 and DG411 series. Note the characterization with a 50 Ω load, common in RF circuits, but much lower than normal for low-frequency signals, for which a typical load impedance is 10k or more. Even with a 50 Ω load, the feedthrough becomes

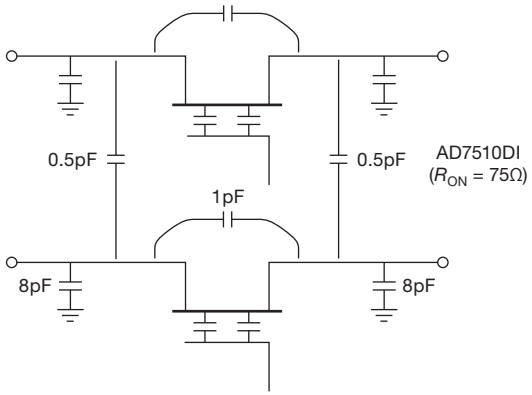


Figure 3.73. Capacitances between isolated sections of the AD7510 quad analog switch cause signal crosstalk.

significant at high frequencies (1 pF has a reactance of 5k at 30 MHz, giving −40 dB of feedthrough). And, of course, there is significant attenuation (and nonlinearity) driving a 50 Ω load, because for these parts R_{ON} is typically 45 Ω and 17 Ω, respectively. With a 10k load the feedthrough situation is much worse, of course.

Exercise 3.5. Calculate the feedthrough into 10k at 1 MHz, assuming $C_{DS} = 1$ pF.

In most low-frequency applications capacitive feedthrough is not a problem. If it is, the best solution is to use a pair of cascaded switches (Figure 3.75) or, better still, a combination of series and shunt switches, enabled alternately (Figure 3.76). The series cascade doubles the attenuation (in decibels) at the expense of additional R_{ON}, whereas the series-shunt circuit (effectively an SPDT configuration) reduces feedthrough by dropping the effective load resistance to R_{ON} when the series switch

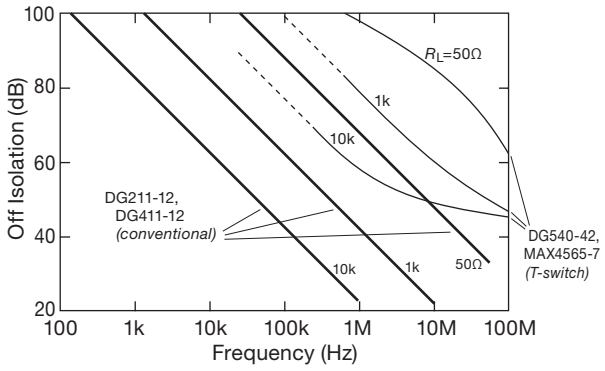


Figure 3.74. High-frequency feedthrough in analog switches. There is less feedthrough with a low load resistance and less still with a "T-switch" configuration.

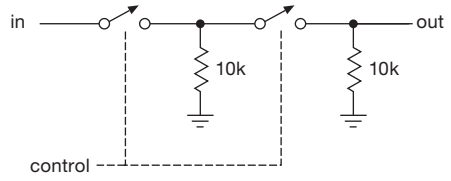


Figure 3.75. Cascaded analog switches for reduced feedthrough.

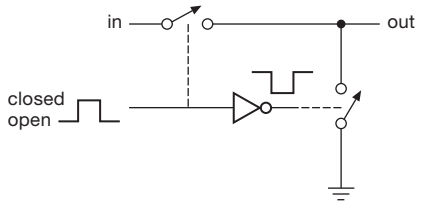


Figure 3.76. SPDT analog switch configuration for reduced feedthrough.

is off. Some commercial analog switches are built with a T-network of three switches (Figure 3.77) to achieve low feedthrough for signals going in either direction; from the outside you can't even tell that they've used this trick, except by noticing the excellent isolation specifications as in Figure 3.74 (unless, of course, they brag about it on the datasheet).

Exercise 3.6. Recalculate switch feedthrough into 10k at 1 MHz, assuming C_{DS}= 1 pF and R_{ON}=50Ω, for the configuration of Figure 3.76.

Most CMOS SPDT switches have controlled break-before-make (BBM) characteristics so that the signal sources are not momentarily connected during switching. In some cases, however, you need the reverse, i.e., make-before-break (MBB), for example in a gain-selecting feedback circuit like Figure 3.84B. To deal with this, some CMOS switches are available in both flavors, for example the ADG619 and ADG620 (BBM and MBB, respectively, as noted in Table 3.3).

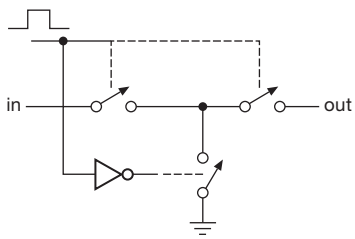


Figure 3.77. A T-switch further reduces high-frequency feedthrough.

C_D, C_S: capacitance to ground

Shunt capacitance to ground leads to the high-frequency rolloff mentioned earlier. The situation is worst with a high-impedance signal source, but even with a stiff source the switch's R_{ON} combines with the shunt capacitance at the output to make a lowpass filter. The following problem shows how it goes.

Exercise 3.7. An AD7510 (here chosen for its complete capacitance specifications, shown in Figure 3.73) is driven by a signal source of 10k, with a load impedance of 100k at the switch's output. Where is the high-frequency -3 dB point? Now repeat the calculation, assuming a perfectly stiff signal source and a switch R_{ON} of 75 ohms.

Capacitance from gate to channel

Capacitance from the controlling gate to the channel causes a different effect, namely, the coupling of nasty little transients into your signal when the switch is turned on or off. This subject is worth some serious discussion, so we'll defer it to the next section on glitches.

C_{DD}, C_{SS}: capacitance between switches

If you package several switches on a single piece of silicon the size of a grain of rice, it shouldn't surprise you if there is some coupling between channels ("crosstalk"). The culprit, of course, is cross-channel capacitance. The effect increases with frequency and with signal impedance in the channel to which the signal is coupled. Here's a chance to work it out for yourself.

Exercise 3.8. Calculate the coupling, in decibels, between a pair of channels with $C_{DD} = C_{SS} = 0.5$ pF (Figure 3.73) for the source and load impedances of the previous exercise. Assume that the interfering signal is 1 MHz. In each case calculate the coupling for (a) OFF switch to OFF switch, (b) OFF switch to ON switch, (c) ON switch to OFF switch, and (d) ON switch to ON switch.

It should be obvious from this example why most broadband RF circuits use low signal impedances, usually 50 Ω. If crosstalk is a serious problem, don't put more than one signal on one chip.

E. Glitches and charge injection

During turn-on and turn-off transients, FET analog switches can do nasty things. The control signal being applied to the gate(s) can couple capacitively to the channel(s), putting ugly transients on your signal. The situation is most serious if the signal is at high impedance levels. Multiplexers can show similar behavior during transitions of the input address as well as a momentary connection between inputs if the turn-off delay exceeds the turn-on delay (i.e., MBB). A related bad habit is the propensity of some switches (e.g., the 4066) to short the input to ground momentarily during changes of state.

Let's look at this problem in a bit more detail. Figure 3.78 shows a typical waveform you might see at the output of an *n*-channel MOSFET analog switch circuit, similar to Figure 3.59, with an input signal level of zero volts and an output load consisting of 10k in parallel with 20 pF, realistic values for an analog switch circuit. The handsome transients are caused by charge transferred to the channel, through the gate-channel capacitance, at the transitions of the gate. The gate makes a sudden step from one supply voltage to the other, in this case between ± 15 V supplies, transferring a slug of charge

$$Q = C_{GC}[V_G(\text{finish}) - V_G(\text{start})].$$

Here, C_{GC} is the gate-channel capacitance, typically around 5 pF. Note that the amount of charge transferred to the channel ("charge injection") depends only on the total voltage change at the gate, not on its rise time. Slowing down the gate signal gives rise to a smaller-amplitude glitch of longer duration, with the same total area under its graph. Lowpass filtering of the switch's output signal has the same effect. Such measures may help if the peak amplitude of the glitch must be kept small, but in general they are ineffective in eliminating gate feedthrough. In some cases the gate-channel capacitance may be predictable enough for you to cancel the spikes by coupling an inverted version of the gate signal through a small adjustable capacitor.

The gate-channel capacitance is distributed over the length of the channel, which means that some of the charge is coupled back to the switch's input. As a result, the size of the output glitch depends on the signal-source impedance and is smallest when the switch is driven by a voltage source. Of course, reducing the size of the load impedance will reduce the size of the glitch, but this also loads the source and introduces error and nonlinearity because of finite R_{ON}. Finally, all other things being equal, a switch with smaller gate-channel capacitance will introduce smaller

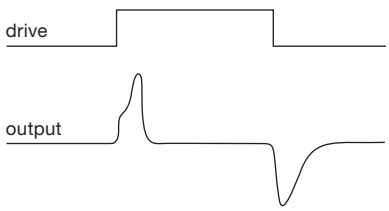


Figure 3.78. Charge-transfer glitches, on a greatly expanded scale.

switching transients, although you pay a price in the form of increased R_{ON}.

Figure 3.79 shows an interesting comparison of gate-induced charge transfers for a collection of analog switches, including JFETs. For the CMOS switches the internal gate signals are making a full rail-to-rail swing (e.g., $\Delta V = 30\,V$ for switches running from $\pm 15\,V$); for the JFET switch the gate swings from $-15\,V$ to the signal voltage. The JFET switch shows a strong dependence of glitch size on signal, because the gate swing is proportional to the level of the signal above $-15\,V$. Well-balanced CMOS switches have relatively low feedthrough because the charge contributions of the complementary MOSFETs tend to cancel out (one gate is rising while the other is falling). Just to give scale to these figures, it should be pointed out that 30 pC corresponds to a 3 mV step across a 0.01 μF capacitor. That's a rather large filter capacitor, and you can see that this is a real problem, since a 3 mV glitch is pretty large when dealing with low-level analog signals.

In Figure 3.80 we've plotted, on an expanded scale, the charge injection scene for a selection of analog switches that exhibit particularly low charge injections. Switches optimized for low charge injection will usually brag about it on the datasheet's headline. For example, the Analog

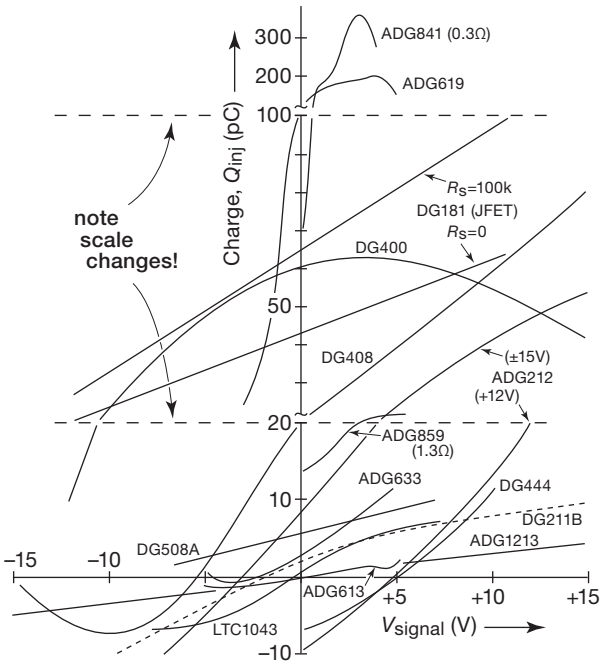


Figure 3.79. Charge transfer for various FET linear switches as a function of signal voltage, taken from the respective datasheets.

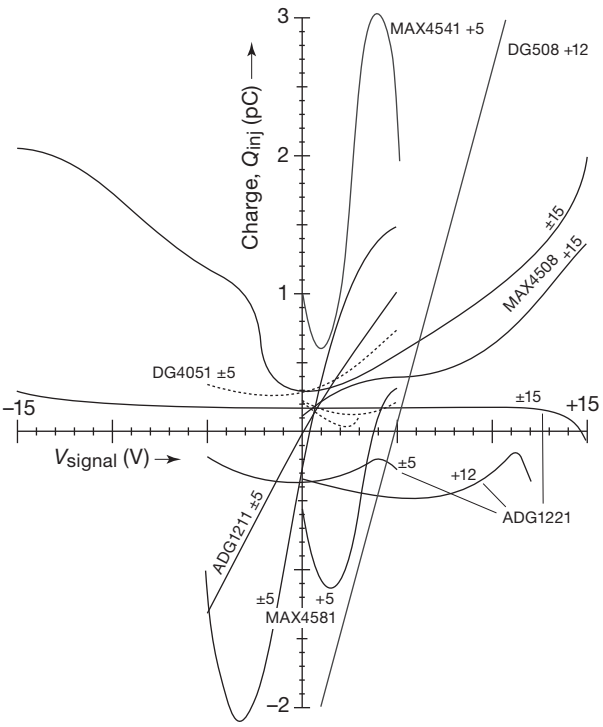


Figure 3.80. Need an analog switch with low charge injection? Here are some candidates, plotted on a greatly expanded scale. The three dotted curves are for the DG4051 with ±5V, +5V, and +3V supplies. Check the datasheet for analogous plots for the DG4053 triple SPDT switch.

Devices ADG1221-series datasheet shouts, in bold letters, "**Low Capacitance, Low Charge Injection, ±15 V/+12 V *i*CMOS® Dual SPST Switches**"; quite a mouthful, but quite a switch!

As might be expected, switches with lower ON-resistance generally exhibit greater charge injection. Figure 3.81 shows this trend, in a scatterplot of datasheet values of Q_{inj} versus R_{ON} for the low-voltage CMOS analog switches currently offered by Analog Devices.

F. Other switch limitations

Some additional characteristics of analog switches that may or may not be important in any given application are switching time, settling time, BBM delay, channel leakage current (both ON and OFF), device quiescent current, input current during overvoltage, R_{ON} matching among multiple channels, and temperature coefficient of R_{ON}. We'll show unusual restraint by ending the discussion at this point, leaving the reader to look into these details when the circuit application demands it.

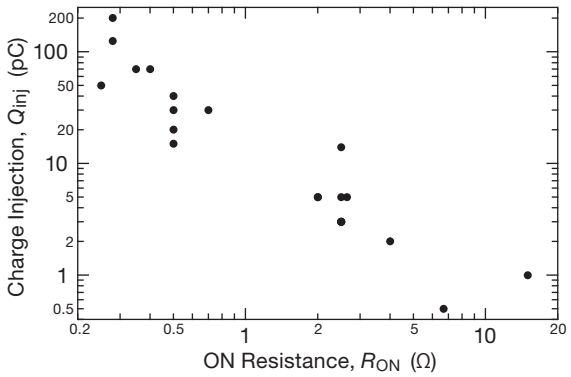


Figure 3.81. Scatterplot of specified charge injection versus ON-resistance for Analog Devices' low-voltage analog switches, illustrating the R_{ON} versus Q_{inj} tradeoff.

3.4.3 Some FET analog switch examples

As we indicated earlier, many of the natural applications of FET analog switches are in op-amp circuits, which we treat in the next chapter. In this section we show a few switch applications that do not require op-amps, to give a feeling for the sorts of circuits you can use them in.

A. Switchable *RC* lowpass filter

Figure 3.82 shows how you could make a simple *RC* lowpass filter with selectable 3 dB points. We've used a multiplexer to select one of four preset resistors, using a 2-bit (digital) address. We chose to put the switch at the input, rather than after the resistors, because there is less charge injection at a point of lower signal impedance. Another possibility, of course, is to use FET switches to select the *capacitor*. To generate a very wide range of time constants you might have to do that, but the switch's finite R_{ON} would limit attenuation at high frequencies, to a maximum of R_{ON}/R_{series}. We've also indicated a unity-gain buffer, following the filter, since the output impedance is high. You'll see how to make "perfect" followers (precise gain, high Z_{in}, low Z_{out}, and no V_{BE} offsets, etc.) in the next chapter. Of course, if the amplifier that follows the filter has high input impedance, you don't need the buffer.

Figure 3.83 shows a simple variation in which we've used four independent switches, rather than a 4-input multiplexer. With the resistors scaled as shown, you can generate 16 equally spaced 3 dB frequencies by turning on binary combinations of the switches.[76]

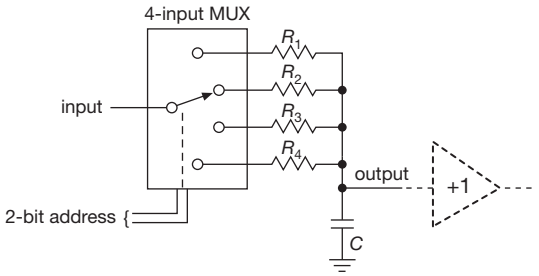


Figure 3.82. Analog-MUX selectable *RC* lowpass filter.

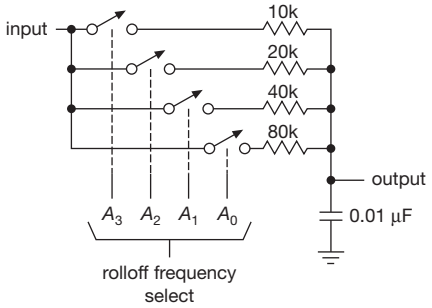


Figure 3.83. *RC* lowpass filter with choice of 15 equally spaced cutoff frequencies.

Exercise 3.9. What are the 3 dB frequencies for this circuit? Estimate the gain-switching glitch amplitude, assuming a charge injection specification of 20 pC, distributed equally to the input and output switch terminals, and a signal source of low impedance.

B. Switchable gain amplifier

Figure 3.84 shows how you can apply the same idea of switching resistors to produce an amplifier of selectable gain. Although this idea is a natural for op-amps, we can use it with the emitter-degenerated amplifier. We used a constant-current sink as emitter load to permit gains much less than unity. We then used the multiplexer to select one of four emitter resistors. Note the blocking capacitor, needed to keep the quiescent current independent of gain.

In Circuit A the switch's R_{ON} value is part of the gain equation. By contrast, in Circuit B the switch selects a voltage-divider tap and presents it to a high impedance op-amp input, so the switch's R_{ON} doesn't affect the gain accuracy. Other (more complex) examples of this approach are found in Figures 5.59, 5.62, and 5.80.

[76] An easy way to see that the 3 dB frequencies are integral multiples of the lowest setting is to rewrite f_{3dB} in terms of the *conductance* of the parallel resistance R_p of the selected resistors: $f_{3dB}=1/2\pi R_pC=G_p/2\pi C$. Then it's easy, because the conductance of resistors in parallel is the sum of their individual conductances. So for this circuit $f_{3dB}=nG_{80k}/2\pi C=199n$ Hz, where $G_{80k}=12.5\mu$S, $C=10$ nF, and n is the integer [1..15] represented by the selected switches A_n.

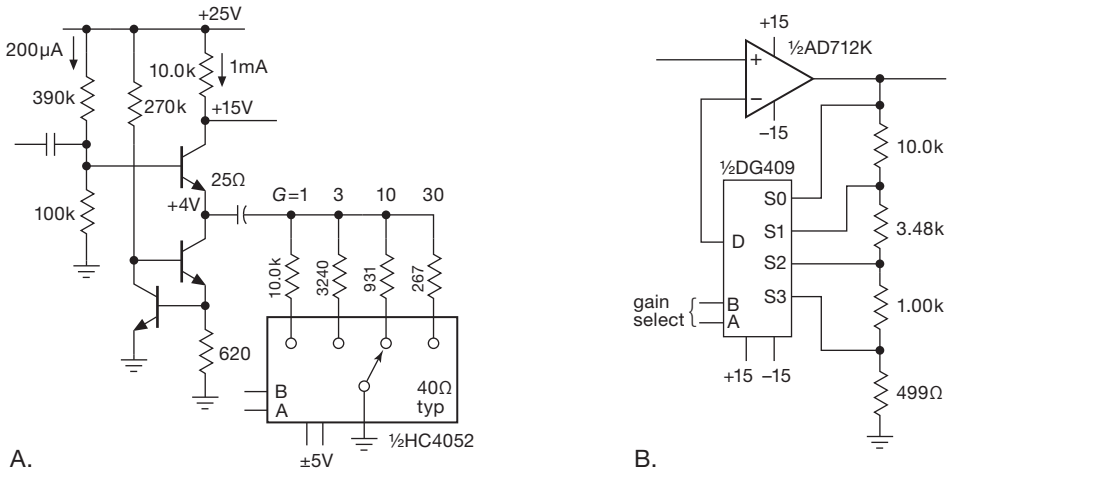


Figure 3.84. A. An analog multiplexer selects appropriate emitter degeneration resistors to achieve decade-switchable gain. B. A similar technique, but with the versatile "op-amp" building block (the hero of Chapter 4).

C. Sample-and-hold

Figure 3.85 shows how to make a "sample-and-hold" (S/H) circuit, which comes in handy when you want to convert an analog signal to a stream of digital quantities ("analog-to-digital conversion") – you've got to hold each analog level steady while you figure out how big it is. The circuit is simple: a unity-gain input buffer generates a low-impedance copy of the input signal, forcing it across a small capacitor. To hold the analog level at any moment, you simply open the switch. The high input impedance of the second buffer (which should have FET input transistors to keep input current near zero) prevents loading of the capacitor, so it holds its voltage until the FET switch is again closed.

Exercise 3.10. The input buffer must supply current to keep the capacitor following a varying signal. Calculate the buffer's peak output current when the circuit is driven by an input sinewave of 1 V amplitude at 10 kHz.

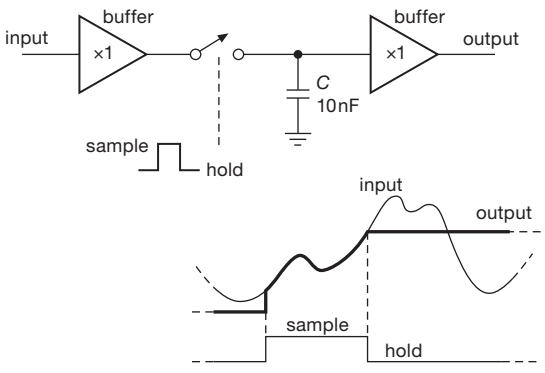


Figure 3.85. Sample-and-hold.

You can do considerably better by closing a feedback loop around the S/H circuit; take a look at §4.5.2. Better still, buy a complete integrated circuit S/H (e.g., the AD783 has an internal hold capacitor, settles to 0.01% in 0.25μs, and droops less than 0.02μV/μs) – let someone else do the hard work!

D. Flying-capacitor voltage converter

Here's a nice way (Figure 3.86) to generate a needed negative power-supply voltage in a circuit that is powered by a single positive supply. The pair of FET switches on the left connects C_1 across the positive supply, charging it to V_{in}, while the switches on the right are kept open.[77] Then the input switches are opened, and the switches on the right are closed, connecting charged C_1 across the output, transferring some of its charge onto C_2. The switches are diabolically arranged so that C_1 gets turned upside down, generating a *negative* output! This particular circuit, often referred to as a *charge-pump dc–dc converter*, originated as the Intersil 7660 voltage converter chip, and is widely available in improved variants, including voltage-doubling versions and regulated versions. You find them, also, as built-in portions of larger integrated circuits that require dual supply

[77] The device labeled "inverter" turns a logic HIGH voltage into a logic LOW voltage, and vice versa. We'll show you how to make one in the next section (and we'll really get you up to speed on them in Chapters 10–14!).

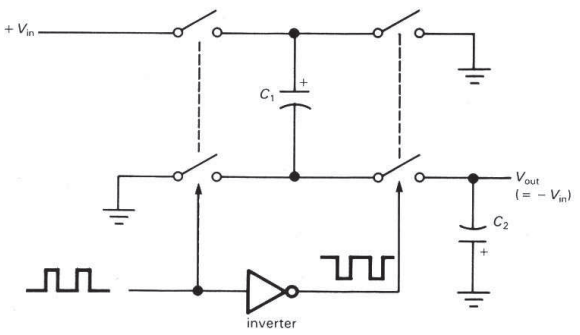


Figure 3.86. Flying-capacitor voltage inverter.

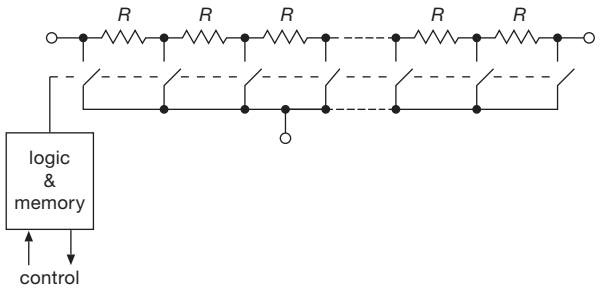


Figure 3.87. An IC "digital potentiometer." Internal digital logic turns on one of the n analog switches to select a tap along the chain of $n-1$ fixed resistors.

voltages, for example RS-232C serial-port drivers. We'll visit these devices in more detail in §9.6.3.

E. Digital potentiometer

It's nice to be able to turn a pot electrically – for example, to adjust the volume control of a television set by the remote control "clicker." This kind of application is common, and the semiconductor industry has responded with a variety of electrically settable pots, known variously as *EEPOT*, *E*^{2}*POT*, or just plain *digital pot*. A digital pot consists of a long resistor chain, with an array of FET switches that connects the selected tap to the output pin (Figure 3.87); the tap is selected by a digital input (Chapters 10 and onward).[78] Digital pots come in single, dual, and multiple units; many have "nonvolatile" memory, to retain the position of the pot after power has been turned off. Some have nonlinear taps, for example for audio volume controls, for which it's best to have equal step sizes in decibels (that is, each step produces the same *fractional* increase in voltage-divider ratio). Note that, whatever the configuration, the switch's R_{ON} appears as a series resistance at the output ("wiper") pin.

As an example, the Analog Devices website lists some 50 digital potentiometers, with from 32 to 1024 steps (256-step models seem to be the most popular), and with one to six channels (singles and duals seem most popular); they use a serial data connection (only two or three pins are needed, regardless of the length of the control data), and they average about $1 (1000-piece quantity). The selection from Maxim/Dallas includes linear and log taper units (the term *taper* predates digital pots and refers to the resistance versus shaft rotation characteristic of a pot; the log taper is for audio applications), again in single and multiple units, and with up to 1024 steps per unit. And at the bottom of Intersil's selection table, after all the usual digital pots, you'll even find digital *capacitors*![79]

3.4.4 MOSFET logic switches

The *other* kinds of FET switch applications are *logic* and *power switching* circuits. The distinction is simple: in analog signal switching you use a FET as a series switch, passing or blocking a signal that has some range of analog voltage. The analog signal is usually a low-level signal at insignificant power levels. In logic switching, on the other hand, MOSFET switches open and close to generate full swings between the power supply voltages. The "signals" here are really digital, rather than analog – they swing between the power supply voltages, representing the two states HIGH and LOW. In-between voltages are not useful or desirable; in fact, they're not even *legal!* Finally, "power switching" refers to turning on or off the power to a load such as a lamp, relay coil, or motor winding; in these applications, both voltages and currents tend to be large. We'll take logic switching first.

Figure 3.88 shows the simplest kind of logic switching with MOSFETs: both circuits use a resistor as load and perform the logical function of *inversion* – a HIGH input generates a LOW output, and vice versa. The *n*-channel version pulls the output to ground when the gate goes HIGH, whereas the *p*-channel version pulls the resistor HIGH for grounded (LOW) input. Note that the MOSFETs in these circuits are used as common-source inverters rather than as source followers. In digital logic circuits like these, we are usually interested in the output voltage ("logic level") produced by a certain input voltage; the resistor serves merely

[78] There are two varieties here: one uses a serial digital data protocol, so that the desired tap position can be sent, as a number, from a controlling microprocessor; the other kind has UP and DOWN pins, with internal memory to hold the current tap position.

[79] The tapped-resistor trick is used in digital-to-analog converters (see §13.2.1), and many ADCs use digital capacitors (see §13.7.

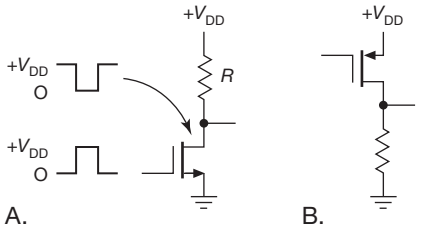


Figure 3.88. nMOS and pMOS logic inverters, with resistive "pullups".

as a passive drain load, to make the output swing to the drain supply when the FET is off. If, on the other hand, we replace the resistor with a lightbulb, relay, print-head hammer, or some other hefty load, we've got a power-switching application (Figure 3.3). Although we're using the same "inverter" circuit, in the power-switching application we're interested instead in turning the load on and off.

A. CMOS inverter

The nMOS and pMOS inverters of the preceding circuits have the disadvantage of drawing current in the ON state and having relatively high output impedance in the OFF state. You can reduce the output impedance (by reducing *R*), but only at the expense of increased dissipation, and vice versa. Except for current sources, of course, it's never a good idea to have high output impedance. Even if the intended load is high impedance (another MOSFET gate, for example), you are inviting capacitive noise pickup problems, and you will suffer reduced switching speeds for the ON-to-OFF ("trailing") edge (because of stray loading capacitance). In this case, for example, the nMOS inverter with a compromise value of drain resistor, say 10k, would produce the waveform shown in Figure 3.89.

The situation is reminiscent of the single-ended emitter follower in §2.4.1, in which quiescent power dissipation and power delivered to the load were involved in a similar compromise. The solution there – the push–pull configuration – is particularly well suited to MOSFET switching. Look at Figure 3.90, which you might think of as a push-pull switch: input at ground cuts off the bottom transistor and turns on the top transistor, pulling the output HIGH. A HIGH input (+V_{DD}) does the reverse, pulling the output to ground. It's an inverter with low output impedance in *both* states, and no quiescent current whatsoever. It's called a CMOS (complementary MOS) inverter, and it is the basic structure of all digital CMOS logic, the logic family that has become universal in large and very-large scale integrated circuits (LSI, VLSI), and has largely replaced earlier logic families (with names like transistor–transistor logic,

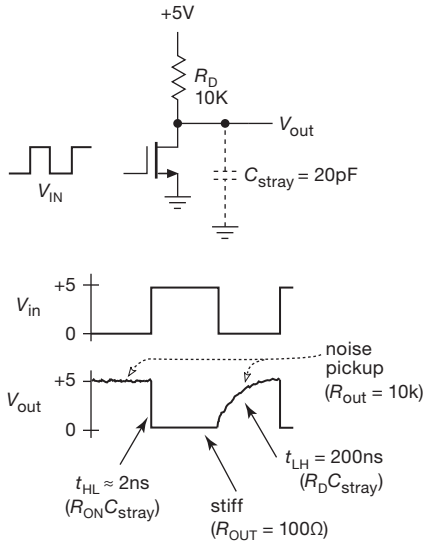


Figure 3.89. High off-impedance in the nMOS inverter causes long rise times and susceptibility to capacitively-coupled noise.

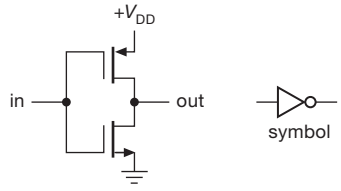


Figure 3.90. CMOS logic inverter and circuit symbol.

TTL) based on bipolar transistors. Note that the CMOS inverter is two complementary MOSFET switches *in series*, alternately enabled, whereas the CMOS analog switch (treated earlier in the chapter) is two complementary MOSFET switches *in parallel*, enabled simultaneously.

Exercise 3.11. The complementary MOS transistors in the CMOS inverter are both operating as common-source inverters, whereas the complementary bipolar transistors in the push-pull circuits of §2.4.1 (e.g., Figure 2.69) are (noninverting) emitter followers. Try drawing a "complementary BJT inverter" analogous to the CMOS inverter. Why won't it work?

B. CMOS gates

We'll be seeing much more of digital CMOS in the chapters on digital logic and microprocessors (Chapters 10–14). For now, it should be evident that CMOS is a low-power logic family (with *zero* quiescent power) with high-impedance inputs, and with stiff outputs that swing the full supply range. Before leaving the subject, however, we can't resist the temptation to show you one additional CMOS

circuit (Figure 3.91). This is a logic NAND *gate*, whose output goes LOW only if input A *and* input B are both HIGH. The operation is surprisingly easy to understand: if A and B are both HIGH, series nMOS switches Q_1 and Q_2 are both ON, pulling the output stiffly to ground; pMOS switches Q_3 and Q_4 cooperate by being OFF; thus, no current flows. However, if either A or B (or both) is LOW, the corresponding pMOS transistor is ON, pulling the output HIGH; since one (or both) of the series chain Q_1Q_2 is OFF, no current flows.

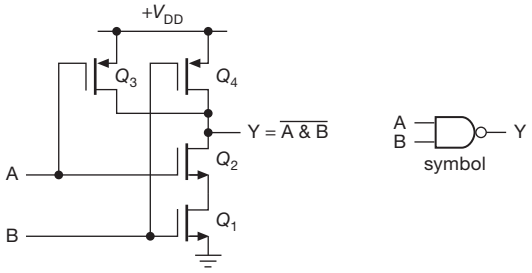


Figure 3.91. CMOS NAND gate and circuit symbol.

This is called a NAND gate because it performs the logical AND function, but with inverted ("NOT") output – it's a NOT–AND, abbreviated NAND. Although gates and their variants are properly a subject for Chapter 10, you will enjoy trying your hand at the following problems.

Exercise 3.12. Draw a CMOS AND gate. *Hint*: AND = NOT NAND.

Exercise 3.13. Draw a NOR gate: the output is LOW if either A *or* B (or both) is HIGH.

Exercise 3.14. You guessed it – draw a CMOS OR gate.

Exercise 3.15. Draw a 3-input CMOS NAND gate.

The CMOS digital logic we'll be seeing later is constructed from combinations of these basic gates. The combination of very low-power dissipation and stiff rail-to-rail output swing makes CMOS logic the family of choice for most digital circuits, accounting for its popularity. Furthermore, for micropower circuits (such as wristwatches and small battery-powered instruments), it's the only game in town.

Lest we leave the wrong impression, however, it's worth noting that CMOS logic is not *zero* power. There are two mechanisms of current drain:

(a) During transitions, a CMOS output must supply a transient current $I = CdV/dt$ to charge any capacitance it sees (Figure 3.92). You get load capacitance both from wiring ("stray" capacitance) and from the input capacitance of additional logic that you are driving. In fact, because a complicated CMOS chip contains many internal gates, each driving some on-chip internal capacitance, there is some current drain in any CMOS circuit that is making transitions, even if the chip is not driving any external load. Not surprisingly, this "dynamic" current drain is proportional to the rate at which transitions take place.

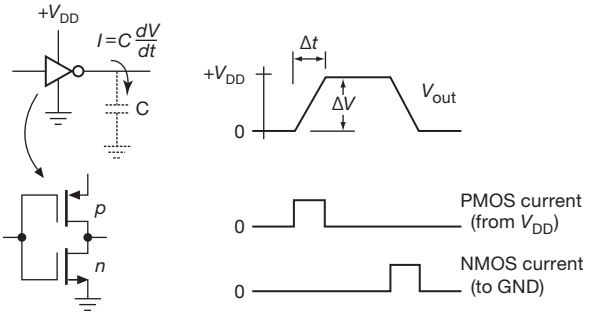


Figure 3.92. Capacitive charging current. The average supply current is proportional to switching rate, and equals CVf.

(b) The second mechanism of CMOS current drain is shown in Figure 3.93: as the input jumps between the supply voltage and ground, there is a region where both MOSFETs are conducting, resulting in large current spikes from V_{DD} to ground. This is sometimes called "class-A current," "shoot-through," or "power-supply crowbarring." You will see some consequences of this in Chapters 10–12. As long as we're dumping on CMOS, we should mention that an additional disadvantage of CMOS (and, in fact, of

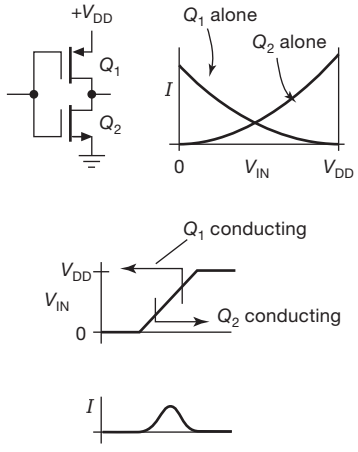


Figure 3.93. When the input gate voltage of a CMOS inverter is intermediate between V_{DD} and ground, both MOSFETs are partially conducting, causing "class-A conduction," also known as "shoot-through" current.

all MOSFETs) is their vulnerability to damage from static electricity. We'll have more to say about this in §3.5.4H.

3.5 Power MOSFETs

MOSFETs work well as saturated switches, as we suggested with our simple circuit in §3.1.1B. Power MOSFETs are now available from many manufacturers, making the advantages of MOSFETs (high input impedance, easy paralleling, absence of "second breakdown") applicable to power circuits. Generally speaking, power MOSFETs are easier to use than conventional bipolar power transistors. However, there are some subtle effects to consider, and cavalier substitution of MOSFETs in switching applications can lead to prompt disaster. We've visited the scenes of such disasters and hope to avert their repetition. Read on for our handy guided tour.

FETs were feeble low-current devices, barely able to run more than a few tens of milliamps, until the late 1970s, when the Japanese introduced "vertical-groove" MOS transistors. Power MOSFETs are now made by all the manufacturers of discrete semiconductors (e.g., Diodes-Inc, Fairchild, Intersil, IR, ON Semiconductor, Siliconix, Supertex, TI, Vishay, and Zetex, along with European companies like Amperex, Ferranti, Infineon, NXP, and ST, and many of the Japanese companies such as Renesas and Toshiba); they are called, variously, VMOS, TMOS, vertical DMOS, and HEXFET. Even in conventional transistor power packages such as TO-220, TO-247, and D-PAK they can handle surprisingly high voltages (up to 1500 V or more), and peak currents over 1000 amps (continuous currents of 200 A), with R_{ON} below 0.001 Ω. Small-power MOSFETs sell for much less than a dollar, and they're available in all the usual transistor packages. You can also get arrays (multiple MOSFETs) in standard multipin IC packages such as the traditional dual in-line package (DIP) and the smaller surface-mount varieties such as SOT-23, SOIC, and TSOP. Ironically, it is now discrete *low-level* MOSFETs that are hard to find, there being no shortage of power MOSFETs; Table 3.4a on page 188 lists a selection of small *n*-channel MOSFETs up to 250 V, Table 3.4b on pages 189–191 lists other *n*-channel MOSFET sizes and voltages, and Table 3.6 on page 210 has a nice selection of depletion-mode power MOSFETs. There are additional types listed in the MOSFET tables in Chapter *3x*.

3.5.1 High impedance, thermal stability

Two important advantages of the power MOSFET compared with the bipolar power transistor, are (a) its extremely high input impedance (essentially infinite at dc), and (b) its inherent thermal stability. As simple as these might seem, there's more to say, and some important cautions.

A. Input impedance

First, the "infinite" input impedance holds only at dc because of substantial input capacitance, which can run to 1000–10,000 pF in typical power MOSFETs. In addition, for *switching* applications you have also to worry about feedback capacitance, i.e., drain-to-gate capacitance (also called *reverse transfer capacitance*, C_{rss}), because the Miller effect (§2.4.5) boosts the effective value by the voltage gain. In §3.5.4 we'll discuss this further and display some waveforms showing how the Miller effect fights your efforts to bring about rapid switching. Jumping to the bottom line, you may have to supply several *amperes* of gate drive current to switch power loads in the tens of nanoseconds that MOSFETs can inherently achieve – hardly the characteristics of an infinite input-impedance device!

B. Thermal stability

Second, there are two mechanisms affecting thermal stability in MOSFETs, namely an increase in R_{ON} with increasing temperature, and, *at the higher end of the transistor's drain current only*, a decrease in drain current (at constant V_{GS}) with increasing temperature; see Figure 3.14 and Figures 3.115, and 3.116 in §3.6.3. This latter effect is very important in power circuits and is worth understanding: the large junction area of a power transistor (whether BJT or FET) can be thought of as a large number of small junctions in parallel (Figure 3.94), all with the same applied voltages. In the case of a bipolar power transistor, the positive temperature coefficient of collector current at fixed V_{BE} (approximately +9%/°C, see §2.3) means that a local hot spot in the junction will have a higher current density, thus producing additional heating. At sufficiently high V_{CE} and I_C, this "current hogging" can cause local thermal runaway. As a result, bipolar power transistors are limited to a "safe operating area" (on a plot of collector current versus collector voltage) smaller than that allowed by transistor power dissipation alone. The important point here is that the *negative* temperature coefficient of MOS drain current, when operating at relatively high currents, prevents these junction hot spots entirely. MOSFETs also have no second breakdown, and their safe operating area (SOA) is limited only by power dissipation (see Figure 3.95, where we've compared the SOAs of an *npn* and an nMOS power transistor of the same I_{max}, V_{max}, and P_{diss}). This is one reason MOSFETs are favored in linear power applications such as audio power amplifiers.

Table 3.4a MOSFETs — Small *n*-channel (to 250V), and *p*-channel (to 100V)

small n-channel to 250V

nMOS type	Pkg[p]	SMT[x]	V_{DSS} (V)	P_D[c] (W)	I_D[y] (A)	R_{DS}[r]@V_{GS} (mΩ)	V_{GS} (V)	Q_G[s] (nC[t])	C_{iss} (pF[t])	Cost[q] ($US)
ZVN4424	TO-92	•	240	0.7	0.3	4.3Ω	2.5	8	110	0.85
BSP89	SOT-223	•	240	1.5	0.4	2.8Ω	10	–	100	0.48
ZVNL120	TO-92	•	200	0.7	0.2	6Ω	3	2	55	0.53
BS107A	TO-92	–	200	0.4	0.2	5Ω	10	–	60	0.31
FQT4N20L	SOT-223	•	200	2.2	0.7	1.0Ω	4.5	4	240	0.34
FQT7N10L	SOT-223	•	100	2	1.2	300	5	4.6	220	0.37
ZXMN10A08E	SOT-23	•	100	1.1	0.6	200	10	7.8	500	0.57
ZXMN10A08G	SOT-223	•	100	2	1.5	200	10	7.7	405	0.48
VN2222LL	TO-92	–	60	0.4	0.1	7.5Ω	10	–	<60	0.36
VN10KN3	TO-92	–	60	0.7	0.2	6.6Ω	5	1.1	48	**0.18**
RHU002N06T	SOT-323	•	60	0.2	0.1	2.8Ω	4	1	15	0.21
2N7000	TO-92	•	60	0.4	0.2	2.5Ω	5	1	20	**0.17**
2N7002	SOT-23	•	60	0.2	0.1	2.5Ω	4.5	0.9	20	**0.16**
2N7002W	SOT-323	•	60	0.28	0.1	2.5Ω	4.5	0.7	25	**0.15**
NDS7002A	SOT-23	•	60	0.36	0.20	1.3Ω	4.5	0.8	80	**0.27**
Si1330EDL	SOT-323	•	60	0.18	0.15	1.4Ω	4.5	0.4	–	0.38
BSS138	SOT-23	•	50	0.36	0.25	1.0Ω	4.5	0.95	27	**0.15**
ZVN2106A	TO-92	•	60	0.7	0.3	800	10	1.5	75	0.49
ZVN4306A	TO-92	•	60	0.85	1.0	320	5	3.5	350	1.11
NDT3055	SOT-223	•	60	3	1.7	84	10	9	250	0.34
PHT8N06LT	SOT-223	•	55	8.3	5[z]	65	5	11.2	500	0.75
IRF7470	SO-8	•	40	1.0	7	10	4.5	29	3400	0.76
2SK3018	SOT-323	•	30	0.2	0.05	5Ω	4	–	13	0.20
FDV303N	SOT-23	•	25	0.35	0.4	330	2.7	1.1	50	**0.23**
IRLML2030	SOT-23	•	30	1.3	0.9	123	4.5	1	110	0.25
NDS355AN	SOT-23	•	30	0.5	1.0	105	4.5	3.5	195	**0.29**
FDN337N	SOT-23	•	30	0.5	1.3[z]	70	2.5	4.2	300	**0.16**
FDT439N	SOT-223	•	30	1.3	4	55	4.5	10.7	500	0.60
NTR4170N	SOT-23	•	30	0.8	2	50	4.5	4.8	430	0.12
PMV40UN	SOT-23	•	30	1.9	2[u]	45	2.5	5.5	445	0.39
NDT451AN	SOT-223	•	30	3	5	42	4.5	11	720	0.72
IRLML0030	SOT-23	•	30	1.3	2[u]	33	4.5	2.6	380	0.18
NTLJS4114N	WDFN	•	30	1.9	2.5[v]	26	2.5	5	650	0.30
NTMS4800N	SO-8	•	30	0.75	5	20	4.5	7.7	940	0.22
IRF7807Z	SO-8	•	30	2.5	10	14.5	4.5	7.2	770	0.61
FDS6680A	SO-8	•	30	1.0	7	10	4.5	16	1600	0.68
FDS8817NZ	SO-8	•	30	2.5	10	7	4.5	17	1800	0.75
NDS331N	SOT-23	•	20	0.5	0.8	150	2.7	2.2	160	**0.30**
FDG327NZ	SC70-6	•	20	0.42	1.2	90	1.8	2.1	410	0.40
IRLML2502	SOT-23	•	20	1.25	1.25	50	2.5	5	740	**0.30**
Si2312CDS	SOT-23	•	20	0.8	2[u]	35	1.8	3.8	870	0.28
Si2312BDS	SOT-23	•	20	0.8	2[v]	30	2.5	3.8	770	**0.40**
IRF6201	SO-8	•	20	2.5	15	2.1	2.5	130	8600	1.06

p-channel to 100V

pMOS type	Pkg[p]	SMT[x]	V_{DSS} (V)	P_D[c] (W)	I_D[y] (A)	R_{DS}[r]@V_{GS} (mΩ)	V_{GS} (V)	Q_G[s] (nC[t])	C_{iss} (pF[t])	Cost[q] ($US)
small										
FQT5P10	SOT-223	•	100	2	0.5	820	10	6.3	190	0.38
VP0106N3	TO-92	–	60	0.7	0.2	8Ω	5	0.5	45	0.55
BS250P	TO-92	•	45	0.7	0.2	9Ω	10	-	60	**0.61**
ZVP2106A	TO-92	•	60	0.7	0.25	3Ω	10	1.8	100	0.61
BSS84	SOT-23	•	50	0.3	0.13	3Ω	5	1	25	**0.26**
NDS0605	SOT-23	•	60	0.4	0.18	1.3Ω	4.5	0.8	79	**0.27**
FDV304P	SOT-23	•	25	0.4	0.3	1.2Ω	2.7	0.75	63	**0.27**
FDN358P	SOT-23	•	30	0.5	1.5	161	4.5	4	182	0.34
ZXMP4A16G	SOT-223	•	40	2	3	83	4.5	14	1000	0.93
IRF7205	SO-8	–	30	2.5	3	60	10	27	870	**0.37**
NTR4171P	SOT-23	•	30	0.5	1.5	60	4.5	16	720	**0.13**
DMP4050	SO-8	•	40	1.6	3	55	4.5	6.9	670	0.54
Si4435DDY	SO-8	•	30	2.5	6	28	4.5	15	1350	**0.53**
IRF7424	SO-8	•	30	2.5	7	20	4.5	75	4000	0.73
Si4463DY	SO-8	•	30	3	7[z]	13	2.5	28	5800	1.03
LP0701N3	TO-92	–	16.5	1	0.4	1.7Ω	3	1.6	120	0.82
ZXM61P02F	SOT-23	•	20	0.6	0.5	550	2.7	1.8	150	0.29
NDS332P	SOT-23	•	20	0.5	0.7	350	2.7	2.4	195	**0.30**
IRLML6402	SOT-23	•	20	1.3	2.2	80	2.5	5	630	**0.33**
SI3443DV	TSOP-6	•	20	2	3.4	70	2.7	5.5	610	0.42
Si2315BDS	SOT-23	•	20	0.57	2.1	71	1.8	8	715	**0.43**
FDS6575	SO-8	•	20	1.5	6[z]	11	2.5	53	4950	1.19
CSD25401	SON	•	20	2.8	8[z]	14	2.5	5.5	1100	1.96
IRLML6401	SOT-23	•	12	1.3	2	125	1.8	6	830	0.19
IRF7702	TSSOP-8	•	12	1.5	6	15	2.5	30	3500	0.90
IRF7420	SO-8	•	12	2.5	8	15	2.5	24	3500	0.66
IRF7410G	SO-8	•	12	2.5	12	8	2.5	55	8700	**0.97**
IRF7210	SO-8	•	12	2.5	10	7	2.5	115	17200	**1.03**
large										
IRF9540	TO-220	–	100	150	19	120	10	40	1400	**2.20**
IRF9540N	TO-220	–	*	140	20	110	*	60	1300	**1.28**
IRFP9140	TO-247	–	100	180	19	120	10	38	1400	3.07
IRFP9140N	TO-247	–	*	140	18	111	*	60	1300	**1.42**
IRF5210	TO-220	–	100	200	25	50	10	115	2700	1.93
IXTR90P10P	TO-247	–	100	190	55	20	10	120	5800	8.10
IXTK170P10	TO-264	–	100	890	130	10	10	240	12600	14.20
IXTN170P10	SOT-227	–	*	*	170	*	*	*	*	19.37
SUD08P06	DPak	•	60	25	2[z]	158	4.5	7	450	0.75
NTD2955G	DPak	•	60	55	2[z]	155	10	14	500	0.45
NTP2955	TO-220	–	*	62	11	*	*	*	*	**0.69**
MJE2955[b]	*TO-220*	–	*60*	*75*	–	*80[b]*	–	–	–	*0.64*
FQB11P06	D^{2}Pak	•	60	53	2[z]	140	10	13	420	0.61
FQP27P06	TO-220	–	60	120	19	55	10	33	1100	0.72
FQB27P06	D^{2}Pak	•	*	*	5[z]	*	*	*	*	0.92
IRF4905	TO-220	–	55	200	50	16	10	120	3400	**2.04**
STB80PF55	D^{2}Pak	•	55	2.4	7	16	10	190	5500	2.53
STP80PF55	TO-220	–	*	300	55	*	*	*	*	1.78
SUM55P06	D^{2}Pak	•	60	125	7[z]	15	10	76	3500	2.75
IRF9Z34	TO-220	–	55	68	15	10	10	23	620	**0.97**
SUP90P06	TO-220	–	60	250	90	9	4.5	90	9200	3.06
SUP75P05	TO-220	–	55	250	75	8	10	140	8500	5.42
IRFP064V	TO-247	–	60	250	80	5.5	10	175	6800	1.89
IRF9204	TO-220	–	40	143	35	20	4.5	150	7700	1.47
MTP50P03HDL	TO-220	–	30	125	30	20	5	74	3500	3.37
FDD6637	DPak	•	35	57	7[z]	14	4.5	25	2400	0.71
SUP75P03	TO-220	–	30	187	75	5.5	10	140	9000	1.91
IPB80P03P4L-04	D^{2}Pak	•	30	137	16	4.7	4.5	60	8700	1.02

Notes: (*) same as row above. (b) BJT for comparison. (c) P_{diss} for T_{case}=25°C. (m) max. (p) the I^2-PAK (TO-262) is a "sawed-off TO-220" stand-up part with 0.1" pitch (three leads plus tab), while the D^2-PAK (TO-263) is an SMT version (2 leads plus tab); the I-PAK (TO-251) is a smaller version of the I^2-PAK (i.e., stand-up, 3 leads plus tab, 0.09" pitch), with its corresponding D-PAK (TO-252) SMT version (2 leads plus tab). (q) qty 100; inexpensive "jellybean" parts are indicated in **boldface**. (r) R_{DS} typ for T_J=25°C, multiply by 1.5 if hot; BJT value is for I_B=0.3A. (s) total gate charge to V_{GS}; switching loss = $Q_G V_{GS} f$. (t) typical. (u) with 6 cm2 PCB copper. (v) with 0.4 cm^2 PCB copper. (x) surface-mount part, or SMT available if through-hole indicated (e.g., TO-92 or TO-220). (y) guideline conservative estimate, saturated switch at V_{GS}, T_{case}=70°C. (z) with 2–5 cm^2 PCB copper; add heatsink for higher current.

This table shows selected representative MOSFETs. The left nMOS column lists TO-92 and small surface-mount parts. The right column lists all pMOS parts up to 100 V. The lists are sorted by decreasing switch spec, $R_{DS(ON)}$. Ignore parts with inadequate V_{DSS}, and evaluate parts with a good switch I_D spec margins. Study candidate datasheets for viability. Amplifiers and linear regulators rely on the P_D spec. But $R_{\Theta JC}=125^\circ C/P_D$, and the junction temp will be $T_J=T_A+P_D(R_{\Theta JC}+R_{\Theta JA})$, where the latter term is your heatsink thermal path. Both R_Θ terms vary widely for different packages, and the P_D spec is useful only in that context. You may find that high-voltage parts have lower $R_{\Theta JC}$.

Table 3.4b *n*-channel Power MOSFETs, 55V to 4500V[a] (page 1 of 3)

Part #[k]	Package[p]	Manufacturer[n]	Surface mount	V_{DSS} 25°C (V)	P_{diss}[c] (W)	I_D (V_{GS}=10V) pulse (A)	I_D 25°C[b] (A)	I_D 70°C[z] (A)	$R_{\theta JC}$ (°C/W)	$R_{DS(on)}$[e] typ[r] (mΩ)	$R_{DS(on)}$ max (mΩ)	at V_{DS} (V)	Gate zener	Superjunction[k]	Charge Q_G[v,s] typ (nC)[t]	Charge Q_{GD}[s] typ (nC)[t]	Capacitance[s,s2] (V_{DS}=25V[k]) C_{iss} typ (pF)	C_{oss} typ (pF)	C_{rss} typ (pF)	Cost[q] US $	year of intro[y]
2.5 to 4.5kV																					
IXTT02N450	TO-268	Ix	-	4500	113	0.6	0.2	*0.14*	1.1	480Ω	625Ω	10	-	-	10.6	5.5	246	19	5.8	17.05	2013
IXTT1N450HV	TO-268HV	Ix	-	4500	520	3	1	*0.8*	0.24	72Ω	85Ω	10	-	-	40	20	1730	78	28	28.67	2013
IXTL2N450	i5-Pak	Ix	-	4500	220	8	2	*1.3*	0.56	16Ω	20Ω	10	-	-	180	83	6860	267	105	88.80	2013
IXTH02N250	TO-247	Ix	-	2500	83	0.6	0.2	*0.14*	1.5	385Ω	450Ω	10	-	-	7.4	5.3	116	8	3	9.52	2013
1500V																					
2SK1317	TO-3P	R	-	1500	100	7	2.5	*1.5*	1.25	9000	12000	10	-	-	-	-	990	125	60	5.46	2004
STP3N150	TO-220	ST	-	1500	140	10	2.5	1.6	0.89	6000	9000	10	-	-	29.3	17	939	102	13.2	4.64	2008
STP4N150	TO-220	ST	-	1500	160	12	4	2.5	0.78	5000	7000	10	-	-	30	9	1300	120	12	4.15	2003
IXTH6N150	TO-247	Ix	-	1500	540	24	6	*4*	0.23	*2905*	3500	10	-	-	67	36	2230	170	64	6.07	2010
1200V																					
IXTY02N120	D-Pak	Ix	•	1200	33	0.6	0.2	*0.2*	3.80	60Ω	75Ω	10	-	-	4.7	3.2	104	8.6	1.9	2.92	2009
IXTP1N120P	TO-220	Ix	-	1200	63	1.8	1	*0.7*	2	16Ω	20Ω	10	-	-	17.6	10.6	550	25	5.4	2.46	2007
IXTP3N120	TO-220	Ix	-	1200	200	12	3	*2.5*	0.62	*3735*	4500	10	-	-	42	21	1100	110	40	4.65	2003
STP6N120K3	TO-220	ST	-	1200	150	20	6	3.8	0.83	1950	2400	10	•	-	39	23.5	1050	90	3	4.28	2010
IXFH16N120P	TO-247	Ix	-	1200	680	35	16	8	0.19	850	950	10	-	-	120	47	6900	390	48	11.02	2007
IXFX26N120P	TO-264	Ix	-	1200	960	60	26	*19*	0.13	*380*	460	10	-	-	225	96	16000	735	58	19.28	2007
IXFN32N120	SOT-227	Ix	-	1200	780	128	32	*20*	0.16	*290*	350	10	-	-	400	188	15900	1000	260	27.15	2001
CMF20120D[x]	TO-247	Cr	-	1200	215	90	42	24	0.44	80	100	20	-	-	91	43	1915	120	13	32.05	2011
800-1000V																					
IXTY01N100	D-Pak	Ix	-	1000	25	0.4	0.1	*0.1*	5	60Ω	80Ω	10	-	-	6.9	3	54	6.9	2	0.93	2004
IXTA05N100	D-Pak	Ix	•	1000	40	3	0.75	*0.5*	3.1	15Ω	17Ω	10	-	-	7.8	4.1	260	22	8	1.68	2006
IXTP1N100	TO-220	Ix	-	1000	54	6	1.5	*0.9*	2.3	8300	11000	10	-	-	14.5	7.5	400	37	13	0.93	2002
FQD2N100	D-Pak	F	•	1000	50	6.4	1.6	1	2.5	7100	9000	10	-	-	12	6.5	400	40	5	0.79	2001
STD2NK100Z	D-Pak	ST	•	1000	70	7.4	1.85	1.16	1.8	6250	8500	10	•	-	16	9	499	53	9	2.19	2006
STP2NK100Z	TO-220	ST	-	1000	70	7.4	1.85	1.16	1.8	6250	8500	10	•	-	16	9	499	53	9	2.16	2006
IRFBG20	TO-220	IR	-	1000	54	5.6	1.4	0.86	2.3	8000	11000	10	-	-	27	15	500	52	17	1.02	1993
IRFBG30	TO-220	IR	-	1000	125	12	3.1	2	1	4000	5000	10	-	-	50	30	980	140	50	1.31	1993
STP5NK100Z	TO-220	ST	-	1000	125	14	3.5	2.2.	1	2700	3700	10	•	-	42	22	1154	106	21	2.52	2005
FQA8N100	TO-3P	F	-	1000	225	32	8	5	0.56	1200	1450	10	-	-	53	23	2475	195	16	2.28	2005
IXTX24N100	TO-247	Ix	-	1000	568	96	24	16	0.22	333	400	10	-	-	267	142	8700	785	315	15.53	2009
FQP9N90C	TO-220	F	-	900	205	32	8	2.8	1.85	1120	1400	10	-	-	45	18	2100	175	14	1.73	2002
FQD1N80	DPak	F	•	800	45	4	1	0.63	2.78	15000	20000	10	-	-	5.5	3.3	150	20	2.7	0.22	1999
STQ1NK80	TO-92	ST	-	800	3	5	0.25	0.16	40	13000	16000	10	•	-	7.7	4.5	160	26	7	0.35	2004
STN1NK80	SOT-223	ST	•	800	2.5	5	*	*	50	*	*	*	•	-	*	*	*	*	*	0.75	2004
STD1NK80	DPak	ST	•	800	45	5	*	*	2.78	*	*	*	•	-	*	*	*	*	*	0.64	2004
SPP02N80C3	TO-220	Inf	-	800	42	6	2	1.2	3	2400	2700	10	-	-	12	6	290	130	6	1.04	2005
FQP7N80C	TO-220	F	-	800	167	26	6.6	4.2	0.75	1570	1900	10	-	-	27	11			10	1.25	2000
SPP11N80C3	TO-220	Inf	-	800	156	33	11	7.1	0.8	390	450	10	-	-	50	25	1600	800	40	2.72	2003

Notes: **(a)** sorted by voltage groups, then by descending $R_{DS(on)}$; parts with rated below 55V are not included; many MOSFETs have an alphanumeric part-number scheme, e.g., 10N60, where 10 is the continuous current capability, N means *n*-channel, and the last number is V_{DSS}/10, so 60 means 600 volts; for these parts a prefix letter usually indicates the package type. **(•)** same as row above. **(b)** the maximum continuous current with T_C = 25°C (manifestly impossible), assuming T_J = 150°C; in some cases the current is limited by the package (bonding wires). **(c)** P_{diss} for T_{case}=25°C from datasheet. **(e)** at T_J = 25°C. **(k)** parts in *italics* have super-junction technology, specs are at V_{DS} = 50 or 100V. **(m)** max. **(n)** Cr = Cree, F = Fairchild, Inf = Infineon, IR = International Rectifier, Ix = IXYS, N = NXP, R = Renesas, ST = STMicroelectronics, Su = Supertex, To = Toshiba, V = Vishay. **(p)** package types (check datasheet to see which pkgs the mfgr offers for that part); plastic power types with three leads are (largest first): TO-264 with 0.215" lead spacing (to match TO-3 center hole and insulated side mounting notches), TO-247 (same spacing, optional center hole), TO-3P (same spacing and center hole, both to match TO-3), TO-220 (smaller with 0.2" tab with hole, 0.1" lead spacing, very popular); the I2-PAK (TO-262) is a "sawed-off TO-220" stand-up part with 0.1" pitch (three leads plus tab), while the D2-PAK (TO-263) is an SMT version (2 leads plus tab); the I-PAK (TO-251) is a smaller version of the I2-PAK (i.e., stand-up, 3 leads plus tab, 0.09" pitch), with its corresponding D-PAK (TO-252) SMT version (2 leads plus tab); SOT-223 three leads 0.09" spacing with short tab; power SO-8P is similar to SOIC-8 with a metal "power pad"; LFPAK=SOT669 has SO-8 footprint replacing 4 pins with drain tab; the SOT-227 package measures 1x1.5 in, has four #4 captive-nut spade-lug lead connections, and a super-useful insulated metal heat-sink plate. **(q)** qty 100. **(r)** R_{DS} typ for T_J = 25°C; if hot, scale by 1.5 to 2x for low-voltage parts, or by 2.2 to 3.5x for high-voltage parts, see Chapter 3x. **(s)** total gate charge to VGS; gate switching loss = $Q_G V_{GS} f$. **(s2)** drain capacitive switching $I_{oss} = C_{oss} V_{DS}^2 f$. **(t)** typical. **(v)** at the V_{GS} at which $R_{DS(on)}$ is specified. **(w)** newer -N version costs less, but has higher $R_{\theta JC}$. **(x)** this is a silicon-carbide (SiC) rather than silicon MOSFET; these have lower capacitance but require higher gate voltages. **(y)** est'd year-of-introduction; see "A 30-year MOSFET Saga" in chapter 3x. **(z)** guideline conservative estimate, saturated switch at V_{GS}, T_C = 70°C.

Usage comments: • when scanning the list for low C_{oss}, or maximum power, or low R_{on}, etc., you may find a better part for your spec, with much higher V_{DSS} than required -- especially true for high-power capability, a linear servo or pass transistor. • 500V parts have been superceded by 600V parts by many manufacturers. • linear apps often need power dissipation capability, this is often better with older large-die parts; newer designs use narrow V-grooves to achieve low $R_{DS(on)}$, and therefore require much less die area. There are good candidates in the table.

Table 3.4b *n*-channel Power MOSFETs, 55V to 4500V[a] (page 2 of 3)

Part #[k]	Package[p]	Manufacturer[n]	Surface mount	V_{DSS} 25°C (V)	P_{diss}[c] (W)	I_D (V_{GS}=10V) pulse (A)	I_D 25°C[b] (A)	I_D 70°C[z] (A)	$R_{\theta JC}$ (°C/W)	$R_{DS(on)}$[e] typ[r] (mΩ)	$R_{DS(on)}$ max (mΩ)	at V_{DS} (V)	Gate zener	Superjunction[k]	Charge Q_G[v,s] typ (nC)[t]	Charge Q_{GD}[s] typ (nC)[t]	Capacitance[s,s2] (V_{DS}=25V[k]) C_{iss} typ (pF)	C_{oss} typ (pF)	C_{rss} typ (pF)	Cost[q] US $	year of intro[y]
600V																					
VN2460N3, N8	TO-92	Su	•	600	1	0.5	0.16	*0.12*	125	25000	25000	5	-	-	5.5	4	120	10	5	0.84	2000
FQN1N60C	TO-92	F	-	600	3	1.2	0.3	0.18	50	9300	11500	10	-	-	4.8	2.7	130	19	3.5	0.31	2003
FQD1N60C	DPak	F	•	600	28	4	1	1.6	4.5	*	*	*	-	-	*	*	*	*	*	0.47	2003
SPD01N60c3	DPak	IR	•	600	11	1.6	0.8	0.5	11	5600	6000	10	-	-	2.2	0.9	100	40	2.5	0.60	2003
IXTP2N60P	TO-220	Ix	-	600	55	4	2	*1.4*	2.25	4200	5500	10	-	-	7	2.1	240	28	3.5	0.79	2005
FQP2N60C	TO-220	F	-	600	54	8	2	1.35	2.32	3600	4700	10	-	-	8.5	4.1	180	20	4.3	0.60	2002
FQP3N60C	TO-220	F	-	600	75	12	3	1.8	1.67	2800	3400	10	-	-	10.5	4.5	435	45	5	0.70	2005
FQP5N60C	TO-220	F	-	600	100	18	4.5	2.6	1.25	2000	2500	10	-	-	15	6.6	515	55	6.5	0.75	2003
FDP5N60NZ	TO-220	F	-	600	100	18	4.5	2.7	1.25	1650	2000	10	•	-	10	4	450	50	5	0.82	2011
SPD03N60	DPak	Inf	•	600	38	9.6	3.2	2	3.3	1280	1400	10	-	-	13	6	400	150	5	0.84	2002
FDP7N60NZ	TO-220	F	-	600	147	26	6.5	3.9	0.85	1050	1250	10	-	-	13	5.6	550	70	7	1.10	2010
STP7NM60	TO-220	ST	-	600	45	20	5	3	2.8	840	900	10	-	•	14	7.7	363	*25*	1.1	1.31	2008
FQB7N60	D2Pak	F	•	600	142	30	7.4	4.7	0.88	800	1000	10	-	-	29	14.5	1100	135	16	1.14	2000
FDP10N60NZ	TO-220	F	-	600	185	40	10	6	0.68	640	750	10	•	-	23	8	1110	130	10	1.49	2010
STP8N65M5	TO-220	ST	-	650	70	28	7	4.4	1.8	560	600	10	-	•	15	6	690	*18*	2	4.48	2009
FCP7N60N	TO-220	F	-	600	64	20	6.8	4.3	1.95	460	520	10	-	•	18	6	719	*30*	2.1	1.60	2009
STP11N65M5	TO-220	ST	-	650	85	36	9	5.6	1.5	430	480	10	-	•	17	8.5	644	*18*	2.5	1.67	2012
FCP9N60N	TO-220	F	-	600	83	27	9	5.7	1.5	330	385	10	-	•	22	7.1	930	*35*	2	2.37	2009
TK10E60W	TO-220	To	-	600	100	39	9.7		1.25	327	380	10	-	•	20	9.5	700	*20*	2.3	2.04	2012
FQA19N60	TO-3P	F	-	600	300	74	18.5	11.7	0.42	300	380	10	-	-	70	33	2800	350	35	2.71	2000
SPP15N60CFD	TO-220	Inf	-	600	156	33	13	8.4	0.80	280	330	10	-	-	63	38	1820	520	21	2.87	2006
STP15NM60ND	TO-220	ST	-	600	125	56	14	9	1	270	299	10	-	•	40	22	1250	*65*	5	3.42	2007
IPP60R280C6	TO-220	Inf	-	600	104	40	14	8.7	1.2	250	280	10	-	•	43	22	950	*60*		1.95	2009
SPP15N60C3	TO-220	Inf	-	600	156	45	15	9.4	0.8	250	280	10	-	-	63	29	1660	540	40	2.33	2002
FCP13N60N	TO-220	F	-	600	116	39	13	8.2	1.07	244	258	10	-	•	30	9.5	1325	*50*	3	2.41	2009
SiHP15N60E	TO-220	V	-	600	180	39	15	9.6	0.7	230	280	10	-	•	38	17	1350	*70*	5	1.77	2011
STP16N65M5	TO-220	ST	-	650	90	48	12	7.3	1.4	230	279	10	-	•	31	12	1250	*30*	3	2.96	2011
IPP60R250CP	TO-220	Inf	-	600	104	40	12	8	1.2	220	250	10	-	•	26	9	1200	*54*		2.56	2010
STP22NM60N	TO-220	ST	-	600	125	64	16	10	1	200	220	10	-	•	44	25	1330	*84*	4.6	3.82	2009
IPP60R199CP	TO-220	Inf	-	600	139	51	16	10	0.9	180	199	10	-	•	32	11	1520	*72*	2	2.88	2006
IRFP27N60K	TO-247	IR	-	600	500	110	27	18	0.29	180	220	10	-	-	105	50	4660	460	41	5.53	2002
TK16E60W	TO-220	To	-	600	130	63	16.8	*11*	0.96	160	190	10	-	•	38	16	1350	*35*	4	2.86	2012
STP21N65M5	TO-220	ST	-	650	125	68	17	10.7	1	150	179	10	-	•	50	23	1950	*46*	3	2.33	2010
SiHP24N65E	TO-220	V	-	700	250	70	24	16	0.5	120	145	10	-	-	81	37	2740	122	4	2.45	2009
IXFH50N60P3	TO-247	Ix	-	600	1040	125	50	*35*	0.12	118	145	10	-	-	94	23	6300	630	2.5	4.95	2011
IPP60R099CP	TO-220	Inf	-	600	255	93	31	19	0.5	90	99	10	-	•	60	20	2800	*130*	100	4.68	2005
SPW47N60C3	TO-247	Inf	-	600	415	141	47	30	0.3	60	70	10	-	-	252	121	9800	2200	145	5.00	2004
FCH47N60N	TO-247	F	-	600	368	141	47	30	0.34	52	62	10	-	•	115	34	5037	*200*	2.5	9.97	2010
IXKN75N60C	SOT-227	Ix	-	600	568	-	75	60	0.22	30	36	10	-	-	500	220	15000	6000	300	33.33	2003
FCH76N60N	TO-247	F	-	600	543	228	76	48	0.23	28	36	10	-	•	218	66	9310	*370*	3.3	18.80	2010
500V																					
IRF820	TO-220	IR	-	500	50	8	2.5	1.6	2.5	2500	3000	10	-	-	16	8.2	360	92	37	0.43	1980
IRF830	TO-220	IR	-	500	74	18	4.5	2.9	1.7	1200	1500	10	-	-	26	12	610	160	68	0.59	1980
IRF840	TO-220	IR	-	500	125	32	8	5.1	1	800	850	10	-	-	52	22	1300	310	120	0.70	1982
FDP5N50NZ	TO-220	F	-	500	78	18	4.5	2.7	4.1	1380	1500	10	•	-	9	4	330	50	4	0.95	2010
STD5NK50Z	DPak	ST	•	500	70	17.6	4.4	2.7	1.78	1220	1500	10	•	-	20	10	535	75	17	1.06	2002
FDP7N50	TO-220	F	-	500	89	28	7	4.2	1.4	760	900	10	-	-	12.8	5.8	720	95	9	1.07	2006
FQP9N50	TO-220	F	-	500	147	36	9	5.7	0.85	580	730	10	-	-	28	12.5	1100	160	20	0.72	2002
IRFP450	TO-247	IR	-	500	190	56	14	9	0.65	350	400	10	-	-	110	55	2600	720	340	2.12	1993
IRFP460	TO-247	IR	-	500	280	80	20	13	0.45	230	270	10	-	-	150	70	4200	870	350	2.58	1993
STW20NK50Z	TO-247	ST	-	500	190	68	20	12.6	0.66	230	270	10	•	-	85	42	2600	328	72	3.15	2002
FDA28N50	TO-3P	F	-	500	310	112	28	17	0.4	122	155	10	-	-	80	32	3866	576	42	3.81	2007
FCP22N60N	TO-220	F	-	500	205	66	22	13.8	0.61	140	165	10	-	•	45	14.5	1950	*76*	3	4.67	2007
FDH44N50	TO-247	F	-	500	750	176	44	32	0.2	110	120	10	-	-	90	31	5335	645	40	6.80	1998
IXFN80N50P	SOT-227	Ix	-	500	700	200	80	*45*	0.18	54	65	10	-	-	195	64	12700	1280	120	18.77	2004
IXFB100N50P	TO-264	Ix	-	500	1250	250	100	75	0.10	44	49	10	-	-	240	78	20000	1700	140	16.18	2005

Table 3.4b *n*-channel Power MOSFETs, 55V to 4500V[a] (page 3 of 3)

Part #[k]	Package[p]	Manufacturer[n]	Surface mount	V_{DSS} 25°C (V)	P_{diss}[c] (W)	I_D (V_{GS}=10V) pulse (A)	I_D 25°C[b] (A)	I_D 70°C[z] (A)	$R_{\theta JC}$ (°C/W)	$R_{DS(on)}$[e] typ[r] (mΩ)	$R_{DS(on)}$ max (mΩ)	at V_{DS} (V)	Gate zener	Superjunction[k]	Charge Q_G[v,s] typ (nC)[t]	Charge Q_{GD}[s] typ (nC)[t]	Capacitance[s,s2] (V_{DS}=25V[k]) C_{iss} typ (pF)	C_{oss} typ (pF)	C_{rss} typ (pF)	Cost[q] US $	year of intro[y]
200-400V																					
IRF710	TO-220	IR	-	400	36	6	2	1.2	3.5	3100	3600	10	-	-	5.7	2.2	170	34	6.3	0.38	1980
IRF720	TO-220	IR	-	400	50	13	3.3	2.1	2.5	1300	1800	10	-	-	15	7	410	120	47	0.28	1980
IRF730	TO-220	IR	-	400	74	22	5.5	3.5	1.7	740	1000	10	-	-	24	13	700	170	64	0.53	1980
IRF740	TO-220	IR	-	400	125	40	10	6.3	1.0	435	550	10	-	-	43	21	1400	330	120	0.67	1981
STP7NK40Z	TO-220	ST	-	400	70	22	5.4	3.4	1.78	850	1000	10	-	-	19	10	535	82	18	1.13	2002
STP11NK40Z	TO-220	ST	-	400	110	36	9	5.67	1.14	490	550	10	-	-	32	18.5	930	140	30	1.45	2003
FQP17N40	TO-220	F	-	400	170	64	16	10.1	0.74	210		10	-	-	45	21.7	1800	270	30	1.45	2006
IRFP244	TO-220	V	-	250	150	60	15	9.7	0.83	180	280	10	-	-	63.0	39	1400	320	73	2.99	1997
FQP16N25	TO-220	F	-	250	250	64	16	10	0.88	180	230	10	-	-	27	15	920	190	23	1.36	2000
FQA30N40	TO-3P	F	-	400	290	120	30	19	0.43	107	140	10	-	-	90	46	3400	580	60	3.53	1999
FDP33N25	TO-220	F	-	250	235	132	33	20	0.53	77	94	10	-	-	36.8	17	1640	30	39	1.23	2006
FDP2710	TO-220	F	-	250	403		50	31	0.48	36	42	10	-	-	78	18	5470	426	97	2.92	2007
FDA69N25	TO-3P	F	-	250	480	276	69	44	0.28	34	41	10	-	-	77	37	3570	780	84	2.50	2006
IXTK120N25P	TO-264	Ix	-	250	700	300	120	*80*	0.18	19	24	10	-	-	185	80	8000	1300	220	8.16	2004
IRF610	TO-220	IR	-	200	36	10	3.3	2.1	3.5	1250	1500	10	-	-	6.3	3.2	140	53	15	0.28	1980
IRL620	TO-220	IR	-	200	50	21	5.2	3.3	2.5	630	800	5	-	-	8.2	5.5	360	91	27	1.03	1993
IRF620	TO-220	IR	-	200	50	18	5.2	3.3	2.5	550	800	10	-	-	10.6	5	260	100	30	0.38	1980
IRL630	TO-220	IR	-	200	74	36	9	5.7	1.7	290	400	5	-	-	24	24	1100	220	70	1.08	1989
IRF630	TO-220	IR	-	200	74	36	9	5.7	1.7	220	400	10	-	-	27	14	800	240	76	0.46	1980
IRL640	TO-220	IR	-	200	125	68	17	11	1.0	125	180	5	-	-	42	24	1800	400	120	0.78	1992
IRF640	TO-220	IR	-	200	125	72	18	11	1.0	130	180	10	-	-	45	24	1300	430	130	0.59	1981
PSMN102-200Y	LFPak	N	•	200	113	65	21.5	13.6	1.1	86	102	10	-	-	31	10	1568	170	55	0.92	2008
IRFP260N	TO-247	IR	-	200	300	200	50	35	0.50	35	40	10	-	-	234	110	4057	603	161	2.15	2009
IRFP4668	TO-247	IR	-	200	520	520	130	92	0.29	8	9.7	10	-	-	161	52	10720	810	160	4.88	2008
55-100V																					
IRF510	TO-220	IR	-	100	43	20	5.6	4	3.5	410	540	10	-	-	5.2	2.2	180	81	15	0.35	1980
IRF520	TO-220	IR	-	100	60	37	9.2	6.5	2.5	200	270	10	-	-	10.3	3.9	360	150	34	0.38	1980
IRF530	TO-220	IR	-	100	88	56	14	10	1.7	100	160	10	-	-	16.2	7	670	250	60	0.44	1980
IRF540	TO-220	IR	-	100	130	110	28	20	1.0	50	77	10	-	-	47	17	1700	560	120	0.60	1980
FQP33N10	TO-220	F	-	100	127	132	33	23	1.18	40	52	10	-	-	38	18	1150	320	62	0.89	1995
PSMN039-100YS	LFPak	N	•	100	74	112	28	20	1	31	40	10	-	-	23	8	1847	86	64	0.57	2010
FQP44N10	TO-220	F	-	100	146	174	43	31	1	30	39	10	-	-	48	24	1400	425	85	1.02	2000
SUP85N10	TO-220	V	-	100	250	240	85	60	0.6	10	12	5	-	-	105	23	6550	665	265	4.88	2000
HUF75652G3	TO-247	F	-	100	515	1200	75 h	75	0.29	6.7	8	10	-	-	393	74	7585	2345	630	4.87	1998
IRFB4110	TO-220	IR	-	100	370	670	180	120	0.4	3.7	4.5	10	-	-	150	43	9620	670	250	2.84	2005
IRFZ14	TO-220	IR	-	60	43	40	10	7.2	3.5	135	200	10	-	-	9.7	4.7	300	160	29	0.44	1986
IRFZ24	TO-220	IR	-	60	60	68	12	17	2.5	68	100	10	-	-	19	8	640	360	79	0.53	1986
IRFZ34	TO-220	IR	-	60	88	120	30	21	1.7	42	50	10	-	-	30	15	1200	600	100	0.65	1986
IRFZ44[w]	TO-220	IR	-	60	150	200	50	36	1.0	24	28	10	-	-	42	17	1900	920	170	0.77	1986
IRLZ44N	TO-220	IR	-	55	110	160	47	33	1.4	20	25	5	-	-	28	17	1700	400	150	0.47	1992
NDP6060L	TO-220	F	-	60	100	144	48	*24*	1.5	20	28	5	-	-	43	21	1630	460	150	1.63	1995
IRL3705N	TO-220	IR	-	55	170	310	89	63	0.9	11	12	5	-	-	95	49	3600	870	320	1.30	2004
IRFP054N	TO-247	IR	-	60	170	290	81	57	0.9	10	12	10	-	-	130	53	2900	880	330	1.36	1996
IRL2505	TO-220	IR	-	55	200	360	104	74	0.75	9	10	5	-	-	130	67	5000	1100	390	1.65	1996
STP80NF55-08	TO-220	ST	-	55	300	320	80 h	80	0.5	6.5	8	10	-	-	112	40	3740	830	265	2.57	2007
IRF3205Z	TO-220	F	-	55	170	440	110	78	0.9	4.9	6.5	10	-	-	76	30	3450	550	310	1.26	2001
IRF1405	TO-220	IR	-	55	330	680	169	118	0.45	4.6	5.3	10	-	-	170	62	5480	1210	280	1.76	2001
FDP025N06	TO-220	F	-	60	395	1060	265	120	0.38	1.9	2.5	10	-	-	174	50	11190	1610	750	3.41	2006
FDP020N06B	TO-220	F	-	60	333	1252	313	221	0.45	1.65	2.0	10	-	-	87	34	16100	3640	127	3.79	2011

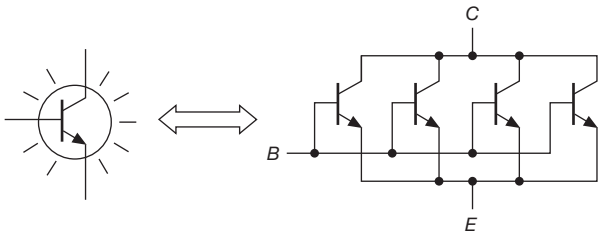


Figure 3.94. A large-junction-area transistor can be thought of as many paralleled small-area transistors.

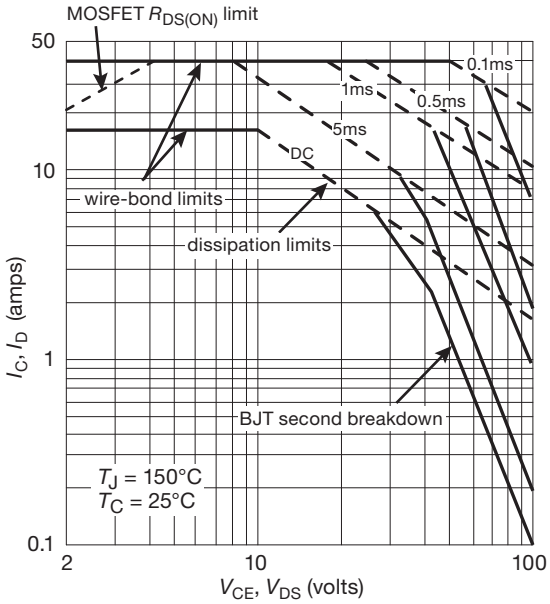


Figure 3.95. Power MOSFETs do not suffer from second breakdown: comparing safe operating areas (SOAs) of a 160 W BJT (MJH6284) and MOSFET (RFP40N10).

This negative temperature coefficient ("tempco") of I_D (at fixed V_{GS}) has produced some bad advice in the amplifier community, in particular the statement that it's always OK to connect a set of power MOSFETs in parallel without the current-equalizing "emitter-ballasting" resistors that are necessary with bipolar transistors.[80] You *could* do that, if the MOSFETs were to be operated in the high-current regime where you get the stabilizing negative tempco. But in practice you usually can't operate up there anyway, because of power-dissipation limitations that we'll see in §9.4.1A. And at the lower currents where the tempco is positive and destabilizing, one of a set of paralleled MOSFETs will tend to hog the current and suffer excessive power dissipation, often leading to early failure. The solution is to use a small source-ballasting resistor in each of the paralleled MOSFETs (which should be of the same type, and from the same manufacturer), chosen to drop about a volt at the operating current.

By contrast, you *can* parallel power MOSFETs in *switching* applications. That's because the MOSFET is here operated in the ohmic region of low V_{DS} (characterized by approximately constant resistance R_{ON}, as opposed to the higher-voltage "current-saturation" region where the transistor is characterized by approximately constant I_D): it is the positive tempco of R_{ON} that stabilizes the current sharing in paralleled power MOSFETs. No ballasting resistors are needed, or even desirable. We'll say more about this in §3.6.3.

3.5.2 Power MOSFET switching parameters

Most power MOSFETs are enhancement type, available in both *n*- and *p*-channel polarities. Relevant parameters are the breakdown voltage V_{DSS} (ranging from 12 V to 4.5 kV for *n*-channel, and to 500 V for *p*-channel); the channel on-resistance $R_{DS(on)}$ (as low as 0.8 mΩ); the current- and power-handling ability (as much as 1000 A and 1000 W); and the gate capacitances C_{rss} and C_{iss} (as much as 2000 pF and 20,000 pF, respectively).

We need to rain on this parade! These impressively high current and power ratings are usually specified at 25°C case temperature, allowing the junction temperature to rise to 175°C (while the impressively low R_{on} is specified at 25°C *junction* temperature!). Unless you're resident at the South Pole, these are completely unrealistic conditions during continuous high-power switching.[81] See the discussion in §3.5.4D.

3.5.3 Power switching from logic levels

You often want to control a power MOSFET from the output of digital logic. Although there are logic families that generate swings of 10 V or more (the "legacy" 4000-series CMOS), the most common logic families (known generically as CMOS) use supply voltages of +5, +3.3, or +2.5 V, and generate output levels close to that voltage or to ground (HIGH and LOW, respectively).[82] Figure 3.96

[80] Because of their *positive* tempco of I_C at constant V_{BE}, see §§2.3 and *2x.2*.

[81] In their defense, datasheets do provide derating coefficients, but the derated current and power somehow are neglected when the front-page banner is typeset.

[82] The family known as TTL operates from +5 V, but its HIGH output can be as low as +2.4 V, a characteristic shared by some other +5 V parts.

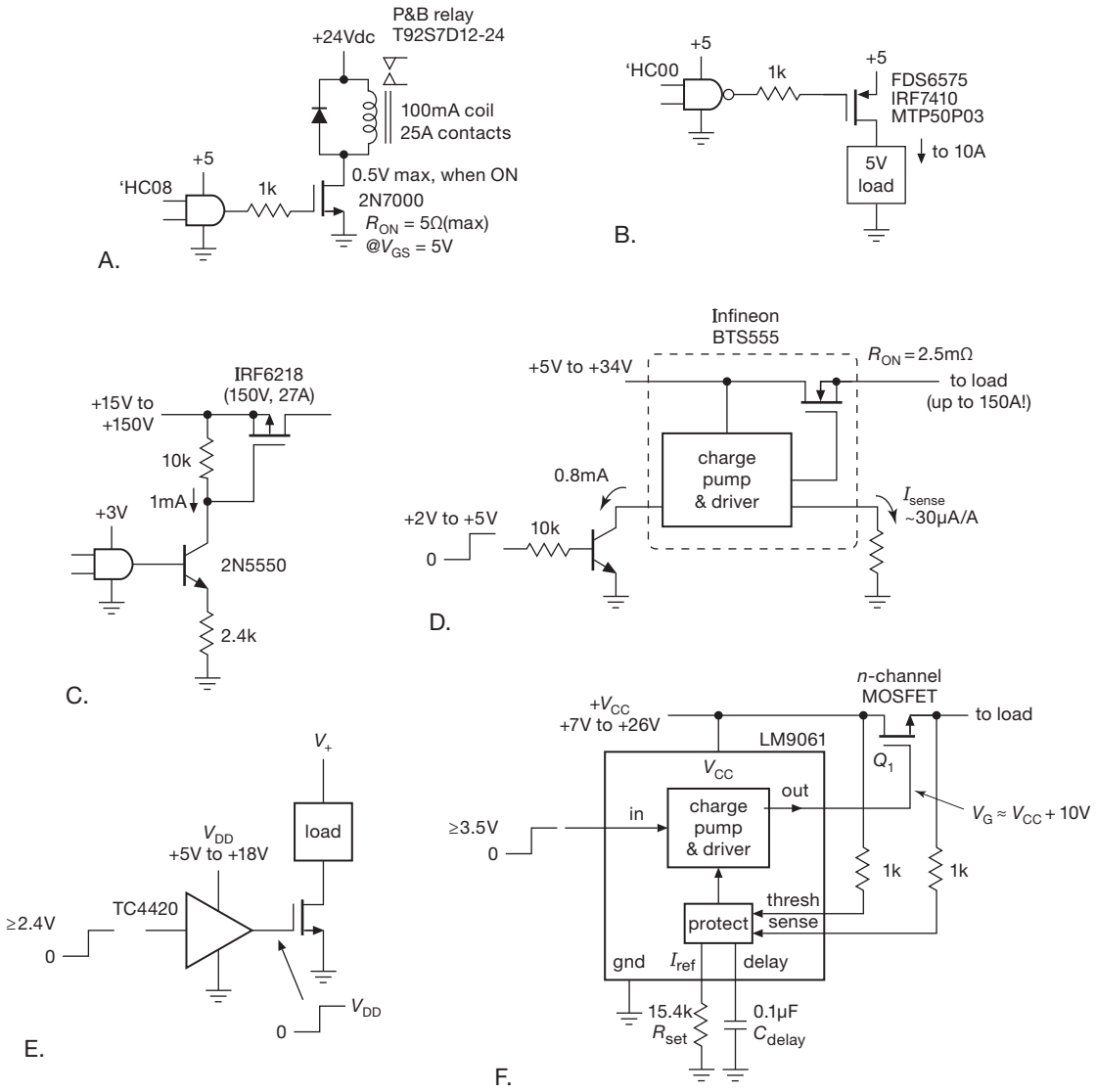


Figure 3.96. MOSFETs can switch power loads when driven from digital logic levels. See also Figure 3.106 on page 204.

shows how to switch loads from these logic families. In the first circuit (Figure 3.96A), the +5 V gate drive will fully turn on a garden-variety MOSFET, so we chose the 2N7000, an inexpensive transistor ($0.04 in quantity!) that specifies $R_{ON} < 5\,\Omega$ at $V_{GS} = 4.5$ V. The diode protects against inductive spike (§1.6.7). The series gate resistor, though not essential, is a good idea, because MOSFET drain-gate capacitance can couple the load's inductive transients back to the delicate CMOS logic (more on this soon).

For variety, in the second circuit (Figure 3.96B) we've used the *p*-channel MTP50P03HDL, driving a load returned to ground. In a commonly used technique called *power switching*, the "load" could be additional circuitry, powered up electrically on command. The '50P03 specifies a maximum R_{ON} of 0.025 Ω at V_{GS}=−5 V and can handle a 50 amp load current; for lower R_{ON} you could select the IRF7410 (0.007 Ω, 16 A, $1.50); see Table 3.4a on page 188.

Lower voltage logic is increasingly popular in digital circuits. The switching configurations of Figures 3.96A & B can be used for lower voltages, but be sure to use MOSFETs with specified "logic thresholds." For example the 20 V 16 A *n*-channel FDS6574A from Fairchild specifies a maximum R_{ON} of 0.009 Ω at a paltry V_{GS} of

1.8 V, and its *p*-channel FDS6575 sibling specs 0.017 Ω maximum at a V_{GS} of −2.5 V; they cost about $1.25 in small quantities. When choosing low-threshold MOSFETs, watch out for misleading specifications. For example, the IRF7470 MOSFET specifies "$V_{GS(th)}$=2 V(max)," which sounds good until you read the fine print ("at I_D=0.25 mA"). It takes considerably more gate voltage than $V_{GS(th)}$ to turn a MOSFET on fully (see Figure 3.115 on page 212). However, the circuit will work OK, because the IRF7470 further specifies "$R_{ON}(\max) = 30\,\text{m}\Omega$ at $V_{GS} = 2.8$ V."

The next two circuits show another way to handle the lower drive voltages from low-voltage logic. In Figure 3.96C we've used a switched *npn* current sink to generate a "high-side" gate drive for a *p*-channel power-switching MOSFET. Note that if the current sink were replaced with a bipolar *switch*, the circuit would fail immediately for switched voltages greater than the gate-source breakdown voltage. In Figure 3.96D an integrated MOSFET and high-side driver (from Infineon's "PROFET" line of smart high-side power switches) is used to switch truly prodigious currents – up to 165 A for this particular device.[83] They make it easy to drive from logic levels, by including internal voltage level translation circuitry, and a charge pump (§9.6.3) for the high-side gate drive. You can also get high-side driver ICs for use with an *external n*-channel MOSFET, for example the LM9061 shown in Figure 3.96F. This particular driver also has an internal charge pump to generate the gate voltage for the external nMOS power switch Q_1; the gate currents are modest, so the switching speeds are relatively slow. This driver also includes a protection scheme for Q_1 that senses $V_{DS(ON)}$, shutting off drive if the MOSFET's forward drop exceeds a threshold (set by R_{set}), with a delay (to accommodate higher inrush currents) set by C_{delay}.

Finally, Figure 3.96E shows how you can sidestep this issue entirely and ensure healthy gate drive voltage *and* current, by using a "MOSFET gate driver" chip like the TC4420. It accepts logic-level input (threshold guaranteed less than +2.4 V), and produces a muscular full-swing output with internal push–pull MOSFETs of its own (Figure 3.97). It can source or sink several amperes of gate current, ensuring fast switching with the large capacitive loads that power MOSFETs present (see §3.5.4B). The trade-off here is cost (about $1) and complexity. Table 3.8 on page 218 lists a selection of nice gate-driver ICs. We'll see these again in detail in Chapter *9x* ("High-side Current Sensing") and in Chapter 12 (§12.4) in the context of controlling external loads from logic-level signals, and in §12.7 in connection with opto-isolated MOSFET drivers.

[83] One of our acquaintances uses the powerful BT555 switch in their "Jaws of Life"-type rescue device.

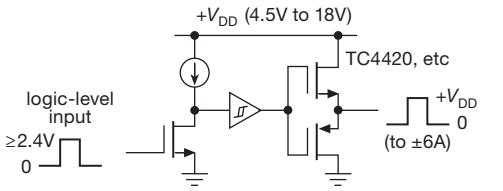


Figure 3.97. A MOSFET driver like the TC4420 accepts logic-level input signals and generates a fast (∼25 ns), high-current (±6 A), full-swing output. The TC4429 is similar, but with inverted output.

A. Some more MOSFET switching examples

The more, the merrier: let's look at a few more circuits that take advantage of the MOSFET's combination of low R_{ON} and negligible gate current. We'll keep the descriptions mercifully brief.

Pulse stretcher

Figure 3.98A is drop-dead simple: MOSFET switch Q_1, driven with a short positive pulse, discharges capacitor C_1, making the output go to the positive rail (here +5 V); when the capacitor charges up to the switching threshold of the output inverter (which can be another nMOS transistor or a logic inverter) the output returns to zero. Note that the timing begins at the trailing edge of the input. See §2.2.2 for some fancier discrete circuits that generate an output pulse width insensitive to the input pulse duration. And in Chapter 7 we go into greater detail (§§7.2.1 and 7.2.2), including integrated timing circuits such as the monostable multivibrator (also called a "one-shot").

Relay driver

An electromechanical relay (more detail in Chapter *1x*) switches its contacts in response to an energizing current in the coil. The latter has some rated voltage that is guaranteed to switch the contacts and hold them in the energized position. For example, the relays specified in Figure 3.98B have a coil rating of +5Vdc, at which they draw 185 mA (i.e., a coil resistance of 27 Ω).[84] In some sense the rated voltage is a compromise: enough to operate the relay reliably, but without excessive current. But you can cheat a bit and get faster closure if you overdrive the coil momentarily,

[84] There's more: it specifies a "must operate" coil voltage of 3.75 V and a "must release" voltage of 0.5 V. Lots of specs, too, about the switched contacts: configuration, voltage and current ratings, endurance, and so on.

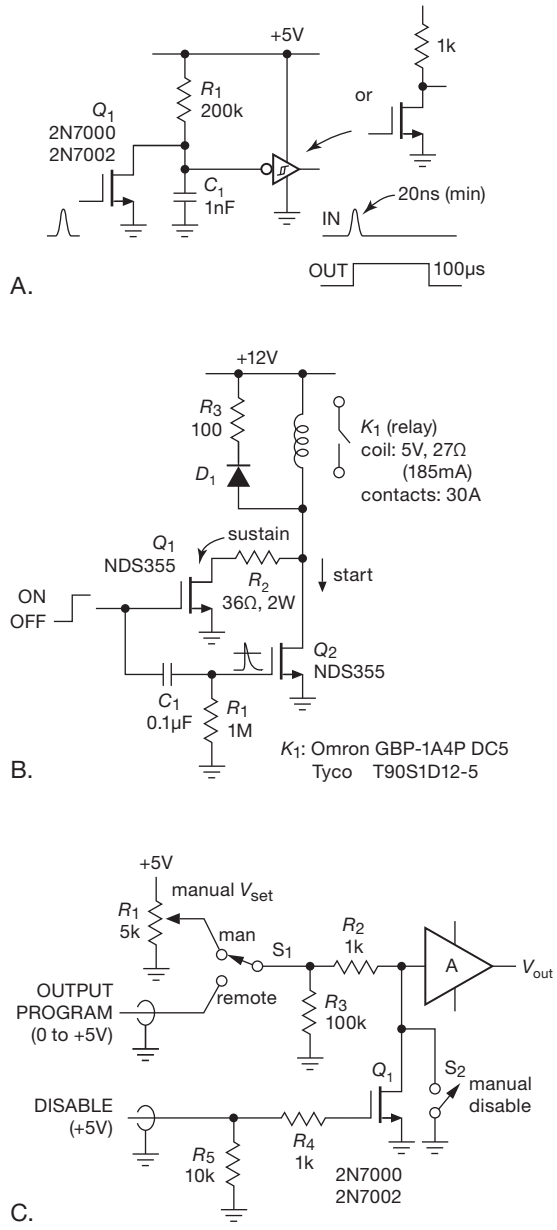


Figure 3.98. Useful – and simple – MOSFET applications: A. pulse stretcher; B. relay driver, with initial pulsed overdrive; C. programmable voltage supply with disable control.

as shown in the figure. Here Q_2 applies 12 V for an initial $\sim$0.1 s, after which Q_1 sustains the by-now firmly closed contacts with the rated 5 V across the coil. Diode D_1 provides a conduction path for the inductive current at release, with series resistor R_3 allowing $\sim$20 V during current decay for faster release (see the discussion in Chapter *1x* to to understand why this is a good thing).

Control of a programmable supply

It's nice to control things remotely, with a brainy computer in charge. You might assemble (or buy) a voltage source that accepts a low-level analog input, in the manner of Figure 3.98C, in which the "A" symbol represents a dc amplifier that outputs a voltage $V_{out}=AV_{in}$, perhaps capable of substantial output current as well. But it's always good to provide a way to disable external control, so things don't go nuts when the computer crashes or is booting up (or gets taken over by an evildoer). The figure shows a simple way to implement a manual DISABLE control (which duplicates the external DISABLE input); might as well have a manual voltage mode as well, as shown.

On/Off battery control

It's convenient to power battery-operated instruments with a 9 volt battery: easy to get; provides plenty of voltage headroom; can be used to create a split supply, see §4.6.1B; but you've got to conserve the battery's vital juices, which run out after about 500 mAh.

Figure 3.99 shows some ways to implement power switching with MOSFETs. Circuit A is the classic flip-flop (two buttons: SET, and RESET; it's called an "SR flip-flop"). The OFF button shuts off Q_2, whose drain goes HIGH, holding Q_1 on and simultaneously holding pass transistor Q_3 off; you can easily make yourself believe that the ON button does the opposite (and you would be right). Two buttons are OK (though we can do better, stay tuned), but this circuit has the disadvantage of drawing current in either state. You could minimize the standby current by using a 10 MΩ resistor for R_1, say; then it draws 1 μA when off, which calculates out to a 50-year battery lifetime – far longer than its $\sim$5 year shelf life.

But there's a better way. Look at Circuit B, where the complementary pair flip-flop Q_1Q_2 draws *no* current (other than nanoamp-scale leakage) in the OFF state. The next step is Circuit C (also zero power when OFF), which achieves the minimum button count, in which a single pushbutton acts as an ON/OFF "toggle." This circuit is a bit tricky, because you have to juggle several time constants appropriately.[85] But the basic concept is simple, and elegant: charge a capacitor from the inverted output of the flip-flop's control input, then momentarily connect the charged capacitor to the control input to make it toggle.

[85] The capacitor's charging time to the new state $\tau_C=R_1C_1$ should be $\sim$100 ms to allow for switch bounce; R_3C_1 should be much shorter, and the discharge time constant of output transistor's gate capacitance $(R_3+R_4)C_g$ should be faster still. Here we chose 100 ms, 2 ms, and $\sim$0.4 ms, respectively.

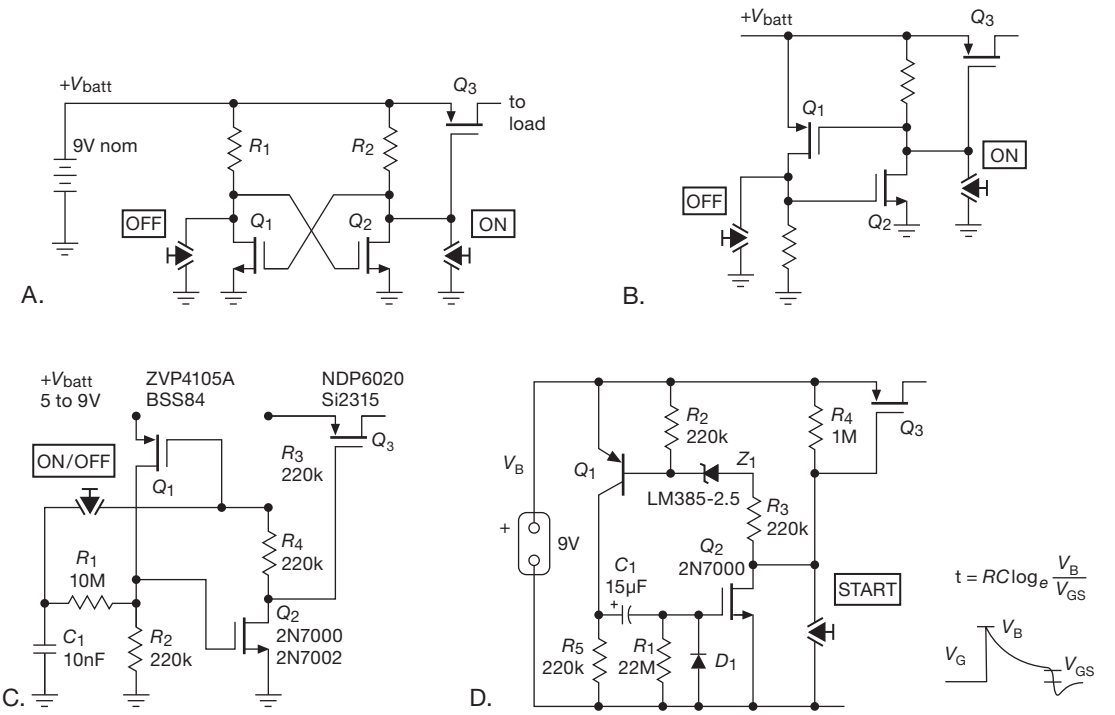


Figure 3.99. ON/OFF battery power control with MOSFETs (no fancy integrated circuits allowed!): A. Classic flip-flop enables *p*-channel series switch; separate ON and OFF buttons. B. Ditto, but zero power when off. C. Single button toggles power ON/OFF; zero power when off. For each MOSFET two choices are listed: the upper is a TO-92 through-hole type, the lower a SOT23 surface-mount type. D. "Five minutes of power" (approximately); zero power when off. The *p*-channel MOSFET pass transistors can be as large as you need, see Table 3.4a.

Finally, for the forgetful we suggest something like Circuit D, a one-button control that turns off automatically (to zero power) after roughly five minutes. Here Q_1Q_2 form a complementary flip-flop, held in the ON state by ac coupling through C_1; once ON the latter discharges through R_1, with a time constant of 330 s. This is an approximation to the actual shutoff time interval, which is set in detail by the ratio of Q_2's gate switching voltage to the actual battery voltage.[86] There's a bit of complexity here, in the form of the zener Z_1, which was forced upon us by the need to keep Q_3 fully ON during the critical interval when the flip-flop is deciding to switch OFF. We chose a BJT (rather than a MOSFET) for Q_1 because of its well-defined turn-on voltage; even so, there are the usual headaches here, caused by the uncertainty in gate threshold voltages of Q_2 and Q_3.

In all four of the circuits in Figure 3.99 you can use as large a *p*-channel pass transistor as needed; see Table 3.4a on page 188 for suggestions. But remember that large MOSFETs have high input capacitance C_{iss}, which slows down switching. As an example, if you were to use an SUP75P05 *p*-channel MOSFET in circuit D, you'd benefit from a very low 8 mΩ on-resistance, but you would have to contend with its substantial 8500 pF value of C_{iss} (thus nearly 10 ms gate turn-off time constant R_4C_{iss}). In these circuits that would not matter. Note that a transistor like the SUP75P05 can switch 50 A, and dissipates about one watt when passing 10 A with a full 10 V gate drive (recall R_{ON} rises with temperature, at 75°C it's about 10 mΩ); at most a small 2 W clip-on heatsink is needed (see §9.4.1).[87]

3.5.4 Power switching cautions

As nice as MOSFETs are, designing circuits with them is not entirely simple, owing to numerous details that can bedevil you. We simply summarize some of the important issues here and delve further into power switching in Chapter *3x* and Chapter 9.

[86] A 9 V alkaline battery starts life at about 9.4 V, and reaches old age at 6 V (1V/cell), or very old age at 5.4 V (0.9V/cell).

[87] If you want to work backward from your available heatsinking capability, use $I=\sqrt{P}/R_{ON}$.

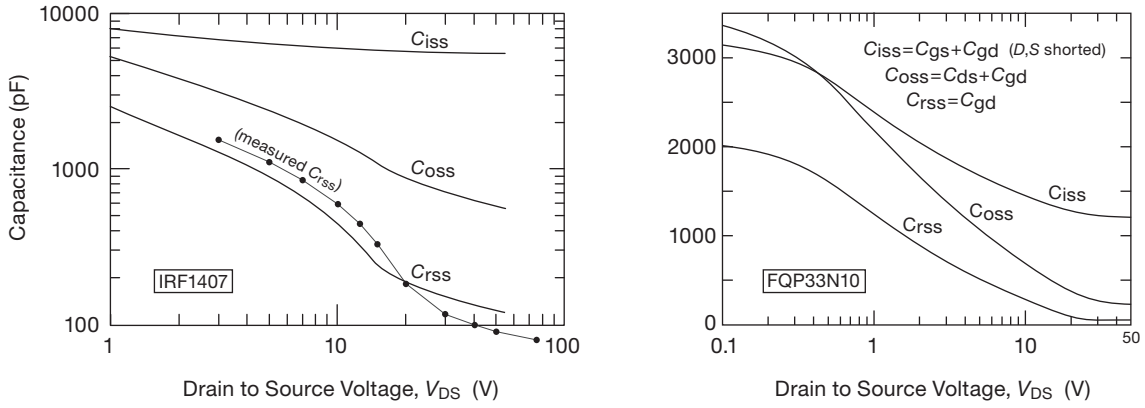


Figure 3.100. Interelectrode capacitances in two power MOSFETs, from graphical data in their respective datasheets. The feedback capacitance C_{rss}, though smaller than the input capacitance C_{iss}, is effectively multiplied by the Miller effect and typically dominates in switching applications.

A. MOSFET gate capacitance

Power MOSFETs have essentially infinite input *resistance*, but they have plenty of input capacitance and also feedback capacitance, such that fast switching may require literally amperes of gate drive current.[88] Although you may not care about speed in many applications, you still have to worry, because low gate drive current causes dramatically higher power dissipation (from the $VI\Delta t$ product during extended switching transitions); it may also permit oscillations during the slow transition. The various interelectrode capacitances are *nonlinear* and increase with decreasing voltage, as shown in Figure 3.100. The capacitance from gate to ground (called C_{iss}) requires an input current of $i{=}C_{iss}\,dV_{GS}/dt$, and the (smaller) feedback capacitance (called C_{rss}) produces an input current $i{=}C_{rss}\,dV_{DG}/dt$. The latter usually dominates in a common-source switch, because ΔV_{DG} is usually much larger than the ΔV_{GS} gate drive, effectively multiplying the feedback capacitance by the voltage gain (Miller effect). A nice way to look at this is in terms of gate *charge*, next.

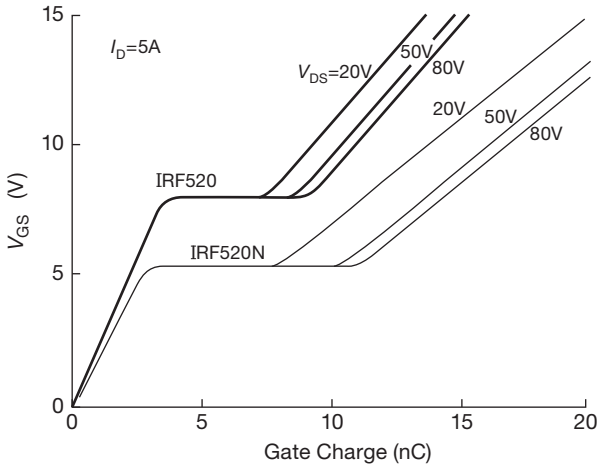


Figure 3.101. Gate charge versus V_{GS}. The newer small-geometry ("shrunk-die") IRF520N has a lower threshold voltage, but comparable gate charge. Note in all cases the larger capacitance (reduced slope of V_{GS} versus Q_g) to the right of the "Miller shelf," the result of larger interelectrode capacitances at low V_{DS} (Figure 3.100).

B. Gate charge

In a common-source switch, charging of the gate-source and gate-drain capacitances require an input gate driving current whenever the gate voltage is changing. Additionally, during drain voltage transitions the Miller effect contributes additional gate current. These effects are often plotted as a graph of "gate charge versus gate-source voltage," as in Figure 3.101.

The initial slope is the charging of C_{iss}. The horizontal portion begins at the turn-on voltage, where the rapidly falling drain forces the gate driver to supply additional charge to C_{rss} (Miller effect). If the feedback capacitance were independent of voltage, the length of the horizontal portion would be proportional to initial drain voltage, after which the curve would continue upward at the original slope. In fact, the "nonlinear" feedback capacitance C_{rss} rises rapidly at low voltage (Figure 3.100), which means that most of the Miller effect occurs during the low-voltage

[88] Newer-generation MOSFETs generally exhibit somewhat lower capacitances, but their smaller size allows less power dissipation, so for high-power applications you may be forced to a larger part, thus giving up the capacitance advantage.

portion of the drain waveform.[89] This explains the change in slope of the gate charge curve, as well as the fact that the length of the horizontal portion is almost independent of initial drain voltage.[90]

We hooked up a common-source MOSFET switch and drove the gate with a constant-*current* drive step, producing the "textbook" traces in Figure 3.102 (hey, this is a textbook, right?). With constant gate drive current, the horizontal (time) axis is proportional to gate charge, in this case 3 nC/div. Here you can see clearly the three regions of gate activity: in region 1 the gate is charging up to the threshold voltage; in region 2 the gate voltage is clamped at the voltage that produces drain currents from 0 to 40 mA (40 V positive rail, 1k load resistor); after the drain is brought to ground the gate resumes its upward voltage ramp, but with reduced slope (owing to increased input capacitance at zero drain voltage).

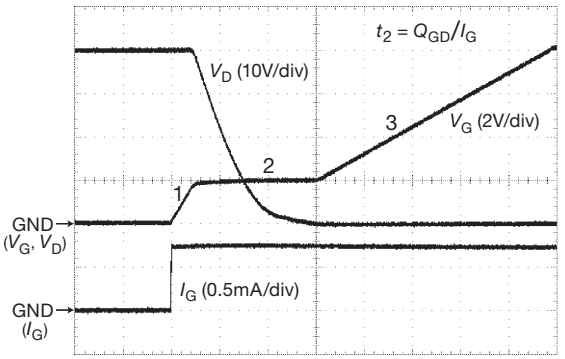


Figure 3.102. Gate charge. Waveforms of an IRLZ34N *n*-channel MOSFET, wired as a common-source switch (1k load to +40 V), with 0.75 mA gate drive. The horizontal scale of 4 μs/div therefore corresponds to 3 nC/div of gate charge.

Notice also that the drain voltage trace is curved, caused by the increasing drain-gate capacitance as it heads toward ground: with constant applied input I_G, the increasing C_{rss} mandates decreasing dV_D/dt (to keep the product, i.e., the feedback current, equal to the input current).

The Miller effect and gate charge in MOSFET switches are no laughing matter – they seriously limit switching speed, and you may need to supply hundreds of milliamps, or even amps, to get fast transitions times in a hefty power switch. For example, the robust IRF1405 featured earlier has $Q_g \sim 100$ nC; so, to switch it on in 10 ns requires $I = Q_g/t = 10$ amperes![91]

On a more modest scale, imagine driving a humble 2N7000 switch from a 0→5 V square wave, perhaps from the output of some digital logic. Figure 3.103 shows what happens if you drive the gate through a 10k resistor. Miller is the killer here, causing $\sim 2\,\mu$s transition times in a transistor which the datasheet says can switch 200 times faster (10 ns). Of course, the datasheet also says R_{GEN}=25 Ω. . . and they mean it!

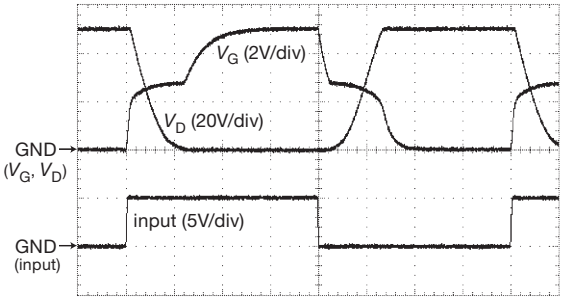


Figure 3.103. A 2N7000 MOSFET switch (1k load to +50 V), whose gate is driven by a 5 V (logic-level) voltage step through a 10k series resistor. The Miller effect stretches the switching time to $\sim 2\,\mu$s. Horizontal scale: 2 μs/div.

There's plenty more to say about gate charge in MOSFETs: dependence on load current, the shape of the "Miller plateau," variations among MOSFET types, and measurement techniques. This is discussed in detail in the advanced material in Chapter *3x*.

C. MOSFET drain capacitances

In addition to the gate-to-ground capacitance C_{iss}, MOSFETs also have a gate-drain feedback capacitance C_{dg} (usually called C_{rss}), and an output capacitance (called C_{oss}) that is the combined capacitances from drain to gate C_{dg} and drain to source C_{ds}. As just seen, the effect of the feedback capacitance C_{rss} is evident in the gate-charge waveforms in Figure 3.102. The output capacitance is important, too: it is the capacitance that must be charged and discharged

[89] This effect can be rather abrupt in power MOSFETs, as seen in the C_{rss} versus V_{DS} plot for the IRF1407 (Figure 3.100). In fact, our measured data shows even steeper behavior than that of the datasheet's plot. As explained in Chapter *3x*, this is due to the effective formation of a cascode within the MOSFET, wherein a depletion-mode JFET acts to clamp the drain of the active MOSFET, isolating the latter's gate and thus greatly reducing the feedback capacitance.

[90] The height (V_{GS}) of the horizontal portion depends modestly on drain current; see the figures in §*3x.12*.

[91] Quite often it is the output *transition* time that you care about; that is, the time spent just in region 2 (apart from the delay time in region 1, or the gate overcharging time in region 3); for this reason MOSFET datasheets separately specify Q_{gd}, the gate-to-drain "Miller" charge. For the IRF1405, for example, $Q_{gd} = 62$ nC, thus requiring a gate input current of 6.2 A to bring about a 10 ns transition time.

each switching cycle, which, if not reactively recycles, consumes power $P=C_{oss}fV_{DD}^2$, which can become significant at high switching frequencies. See §§9.7.2B for details.

D. Current and power ratings

MOSFET datasheets specify a maximum continuous drain current, but this is done assuming an unrealistic 25°C case temperature. It is calculated from $I_{D(max)}^2 R_{DS(ON)}=P_{max}$, substituting a maximum power (see §9.4) $P_{max}R_{\Theta JC}=\Delta T_{JC}=150°C$, where they have assumed $T_{J(max)}=175°C$ (thus a 150°C ΔT_{JC}), and they use the value of $R_{DS(ON)}$ (max) at 175°C from an R_{DS} tempco plot (e.g., see Figure 3.116). That is, $I_{D(max)}=\sqrt{\Delta T_{JC}/R_{\Theta JC}R_{ON}}$. Some datasheets list power and drain current at a more realistic 75°C or 100°C case temperature. This is better, but you don't really want to run your MOSFET junction at 175°C, so we recommend using a still lower maximum I_D rating for continuous current, and corresponding dissipated power.

E. Body diode

With rare exceptions[92] power MOSFETs have the body connected to the source terminal. Because the body forms a diode with the channel, this means that there is an effective diode between drain and source (Figure 3.104) (some manufacturers even draw the diode explicitly in their MOSFET symbol so that you won't forget). This means that you cannot use power MOSFETs bidirectionally, or at least not with more than a diode drop of reverse drain-source voltage. For example, you couldn't use a power MOSFET to zero an integrator whose output swings both sides of ground, and you couldn't use a power MOSFET as an analog switch for bipolarity signals. This problem does not occur with *integrated circuit* MOSFETs (analog switches, for example), where the body is connected to the most negative power-supply terminal.

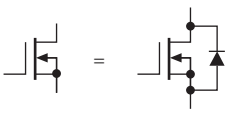


Figure 3.104. Power MOSFETs connect body to source, forming a drain-source diode.

The MOSFET's body diode exhibits the same reverse-recovery effect as ordinary discrete diodes. If biased into forward conduction, it will require some duration of reversed current flow to remove the stored charge, ending with a sharp "snap-off." This can cause curious misbehavior, analogous to the rectifier snap-off transients discussed in the colorfully-named section "Transformer + rectifier + capacitor = giant spikes!" in Chapter *9x*.[93]

F. Gate-source breakdown

Another trap for the unwary is the fact that gate-source breakdown voltages (±20 V is a common figure) are lower than drain-source breakdown voltages (which range from 20 V to more than 1000 V). This doesn't matter if you're driving the gate from the small swings of digital logic, but you get into trouble immediately if you think you can use the drain swings of one MOSFET to drive the gate of another.

G. Gate protection

As we discuss next, all MOSFET devices are extremely susceptible to gate oxide breakdown, caused by electrostatic discharge. Unlike JFETs or other junction devices, in which the junction avalanche current can safely discharge the overvoltage, MOSFETs are damaged irreversibly by a single instance of gate breakdown. For this reason it is a good idea to use a series gate resistor of 1k or so (assuming that speed is not an issue), particularly when the gate signal comes from another circuit board. This greatly reduces the chances of damage; it also prevents circuit loading if the gate is damaged, because the most common symptom of a damaged MOSFET is substantial dc gate current.[94] You can get additional protection by using a pair of clamp diodes (to $V+$ and to ground), or a single clamp zener to ground, downstream of the gate resistor (which can then be of much lower resistance, or omitted altogether); but note that a zener clamp adds some input capacitance.[95] It's also a good idea to avoid floating (unconnected) MOSFET

[92] Such as the 2N4351 and the SD210 series of lateral MOSFETs. These are available from Linear Systems, Fremont, CA.

[93] This can be a serious problem in certain types of switching circuits, where inductive currents continue to flow after termination of switch conduction. It can be addressed by adding an external diode between drain and source, which will be in parallel with the MOSFET's intrinsic drain-source diode. You can use a Schottky diode for voltages below roughly 60 V, but at higher voltages even Schottky diodes have too much voltage drop and thus fail to take over from the FET's intrinsic diode. To deal with this there are power MOSFETs available that include special soft-recovery diodes in their design. These diodes have low reverse-recovery-charge Q_{rr}, and hence faster recovery time t_{rr}, so they don't get stuffed with as much charge from this persistent inductive current. They may also have a slower snapoff, further reducing the spike energy.

[94] A MOSFET with damaged gate may exhibit drain conduction when it should be in a non-conducting state: leakage current from drain to (damaged) gate brings the drain down to a voltage that produces the V_{GS} corresponding to the drain current.

[95] Power MOSFETs used to incorporate internal zener protection, but now it's rare: the zener itself became a dominant failure mechanism! MOSFETs with internal gate zener diodes are marked in the "Gate zener" column of Table 3.4b on page 189.

gates, which are susceptible to damage when floating (there is then no *circuit* path for static discharge, which otherwise provides a measure of safety). This can happen unexpectedly if the gate is driven from another circuit board. A good practice is to connect a pull-down resistor (say 100k to 1M) from gate to source of any MOSFETs whose gates are driven from an off-card signal source. This also ensures that the MOSFET is in the off state when disconnected or unpowered.

H. MOSFET handling precautions

The MOSFET gate is insulated by a layer of glass (SiO_2) a hundred nanometers thick (less than a wavelength of light). As a result, it has very high resistance, and no resistive or junction-like path that can discharge static electricity as it is building up. In a classic situation you have a MOSFET (or MOSFET integrated circuit) in your hand. You walk over to your circuit, stick the device into its socket, and turn on the power, only to discover that the FET is dead. You killed it! You should have grabbed onto the circuit board with your other hand before inserting the device. This would have discharged your static voltage, which in winter can reach thousands of volts.[96] MOS devices don't take kindly to "carpet shock," which is officially called *electrostatic discharge* (ESD). For purposes of static electricity, you can be approximated by the "human body model" (HBM), which is 100 pF in series with 1.5k;[97] in winter your capacitance may charge to 10 kV or more with a bit of shuffling about on a fluffy rug, and even a simple arm motion with shirt or sweater can generate a few kilovolts. Here are some scary-looking numbers:

Typical Electrostatic Voltages[a]

	Electrostatic Voltage	
Action	10%–20% humidity (V)	65%–90% humidity (V)
Walk on carpet	35,000	1,500
Walk on vinyl floor	12,000	250
Work at bench	6,000	100
Handle vinyl envelope	7,000	600
Pick up poly bag	20,000	1,200
Shift position on foam chair	18,000	1,500

(a) adapted from Motorola Power MOSFET Data Book.

Although any semiconductor device can be clobbered by a healthy spark, MOS devices are particularly susceptible because the energy stored in the gate-channel capacitance, when it has been brought up to breakdown voltage, is sufficient to blow a hole through the delicate gate oxide insulation. (If the spark comes from your finger, your additional 100 pF only adds to the injury.) Figure 3.105 (from a series of ESD tests on a power MOSFET[98]) shows the sort of mess this can make. Calling this "gate breakdown" gives the wrong idea; the colorful term "gate *rupture*" might be closer to the mark!

The electronics industry takes ESD very seriously. It is probably the leading cause of nonfunctional semiconductors in instruments fresh off the assembly line. Books are published on the subject, and you can take courses on it.[99] MOS devices, as well as other susceptible semiconductors[100] should be shipped in conductive foam or bags, and you have to be careful about voltages on soldering irons, etc., during fabrication. It is best to ground soldering irons, table tops, etc., and use conductive wrist straps. In addition, you can get "antistatic" carpets, upholstery, and even clothing (e.g., antistatic smocks containing 2% stainless steel fiber). A good antistatic workstation includes humidity control, air ionizers (to make the air slightly conductive, which keeps things from charging up), and educated workers. In spite of all this, failure rates increase dramatically in winter.

Once a semiconductor device is safely soldered into its circuit, the chances for damage are greatly reduced. In addition, most small-geometry MOS devices (e.g., CMOS digital integrated circuits, but not power MOSFETs) have protection diodes in the input gate circuits. Although the internal protection networks of resistors and clamping diodes (or sometimes zeners) compromise performance somewhat, it is often worthwhile to choose those devices because of the greatly reduced risk of damage by static electricity. In the case of unprotected devices, for example power MOSFETs, small-geometry (low current) devices tend to be the most troublesome, because their low input capacitance is easily brought to high voltage when it comes in contact with a charged 100 pF human. Our personal experience with the small-geometry VN13 MOSFET

[96] "Smokey, my friend, you are entering a world of pain."

[97] A bit simplistic, though. The HBM, charged to 2.5 kV, peaks at 1.7 A, with a time constant of 150 ns. There are other models, for example the "machine" model (several cycles of 12 kHz, up to 6 A), or the "charge device model" (CDM), which recognizes that a portion of a charged object with less series resistance can discharge directly into the circuit with 6 A pulses 2 ns wide. See also §12.1.5.

[98] The MOSFET is an MTM6M60, for which C_{iss}=1100 pF. This forms a capacitive divider with the HBM's 100 pF, attenuating the 1 kV 150 ns spike to about 80 V. But that's still way over the part's 20 V maximum V_{GS} rating.

[99] Of course, we academics love to give courses on just about anything.

[100] Which includes just about everything: small-geometry RF bipolar transistors are very delicate; and you can zap a plain ol' BJT if you hit it hard enough.

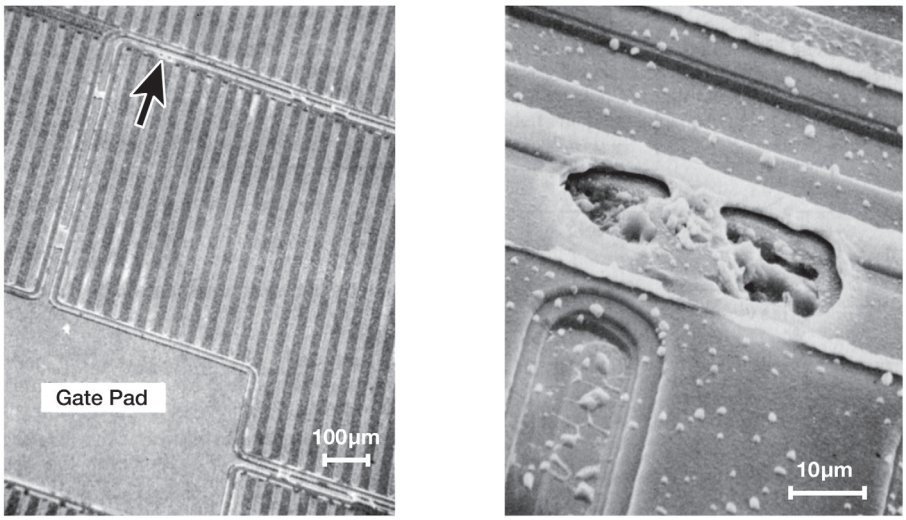


Figure 3.105. Scanning electron micrograph of a 6 amp MOSFET destroyed by 1 kV charge on "human body equivalent" (1.5k in series with 100 pF) applied to its gate. (Courtesy of Motorola, Inc.)

was been so dismal in this regard that we stopped using it in production instruments.

It is hard to overstate the problem of gate damage caused by breakdown in MOSFETs. Chip designers realized the seriousness of the problem, and routinely rate the "ESD tolerance" of their devices. Typically MOS ICs survive 2 kV, applied with the HBM of a charged 100 pF capacitor in series with a 1.5k resistor, and the datasheet will say so. Devices that might be exposed to external impulses (e.g., interface and line drivers) are sometimes rated to 15 kV (for example, Maxim RS-422/485 and RS-232 interface chips with an "E" suffix, and many of the analogous parts from other manufacturers).

See also the discussion of input protection in Chapter 12 (§12.1.5).

I. MOSFETs in parallel

Sometimes you need to use several power transistors in parallel, either to handle greater currents or to be able to dissipate more power, or both. As we discussed earlier, bipolar transistors, owing to their +9%/°C tempco of collector current with temperature, need emitter-ballasting resistors to ensure that the current is distributed equally among the participating transistors. For MOSFETs, as we mentioned in §3.5.1, the situation is different: sometimes you can connect them in parallel without any resistors (e.g., as saturated switches), and sometimes you can't (as linear power devices[101]). There's also the related topic of thermal runaway. These are important topics, and deserve the more extended discussion in §3.6.3.

[101] Exception: "lateral" power MOSFETs, such as the 2SK1058.

3.5.5 MOSFETs versus BJTs as high-current switches

Power MOSFETs are attractive alternatives to conventional power BJTs most of the time. They're comparable in price, simpler to drive, and they don't suffer from second breakdown and consequently reduced safe-operating-area (SOA) constraints (Figure 3.95).

Keep in mind that, for small values of drain voltage, an ON MOSFET behaves like a small resistance (R_{ON}), rather than exhibiting the finite saturation voltage ($V_{CE(sat)}$) of its bipolar transistor cousin. This can be an advantage, because the "saturation voltage" goes clear to zero for small drain currents. There is a general perception that MOSFETs don't saturate as well at high currents, but our research shows this to be largely false. In the following table we've chosen comparable pairs (*npn* versus *n*-channel MOSFET), for which we've looked up the specified $V_{CE(sat)}$ or $R_{DS(ON)}$. The *low-current* MOSFET is comparable to its "small-signal" *npn* cousin, but in the range of 6–10 A and 0–100 V, the MOSFET does better. Note particularly that very large base currents are needed to bring the bipolar power transistor into good saturation – 10% or more of the collector current (thus as much as 1 A!) – compared with the (zero-current) 10 volt bias at which MOSFETs are usually specified. Note also that

high-voltage MOSFETs (say, B$V_{DS} > 200$V) tend to have larger $R_{DS(ON)}$, with larger temperature coefficients, than the lower-voltage units; here IGBTs excel over MOSFETs above 300 to 400 volts. We've listed capacitances in the table, because power MOSFETs traditionally had more capacitance than BJTs of the same rated current. In some applications (particularly if switching speed is important) you might want to consider the product of capacitance and saturation voltage as a figure of merit.

BJT-MOSFET-IGBT Comparison[a]

class	part#	V_{sat} 25°C (V)	V_{sat} 125°C (V)	C_r[b] (pF)	price
60 V, 0.5 A	2N4401[N]	0.75	0.8	8	$0.06
	2N7000[V]	0.6	0.95	25	$0.09
60 V, 6 A	TIP42A[N]	1.5	1.7	50	$0.63
	IRFZ34E[V]	0.25	0.43	50	$1.03
100 V, 10 A	TIP142[D]	3.0	3.8	low	$1.11
	IRF540N[V]	0.44	1.0	40	$0.98
400 V, 10 A	2N6547[N]	1.5	2.5	125	$2.89
	FQA30N40[V]	1.4	3.2	60	$3.85
600 V, 10 A	STGP10NC60[I]	1.75	1.65	12	$0.86

(a) $I_B = I_C/10$, $V_{GS} = 10$ V, except $I_B = I_C/250$ for Darlington.
(b) C_{ob} or C_{rss}.
(D) Darlington. (I) IGBT. (N) *npn* BJT. (V) vertical nMOS.

Remember that power MOSFETs can be used as BJT substitutes for linear power circuits, for example audio amplifiers and voltage regulators (we'll treat the latter in Chapter 9). Power MOSFETs are also available as *p*-channel devices, although there tends to be a much greater variety available among the (better-performing) *n*-channel devices. The available *p*-channel MOSFETs go only to 500 V (or occasionally 600 V), and generally cost more for comparable performance in some parameters ($V_{DS(max)}$ and $I_{D(max)}$, say), with reduced performance in other parameters (capacitance, R_{ON}). Here, for example, are specifications for a pair of complementary MOSFETs from Fairchild, matched in voltage and current ratings, and packaged in the same TO-220 power package.

parameter	*n*-channel FQP9N25	*p*-channel FQP9P25
V_{max}	250 V	250 V
I_{max}	9.4 A	9.4 A
R_{ON}(max)	0.42 Ω	0.62 Ω
C_{rss}(typ)	15 pF	27 pF
C_{iss}(typ)	540 pF	910 pF
Q_g(typ)	15.5 nC	29 nC
T_{JC}(max)	1.39 °C/W	1.04 °C/W
Price (qty 1k)	$0.74	$0.97

Note that the *p*-channel device, having been fabricated with a larger area to achieve comparable $I_{D(max)}$, winds up with inferior (i.e., larger) capacitance, gate charge, R_{ON}, and pricing. It is also slower and has lower transconductance, according to the datasheet. Paradoxically, the *p*-channel device has improved thermal conductivity (see §9.4.1A), presumably resulting from the larger required chip size.

3.5.6 Some power MOSFET circuit examples

Enough theory! Let's look at a few circuit examples with power MOSFETs.

A. Some basic power switches

Figure 3.106 shows six ways to use a MOSFET to control the dc power to some subcircuit that you want to turn on and off. If you have a battery-operated instrument that needs to make some measurements occasionally, you might use Circuit A to switch the power-hungry microprocessor off except during those intermittent measurements. Here we've used a pMOS switch, turned on by a 1.5 V logic swing to ground; the particular part shown is specified for low gate voltage, in particular R_{ON}=17 mΩ (max) at V_{GS}=−1.5 V. The "1.5 V logic" is micropower CMOS digital circuitry, kept running even when the microprocessor is shut off (remember, CMOS logic has zero static dissipation).

An important point: you have to worry about proper switch operation at lower voltages, if the "1.5 V supply" is in fact an alkaline battery, with an end-of-life voltage of ∼1.0 V. In that case you may be better off using a *pnp* transistor – see the discussion in "Low-voltage switching: MOS versus BJT" in Chapter *3x*.

In the second circuit (B), we're switching dc power to a load that needs +12 V at considerable current; maybe it's a radio transmitter, or whatever. Because we have only a 3.3 V logic swing available, we've used a small *npn* current sink to generate an 8 V negative-going swing (relative to +12 V) to drive the pMOS gate. Note the high-value

collector resistor, perfectly adequate here because the pMOS gate draws no dc current (even a beefy 10 A brute), and we don't need high switching speed in an application like this.

The third circuit (C) is an elaboration of circuit B, with short-circuit current limiting courtesy of the *pnp* transistor. That's always a good idea in power-supply design – for example, it's impressively easy to slip with the oscilloscope probe. In this case, the current limiting also prevents momentary short-circuiting of the +12 V supply by the initially uncharged bypass capacitor. See if you can figure out how the current limiting circuit works.

Exercise 3.16. How does the current-limiting circuit work? How much load current does it allow?

An interesting detail: in Circuits B and C we could have hooked up the driver transistor as a *switch* (instead of a current source), omitting the emitter resistor and adding a current-limiting base resistor of 100k or so. But that circuit would create problems if you attempted to operate the circuit from a higher supply voltage, owing to the limited gate breakdown voltages of MOSFETs (±20 V or less). It would also defeat the current-limit scheme of Circuit C. You could fix those problems by adding a resistor directly in series with the collector, tailoring its value for correct gate drive; but the current-source scheme we've used solves these problems automatically, and it can be used to switch 24 or 48 V with no component changes.

Exercise 3.17. You have a dc source using full-wave rectified 120 Vac. Design a 155 to 175 V version of Figure 3.106C to pulse 0.5 A into a flash string consisting of 38 white LEDs in series. Explain your choice for R_1 and R_2 and the ratio R_2/R_1. Select Q_1 and Q_2 and evaluate their power dissipation. Use Table 2.1 on page 74, along with the MOSFET tables in this chapter. *Extra credit*: evaluate Q_2's worst-case heating with a 10 ms maximum flash length (hint: use the datasheet's plot of "Transient Thermal Impedance").

There is still a problem (uh, an "issue"?) with Circuit C, namely the large power dissipation in pass transistor Q_2 under fault conditions such as a shorted output. The brute-force approach (which we've adopted more often than we care to admit) is to use a husky MOSFET with enough heatsinking to handle $P=V_{IN}I_{lim}$; that works OK for modest voltages and currents. Better to add foldback current limiting, as in Figure 12.45C on page 823. But ideally we'd want something like a pass transistor with internal thermal limiting.[102] That's one benefit of devices like that in Circuit E.

[102] If instead we were switching the *low* side (with an *n*MOS switch), we could use a protected MOSFET; see Table 12.4 on page 825.

A popular alternative, at least for low voltage switching, is the use of a low R_{ON} analog switch (recall Table 3.3 on page 176), as in Circuit D. The switch listed there operates with power-supply voltages from 1.1 V to 4.5 V, with a worst-case R_{ON} that is plenty good enough to power loads up to 100 mA or so. It may seem strange to be using an analog switch, designed with complementary *n*-channel and *p*-channel MOSFETs for good signal properties over the full rail-to-rail range, as a simple positive-voltage power switch; but these things are inexpensive, and they take care of the logic interfacing and other details for you, so why not?

In Circuit E we've shown the interesting alternative of an *n*-channel MOSFET switch, for which you need to generate a gate drive that is more positive than the input supply voltage, preferably by a healthy 10 volts or so. You can get "high-side driver" ICs for this job, in varieties that cope with speed and voltage tradeoffs (for example the LM9061 in Figure 3.96; see also §§3.5.3, 12.4.2, 12.4.4, and Table 12.5 on page 826). Here we've gone a step further, using a high-side driver that includes the power MOSFET as well. It gets its gate drive signal with an internal oscillator and charge-pump converter (of the sort we saw in §3.4.3D). This particular device is intended for low-voltage operation, and includes internal current limiting and overtemperature protection.

Why bother with all this, when a *p*-channel MOSFET is easier to drive? Although the use of an nMOS switch with high-side drive does add complexity, it benefits from the better characteristics and much wider variety of *n*-channel MOSFETs; it is generally the preferred scheme.

Finally, Circuit F shows how to switch a negative supply rail to a load; it's analogous to Circuit B, but with an *n*-channel switch and a grounded-base *pnp* transistor to convert a positive logic level into a sourcing current that creates a 10 V gate swing across R_2. You can (and probably should) add current limiting in the manner of Circuit C.

B. Floating power switches

Sometimes you need to switch a voltage (and its load) that's "floating" far from ground. For example, you might want to test the pulsed power capability of a resistor while sensing the current at the low side; or you might want to make millisecond-scale pulsed measurements of a transistor to circumvent heating effects; or you might want a general-purpose floating two-terminal switch that can handle ac or dc. In such situations you can't use the basic ground-referenced schemes of Figure 3.106. Figure 3.107 shows two straightforward approaches, both using an optoisolator (§12.7) to convey the switching command from its ground-referenced home to the floating switch circuit.

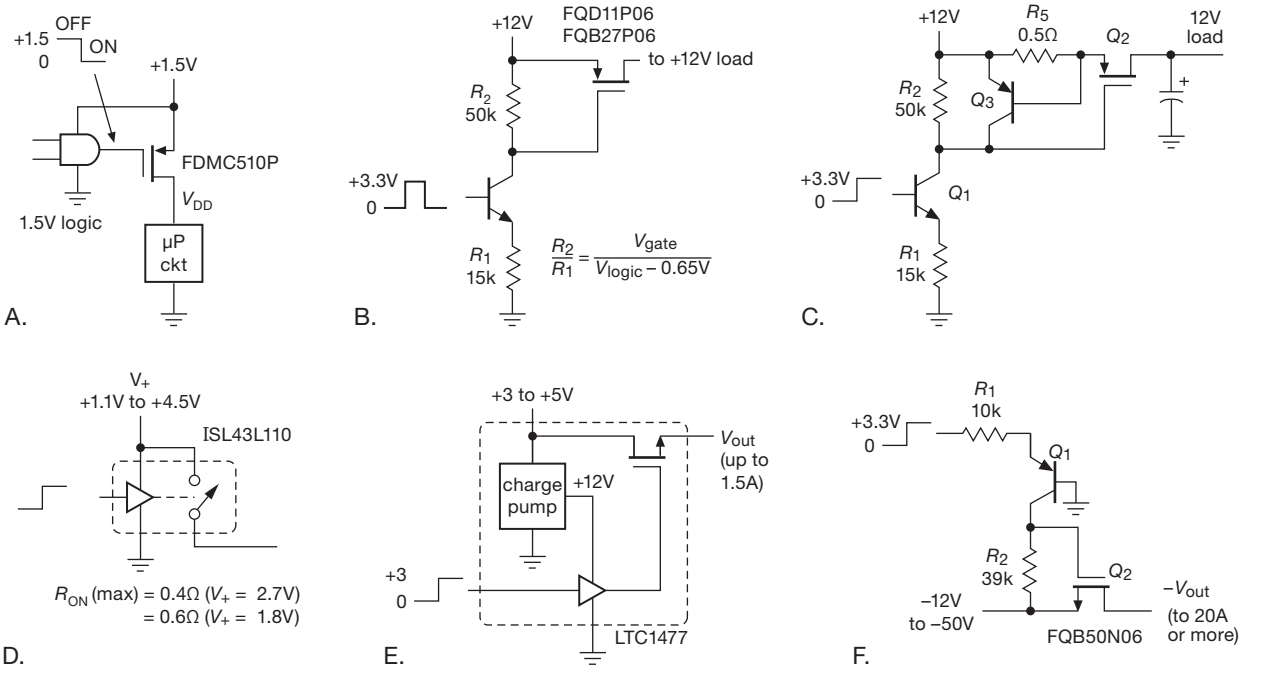


Figure 3.106. dc power switching with MOSFETs.

In Circuit A the gates of a pair of series-connected *n*-channel power MOSFETs are driven by a push–pull BJT follower that receives its base drive signal from a self-generating ("photovoltaic") optocoupler, U_2. The latter uses a series-connected photovoltaic stack to generate a floating ∼8 V signal in response to a 10 mA LED input drive current (see Figure 12.91, and discussion in §*1x.7*), with some internal circuitry to enhance the turn-off time. The gate driver pair Q_1Q_2 could be omitted at the expense of greater switching time (see below). These drivers reduce the effective load capacitance of the MOSFETs by a factor of beta, so that the resulting switching times (with typical power MOSFETs for Q_3 and Q_4) are limited by the optocoupler's intrinsic speed, of order 200 µs.

Of course the gate drivers Q_1 and Q_2 need a floating voltage source, here provided by a second inexpensive optovoltaic generator U_1, which need not be fast (assuming the circuit is not operated at a rapid switching rate) since it serves only to keep C_1 charged to ∼8 V. You can substitute a floating 9 V battery for U_1: it can provide lots more current than U_1's feeble ∼20 µA output, but of course you have to replace it from time to time (an alkaline 9 V "1604"-style battery is good for about 500 mAh, and has a shelf life of 5 years or so). This circuit can switch either polarity – when ON, the series MOSFETs sum to $2R_{ON}$ (the body diodes conduct only during ON–OFF transitions, or at very high currents). Note that this circuit is an "unprotected" switch – there's no provision for current or power limiting of the output transistors.

Circuit B addresses this vulnerability and leverages the benefits of the BTS555 integrated protected switch. Here we've taken the simple approach of a floating 9 V battery to supply its internal circuitry's operating power (15 µA typical when off, 1 mA on). This thing is protected against pretty much anything bad that you can throw at it. Its switching speed is comparable to that of Circuit A (typically 300 µs on, 100 µs off), and it's good for lots of current (100 A or more); but it's limited to 34 V across its switch terminals. See §12.4.4 for more details, and Table 12.5 on page 826 for additional component suggestions.

Returning to Circuit A in Figure 3.107, what sort of performance can you get with readily available *n*-channel MOSFETs? Here's a selection of candidates (see Table 3.5 on page 206), culled from the many thousands of possibilities,[103] spanning the full range of voltages.[104]

Several trends are clearly evident:

[103] A search today for *n*-channel discrete MOSFETs finds 20,330 types at Digi-Key, 11,662 at Mouser, and 4,607 at Newark. A bit of an overestimate, because different package options are listed separately – but you get the point.

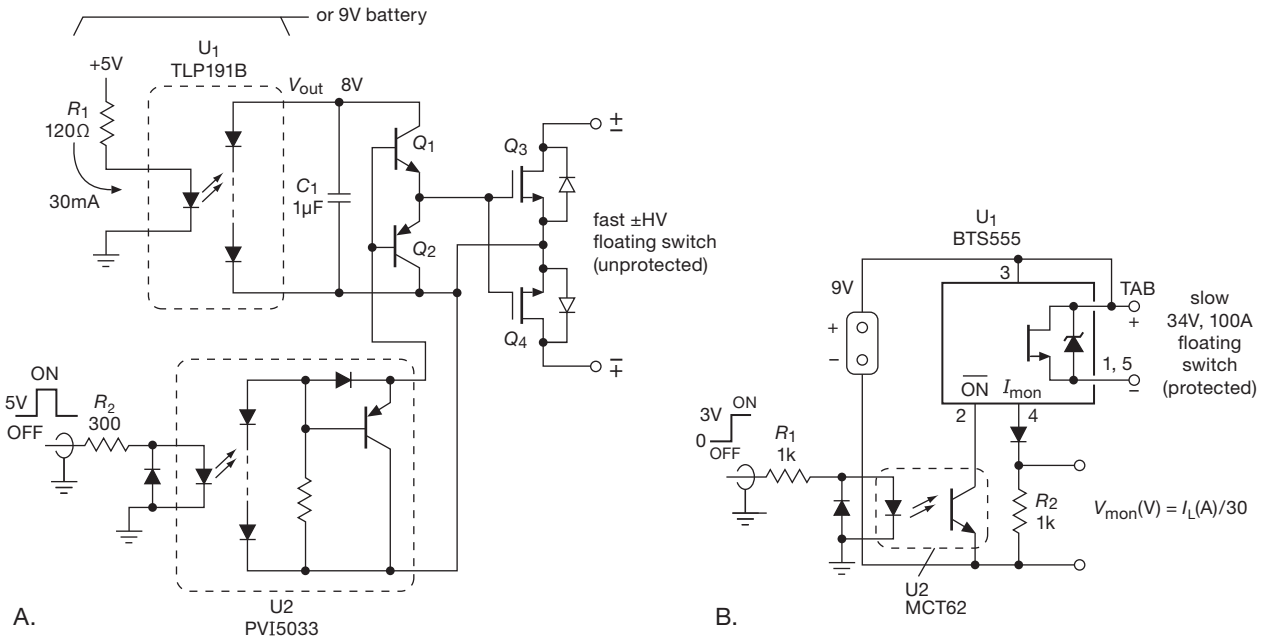


Figure 3.107. Floating MOSFET power switches: A. bipolarity, unprotected; B. unipolarity, limited voltage, protected.

(a) There's a severe tradeoff of R_{ON} versus voltage rating – for the listed types the ON-resistance spans a ∼100,000:1 range over the ∼100:1 voltage range.
(b) You also pay a price, literally, for very high-voltage parts; for example, the listed 4.5 kV part costs $22.
(c) Higher current parts have greater output capacitance (which is what you see across the switch terminals when OFF), even here where we've carefully selected parts to minimize the $R_{ON}C_{oss}$ figure-of-merit tradeoff. They also have larger input capacitance and gate charge, which are relevant to switching speed.
(d) And, important data is *missing*! You need to go to the datasheets for important information such as thermal resistance, pulse current and pulse energy specifications, gate charge, and the like. The data presented here are at most advisory, and you need to use detailed specifications in the circuit context to predict actual performance. For example, the "maximum pulsed current" specs generally apply to pulse lengths somewhat shorter than this circuit can produce; and the R_{ON} spec assumes 10 V of gate drive, also somewhat greater than we have here.

Let's finish this example by estimating the switching speed of the circuit of Figure 3.107A. Imagine we want 600 V capability and choose the middle-of-the-road FCP22N60N, a MOSFET that delivers a good combination of ON-resistance and capacitance at modest cost (about $5, qty 100). For switching speed the relevant parameter here is gate charge ($Q_{GS}+Q_{GD}$), approximately 25 nC according to tabulated and graphical data. That must be supplied by the isolating driver U_2, boosted by the current gain β of Q_1 and Q_2. From U_2's datasheet we can estimate the output sourcing current (from its "Typical Response Time" plot) as approximately 3 μA. If for the moment we imagine that Q_1 and Q_2 were omitted, with U_2 driving the MOSFET gates directly, the turn-on time would be $t \approx Q_{gate}/I_{U2}$, or 8.3 ms. Now magically restore the BJT drivers, and the estimated switching time drops by a factor of beta; for typical $\beta \sim 200$ it becomes $\sim 40\,\mu$s.[105]

Not so fast! Look again at the datasheet for U_2, you'll find that the turn-on time bottoms out at about 100 μs; likewise, its intrinsic turn-off time is about 350 μs, even with small load capacitance. Those numbers dominate the performance of Circuit A, for nearly every MOSFET listed, assuming of course that the BJT driver Q_1Q_2 is included. If you can tolerate slower switching, you can simplify things by omitting the drivers and their floating power source.

If you need *faster* switching, there are many integrated high-side driver chips that can do the job, for example the series of "high-voltage gate-driver ICs" from Interna-

[104] See the MOSFET tables in this chapter and in Chapter *3x* for additional data on these and other power MOSFETs.

[105] If you look at the MOSFET drivers in Table 3.8 on page 218, you'll see that the ZXGD3002–04 are simply a pair of very-high-gain *npn* and *pnp* BJTs, in SOT23-6 packages, perfect for Q_1 and Q_2.

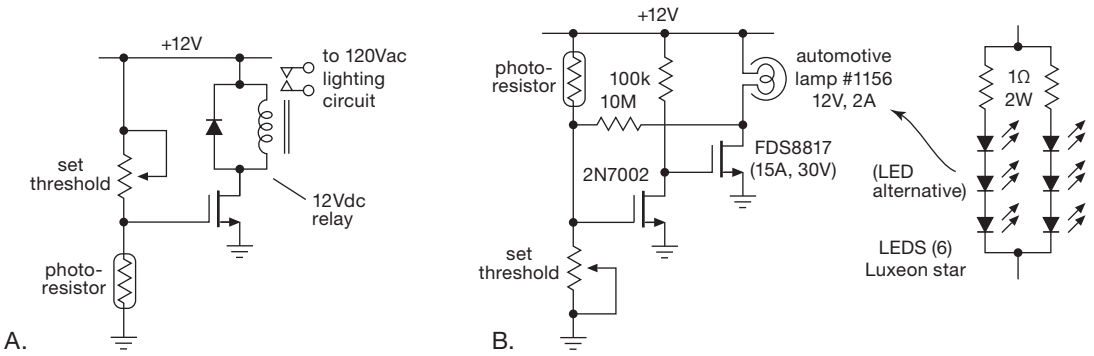


Figure 3.108. Ambient-light-controlled power switch.

Table 3.5 MOSFET switch candidates[a]

Part #	BV_{DS} (V)	$I_{D(max)}$[b] *pulse* (A)	$I_{D(max)}$[b] cont (A)	$2R_{ON}$[c] (Ω)	C_{oss}[t] (pF)
IXTT02N450[$$]	4500	*0.6*	**0.2**	960	19
IXTH02N250[$]	2500	*0.6*	**0.2**	770	9
STW4N150	1500	*12*	**2**	10	120
IXTP3N120	1200	*12*	**3**	6.5	100
IXFH16N120P[$]	1200	*35*	**10**	1.7	390
IRFBG20	1000	*5.6*	**1**	16	52
IRFBG30	1000	*12*	**2**	8	140
IXFH12N100[$]	1000	*48*	**5**	2	320
IPP60R520CP	650	*17*	**4**	1	32
FCP22N60N	600	*66*	**12**	0.28	76
FCH47N60N[$]	600	*140*	**30**	0.1	200
IRF640N	200	*72*	**12**	0.24	190
FQP50N06L	60	*210*	**25**	0.08[d]	450
IRLB3034	40	*1400*	**125**	0.003[e]	2000
FDP8860	30	*1800*	**100**	0.004[d]	1700

Notes: (a) all are in TO-220 or TO-247 pkgs. (b) *italics* designate maximum pulsed drain current, for pulse width specified in the part's datasheet (e.g., 80μs); **boldface** designates maximum continuous drain current at T_J=70°C. (c) at V_{GS}=10V, unless marked otherwise. (d) at V_{GS}=5V. (e) at V_{GS}=4.5V. (t) typ. ($) not inexpensive. ($$) expensive.

tional Rectifier. These use internal high-voltage transistors to send the control signals up to the high side, with maximum voltage ratings most commonly of 600 V. These typically have switching times in the range of 100 ns to 1 μs. They are intended for cyclic applications such as pulse-width-modulated bridge drivers and use high-side charge pumps to develop the over-the-rail gate drive voltage; but you can adapt them for pulsed applications by substituting a flying 9 V battery, as we've done here. See Chapter *3x*'s table "High-voltage Half-Bridge Drivers," and the section on "High-side high-voltage switching" in Chapter *9x*.

Another class of ICs that can be used in applications like this is typified by Avago's ACPL-300 series of "gate drive optocouplers," which combine an optocoupler and isolated push-pull output stage. For example, the ACPL-W343's output stage can source or sink 3 A (minimum), with 40 ns rise and fall times (into a load of 25 nF in series with 10 Ω), and isolation good to 2 kV. You have to supply 15–30 V isolated dc for the output stage,[106] in the manner of Figure 3.107B, with the usual bypass capacitor (for peak output currents); the quiescent current is 2 mA, good for 200 hours of operation if you use a pair of 9 V batteries. See §12.7.3 and Figure 12.87 for additional circuit discussion and suggestions.

C. Some unusual switching examples

Light-at-night

Figure 3.108A shows a simple MOSFET switching example, one that takes advantage of the high gate impedance. You might want to turn on exterior lighting automatically at sunset. The photoresistor has low resistance in sunlight, high resistance in darkness. You make it part of a resistive divider, driving the gate directly (no dc loading!). The light goes on when the gate voltage reaches the value that produces enough drain current to close the relay. Sharp-eyed readers may have noticed that this circuit is not particularly precise or stable; that's OK, because the photoresistor undergoes an enormous change in resistance (from 10k to 10M, say) when it gets dark. Note that the MOSFET may have to dissipate some power during the time the gate bias is inching up, since we're operating in the linear region;

[106] A bit high for MOSFET gate driving (these are targeted at IGBTs); but you can get parts with lower minimum output supply voltage, for example the HCPL-3180 or the PS9506 from Renesas (both can operate with 10 V min).

but it's switching only a relay, not the power load, so this is of little concern. The circuit's lack of a precise and stable threshold means that the light may turn on a few minutes early or late – again, no big deal. But an additional worry is the behavior of the relay, which is not happy with a marginal coil-driving voltage (which holds the contacts closed with less than the rated mechanical force, thus potentially shortening the relay's life; see the discussion of relays in Chapter *9x*).

These problems are remedied in Figure 3.108B, where a pair of cascaded MOSFETs delivers much higher gain, augmented by some positive feedback through the 10M resistor; the latter adds hysteresis, which causes the circuit to snap on regeneratively as it reaches threshold.

Hefty piezo driver

Figure 3.109 shows a real power MOSFET job: a 200 watt amplifier to drive a piezoelectric underwater transducer at 200 kHz. We've used a pair of hefty nMOS transistors, driven alternately to create ac drive in the (high-frequency) transformer primary. The series inductor in the secondary resonates with the transducer's capacitance to step up the voltage across the piezo to several kilovolts. The TC4425A is a handy "3A Dual High-Speed Power MOSFET Driver" (like the TC4420 in Figure 3.97), which takes a logic-level input (0 V=LOW, $\geq$2.4 V=HIGH), and generates a full-swing (0 to $+V_{DD}$) output pair, one inverted and the other noninverted; see Table 3.8 on page 218. It's needed to overcome capacitive loading, since the MOSFETs must be turned on fully in a fraction of a microsecond. The diodes shunted across the series gate resistors cause a rapid turn-off to prevent undesirable conduction overlap of the power transistors.

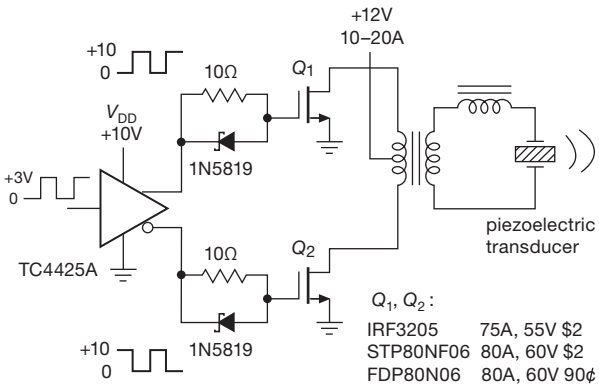


Figure 3.109. MOSFET piezo power driver.

3.5.7 IGBTs and other power semiconductors

The contemporary power MOSFET is a versatile transistor for both power switching applications (e.g., dc power control, or dc–dc switching converters), and for linear power applications (such as audio amplifiers). But there are some drawbacks and some useful alternatives.

A. Insulated-gate bipolar transistor (IGBT)

The IGBT is an interesting MOSFET-bipolar hybrid, most simply described as an integrated complementary-Darlington-like (Sziklai) connection of an input MOSFET with a power bipolar transistor (Figure 3.110). So it has the input characteristics of a MOSFET (zero dc gate current), combined with the output characteristics of a power bipolar transistor; note, however, that it cannot saturate to less than V_{BE}. Unlike MOSFETs, IGBTs do not have an intrinsic reverse diode, so inductive ringing, etc., can easily exceed the reverse voltage rating (e.g., 20 V). Many IGBTs include an internal "anti-parallel" diode to protect against this problem.[107]

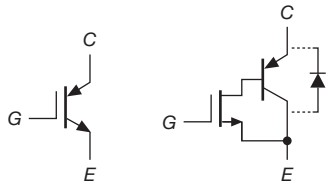


Figure 3.110. IGBT symbol and simplified equivalent circuit showing the optional "anti-parallel" diode.

Nearly all available IGBTs come in the nMOS-*pnp* polarity only, and thus behave as *n*-type devices.[108] They are generally high-voltage and high-power devices, available in discrete transistor power packages like the TO-220, TO-247, and in surface-mount packages like the D^{2}PAK and SMD-220, with ratings to 1200 V and 100 A. For higher currents you can get them in larger rectangular power "modules," with higher voltage ratings and with current ratings to 1000 A or more.

IGBTs excel in the arena of high voltage switching, because high-voltage MOSFETs suffer from greatly increased R_{ON}: an approximate rule-of-thumb for MOSFETs is that the R_{ON} increases as the square of the voltage

[107] Some parts are available with and without the added diode, indicated for example by a -D suffix on the part number.

[108] Currently the only *p*-type IGBTs we know of are the Toshiba GT20D200 series.

rating.[109] For example, let us compare two power products from International Rectifier (along with a BJT of comparable ratings):

		MOSFET	IGBT	BJT
Type		IRFPG50	IRG4PH50S	TT2202
V_{max}		1000 V	1200 V	1500 V
I_{max}	dc	6.1 A	57 A	10 A
	pulse	24 A	114 A	25 A
R_{ON} (typ)	25°C	1.5 Ω	–	–
	150°C	4 Ω	–	–
V_{ON}	25°C	23 V	1.2 V	1 V (@8 A)
(typ, 15 A)	150°C	60 V	1.2 V	1 V (@8 A)

These are comparably priced (about $5) and packaged (TO-247), have similar input characteristics (2.8 nF and 3.6 nF input capacitance), and the resulting saturation voltages V_{ON} when switching 15 A are shown for the same full input drive of $V_{in} = +15$ V. The IGBT is the clear winner in this high-voltage and high-current regime.[110] And, when compared with a power BJT, it shares the MOSFET advantage of high static input impedance (though still exhibiting the drastically reduced dynamic input impedance during switching, as we saw in §3.5.4B). The BJT does have the advantage of lower saturation voltage (the IGBT's V_{ON} is at least V_{BE}) and lower drive voltage (see the figure in Chapter *3x*'s section "Power transistors for linear amplifiers"), at the expense of a high static driving current; the latter drawback is exacerbated at high currents, where BJT beta drops rapidly. A saturated BJT also suffers from slow recovery owing to stored charge in the base region.

With the very high voltages and currents encountered with IGBTs, it is mandatory to include fault protection in the circuit design: an IGBT that may be required to switch up to a 50 A load from a 1000 V supply will be destroyed in milliseconds if the load becomes short-circuited, owing to the 50 kW (!) power dissipation. The usual method is to shut off the drive if V_{CE} has not dropped to just a few volts after 5 μs or so of input drive (see Figure 12.87B). We'll revisit these three power transistor technologies in Chapter *3x*.

B. Thyristors

For the utmost in *really* high-power switching (we're talking kiloamperes and kilovolts), the preferred devices are the *thyristor* family, which include the unidirectional "silicon controlled rectifiers" (SCRs) and the bidirectional "triacs." These three-terminal devices behave somewhat differently from the transistors we've seen (BJTs, FETs, and IGBTs): once triggered into conduction by a small control current (a few milliamps) into their control electrode (the *gate*), they remain ON until external events bring the controlled current (from *anode* to *cathode*) to zero. They are used universally in house-current lamp dimmers, where they are switched on for a fraction of each half-cycle of ac line voltage, thus varying the *conduction angle*.

Thyristors are available in ratings from 1 A to many thousands of amperes, and voltage ratings from 50 V to many kilovolts. They come in small transistor packages, the usual transistor power packages, larger modules, and really scary "hockey puck" packages that are capable of switching megawatts. These are hefty devices; you can hurt yourself just by dropping one on your foot.

3.6 MOSFETs in linear applications

Although we dealt extensively with linear applications in this chapter's treatment of *JFETs*, our discussion of MOSFETs has concentrated almost entirely on switching applications. Lest we leave the wrong impression, we address in this section some applications of discrete power MOSFETs to linear applications, particularly those that benefit from their unique properties. See also additional applications to linear voltage regulators in Chapter 9 (e.g., Figures 9.17, 9.20, 9.104, 9.110, and 9.113), and advanced material in §*3x.5*.

3.6.1 High-voltage piezo amplifier

A nice application of MOSFETs as linear amplifiers takes advantage of available types with high voltage ratings, and their immunity to second breakdown. Ceramic *piezoelectric* transducers are often used in optical systems to produce controlled small motions; for example, in *adaptive optics* you might use a piezoelectrically controlled "rubber mirror" to compensate for local variations in the index of refraction of the atmosphere. Piezo transducers are nice to use, because they're very stiff. Unfortunately, they may require as much as a kilovolt of drive to produce significant motions. Furthermore, they're highly capacitive – typically 0.01 μF or more – and have mechanical resonances in the kilohertz range, thus presenting a nasty load. We needed dozens of such driver amplifiers, which for some reason cost a few thousand dollars apiece if you buy them commercially.

[109] You find exponents from 1.6 to 2.5 in the literature; the lower end of this range is likely to be more accurate; see our plots in Chapter *3x*.

[110] Where it excels also in maintaining high transconductance, compared with the MOSFET. The advantage goes to the IGBT, starting around 200 V. See also the comparison table in §3.5.5.

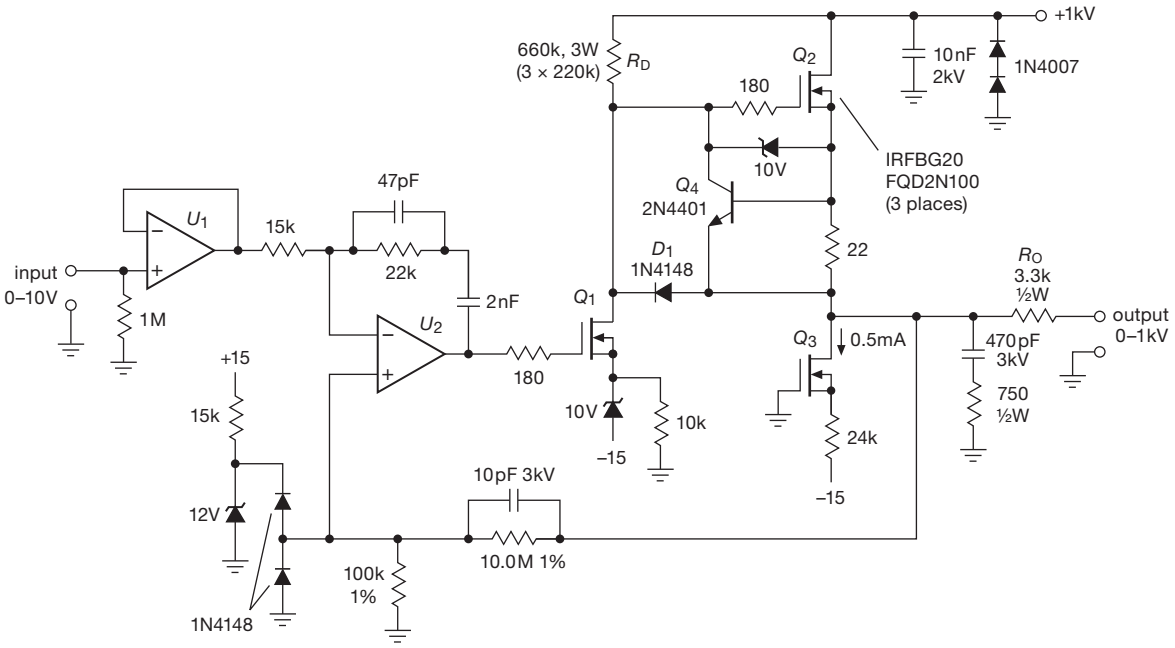


Figure 3.111. 1 kV low-power piezo driver with totem-pole output stage. A similar design is used for the high-voltage regulated dc supply shown in Figure 9.110.

We solved our problem with the circuit shown in Figure 3.111. The IRFBG20 is an inexpensive (∼$2) MOSFET, good for 1 kV and 1.4 A; the similar FQD2N100 (1 kV, 1.6 A) costs about $0.85. The first transistor is a common-source inverting amplifier, driving a source follower with active current-sink load. The *npn* transistor is a current limiter and can be a low-voltage unit, since it floats on the output. One subtle feature of the circuit is the fact that it's actually push–pull, even though it looks single-ended: you need plenty of current (20 mA) to push 10,000 pF around at 2 V/μs; the output transistor can *source* current, but the pulldown resistor can't sink enough (look back to §2.4.1, where we motivated push–pull with the same problem). In this circuit the driver transistor is the pull-down, with conduction through the gate-source diode![111] The rest of the circuit involves feedback with op-amps, a forbidden subject until the next chapter; in this case the magic of feedback makes the overall circuit linear (100 V of output per volt of input), whereas without it the output voltage would depend on the (nonlinear) I_D-versus-V_{GS} characteristic of the input transistor. A nice improvement to this circuit consists of replacing the 660kΩ 3 W pullup resistor (whose current drops at high output voltages, e.g., to 0.15 mA at 900 V) with a depletion-mode MOSFET current source set to, say, 0.25 mA. See the discussion below (§3.6.2C, and also Figure 3.23, Table 3.6, and §§9.3.14C and *3x.6*).

For a detailed analysis of a precision high-voltage amplifier with bipolarity output capability, see the section with that name in Chapter *4x*. For an analysis of relevant issues, such as MOSFET transconductance at low currents, see §*3x.5.2*, and for the response of a MOSFET source follower into a capacitive load see §*3x.8*.

Exercise 3.18. Modify this circuit so the high-voltage output can be turned on and off, under control of an input signal (0 V for off, +3V for on).

3.6.2 Some depletion-mode circuits

Depletion-mode MOSFETs are the neglected siblings of the far more popular enhancement-mode MOSFETs. They can do some nice tricks, though, that are worth knowing about. And they are available in high voltage (to 1 kV) and high current (to 6 A) varieties. Table 3.6 on the following page lists nearly all available parts of this somewhat rarefied species. Here are some applications that exploit their property of conduction at zero gate voltage.

[111] This is called a "totem-pole" output stage, and became popular in the early 1970s in bipolar TTL logic, see Figure 10.25A.

Table 3.6 Depletion-mode *n*-channel MOSFETs

		Available packages																					
		TO-92	SOT-23	TO-243	SOT-223	TO-220	TO-247	D2-Pak	D-Pak	BV_{DS}	P_D[c]	I_{DSS} min	I_{DSS} max	R_{ON}[t] @V_{GS}=0	g_m[t] @	I_D	$V_{GS(th)}$ min	$V_{GS(th)}$ max	C_{iss}	C_{oss}	C_{rss}	Cost[y]
Part #	Manuf									(V)	(W)	(mA)	(mA)	(Ω)	(mS)	(A)	(V)	(V)	typ (pF)	typ (pF)	typ (pF)	$US
small, RF																							
BF998	NXP	-	f	-	-	-	-	-	-	12	-	2	18	-	24	0.01	-	−2	2.1	1.1	0.025	-
BF999	Vishay	-	•	-	-	-	-	-	-	20	-	5	18	-	16	0.01	-	−2.5	2.5	1.0	0.025	-
SKY65050[s]	Skyworks	-	g	-	-	-	-	-	-	6	-	40	70	-	80	0.12	−1	−0.7	really small			-
small																							
DN2470K4	Supertex	-	-	-	-	-	-	-	•	700	2.5	500[t]		42[m]	100[n]	0.1	−1.5	−3.5	540	60	25	0.81
BSS126	Infineon	-	•	-	-	-	-	-	-	600	0.5	7	-	320	17	0.01	−2.7	−1.6	21	2.4	1	0.13
BSP135	Infineon	-	-	-	•	-	-	-	-	600	1.8	20	-	30	160	0.1	−1.8	−1.0	98	8.5	3.4	1.38
LND150	Supertex	◊	k	•	-	-	-	-	-	500	0.7	1	3	850	2	0.001	−1.0	−3	7.5	2	0.5	0.58
DN3145	Supertex	-	-	•	-	-	-	-	-	450	1.3	120	-	60[m]	140[n]	0.1	−1.5	−3.5	120	15	10	0.68
DN3545	Supertex	•	-	◊	-	-	-	-	-	450	1.6	200	-	20[m]	150[n]	0.1	−1.5	−3.5	360	40	15	0.74
DN2540	Supertex	•	-	◊	-	•	-	-	-	400	1.6	150	-	17	325	0.1	−1.5	−3.5	200	12	1	0.81
DN3135	Supertex	-	•	◊	-	-	-	-	-	350	1.3	180	-	35[m]	140	0.1	−1.5	−3.5	60	6	1	0.62
CPC3720C	Clare	-	-	•	-	-	-	-	-	350	1.6	130	-	22[m]	225	0.1	−1.6	−3.9	70	20	10	0.37
CPC5603C	Clare	-	-	-	•	-	-	-	-	415	2.5	130	-	14[m]	-	-	−3.6	−2	300	-	-	0.69
CPC5602C	Clare	-	-	-	•	-	-	-	-	350	2.5	130	-	14[m]	-	-	−3.6	−2	300	-	-	0.60
DN3535	Supertex	-	-	•	-	-	-	-	-	350	1.6	200	-	10[m]	200[n]	0.1	−1.5	−3.5	360	40	10	0.68
BSS139	Infineon	-	•	-	-	-	-	-	-	250	0.4	30	-	12.5	130	0.08	−1.4	-	60	6.7	2.6	0.57
BSP129	Infineon	-	-	-	•	-	-	-	-	240	1.8	50	-	6.5	360	0.28	−2.1	−1.0	82	12	6	0.47
DN3525N8	Supertex	-	-	•	-	-	-	-	-	250	1.6	300	-	6[m]	225[n]	0.15	−1.5	−3.5	270	20	5	0.66
CPC3703	Clare	-	-	•	-	-	-	-	-	250	1.6	300	-	4[m]	225[n]	0.1	−1.6	−3.9	327	51	27	0.57
BSP149	Infineon	-	-	-	•	-	-	-	-	200	1.8	140	-	1.7	800	0.48	−1.8	−1.0	326	41	17	0.80
BSS159	Infineon	-	•	-	-	-	-	-	-	50	0.4	70	-	4	160	0.16	−2.5	0.0	70	15	6	0.31
large						see note x																	
IXTx01N100D[x]	IXYS	-	-	-	-	P	-	-	Y	1000	25	100[t]		90	150	0.1	−2.5	-	120	25	5	0.75
IXTx08N100D2[x]	IXYS	-	-	-	-	P	-	-	Y	1000	60	100[t]		21[m]	560	0.4	−2	−4	325	24	6.5	0.69
IXTx1R6N100D2[x]	IXYS	-	-	-	-	P	-	-	Y	1000	100	100[t]		21[m]	1100	0.8	−2.5	−4.5	645	43	11	1.66
IXTx3N100D2[x]	IXYS	-	-	-	-	P	-	A	-	1000	125	3000	-	5.5[m]	4200	3	−2.5	−4.5	1020	68	17	2.11
IXTx6N100D2[x]	IXYS	-	-	-	-	P	H	A	-	1000	300	6000	-	5.5[m]	4200	3	−2.5	−4.5	2650	167	41	4.66
IXTx02N50D2[x]	IXYS	-	-	-	-	P	-	-	Y	500	25	250[t]		20	150	0.2	−2	−5	120	25	5	1.05
IXTx08N50D2[x]	IXYS	-	-	-	-	P	-	A	-	500	60	800	-	4.6[m]	570	0.4	−2	−4	312	35	11	1.62
IXTx1R6N50D2[x]	IXYS	-	-	-	-	P	-	-	Y	500	100	1600[t]		2.3	1750	0.8	−2	−4	645	65	17	1.66
IXTx3N50D2[x]	IXYS	-	-	-	-	P	-	A	-	500	125	1600[t]		1.5	2100	1.5	−2	−4	1070	102	24	2.13
IXTx6N50D2[x]	IXYS	-	-	-	-	P	H	A	-	500	300	6000	-	0.5[m]	4500	3	−2	−4	2800	255	64	4.66
IXTH20N50D[x,e]	IXYS	-	-	-	-	-	H	-	-	500	400	1500[t,e]		0.5[m]	7500	10	−1.5	−3.5	2500	400	100	8.61

Notes: (c) P_D at T_C=25°C, for package marked ◊. (e) the IXTH20N50D delivers most of its current capability in the enhancement-mode region. (f) has two gates, 4-lead SOT-143 package. (g) SC-70 package. (k) unusual pinout; also try LND250K1. (m) maximum. (n) minimum. (s) SKY part's full name: SKY65050-372LF. (t) typical. (x) substitute the letter listed in the package column for the "x" in the part number; for example, IXTP01N100D is a TO-220 package. (y) quantity 100.

A. Input protection

Low-level circuits (such as sensitive amplifiers) don't like to be driven beyond their power supply rails. One simple protection scheme uses a series resistor at the input, with a pair of downstream clamp diodes to the amplifier's supply rails. That's OK for small overdrive; but it's unsatisfactory if the input may go to a few hundred volts (think *powerline!*), because the large resistor value (∼100k or so, to limit the fault current and dissipated power) compromises the signal bandwidth and noise. Figure 3.112 shows how to use a pair of depletion-mode MOSFETs (instead of a large resistor) as the series element. The particular part shown

is small (SOT23, SOT89, or TO-92), inexpensive ($0.60), and able to withstand momentary inputs to ±500 V. The pair looks like a series resistance of ∼1.7k (twice $R_{DS(ON)}$) until the input goes beyond the amplifier's rails, whereupon it limits the current through the clamp diodes to ∼2 mA. See some subtleties in "A Riff on Robust Input Protection" (§5.15.5).

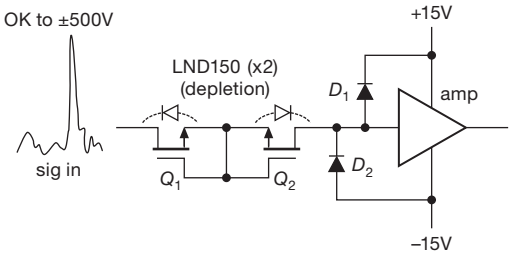


Figure 3.112. Protecting a low-level input from outrageous overvoltage "faults." Under normal conditions the series-connected depletion-mode MOSFETs Q_1 and Q_2 (with their intrinsic body diodes shown) conduct with an effective series resistance of R_{ON}∼1 kΩ each. An input signal beyond the amplifier's ±15 V rails is clamped by diodes D_1 or D_2, with Q_1 and Q_2 safely limiting the clamp current to their I_{DSS}∼2mA. See also Figure 5.81.

B. HV capacitor discharge

Human contact with a circuit at a few hundred volts can be, well, a *shocking* experience. That's why it's considered good manners to arrange things so that storage capacitors charged to such voltages are discharged promptly after power is removed. Capacitors, after all, have pretty good memory – they can stay charged for hours, or even years (that's how bits are stored in "flash memory," see §14.4.5).

The traditional approach is to put a "bleeder" resistor across the storage capacitor, sized to discharge it in ∼10 s or so. Good enough. But it is not really satisfactory when you have a large-value capacitor, for example one used to store the energy for a short-duration high-voltage pulse generator. Figure 3.113 shows such an application, with a 100 μF storage capacitor charged to +400 V by a low-power dc–dc converter (say 10 W), the latter powered from a low-voltage dc supply that also powers the other pulse-generator circuitry.

What you'd like is a bleeder resistor that is connected only when the external power is removed. Here depletion-mode MOSFET Q_1 is held in the nonconducting state when the supply is powered (V_{GS}=−9 V), but is sent into conduction (V_{GS}≈0.6V) when the +12 V is absent. It's rated at 500 V, 3 A I_{DSS}(min), and costs about $2. (You can get depletion-mode MOSFETs up to 1 kV.) We don't need 3 A (which would discharge 100 μF in just 13 ms); but we *do* need a MOSFET large enough to absorb the stored energy,

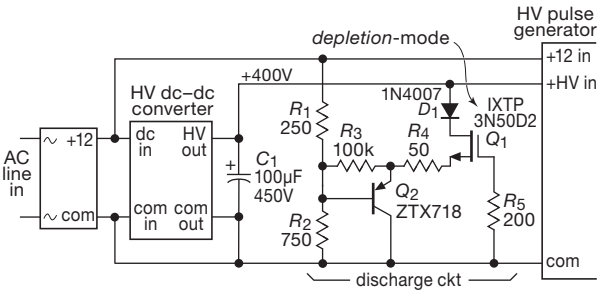


Figure 3.113. Depletion-mode MOSFET Q_1 discharges the 100 μF high voltage capacitor C_1 when power is removed; when powered it is inactive.

here 8 joules – this part can absorb a 25–50 J pulse without exceeding $T_{J(max)}$ (see Chapter *9x*). Follower Q_2 boosts the discharge current, which otherwise would be just a few milliamps (set by bleeder R_2).

C. Current source

Depletion-mode power MOSFETs make excellent 2-terminal current sources, capable of high voltages (to 1000 V for some parts, see Table 3.6) and many watts of power dissipation. They extend the basic idea, seen earlier with JFETs (§3.2.2, Figures 3.20 and 3.23), to higher voltage and power levels. Because these applications are associated with power, we defer the discussion to Chapter 9 (§9.3.14C), where you can see that the circuits are the same as with JFETs (Figure 9.36), and you can delight in curves of measured current versus voltage (Figures 9.40 and 9.41).[112] Such a depletion-mode MOSFET current source is ideal for an application like the high-voltage piezo driver we just saw (§3.6.1), where it can replace the primitive 660k power-resistor pullup and thereby supply approximately constant driver-stage drain current over the signal swing.

D. Extending regulator V_{IN}

Sometimes you need to extend the permissible range of dc input voltage to some low-voltage device. Figure 3.114A shows an example: a linear voltage regulator (§9.3) that provides +3.3 V (for example) from a higher dc input. Such regulators have limited maximum input voltage range – perhaps +20 to +30 V (if made with BJTs), or as little as +6 V (if made with CMOS). Here the *n*-channel depletion-mode MOSFET Q_1 is wired as a follower, providing at the regulator's input a voltage that is greater than V_{OUT} by the

[112] There's plenty more detail fleshed out in Chapter *3x* (§*3x.6*), where we show tricks for raising the output impedance (i.e., constancy of current) with the ever-wonderful cascode, raising operating voltage with a series stack, and reducing power dissipation.

magnitude of its V_{GS}; for the IXTP08N50 V_{GS} is between −2 and −4 V, so the regulator's input is held between 2 and 4 V above its output. The circuit's input voltage can go to +500 V (Q_1's rated maximum), with due regard to heat dissipation of course. The resistor R_{CL} protects Q_1 by setting a rough current limit. You can play the same trick with a constant-*current* regulator (Figure 3.114B). For details see the fuller discussion in Chapter 9 (§9.13.2). See also §*3x.6* for a discussion of the use of depletion-mode MOSFETs as current sources, especially at high voltages.

3.6.3 Paralleling MOSFETs

You sometimes hear the statement that power MOSFETs can be paralleled directly (without ballasting resistors in the source leads), because their negative temperature coefficient of I_D at fixed V_{GS} guarantees automatic redistribution of drain currents in a paralleled array. Furthermore, the story goes, the same property prevents thermal runaway.

A. As *switches* – Yes!

Power MOSFETs exhibit negative tempco of I_D – but only at high drain currents (or, more accurately, at relatively large values of V_{GS}), as seen in Figure 3.115. For *switching* applications, in which you are operating at essentially zero V_{DS} (limited by R_{ON}), the large gate drive puts the device into the region of negative I_D tempco, so you *can* (and should) simply tie multiple MOSFETs in parallel, with no ballasting resistors.[113] Here R_{ON} increases with increasing temperature (Figure 3.116), and the parallel connection shares drain current (and power) properly.

The positive tempco of R_{ON}, though helpful for parallel operation of MOSFET switches, creates a new problem, namely the possibility of *thermal runaway*. See §3.6.4.

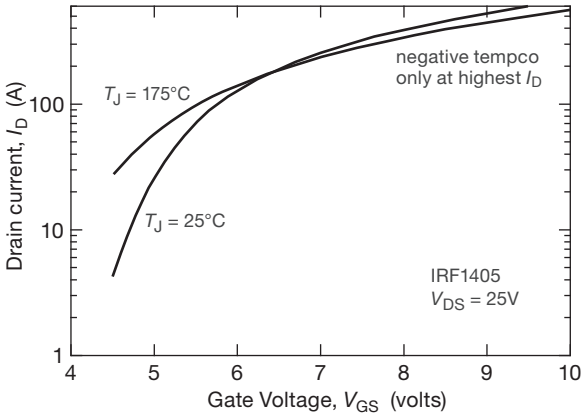


Figure 3.115. Transfer characteristics (I_D versus V_{GS}) for the IRF1405 n-channel power MOSFET. Note that the temperature coefficient is *positive* except at the highest drain currents (>175 A); for *linear* applications you would rarely exceed 10 A of drain current.

B. In linear power circuits – No!

Here the situation is more complicated: in most *linear* applications (e.g., audio power amplifiers, in which there is substantial voltage V_{DS} across the transistor) you are operating in the positive tempco region of relatively low drain currents – because otherwise the power dissipation ($I_D V_{DS}$) would be much greater than allowable by thermal considerations (i.e., excessive junction temperature; see §9.4.1A). For example, the transistor of Figure 3.115 is limited to 200 watt dissipation at a case temperature of 75°C; so, in a circuit with 25 V across it, the average drain current is limited to 8 A, in which I_D has a large positive tempco. So, for practical linear applications – in which you operate

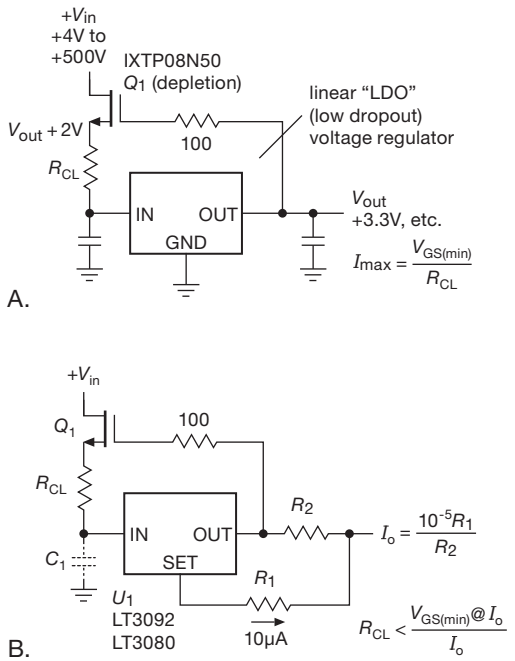


Figure 3.114. A. A high voltage depletion-mode MOSFET extends the input voltage range of a series voltage regulator. B. An analogous circuit for a current source. For good performance at high frequencies C_1 should be small, or even eliminated altogether. See Figure 9.104, where the *i*s are dotted, *t*s are crossed, and all is explained.

[113] However, each FET should have its own series gate resistor to prevent oscillation during the switching transitions; these are typically in the range of a few ohms to a few tens of ohms, and they should ordinarily be used as well for single switching MOSFETs. Ferrite beads on the gate or drain leads can also be helpful to tame oscillations.

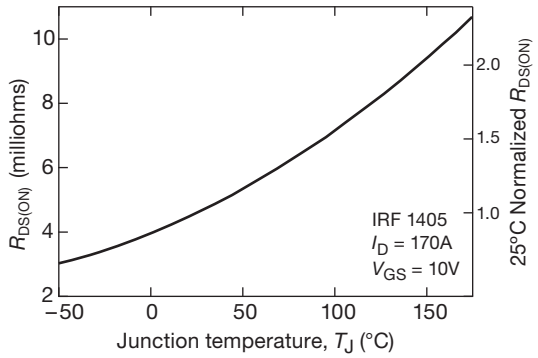


Figure 3.116. ON-resistance increases with increasing temperature: R_{ON} versus temperature for the IRF1405 *n*-channel power MOSFET.

with substantial V_{DS} – unequal current sharing of paralleled MOSFETs is in fact exacerbated. And, because you're using multiple transistors exactly because a single one won't handle the power, the circuit is in serious trouble; a single transistor will likely hog an excessive share of the current, putting its dissipation well over the limit set by thermal resistance and heatsinking.

Source-ballast resistors

The solution is to add small ballasting resistors in the individual source leads, chosen roughly so that the voltage drop across them is at least comparable to the scatter in gate-source operating voltages (Figure 3.117A). We've found that a few tenths of a volt drop is frequently adequate for MOSFETs of a given type, from a single manufacturing batch or from transistors selected for matched V_{GS};[114] however, the datasheet specs would suggest (conservatively) larger drops – a volt or two at full operating current. Unless you are willing to worry about matched transistors (both during initial construction, and later replacement), you should take the conservative approach to produce a robust design, with source-ballasting resistors sized to drop a volt or two at the currents where power dissipation becomes important.

This example illustrates a frequent designer's quandary, namely a choice between a conservative circuit that meets the strict worst-case design criterion, and is therefore *guaranteed* to work, and a better-performing circuit design that doesn't meet worst-case specifications, but is overwhelmingly likely to function without problems. There are times when you will find yourself choosing the latter, ignoring the little voice whispering into your ear.

[114] In the example in §3.6.3A you might put four IRF1405s in parallel, with 0.1 Ω 10 W resistors in each source lead to handle a total current of 25A.

Active feedback

This current-matching problem exemplifies a typical circuit tradeoff of robustness versus performance: a conservatively large ballast drop produces increased R_{ON} and power dissipation. As is often the case, a clever circuit can recover the lost benefits.

Figure 3.117B shows a nice solution, another in our "Designs from the Masters" series; just a snippet this time, but a valuable snippet. The small current-sense resistors in the MOSFET source leads provide active feedback through a primitive differential amplifier. Compared with a conservative source-ballasting circuit (Figure 3.117A), in which the source resistors are chosen to provide a 2 V drop (at a nominal 1 A per transistor operating current), the active circuit uses much smaller 0.1 Ω sense resistors, providing 100 mV of drop, which is applied to the *npn* differential pair to adjust the gate voltages as needed to equalize the source currents. This circuit requires larger gate drive voltage, which is seldom a problem; in exchange, it minimizes the voltage drop and impedance in the MOSFET's high-current path. This scheme is well suited to relatively slow circuits, for example the series pass element in a linear power supply. Note that this arrangement is easily expanded to any number of MOSFETs.[115]

There's a pleasant exception to this general characteristic of positive I_D tempco in power MOSFETs: *lateral* devices (as opposed to the *vertical* structure of nearly all power MOSFETs) exhibit negative tempco beginning at very low gate voltage (and very low I_D); see Figure 3.118. Lateral power MOSFETs do not attain the high breakdown voltage and low R_{ON} ratings of vertical power MOSFETs, but they are favored in linear power applications such as audio amplifiers for their linearity and thermal stability. A popular choice is the 2SK1058 (*n*-channel) and 2SJ162 (*p*-channel) complementary pair from Renesas (Hitachi), limited to 160 V and 7 A; their R_{ON} is an unimpressive ∼1 Ω. This is not of great concern in the linear amplifier context, where they do not operate near voltage-saturation; but it's high enough that you often see several of them used in parallel. See the section "Power transistors for linear amplifiers" in Chapter *3x* for further discussion.

The positive tempco of I_D in power MOSFETs creates an additional problem, namely the possibility of *thermal runaway*.

[115] We found this cute circuit trick used in some HP (subsequently Agilent, now Keysight) E3610-series linear power supplies. It's much simpler than using individual op-amps to bias each transistor, as some MOSFET manufacturers suggest. A different way to benefit from the higher power-dissipation capability of multiple transistors is to wire them in *series*, see for example Figure 9.111. A series connection guarantees equal current distribution.

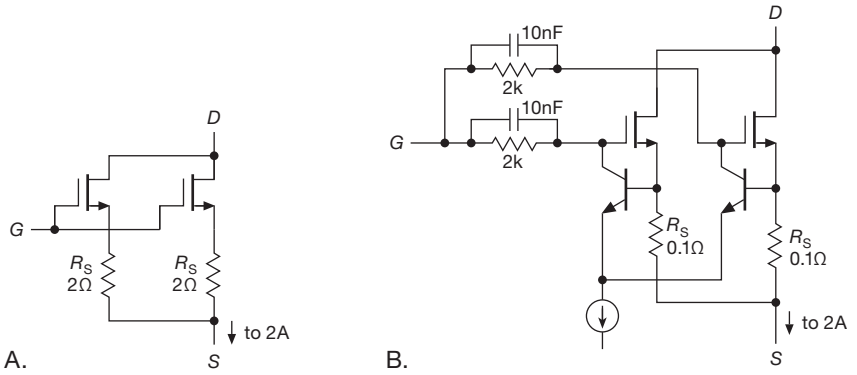


Figure 3.117. Paralleling power MOSFETs: A. with source ballasting resistors; B. with sense resistors and active feedback.

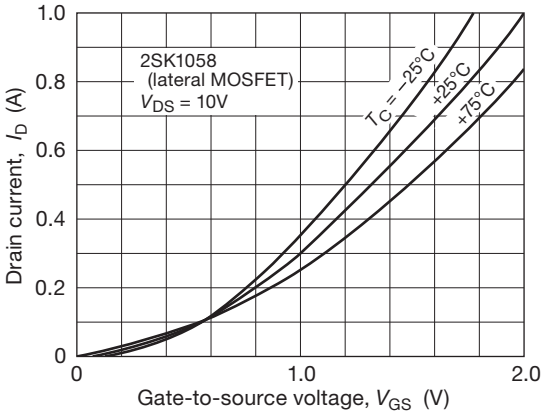


Figure 3.118. Transfer characteristics (I_D versus V_{GS}) for the 2SK1058 *lateral* n-channel power MOSFET, popular for use in high fidelity audio power amplifiers. Here the temperature coefficient is *negative* over most of the operating region.

3.6.4 Thermal runaway

Up to now, we've avoided the R-word, because "thermal runaway" is quite independent of whether transistors are used in parallel; it refers particularly to circuit configurations in which the power dissipation produces a rise in temperature that in turn raises the power that must be dissipated. Two important examples are the push–pull linear amplifier and the saturated power switch.

A. Push–pull power amplifier

In the class-AB push–pull power amplifier, commonly used in audio output stages, the push–pull pair is biased with substantial quiescent current (typically ∼100 mA) to preserve linearity during waveform crossover. The quiescent current varies with temperature because both I_D (with MOSFETs) and I_C (with bipolar transistors) have positive temperature coefficients at constant drive voltage. Depending on the circuit configuration and the degree of heatsinking, the output transistors may or may not reach a stable temperature; if they don't, you've got thermal runaway (independent of whether or not you've paralleled multiple transistors).

We saw this earlier in §2.4.1B, where we introduced the push–pull audio power amplifier built with complementary *bipolar* transistors. Because bipolar transistors have a positive tempco of collector current at fixed V_{BE},[116] the usual approach is to bias the bases apart with a voltage source that tracks the tempco of the output stage V_{BE}'s – typically by using diodes or transistor base–emitter junctions, thermally coupled to the output stage heatsink – often in conjunction with small emitter resistors in the output stage (Figure 3.119B).

Power MOSFETs in linear push–pull amplifiers present the same problem, because they are operated in the region of positive tempco of I_D (§3.6.3B). You can use the same trick (bias generator with tracking negative tempco, perhaps in combination with small-value output-stage source resistors; see the section "Power transistors for linear amplifiers" in Chapter *3x*). However, the problem is nicely finessed by using lateral power MOSFETs, whose negative tempco of I_D (Figure 3.118) beginning at $I_D \approx 100$ mA guarantees no thermal runaway. The usual approach is to bias the output stage gates apart with a (settable) constant dc voltage, as shown in Figure 3.119, bypassed at signal frequencies.[117] The bias is typically set for a quiescent current I_Q close to the zero-tempco crossover (100 mA for

[116] Or, alternatively, a negative tempco of V_{BE} for constant I_C.

[117] The figure shows bare-bones circuitry. In practice the bipolar transistors would be configured as Darlington or Sziklai pairs, and the single-ended driver stage might be replaced with a symmetrical pair of drivers, driven from the differential input stage. For a 150 W amplifier you would probably use paralleled pairs of transistors to stay within

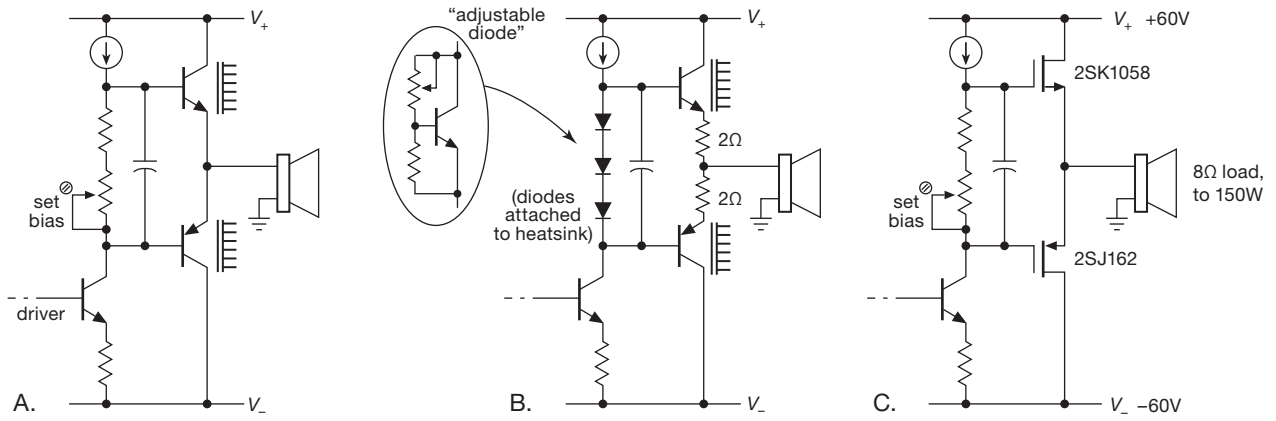


Figure 3.119. Thermal stability in push–pull power amplifiers – simplified output stage configurations. A. Fixed V_{BE} biasing promotes runaway, owing to positive I_C tempco in bipolar output stage. B. Tracking thermally-coupled bias generator tames runaway. C. Stable quiescent current in lateral MOSFETs biased at fixed V_{GS}; no thermal compensation is needed.

the 2SK1058/2SJ162 complementary pair), ensuring that I_Q remains relatively constant as the amplifier warms up.[118]

B. Saturated switch

It's widely believed that MOSFETs are immune to thermal runaway when used for power *switching*. The thought process goes like this: "These puppies have really low R_{ON} when driven to full conduction, so they hardly need any heatsinking; besides, if they do heat up (while carrying some large but bounded current), the thing will stabilize at some elevated temperature, because the power carried off by the heatsink increases roughly proportional to the rise above ambient temperature, and eventually catches up to the power being dissipated; plus, hey, these things are *tough*!"

Nice thoughts. But the reality can be different. That's because R_{ON} isn't constant, but increases with temperature (Figure 3.116); so the switch dissipates more power as it heats up, and, if the heatsink is too small, the heat it carries off may never catch up – in which case the process runs away!

To lend some perspective: you don't need to have actual thermal *runaway* to cause overheating and destruction – an undersized heatsink[119] will do the job just fine by allowing the junction temperature to soar above $T_{J(max)}$. And, as we'll see shortly, the better approach is to reduce power dissipation by reducing R_{ON}, rather than by piling on larger heatsinks. With this cautionary comment, let's see how actual thermal runaway could occur in an ill-advised design.

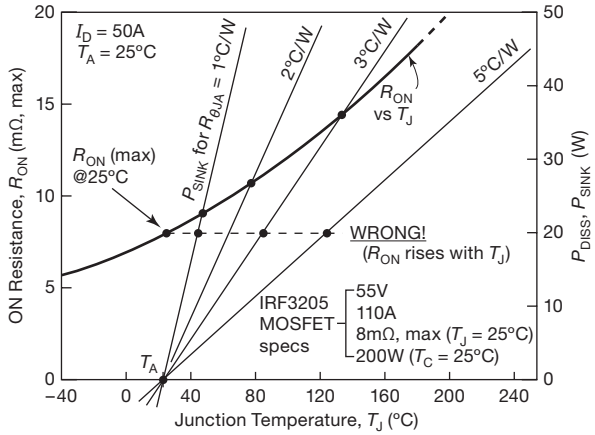


Figure 3.120. Thermal runaway in a MOSFET switch. The curved line plots maximum ON-resistance and corresponding power dissipation at 50 A for an IRF3205 *n*-channel power MOSFET. The straight lines plot power removed by three choices of heatsink thermal resistance $R_{\Theta JA}$. Thermal runaway occurs with the smallest heatsink, where there is no graphical intersection.

Figure 3.120 shows an easy graphical way to see what's happening and to figure out how much heatsink you need to

allowable junction temperature; no ballasting resistors would be needed for the MOSFET version.

[118] Because lateral MOSFETs may be hard to get, a regular power MOSFET can be used in place of the BJT "V_{BE} diode" (as in Figure 3.119B) to bias a complementary pair of ordinary power MOSFETs. This approach prevents thermal runaway because the tempco of a MOSFET is higher at low currents than at high currents, see Figure 3.115.

[119] Or none at all! The impressive $I_{D(max)}$ ratings on the datasheet could tempt you to omit the heatsink entirely, even in a power-switching circuit operating at substantial drain currents.

prevent runaway (and, by the way, to keep the junction temperature T_J below the specified $T_{J(max)}$). We begin by plotting the datasheet's R_{ON} versus temperature for an inexpensive power MOSFET (their graph tops out at 175°C, because that is the maximum rated junction temperature; we took the liberty of extending the graph another 75°). Then we use that to evaluate the power dissipation, as $P_{diss} = I^2 R_{ON}$; for our chosen 50 A drain current we got the values marked on the right-hand axis. Finally, we plot separately the power carried off by four values of heatsink "thermal resistance" $R_{\Theta JA}$ (given by $P_{diss} = (T_J - T_A)/R_{\Theta JA}$), assuming an ambient temperature of $T_A = 25°C$ (plenty of discussion of this in §9.4.1A).

The heatsinks carry off an amount of power proportional to the temperature rise above ambient, as plotted; the transistor generates power according to its graph. The intersection (if any!) is the equilibrium temperature, which in this case is about 45°C or 75°C, for the two larger heatsinks. But the smallest heatsink has no intersection – it cannot carry off as much heat as the transistor generates, at any temperature: *thermal runaway!* In real life you should assume that the ambient temperature will be higher (equipment is put into racks, or stacked with other equipment; and, you can have hot weather!): you would do this by sliding the heatsink curves to the right.

From this simple example you might conclude that larger heatsinks are the right cure for thermal runaway in saturated switching applications. But look again at the numbers: we're switching 50 A with an R_{ON} of order 10–15 mΩ – that's a 0.5 to 0.75 V drop, and 25 to 40 W dissipation. At these sorts of currents we really should be using a larger transistor, or several in parallel, to reduce R_{ON} (and therefore the dissipated power). The "maximum I_D" specification (here 110 A) looks pretty good on a datasheet, but is not a realistic guide to appropriate dc operation of the part. In this example a better choice would be a low R_{ON} part like the FDB8832[120] with R_{ON}=2.3 mΩ (max) at 25°C, and with a typical ON voltage of 115 mV at 50 A and power dissipation of 5.8 W.[121] This is a 30 V part (high-voltage MOSFETs have higher R_{ON}); if you wanted to switch somewhat higher voltages with low R_{ON} and P_D, your choices are to use a high-power MOSFET module[122] or (less expensive) several conventional MOSFETs in parallel. For voltages above 400 V or so the transistor of choice is the IGBT (insulated-gate bipolar transistor; see §3.5.7A), which has the input properties of a MOSFET and the output properties of a BJT. An example is the Mitsubishi CM1200HC-50H, rated at 2500 V and 1200 A: at full current it saturates at just 3 V (equivalent to a 2.5 mΩ ON-resistance). That's pretty good... but that works out to 3.6 *kilo*watts dissipation! (These things are used for power switching in applications like electric locomotives.)

Lest we leave the wrong impression, we hasten to point out that thermal calculations and heatsink selection need not require the kind of graphical plotting we've done here (in which we were interested primarily in the possibility of true thermal runaway). More simply, you can just apply a safety factor m to the datasheet's 25°C R_{ON} value to get a reasonable estimate of R_{ON} at the maximum junction temperature (150°C); from that you get

$$T_J \approx T_A + I_D^2 \cdot m R_{ON(25C)} \cdot R_{\Theta JA}. \qquad (3.15)$$

The multiplier m varies somewhat with the voltage rating of the MOSFET; based on data from many datasheets (see the graph and discussion in the section "MOSFET ON-resistance versus temperature" in Chapter *3x*), it ranges from roughly $m \approx 1.5$ (for low-voltage MOSFETs) to roughly $m \approx 2.5$ (for high-voltage MOSFETs). As a practical rule-of-thumb, you'll be safe if you use m=2 for MOSFETs rated to 100 V and m=2.5 for those of higher voltages (at least to 1kV).

C. Second breakdown and safe operating area

It's worth emphasizing a related thermal effect ("second breakdown") that was discussed earlier in §3.5.1B: power transistors fail (usually[123]) if operated beyond their maximum voltage, their maximum current, or their maximum junction temperature (the latter dependent on power dissipation, pulse duration, heatsink thermal resistance, and ambient temperature; see §9.4.2). The boundaries define the *safe operating area*, or SOA, for example as shown in Figure 3.95. Bipolar transistors suffer from an additional failure mode known as *second breakdown*, an incompletely understood instability characterized by local heating, reduction of the breakdown voltage, and, often, destruction of the junction. It is second breakdown that imposes the additional constraint on the bipolar SOAs in Figure 3.95. Happily, MOSFETs are less likely to suffer from second breakdown, which contributes to their popularity in power circuits.[124] Note that for both kinds of transistors the maximum current and power limits are higher for short pulses.

[120] From the same manufacturer, Fairchild Semiconductor.

[121] Rising to 3.6 mΩ, 180 mV, and 9 W (maximum) at T_J=150°C.

[122] These come in husky "SOT-227" packages, with screw terminals on the top, an isolated metal base, and with names like "ISOTOP" and "miniBLOC."

[123] Or, perhaps more accurately, they are not guaranteed *not* to fail!

[124] Some newer fine-geometry types *are* susceptible, however; see IR App Note AN-1155.

Table 3.7 Junction Field-Effect Transistors (JFETs)[a]

Part #[b]	N or P-channel	Jellybean?	Prefixes & Pkgs[d]: TO-92: 2N, PN	SOT23: MMBF	SOT23: PMBF	MMBF_LT	V_{DSS} max (V)	I_{DSS} min (mA)	I_{DSS} max (mA)	I_{DSS} meas (mA)	R_{ON} max (Ω)	$V_{GS(off)}$[c] min (V)	$V_{GS(off)}$[c] max (V)	V_{GS} @ I_D measured (V)	V_{GS} @ I_D (mA)	g_m min (mS)	g_m max (mS)	g_m @I_D (mA)	g_m meas (mS)	g_m @I_D (mA)	G_{max}[e]	C_{iss} typ (pF)	C_{rss} typ (pF)
PN4117	N	•	A	C	-	-	40	0.03	0.09	0.07	-	-0.6	-1.8	-0.33	0.03	0.07	0.21	z	0.09	0.03	420[r]	1.2	0.3
'4118	N	-	A	C	-	-	40	0.08	0.24	0.20	-	-1	-3	-1.33	0.1	0.08	0.25	z	0.13	0.1	260[r]	1.2	0.3
'4119	N	-	A	C	-	-	40	0.20	0.60	0.30	-	-2	-6	0.0	0.3	0.10	0.33	z	0.18	0.3	140[r]	1.2	0.3
BFT46	N	-	-	C	-	-	25	0.20	1.5	0.63	-	-	-1.2	-0.16	0.3	1	-	z	1.7	0.3	190[s]	3.5	0.8
BF511	N	-	-	D	-	-	20	0.7	3	4.2	-	-	-1.5	-0.75	1	4	-	z	2.7	1	120	-	0.3
2N5457	N	•	A	C	-	-	25	1	5	3.5	-	-0.5	-6	-0.81	1	1	5	z	2.3	1	220	4.5	1.5
'5458	N	-	A	C	-	-	25	2	9	4.1	-	-1	-7	-0.97	1	1.5	5.5	z	2.2	1	190	4.5	1.5
'5459	N	-	A	C	-	-	25	4	16	9.9	-	-2	-8	-1.82	3	2	6	z	2.9	3	100	4.5	1.5
2N5460	P	•	A	C	-	-	25	-1	-5	3.4	-	0.75	6	+0.97	1	1	4	z	1.9	1	260	4.5	1.2
'5461	P	•	A	C	-	-	25	-2	-9	2.7	-	1	7.5	+0.67	1	1.5	5	z	2	1	210	4.5	1.2
'5462	P	•	A	C	-	-	25	-4	-16	5.9	-	1.8	9	+4.15	1	2	6	z	2.5	3	30	4.5	1.2
MMBF4416	N	•	A	C	*C*	C	30	5	15	5.9	-	-	-6	-0.19	5	4.5	7.5	z	3.9	5	70	4	0.8
2N5484	N	-	A	C	-	C	25	1	5	3.3	-	-0.3	-3	-0.73	1	3	6	z	2.3	1	230	10	2.2
'5485	N	-	A	C	-	-	25	4	10	6.6	-	-0.5	-4	-1.65	1	3.5	7	z	2.1	1	150	10	2.2
'5486	N	-	A	C	-	-	25	8	20	14	-	-2	-6	-2.61	1	4	8	z	2.1	1	75	10	2.2
2SK170BL	N	-	B	-	-	-	40	6	12	6.1	-	-0.2	-1.5	-0.04	5	22[t]		z	29	5	470	30	6
LSK170B	N	-	B	C	-	-	40	6	12	7.6	-	-0.2	-2	-0.17	3	10[t]		1	20	3	160	20	5
LSK170C	N	-	B	C	-	-	40	10	20	13	-	-0.2	-2	-0.26	5	10[t]		1	24	5	90	20	5
BF861B	N	-	-	*C*	-	-	25	6	15	8	-	-0.5	-1.5	-0.47	1	16	25	z	16	5	150	7.5	
BF545C	N	-	-	*C*	-	-	30	12	25	19	-	-3.2	-7.8	-1.80	5	3.0	6.5	z	3.7	5	30	1.7	0.8
BF862	N	-	-	*C*	-	-	20	10	25	12	-	-0.3	-1.2	-0.21	5	35	45[t]	z	26	5	270	10	1.9
PF5103	N	-	A	C	-	-	40	10	40	19	30	-1.2	-2.7	-1.00	5	7.5		2	10	5	160	16	6
PN4391	N	•	A	C	C	C	40	50	150	115	30	-4	-10	-7.15	5	12[t]		5	8.8	5	30	12	3.5
'4392	N	•	A	C	C	C	40	25	75	38	60	-2	-5	-1.67	5	16[t]		10	10	5	130	12	3.5
'4393	N	•	A	C	C	C	40	5	30	16	100	-0.5	-3	-1.25	1	13[t]		10	6.2	1	150	12	3.5
J105	N	-	A	*C*	-	-	25	500	-	-	3	-4.5	-10	-8.39	5	40[t]		5	37	10	60	160[m]	35[m]
J106	N	-	A	*C*	-	-	25	200	-	-	6	-2	-6	-2.42	5	53[t]		5	43	10	230	160[m]	35[m]
J107	N	-	A	*C*	-	-	25	100	-	-	8	-0.5	-4.5	-1.93	5	75[t]		5	48	10	340	160[m]	35[m]
J108	N	•	A	C	C	-	25	80	-	325	8	-3	-10	-5.83	5	37[t]		5	31	10	60	85	15
J109	N	•	A	C	C	-	25	40	-	201	12	-2	-6	-2.85	5	26[t]		5	32	10	160	85	15
J110	N	•	A	C	C	-	25	10	-	122	18	-0.5	-4	-1.80	5	20[t]		5	34	10	220	85	15
J111	N	-	A	C	C	-	35	20	-	115	30	-3	-10	-7.6	5	-	-	-	8.4	5	30	28	5
J112	N	•	A	C	C	-	35	5	-	47	50	-1	-5	-2.8	5	6.7[t]		1	9.5	5	100	28	5
J113	N	-	A	C	C	-	35	2	-	21	100	-0.5	-3	-1.0	5	8[t]		1	11	5	100	28	5
J174	P	-	B	C	C	-	30	-20	-135	26	85	-5	-10	+2.08	5	4.5	-	5	-	-	15	13	6
J175	P	•	B	C	C	C	30	-7	-60	13	125	-3	-6	+1.58	1	-	-	-	-	-	30	13	6
J176	P	-	B	C	C	C	30	-2	-25	6.1	250	-1	-4	+0.86	1	6.3	-	5	-	-	40	13	6
J177	P	•	B	C	C	C	30	-1.5	-20	4.2	300	-0.8	-2.5	+0.62	1	-	-	-	-	-	50	13	6
J308	N	•	A	C	*C*	-	25	12	60	35	-	-1	-6.5	-	-	8	-	10	12	5	120	4	2
J309	N	•	A	C	*C*	C	25	12	30	23	-	-1	-4	-1.2	5	10	20	10	11	5	300	4	2
J310	N	-	A	C	*C*	C	25	24	60	39	-	-2	-6.5	-2.4	5	8	18	10	8.9	5	100	4	2
dual JFETs																							
LS840-42	N		F				60	0.5	5	3.3	-	-1	-4.5	-0.85	1	0.5	1	0.2	2.1	1	180	4	1.2
'843-5	N		F				60	1.5	15	-	-	-1	-3.5	-	-	1	1.5[t]	0.5	-	-	-	8[m]	3[m]
LSK389A	N		F, J				40	2.6	6.5	-	-	-0.15	-2	-	-	8	20[t]	3	-	-	-	25	5.5
'389B	N		F, J				40	6	12	12	-	-0.15	-2	-0.24	5	8	20[t]	3	23	5	170	25	5.5
'389C	N		F, J				40	10	20	-	-	-0.15	-2	-	-	8	20[t]	3	-	-	-	25	5.5
LS5912	N		F, J, K[p]				25	7	40	18	-	-1	-5	-1.75	5	4	10	5	5.7	5	70	5	1.2
IFN146	N		F[v]				40	-	30	6	-	-0.3	-1.2	-0.19	1	30	40[t]	z	25	5	660	75[m]	15[m]

(a) listed generally by increasing I_{DSS}, but also by part number within a family (e.g., J105–J113); see also Table 8.2 for noise parameters. (b) for families of related parts, **boldface** designates the family matriarch. (c) usually specified at I_D=1nA or 10nA, though sometimes at 10μA or even 200μA (e.g., for the J105–J113 "switches"); it doesn't much matter, given the wide range of specified $V_{GS(off)}$. (d) see the accompanying pinout figure; all JFETs appear to be symmetric (source and drain are interchangeable), but *italic* designates a datasheet pinout in which the S and D terminals are interchanged. (e) $G_{max}=g_m/g_{os}$, the maximum grounded-source voltage gain into a current source as drain load; listed values measured at I_D=1mA and V_{DS}=5V, unless noted otherwise. G_{max} is proportional to V_{DS}, and for most JFETs G_{max} is relatively constant over varying I_D. Use tabulated G_{max} to find $g_{os}=g_m/G_{max}$. (m) maximum. (p) several PDIP-8 pkgs available. (r) at I_D=30μA. (s) at I_D=300μA. (t) typical. (v) variant of "F" pinout: G and D terminals interchanged. (z) at I_{DSS}.

Table 3.8 Low-side MOSFET Gate Drivers[a]

Part #	Mfg[d]	# channels	V_{min} (V)	V_{max} (V)	I_{pk} (A)	Speed[b] $t_d + 0.5t_r$ (ns, typ)	Speed[b] C_{load} (nF)	logic thresh[p]	source below gnd?	current limit	UVLO?	enable?	inv?	non-inv?	output rail-to-rail?	Packages: TO220, Dpak	Packages: DIP	Packages: SOIC, MSOP	Packages: SOT23	Packages: smaller	Comments
TC4426-28	MC+	2	4.5	18	1.5	55	1	T	-	-	-	-	n	n	•	-	•	•	-	•	G,H
TC4423-25	MC+	2	4.5	18	3	70	1.8	T	-	-	-	-	n	n	•	-	•	•	-	•	G,H
TC4420,29	MC+	2	4.5	18	6	80	2.5	T	-	-	-	-	•	•	•	•	•	•	-	•	G
TC4421-22	MC+	1	4.5	18	9	85	10	T	-	-	-	-	•	•	•	•	•	•	-	•	G,J
FAN3111	F	1	4.5	18	1	20	0.5	C	-	-	-	c	c	c	•	-	-	-	•	-	
FAN3100C,T	F	1	4.5	18	2	20	1	C,T	-	-	•	c	c	c	•	-	-	-	•	•	A
FAN3180	F	1	5	18	2	30	1	T	-	-	•	-	-	•	•	-	-	-	•	-	B
FAN3216-17	F	2	4.5	18	2	25	2.2	T	-	-	•	-	•	•	•	-	-	•	-	-	D
FAN3226-29C,T	F	2	4.5	18	2	25	1	C,T	-	-	•	c	c	c	•	-	-	•	-	•	C,E
FAN3213-14	F	2	4.5	18	4	20	2.2	T	-	-	•	-	•	•	•	-	-	•	-	-	C
FAN3223-25C,T	F	2	4.5	18	4	25	2.2	C,T	-	-	•	c	c	c	•	-	-	-	•	•	E
FAN3121-22	F	1	4.5	18	9	21	10	T	-	-	•	•	•	•	•	-	-	•	-	•	
IRS44273L	IR	1	12	20	1.5	50	1	T	-	-	•	-	-	•	-	-	-	-	•	-	
IR25600	IR	2	6	20	1.5	75	1	T	-	-	-	-	-	•	-	-	•	•	-	-	
MAX17600-05	MA	2	4	14	4	15	1	C5,T	-	-	-	•	n	n	•	-	-	•	-	•	H
MAX5054-07	MA	2	4	15	4	38	5	C,T	-	-	-	c	c	c	•	-	-	•	-	•	
MAX5048A,B	MA	1	4	12.6	7.6[h]	18	1	C,T	-	-	•	c	c	c	•	-	-	-	•	•	
UCC37323-25[k]	TI	2	4.5	15	4	47	1.8	T	-	-	-	-	•	•	•	-	•	•	-	•	
UCC27517	TI	1	4.7	20	4	17	1.8	T	-	-	•	c	c	c	-	-	-	-	•	-	
UCC27516-19	TI	1	4.7	20	4	17,21	1.8	T,C	-	-	•	•	•	•	•	-	-	-	•	•	
UCC27523-26	TI	2	4.7	20	5	17	1.8	T	-	-	•	•	•	•	-	-	•	•	-	•	E,H
UCC37321-22[k]	TI	1	4	15	9	50	10	T	-	-	-	•	•	•	•	-	•	•	-	•	
MIC44F18-20	MI	1	4.5	13.2	6	24	1	T	-	-	•	•	•	•	•	-	-	•	-	•	
ADP3623-25	A	2	4.5	18	4	28	2.2	T	-	-	•	-	•	•	•	-	-	•	-	-	H,P
LM5110	TI	2	3.5	14	5[f]	38	2	T	•	-	•	-	n	n	•	-	-	•	-	•	H,L
LM5112	TI	2	3.5	14	7[g]	38	2	T	•	-	•	-	n	n	-	-	-	•	-	•	H,L,M
LM5114	TI	1	4	12.6	7.6[h]	16	1	C	-	-	•	c	c	c	•	-	-	-	•	•	
ISL89367	IN	2	4.5	16	6	45	10	F	•	-	•	n	o	o	•	-	-	•	-	•	N
ISL89160-62	IN	2	4.5	16	6	45	10	C5,T	-	-	•	-	•	•	•	-	-	•	-	•	O
MC34151	O	2	6.5	18	1.5	50	1	T	-	-	•	-	-	•	-	-	•	•	-	-	
IR2121	IR	1	12	18	2[e]	200	3.3	T	-	•	-	-	-	•	-	-	•	-	-	-	F
UC3708	TI	2	5	35	3	37	1	T	-	-	-	•	-	•	•	-	•	-	-	-	
IXDD602	IX	1	4.5	35	2	50	1	C5	-	-	-	•	•	•	•	-	•	•	-	•	H,R
IXDD604	IX	1	4.5	35	4	40	1	C5	-	-	-	•	•	•	•	-	•	•	-	•	H,R
IXDD609	IX	1	4.5	35	9	60	10	C5	-	-	-	•	•	•	•	•	•	•	-	•	R
IXDD614	IX	1	4.5	35	14	70	15	C5	-	-	-	•	•	•	•	•	•	•	-	-	R
IXDD630	IX	1	10	35	30	65	5.6	C5	-	-	•	•	•	•	•	•	-	-	-	-	K,R
ZXGD3002-04	D	1	-	20,40	9,5	11	1	-	-	-	-	-	-	•	-	-	-	-	•	-	M,S

Notes: (a) sorted by family, within family sorted by Iout; except for ZXGD3000-series, all devices swing rail-to-rail, or nearly so. (b) into Cload at V_S=12V. (c) input gate with inv and non-inv inputs. (d) A=Analog Devices; D=Diodes,Inc; F=Fairchild; IN=Intersil; IR=International Rectifier; IX=Ixys/Clare; L=LTC; MA=Maxim; MC=Microchip; MI=Micrel; O=OnSemiconductor; S=STMicroelectronics; TI=Texas Instruments. (e) 1A source, 2A sink. (f) 3A source, 5A sink. (g) 3A source, 7A sink. (h) 1.3A source, 7.6A sink. (k) 37xxx for 0 to 70°C, 27xxx for -40°C to 105°C. (n) see part-specific comments. (o) XOR input sets optional invert. (p) C=CMOS; C5=5V CMOS; F=flexible, set by V_{ref-} and V_{ref+} input pins; T=TTL.

Comments: **(A)** suffix specifies logic threshold. **(B)** includes 3.3V LDO output. **(C)** 2ns td channel match. **(D)** 1ns td channel match. **(E)** dual inv+en, dual non-inv+en, dual inputs. **(F)** source-resistor current-sense input terminal, suitable for driving an IGBT. **(G)** industry std, many mfgrs. **(H)** dual inv, dual non-inv, or one each. **(J)** for 8-pin pkgs, *n*- and *p*-ch drains on separate pins. **(K)** t_r, t_f = 50ns into 68nF. **(L)** output swing to neg rail, can be 5V below logic gnd. **(M)** *n*- and *p*-ch drains on separate pins. **(N)** resistor-programmed edge-delay timers; 2-input AND signal inputs. **(O)** ISL89163-65 same, but include enable inputs; ISL89166-68 same, but include resistor-programmed edge-delay timer inputs. **(P)** overtemp protection and output. **(R)** full p/n is IXDx6…, where x = N, I, D and F for non-inv, inv, dual non-inv+en, or one of each. **(S)** series is one each high-current high-gain *npn* and *pnp* transistor emitter-followers for pullup and down.

Review of Chapter 3

An A-to-Z summary of what we have learned in Chapter 3. This summary reviews basic principles and facts in Chapter 3, but it does not cover application circuit diagrams and practical engineering advice presented there.

¶A. FETs.
In Chapter 3 we explored the world of Field-Effect Transistors, or FETs. FETs have a conducting channel with terminals named *Drain* and *Source*. Conduction in the channel is controlled by an electric field created by a third *Gate* electrode (§3.1). As with bipolar transistors (BJTs), FETs are transconductance devices (see ¶G below), which means the drain *current* (assuming sufficient drain-to-source voltage) is controlled by the gate *voltage*.

¶B. *n*-channel and *p*-channel.
Like BJTs with their *npn* and *pnp* types, FETs come in both *n*- and *p*-channel polarities (§3.1.2). In either case the channel conductance increases if the gate voltage is taken toward the drain voltage. For example, for an *n*-channel FET with a positive drain voltage, the channel can be turned on with a sufficient positive-going voltage, and cutoff with a sufficient negative-going voltage. That's not to say the *n*-channel device requires positive and negative voltages to turn on and off. A threshold voltage V_{th} can be defined where the FET is just slightly turned on, and the channel responds to gate voltages above and below V_{th} for control.

¶C. Enhancement and Depletion Modes.
See Figure 3.8. Enhancement-mode devices have a high enough threshold voltage V_{th} that they are nonconducting (i.e., off) when their gate voltage is at $V_{GS}=0$ V. To bring such a FET into conduction, the gate of is brought positive (if *n*-channel) or negative (if *p*-channel). Depletion-mode devices, by contrast, have their threshold voltage well into the "off" direction, thus they are conducting (i.e., on) with their gate-voltage at $V_{GS}=0$ V. Thus for example you must apply a considerable negative gate voltage V_{GS} to turn off an *n*-channel depletion-mode FET. See Figure 3.9 where drain current versus gate voltage is shown for a selection of *n*-channel devices. FETs can be fabricated with the transfer curve shifted left or right (more about this in ¶H below). Figures 3.10 and 3.11 show convenient maps of the FET types.

¶D. MOSFETs and JFETs.
In *metal-oxide* FETs (MOSFETs) the gate electrode is fully insulated from the channel, and can be taken positive or negative, typically up to ±20 V. In junction FETs (JFETs) the semiconductor gate contacts the channel and acts as a diode junction, so it is insulated only in the reverse direction. Therefore JFETs are necessarily depletion mode devices; one cannot make an enhancement-mode JFET. Figures 3.6 and 3.7 show FET symbols.

¶E. FET Characteristics, Gate and Drain.
See Figure 3.13. A FET's channel conductance and current is controlled primarily by its gate voltage, but it's also affected by the drain voltage V_{DS}. At very low drain voltages the channel acts like a resistor, whose value is controlled by the gate (§3.1.2 and §3.2.7); this is called the *linear* region. At higher drain voltages the drain current levels off, being controlled by the gate voltage and only weakly dependent upon drain voltage; this is called the *saturated* region. In the saturated region the FET drain acts like a current source (or sink), and the device is characterized by its transconductance g_m (see ¶G below). MOSFETs are often used as switches. In this mode of operation a large gate voltage (e.g., 10 V) is applied to make the channel resistance low enough to approximate a closed switch. More on FET switches in sections ¶¶O–Q below.

¶F. Square-law.
Over a large region of gate voltages greater than V_{th}, and for drain voltages above a volt or so (i.e., in the saturated region), a FET's drain current behaves like a square-law device; that is, its drain current is proportional to the square of the excess gate-drive voltage $(V_{GS}-V_{th})^2$, see Figure 3.14 and eq'n 3.2. This is sometimes called the *quadratic* region. The threshold voltage V_{th} is generally determined with an extrapolated $\sqrt{I_D}$ plot, as the figure shows. For V_{GS} below threshold the FET is in the subthreshold region; see ¶I below.

¶G. Transconductance and Amplifiers.
Transconductance g_m is the change in output drain current caused by a change in gate voltage: $g_m=i_D/v_{GS}$ (the lowercase i and v signify small signals). Common-source FET amplifiers (§3.2.3, Figures 3.28 and 3.29) have voltage gain $G=-g_mR_D$, where R_D is the drain load resistance. In contrast to BJTs (where $g_m \propto I_C$), the transconductance of FETs rises only as $\sqrt{I_D}$ in the important quadratic region; see Figures 3.53 and 3.54. As a consequence FET amplifiers with resistive drain loads have lower gain when designed to operate at higher current, because R_D is generally chosen inversely proportional to drain current. The FET's internal output resistance also acts as a load resistance, thus

limiting gain ("G_{max}") even with an ideal current-source drain load; see §3.3.2 eq'n 3.13, and Table 3.1.

When used as a *follower*, an FET has an output impedance $r_{out}=1/g_m$, see ¶K below.

¶H. Biasing JFET Amplifiers.
JFETs are well suited for making signal amplifiers (by contrast there are few viable small discrete *MOS*FETs), and they work especially well in low-noise amplifiers. But there's one very painful issue analog designers face: the uncertain value of the gate operating voltage for any given part. Scanning the min and max columns for $V_{GS(off)}$ in the JFET Table 3.1 on page 141, we see values for a particular JFET that range from -1 V to -7 V, or -0.4 V to -4 V. The latter is a 10:1 ratio! Figure 3.17 shows V_{GS} histograms for 300 parts, 100 each for three different JFET types in a family. Here we see gate voltage spreads of about 1 V, which you might rely upon if you buy a batch of parts from one manufacturer and measure them. But, *caution*: Figures 3.51 and 3.52 show how the same part type may vary when purchased from different manufacturers. To deal with the uncertainty, special biasing schemes are often required in FET amplifier circuits. Figures 3.25 and 3.41 show examples of the load-line concept for analyzing amplifier biasing.

¶I. Subthreshold Region.
The simple FET formula of eq'n 3.2 predicts zero drain current when the gate voltage reaches threshold ($V_{GS}=V_{th}$). In reality the drain current is not zero, and transitions smoothly to a subthreshold region (see Figure 3.16) where the FET looks more like a BJT, with its exponential Ebers-Moll characteristic (§2.3.1). In this region (where I_D rises exponentially with V_{GS}) we're glad to see a higher $g_m \propto I_D$; but sadly the FET proportionality constant is usually $2\times$ to $5\times$ smaller than for BJTs, see Figure 3.53.

¶J. Self-biased Amplifiers.
Depletion-mode MOSFETs (and all JFETs) operate with a reverse voltage on their gates, which allows them to be self-biased (§3.2.6A). The source terminal is "higher" than the gate terminal, so a source resistor connected between them sets the drain current to $I_D=V_{GS}/R$. This is also a convenient way to make a 2-terminal current source, but the tolerance will be poor due to the wide variability in V_{GS}, see ¶H. Alternately the V_{GS} voltage available at the source pin may be used to operate a current-setting IC like the LM334.

¶K. Source Followers.
Source followers (§3.2.6), Figure 3.40 have a nominal gain of 1, analogous to the BJT emitter follower. Because of their lower g_m, however, they have considerably higher output resistance, $r_{out}=1/g_m$, so the ideal unity gain is reduced by load resistance, see eq'n 3.7.

¶L. FETs as Variable Resistors.
At low drain voltages ($V_{DS} \ll V_{GS}$) FETs act as variable resistors programmed by the gate voltage. Because the slope varies with V_{DS}, however, the resistance is somewhat nonlinear. But there's a simple trick to linearize this resistance, by exploiting the quadratic behavior of FETs, see Figures 3.46 and 3.47.

¶M. FET Gate Current.
The gate of a JFET forms a diode junction with the channel; it's normally reverse-biased, with some non-zero dc leakage current (§3.2.8). This current roughly doubles for every 10°C temperature increase; furthermore it increases dramatically at high drain currents and drain voltages due to impact ionization, see Figure 3.49. MOSFET gates do not suffer from either of these leakage-current-increasing effects. In contrast to the generally negligible dc gate leakage, the input capacitance C_{iss} of FETs (which can be quite high, many hundreds of pF for large power MOSFETs) often presents a substantial ac load. Use a gate-driver chip (Table 3.8) to provide the high transient currents needed for rapid switching.

¶N. JFET Switches.
JFETs can be used as analog-signal switches, as in the *n*-channel switch of Figure 3.62. The switch is OFF when the gate is taken at least V_{th} below than the most negative input signal. To turn the switch ON the gate voltage must be allowed to equal the source. JFETs are symmetrical, so e.g., for an n-channel part, the "source" would be the most negative pin. Large-die JFETs work well as power switches up to 100 mA; Table 3.1 lists parts with R_{ON} as low as 3 Ω.

¶O. CMOS Switches.
CMOS signal switches are made with a parallel pair of complementary *n*- and *p*-channel MOSFETs. This reduces R_{ON} as shown in Figure 3.61, and beneficially causes cancellation of most of the injected charge transfer (§3.4.2E), see Figure 3.79. The injected charge scales roughly inversely proportional to R_{ON} (Figure 3.81), so there's a tradeoff between desirably low on-resistance and desirably low self-capacitance. As an example, Table 3.3 lists a switch with an impressive $R_{ON}=0.3\,\Omega$ – but it's burdened

with a whopping 300 pF of self-capacitance. A T-switch configuration can be used to reduce the signal feedthrough at high-frequencies, see Figure 3.77.

¶P. CMOS Logic Gates.
See Figure 3.90. A series pair of complementary (*n*- and *p*-channel) small-geometry MOSFETs between the positive rail and ground forms the simplest logic inverter (Figure 3.90); more switches can be arranged to make CMOS logic gates (e.g., Figure 3.91, §3.4.4), with the attractive property of nearly zero static power, except when switching. CMOS logic is covered extensively in Chapters 10 and 12, and is the basis for all contemporary digital processors.

¶Q. MOSFET Power Switches.
Most power MOSFETs (§3.5) are enhancement type, available in both *n*- and *p*-channel polarities. They are very popular for use as high-current high-voltage power switches. A few relevant parameters are the breakdown voltage V_{DSS} (ranging from 20 V to 1.5 kV for *n*-channel, and to 500 V for *p*-channel); the channel on-resistance $R_{DS(on)}$ (as low as 2 mΩ); the power-handling ability (as high as 1000 W with the case held unrealistically at 25°C); and the gate capacitance C_{iss} (as high as 10,000 pF), which must be charged and discharged during MOSFET switching, see ¶S below. Table 3.4a lists representative small-package *n*-channel parts rated to +250 V and *p*-channel parts of all sizes to −100 V; Table 3.4b extends the *n*-channel selection to higher voltage and current; more complete tables are found in Chapter *3x*.

¶R. Maximum Current.
MOSFET datasheets list a maximum continuous rated current, specified however at an unrealistic 25°C case temperature. This is calculated from $I^2_{D(max)}R_{DS(ON)}=P_{max}$, substituting a maximum power $P_{max}R_{\Theta JC}=\Delta T_{JC}=150°C$ (see §9.4), where they have assumed $T_{J(max)}=175°C$ (thus a 150°C ΔT_{JC}), and they use the value of $R_{DS(ON)}$ (max) at 175°C from an R_{DS} tempco plot (e.g., see Figure 3.116). That is, $I_{D(max)}=\sqrt{\Delta T_{JC}/R_{\Theta JC}R_{ON}}$. Some datasheets show the calculation for a more realistic 75°C or 100°C case temperature. Even so, you don't really want to run your MOSFET junction at 175°C, so we recommend using a lower maximum continuous I_D and corresponding P_{diss}.

¶S. Gate Charge.
The capacitances in power MOSFETs that slow down switching are most easily analyzed with gate-charge plots, like Figure 3.101. First consider turn-ON: as current flows into the gate capacitance $C_{iss}+C_{rss}$ (dominated by C_{iss}) the gate voltage rises. There is a switching delay, because the FET's drain remains off until the gate voltage is high enough for the FET to sink the drain current. Then the drain voltage starts to fall, as seen in Figures 3.102 and 3.103. The falling drain creates a reverse gate current $I=C_{rss}\,dV_D/dt$ that prevents further increase in the gate voltage. Put another way, the falling slew-rate $dV_D/dt=I_G/C_{rss}$ is set by the gate current available to charge the feedback (Miller) capacitance C_{rss}. When V_{DS} reaches zero the gate resumes charging, now at a slower rate because the C_{rss} contribution to total gate capacitance is larger at $V_{DS}=0$, see Figure 3.100). The MOSFET does not reach its intended low value of $R_{DS(ON)}$ until the gate attains its full drive voltage. Turn-off proceeds similarly. MOSFET datasheets include values for C_{iss} and C_{rss}, but the latter is typically at $V_{DS}=25$ V, so you need to go to the datasheet plots of capacitances versus drain voltage.

¶T. MOSFET Gate Damage.
MOSFET gates typically have ± 20 V to ± 30 V maximum ratings, beyond which the very thin metal-oxide gate-channel insulator can be permanently damaged, see Figure 3.105. Be sure to discharge static charge before installation of discrete MOSFETs and MOS ICs.

¶U. FET versus BJT for Power Switching.
See §3.5.4H; see also ¶Z below.

¶V. MOSFET Switch Polarity.
Both *n*- and *p*-channel polarities of MOSFETs can be used to switch a voltage, see Figure 3.106 where most of the circuits show a conventional approach with a *p*-channel FET switching a positive voltage. But circuit E shows an *n*-channel FET doing the same task, with an additional voltage source powering the gate (the better-performing *n*-channel FET is preferred if it can be easily used, see §3.1.2). Figure 3.107 illustrates the use of photodiodes to power the high-side gates, to make "floating" switches.

¶W. Power MOSFET Amplifiers.
Unlike bipolar power transistors, power MOSFETs have a wide safe-operating area (SOA) and do not suffer from second breakdown (see Figure 3.95), which is due to a localized thermal-runaway heating problem. Figure 3.119 shows typical class-AB biasing techniques necessary for use in linear power amplifiers.

¶X. Depletion-mode Power MOSFETs.
Although most power MOSFETs are enhancement-mode types, *n*-channel depletion-mode types are available; §3.5.6D shows some applications. See also Table 3.6 on page 210.

¶Y. Paralleling Power MOSFETs.
When used as switches, yes, but when used in power amplifiers, no, at least not without high-value source-ballast resistors! Figure 3.117B shows an elegant active-feedback workaround for use with regulator pass elements.

¶Z. IGBTs.
IGBTs are an alternative to power MOSFETs, see §3.5.7 where we show a comparison between power MOSFETs, IGBTs and BJTs. They're primarily useful at voltages above 300 V and switching rates below 100 kHz, though there are some nice IGBTs for use at RF, for example the IRGB4045, good for 150 W or more at 20 MHz.